

Disruptive IT Impacts Companies, Competition, and Careers

CHAPTER OUTLINE

LEARNING OBJECTIVES

Case 1.1 Opening Case: Uber, Airbnb, and the On-Demand Economy

1.1 Doing Business in the On-Demand Economy

1.1 Describe how the on-demand economy is changing the way that business is conducted.

1.2 Business Process Improvement and Competitive Advantage

1.2 Explain the role of IT in business process improvement. Understand the concepts of business process reengineering and competitive advantage.

1.3 IT Innovation and Disruption

1.3 Describe innovating technologies and explain how they are disrupting enterprises.

1.4 IT and You

1.4 Understand the value of being an “informed user” of IT and the ways in which IT can add value to your career path and performance in the on-demand economy.

Case 1.2 Business Case: The Internet of Things Comes to the NFL

Case 1.3 Video Case: What Is the Value of Knowing More and Doing More?

Introduction

The more digital technology advances, the more it is almost instantly integrated into our daily lives. Many managers and entrepreneurs recognize the need to integrate digital technology into their products and services. For example, it has been estimated that 78% of business

leaders expect their organizations to be a digital business by 2020. Outdated and complex application architectures with a mix of interfaces can delay or prevent the release of new products and services, and maintaining these obsolete systems absorbs large portions of the information technology (IT) budget.

Companies such as Uber, Airbnb, Shyp, TaskRabbit, and other participants in the on-demand economy are leveraging IT to create exciting new business models and revolutionize the way workers, businesses, and customers interact and compete. Peter Hinssen, a well-known business author, university lecturer, and digital consultant, described the change in digital technology as follows:

Technology used to be nice. It used to be about making things a little bit better, a little bit more efficient. But, technology stopped being nice: it's disruptive. It's changing our business models, our consumer markets, our organizations. (MacIver, 2015)

As businesses continue to join the on-demand economy, IT professionals must constantly scan for innovative new technologies to provide business value and help shape the future of the business. For example, smart devices, mobile apps, sensors, and technology platforms—along with increased customer demand for digital interactions and on-demand services—have moved commerce in fresh new directions. We've all heard the phrase “there's an app for that” and that kind of consumer thinking is what drives the on-demand economy.

Business leaders today need to know what steps to take to get the most out of mobile, social, cloud, big data, analytics, visualization technologies, and the Internet of Things (IoT) to move their business forward and enable new on-demand business models. Faced with opportunities and challenges, managers need to know how to leverage IT earlier and more efficiently than their competitors.

A goal of this book is to empower you to improve your use and management of IT at work by raising your understanding of IT terminology, practices, and tools and developing your IT skills to transform you into an informed IT user. Throughout this book, you will learn how digital technology is transforming business and society in the on-demand economy as the IT function takes on key strategic and operational roles that determine an enterprise's success or failure. You will also be provided with an in-depth look at IT trends that have immediate and future capacity to influence products, services competition, and business relationships. Along the way, we'll describe many different ways in which IT is being used and can be used in business and provide you with some of the terminology, techniques and tools that enable organizations to leverage IT to improve growth, performance, and sustainability.

In this opening chapter, you will learn about the powerful impacts of digital technology on people, business, government, entertainment, and society that are occurring in today's on-demand economy. You will also discover how leading companies are deploying digital technology and changing their business models, business processes, customer experiences, and ways of working. We will present examples of innovative products, services, and distribution channels to help you understand the digital revolution that is currently shaping the future of business, the economy and society and changing management careers. And, we'll explain why IT is important to you and how becoming an “informed user” of IT will add significant value to your career and overall quality of life.

Case 1.1 Opening Case



NICOLAS MAETERLINCK / Stringer / Getty Images

THE ON-DEMAND BUSINESS FRAMEWORK



Uber and Airbnb Revolutionize Business Models in the On-Demand Economy

If you've used Uber or Airbnb, then you have participated in the on-demand economy where speed, convenience, and simplicity are key factors in consumer behavior and purchasing decisions. Michael Boland, author of *What's Driving the Local On-Demand Economy*, explains that as consumers, "We're being conditioned to expect everything on-demand as the mobile device increasingly becomes the *remote control for the physical world*" (Boland, 2015). For example, the majority of consumers who tap an Uber app to get a ride would not consider dialing an 800 number for a taxi. With all transactions performed by apps and automated processes, the entire process from hailing to paying for a ride is slick, quick, and easy, without cash or credit cards.

Tech Platforms Enabled On-Demand Services to Take Off

Decades of technological innovation have given us smartphone apps, mobile payment platforms, GPS and map technology, and social authentication. These technologies are being used to build the infrastructure needed for **on-demand services**. This infrastructure—also referred to as a **technology platform** or **technology stack**—supports the exchange and coordination of staggering amounts of data. The term technology stack reflects the fact that the platform is made up of multiple layers (stacks) of hardware, software, network connectivity, and **data analytics** capabilities.

In many consumer markets today, companies that do not have iPhone or Android apps or technology platforms that support the exchange of goods and services—no matter how useful their website—may find themselves losing their competitive edge.

On-Demand Economy Requires a New Business Model

Uber and Airbnb are popular examples of companies that developed on-demand business models to transform slow-to-innovate industries. A simple definition of **business model** is the way a company generates revenue and makes a profit. On-demand business models provide real-time fulfillment of goods and services, which have attracted millions of users worldwide. This model fits best when speed and convenience matter the most. The ground transportation, grocery, and restaurant industries are examples of hyper-growth

categories in the on-demand world. Forward-thinking companies are reshaping these industries.

Uber Business Model

Uber disrupted the taxi industry with a workforce that is essentially any person with a smartphone and a car. Location-aware smartphone apps bring drivers and passengers together, while in-app accounts make the cashless payment process effortless. By simply opening the Uber app and pressing the middle button for several seconds (a long press), customers can order a ride to their current location, selecting the kind of car they want. Payment is automatically charged to the credit card on file with receipts via email.

The Uber concept developed in response to taxi scarcities. It started on a snowy Paris night in 2008 when the two founders could not get a cab. They wanted a dead-simple app that could get them a car with a tap. On June 1, 2015, the entrepreneurs celebrated Uber fifth anniversary and announced that the company had grown into a transportation network covering 311 cities in 58 countries in North and South Americas, Europe, Africa, Asia Pacific, and the Middle East.

Uber has invested in new and developing technologies and partnerships. The company partnered with Carnegie Mellon University to build robotic cars and new mapping software. In March 2015, Uber purchased deCarta, a 40-person mapping start-up to reduce its dependence on Google maps.

Airbnb Business Model

Another disruption to a traditional industry occurred when Airbnb blindsided the hotel industry. Airbnb allows anyone with a spare apartment or room—even if only for a day—to run their own bed and breakfast by giving them a technology platform to market themselves to a global market. By 2016, the Airbnb site had over 1.5 million listings in 190 countries and 34,000 cities. Over 40 million guests have used Airbnb worldwide. For comparison, Hilton, InterContinental, and Marriott, the largest hotel chains in the world, have less than 1 million rooms each.

Uber and Airbnb do not own inventory. Instead, they scale up (expand) by improving their ability to acquire and match customers and service providers.

Business Success in Terms of Company Growth and Valuation

The ride-hailing app Uber and the housing rental app Airbnb are two of the most valuable start-ups, as displayed in **Figure 1.1**. Valuation of a company at its early stages is based heavily on its growth potential and future value. In contrast, the valuation of an established company is based on its present value, which is calculated using traditional financial ratios and techniques related to revenues or other assets.

Uber's massive market value—estimated at \$60 billion—is greater than 80% of all Standard & Poor (S&P) 500 companies, many of which have been around for 25, 50, or 100 years. Investors valued Airbnb at \$24 billion—higher than the value of the hotel giant Marriott

International. These companies would never have been able to grow in the old way as a traditional organization, with their own inventory of products, services, and workforce and traditional forms of technology.

Questions

1. In what ways are the Uber and Airbnb business similar or different?
2. How did Uber achieve its new business model?
3. To what extent do you think changing their business models contributed to the success of Uber and Airbnb?

Sources: Compiled from Primack (2015), Storbaek (2015), Winkler and MacMillan (2015), Jaconi (2014), *Uber.com* (2017), *Airbnb.com* (2017).

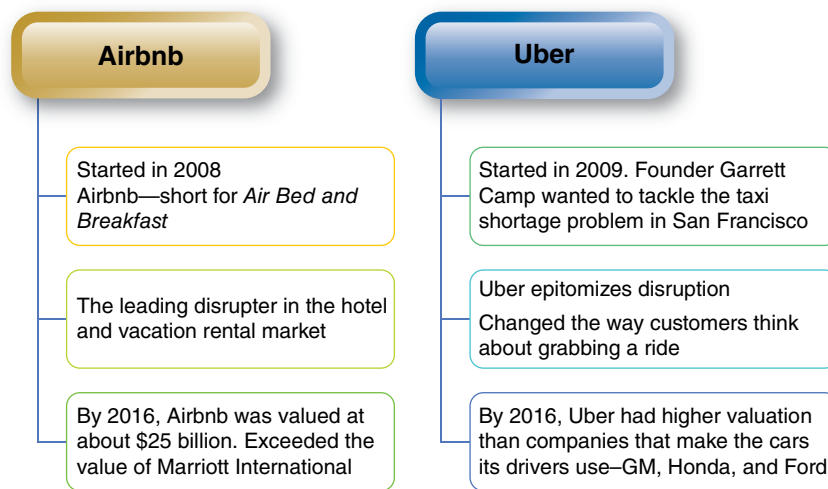


FIGURE 1.1 On-demand business models of Airbnb and Uber have been extremely successful.

1.1 Doing Business in the On-Demand Economy

The on-demand economy is revolutionizing commercial activities in businesses around the world. The businesses in this new economy are fueled by years of technology innovation and a radical change in consumer behavior. As companies become more highly digitized, it becomes more and more apparent that what companies can do depends on what their IT and data management systems can do. For over a decade, powerful new digital approaches to doing business have emerged. And there is sufficient proof to expect even more rapid and dramatic changes due to IT breakthroughs and advances.

In market segment after market segment, mobile communications and technology stacks make it financially feasible for companies to bring together consumers and providers of products and services. These capabilities have created the **on-demand economy**. As Ev Williams, cofounder of Twitter says,

The internet makes human desires more easily attainable. In other words, it offers convenience. Convenience on the internet is basically achieved by two things: speed, and cognitive ease. If you study what the really big things on the internet are, you realize they are masters at making things fast and not making people think.

On-demand economy is the economic activity created by technology companies that fulfill consumer demand through the immediate provisioning of products and services.

The proliferation of smartphone-connected consumers, simple and secure purchase flows, and location-based services are a few of the market conditions and technological innovations that are propelling the explosion of on-demand services.

Just as the rapid growth of online-only Amazon and eBay transformed retail, the even faster growth of app-driven companies, like Uber, Airbnb, and Grubhub, has disrupted the taxi, hotel, and restaurant markets. As you read in the opening case, in six short years, Uber changed the taxi industry as it rose from start-up to the world's most valuable private technology company, and Airbnb tackled the fiercely competitive hotel market and attracted more than 60 million customers to become the third most valuable venture-capital-backed company in the world. Another example is Grubhub who became No. 1 in online food ordering, controlling over 20% of that \$9 billion market. What today's successful technology businesses have in common are platform-based business models. Platforms consist of hardware, software, and networks that provide the connectivity for diverse transactions, such as ordering, tracking, user authentication, and payments. These business models are designed to serve today's on-demand economy, which is all about time (on-demand), convenience (tap an app), and personalized service (my way). For example, millennials want the ease of online payment over cash and insist on efficiency for all aspects of their lives, including shopping, delivery, and travel.

Key strategic and tactical questions that determine an organization's profitability and management performance are shown in **Figure 1.2**. Answers to each question require an understanding of the capabilities of mundane to complex IT, which ones to implement and how to manage them.

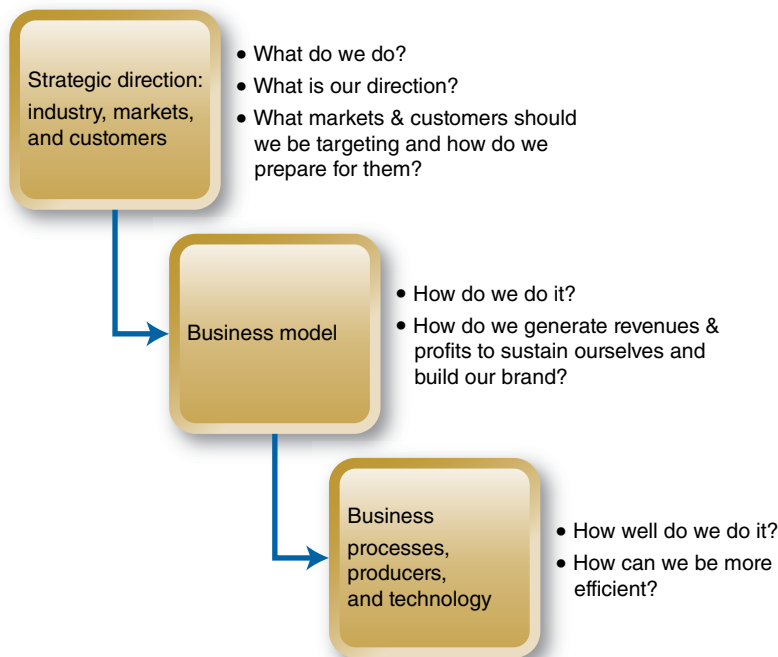


FIGURE 1.2 Key strategic and tactical questions.

Growth of the On-Demand Economy

Whether it is ease of scheduled deliveries or the corresponding time savings, the growth of the on-demand economy is a product of its alignment with consumers' growing appetite for greater convenience, speed, and simplicity. A recent survey reported that 86.5 million Americans have used the services of at least one on-demand start-up company (Chriss, 2016).

The growth of the on-demand economy demonstrates the high level of interest consumers have in on-demand services from dog walking to laundry services, short-term home rentals, massages, and truck hauling. Although just applying a mobile app to an existing service will not ensure a company's success, IT is a vital and integral part of the all businesses that are part of the on-demand economy.

Low Cost of Entry One of the reasons that the on-demand economy has taken off is that it is easier than ever to become an on-demand business. Companies like Dispatch, a software as-a-service company, allow entrepreneurs to move into the on-demand world quickly and inexpensively. For example, Aatlantic Fitness, a fitness equipment repair service company, moved into the on-demand economy using Dispatch, and Handyman Connection, a 20-year-old home repair service company, is using Dispatch's platform to compete with Handy, an on-demand service for house cleaning that has raised \$60 million in venture capital.

Digital Business Models

The on-demand economy is driving the transformation of traditional business models to digital business models to serve customers what they want and where they want it.

Business models are the ways enterprises generate revenue or sustain themselves. Digital business models define how businesses make money via digital technology. Companies that adopt digital business models are better positioned to take advantage of business opportunities and survive, according to the Accenture *Technology Vision* 2013 report (Accenture, 2013). **Figure 1.3** contains examples of new technologies that destroyed old business models and created new ones.

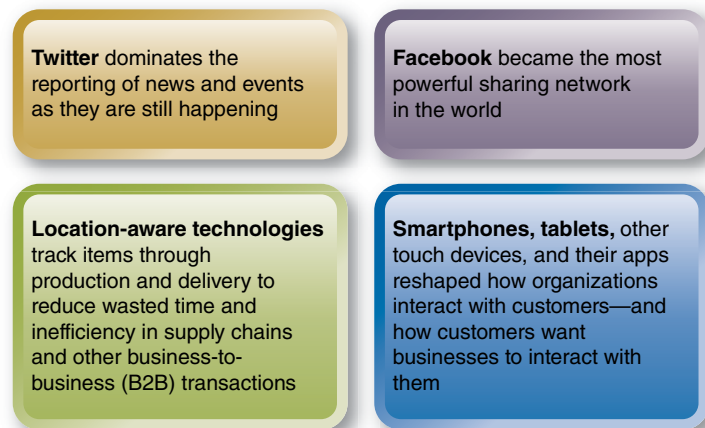


FIGURE 1.3 Digital business models refer to how companies engage their customers digitally to create value via websites, social channels, and mobile devices.

The ways in which market leaders are transitioning to digital business models include the following:

- **NBA talent scouts rely on sports analytics and advanced scouting systems** NBA talent scouts used to crunch players' stats, watch live player performances, and review hours of tapes to create player profiles. Now software that tracks players' performance has changed how basketball and soccer players are evaluated. For example, STATS' SportVU technology is revolutionizing the way sports contests are viewed, understood, played, and enjoyed. SportVU uses six palm-sized digital cameras that track the movement of every player on the court, record ball movement 25 times per second, and convert movements into statistics. SportVU produces real-time and highly complex statistics to complement the traditional play-by-play. Predictive sport analytics can provide a 360-degree view of a player's performance and help teams make trading decisions. Sports analytics bring about small competitive advantages that can shift games and even playoff series.
- **Dashboards keep casino floor staff informed of player demand** Competition in the gaming industry is fierce, particularly during bad economic conditions. The use of manual spreadsheets and gut-feeling decisions did not lead to optimal results. Casino operators facing pressure to increase their bottom line have invested in analytic tools, such as

Tangam's Yield Management solution (TYM). TYM is used to increase the yield (profitability) of blackjack, craps, and other table games. The analysis and insights from real-time apps are used to improve the gaming experience and comfort of players.

Today, a top concern of well-established corporations, global financial institutions, born-on-the-Web retailers, and government agencies is how to design their **digital business models** in order to

- Deliver an incredible customer experience
- Turn a profit
- Increase market share
- Engage their employees

In the digital (online) space, the **customer experience** must measure up to the very best the Web has to offer. Stakes are high for those who get it right—or get it wrong. Forrester research repeatedly confirms there is a strong relationship between the quality of a firm's customer experience and loyalty, which, in turn, increases revenue (Schmidt-Subramanian et al., 2013).

IT's Role in the On-Demand Economy

According to the 2016 survey conducted by the Society of Information Management (SIM), 1,213 IT leaders (including 490 chief information officers (CIOs)) from 801 companies reported companies that are more highly digitized and tightly connected are putting a greater emphasis on the strategic use of IT to enhance growth and improve performance. As a result, IT priorities and spending are changing (Kappelman et al., 2017).

A review of the top 10 IT management priorities reported in the survey results is shown in **Table 1.1**. Along with business-IT alignment and security, Table 1.1 clearly demonstrates a need for companies to focus on strategic and organizational priorities such as innovation, IT and business agility, speed of IT delivery, and business **productivity** and efficiency.

TABLE 1.1 10 Top IT Management Priorities

IT Management Issues	
1	Technology Alignment with the Business
2	Security, Cybersecurity & Privacy
3	Innovation
4	IT Agility & Flexibility
5	Business Agility & Flexibility
6	Business Cost Reduction & Controls
7	IT Cost Reduction & Controls
8	Speed of IT Delivery & IT Time to Market
9	Business Strategic Planning
10	Business Productivity & Efficiency

Adapted from Kappelman et al. (2017).

To address these issues, IT leaders said they need to focus on relationships, meet more frequently with top management, and spend significant amounts of time with functional leaders, customers, and suppliers. Companies also need to emphasize finding, keeping, and developing IT talent and on improving IT to improve business performance. These findings point to one clear message—IT in the on-demand economy is about meeting customer needs.

IT Business Objectives

Now, more than ever, IT must be responsive to the needs of consumers who are demanding a radical overhaul of business processes in companies across diverse industry sectors. Intuitive interfaces, around-the-clock availability, real-time fulfillment, personalized treatment, global consistency, and zero errors—this is the world to which customers have become increasingly accustomed. And, it's not just about providing a superior user or customer experience—when companies get it right, they can also offer more competitive prices because of lower costs, better operational controls, and open themselves up to less risk.

According to Chirantan Basu of Chron (Basu, 2017), to stay abreast of the ever-changing business landscape and customer needs, IT today must concentrate on the following six business objectives:

- 1. Product development** From innovations in microprocessors to efficient drug-delivery systems, IT helps businesses respond quickly to changing customer demands.
- 2. Stakeholder integration** Companies use their investor relations websites to communicate with shareholders, research analysts, and others in the market.
- 3. Process improvement** An ERP system replaces dozens of legacy systems for finance, human resources, and other functional areas, to increase efficiency and cost-effectiveness of internal business processes.
- 4. Cost efficiencies** IT allows companies to reduce transaction and implementation costs, such as costs of duplication and postage of email versus snail mail.
- 5. Competitive advantage** Companies can use agile development, prototyping, and other systems methodologies to bring a product to market cost-effectively and quickly.
- 6. Globalization** Companies can outsource most of their noncore functions, such as HR and finance, to offshore companies and use ICT to stay in contact with its global employees, customers, and suppliers 24/7.

Every technology innovation triggers opportunities and threats to business models and strategies. With rare exceptions, every business model depends on a mix of IT, knowledge of its potential, the requirements for success, and, equally important, its limitations.

Questions

1. What precipitated the on-demand economy?
2. How is IT contributing to the success of the on-demand economy?
3. List the six IT business objectives.
4. What are the key strategic and tactical questions that determine an organization's profitability and management performance?
5. What is a business model?
6. What is a digital business model?
7. Give two examples of how companies are transitioning to digital business models.
8. What factors are driving the move to digital business models?

1.2 Business Process Improvement and Competitive Advantage

Given that a company's success depends on the efficiency of its business processes, even small improvements in key processes can have significant payoff. All functions and departments in the enterprise have tasks they need to complete to produce outputs, or deliverables, in order to meet their **objectives**.

Before you can begin to improve something, you have to understand what it is you are improving. We'll start by defining a business process, looking at its characteristics, and then exploring ways in which a business process can be improved either incrementally or radically through Business Process Reengineering.

What Is a Business Process?

Business processes are series of steps by which organizations coordinate and organize tasks to get work done. In the simplest terms, a **process** consists of activities that convert inputs into outputs by doing work.

Examples of common business processes are as follows:

- **Accounting** Invoicing; reconciling accounts; auditing
- **Finance** Credit card or loan approval; estimating credit risk and financing terms
- **Human resources (HR)** Recruiting and hiring; assessing compliance with regulations; evaluating job performance
- **IT or information systems** Generating and distributing reports and data visualizations; data analytics; data archiving
- **Marketing** Sales; product promotion; design and implementation of sales campaigns; qualifying a lead
- **Production and operations** Shipping; receiving; quality control; inventory management
- **Cross-functional business processes** Involving two or more functions, for example, order fulfillment and product development

Three Components of a Business Process Business processes have the three basic components shown in **Figure 1.4**. They involve inputs, activities, and deliverables.

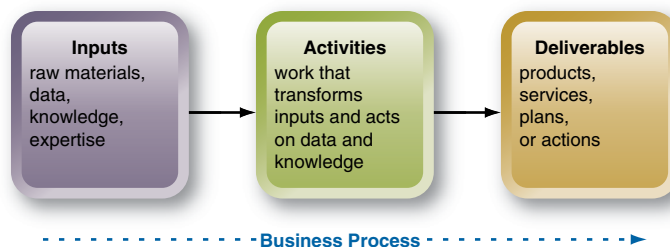


FIGURE 1.4 Three components of a business process.

Processes can be formal or informal. **Formal processes** are documented and have well-established steps. Order taking and credit approval processes are examples. Routine formal processes are referred to as **standard operating procedures (SOPs)**. An SOP is a well-defined and documented way of doing something. An effective SOP documents who will perform the tasks; what materials to use; and where, how, and when the tasks are to be performed. SOPs are needed for the handling of food, hazardous materials, or situations involving safety, security, or compliance. In contrast, **informal processes** are typically undocumented, have inputs that may not yet been identified, and are knowledge-intensive. Although enterprises would prefer to formalize their informal processes in order to better understand, share, and optimize them, in many situations process knowledge remains in people's heads.

Processes range from slow, rigid to fast-moving, adaptive. Rigid processes can be structured to be resistant to change, such as those that enforce security or compliance regulations. Adaptive processes are designed to respond to change or emerging conditions, particularly in marketing and IT.

Improving Business Processes

Designing an effective process can be complex because you need a deep understanding of the inputs and outputs (also known as **deliverables**), how things can go wrong, and how to prevent

Deliverables are the outputs or tangible things that are produced by a business process. Common deliverables are products, services, actions, plans, or decisions, such as to approve or deny a credit application. Deliverables are produced in order to achieve specific objectives.

things from going wrong. For example, Dell had implemented a new process to reduce the time that tech support spent handling customer service calls. In an effort to minimize the length of the call, tech support's quality dropped so much that customers had to call multiple times to solve their problems. The new process had backfired—increasing the time to resolve computer problems and aggravating Dell customers.

The importance of efficient business processes and continuous process improvement cannot be overemphasized. Why? Because 100% of an enterprise's performance is the result of its processes. Maximizing the use of inputs in order to carry out similar activities better than one's competitors is a **critical success factor (CSF)**. Poorly designed, flawed, or outdated business processes waste resources, increase costs, cause delays, and aggravate customers. For example, when customers' orders are not filled on time or correctly, customer loyalty suffers, returns increase, and reshipping increases costs. The blame may not be employee incompetence, but a flawed order fulfillment process.

Don't Automate, Obliterate!

In today's on-demand economy, incrementally improving a business process isn't always sufficient to create the type of change required. Instead, radical changes need to occur to meet higher customer expectations. To do this, companies have to go beyond simply automating an existing process. They must reinvent the entire business process, including reducing the number of steps required, eliminating documents, developing automated decision-making, and dealing with regulatory and fraud issues. Operating models, skills, organizational structures, and roles need to be redesigned to match the reinvented processes. Data models should be adjusted and rebuilt to enable better decision-making, performance tracking, and customer insights.

Leading organizations have come to recognize that it can take a long time to see the benefits of traditional large-scale projects that migrate all current processes to digital and sometimes they don't work. Instead, successful companies are reinventing processes, challenging everything related to an existing process and rebuilding it using cutting-edge digital technology. For example, rather than creating technology tools to help back-office employees type customer complaints into their systems, leading organizations create self-serve options for customers to type in their own complaints.

Business Process Reengineering (BPR) The process by which these types of radical process change can be achieved is referred to as **business process reengineering (BPR)**, its slogan is "Don't automate, obliterate!" (Hammer and Champy, 2006). Consisting of eight stages, shown in **Figure 1.5**, BPR proposes that simply applying IT to a manual or outdated process does not always optimize it. Instead, processes need to be examined to determine whether they are still necessary. After unnecessary processes are identified and eliminated, the remaining ones are redesigned (or reengineered) in order to automate or streamline them. Next, the new process is implemented and put into operation and its performance is evaluated. Finally, the process is reassessed over time to continually improve it.

The goal of BPR is to eliminate unnecessary, non-value-added processes, and simplify and automate the remaining processes to significantly reduce cycle time, labor, and costs. For example, reengineering the credit approval process cuts time from several days or hours to minutes or less. Simplifying processes naturally reduces the time needed to complete the process, which also cuts down on errors.

After eliminating waste, technology can enhance processes by (1) automating existing manual processes; (2) expanding the data flows to reach more functions in order to make it possible for sequential activities to occur in parallel; and (3) creating innovative business processes that, in turn, create new business models. For instance, consumers can scan an image of a product and land on an e-commerce site, such as Amazon.com, selling that product. This process flips the traditional selling process by making it customer-centric.

You will read more about optimizing business processes and role of business process management (BPM) role in the alignment of IT and business strategy in Chapter 13.

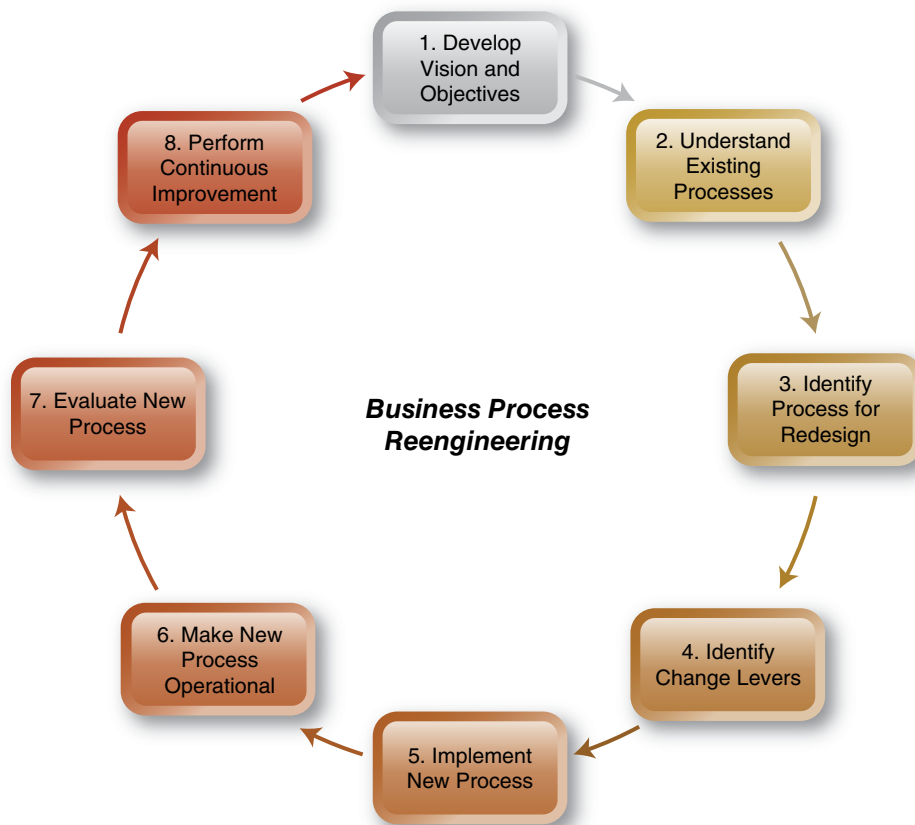


FIGURE 1.5 Eight phases of business process reengineering.

Gaining a Competitive Advantage

Understanding trends that affect the new ways business is being done and getting in front of those trends by changing adding, deleting, and changing existing business processes gives organizations an important **competitive advantage** over their competitors. Helping a company gain, maintain, and sustain a competitive advantage in the market is a very important function of IT. In business, as in sports or politics, companies want to win—customers, market share, and position in the industry. Basically, this requires gaining an edge over competitors by being first to take advantage of market opportunities, providing better customer experiences, offering unique products or services, or convincing customers why your business is a more attractive alternative than your competitors.

Influential industry leaders cite “new competition” as their largest business challenge. Once an enterprise has developed a competitive edge, it can only be sustained by continually pursuing new and better ways to compete. Maintaining a competitive advantage requires forecasting trends and industry changes and figuring out what the company needs to do to stay ahead of the game. It demands continuously tracking competitors and their future plans and promptly taking corrective actions. Competitiveness depends heavily on IT **agility** and **responsiveness**. The benefit of IT agility is that it enables organizations to take advantage of opportunities faster or more effectively.

Closely related to IT agility is **flexibility**. For example, mobile networks are flexible—able to be set up, moved, or removed easily, without dealing with cables and other physical requirements of wired networks. Mass migration to mobile devices from PCs has expanded the scope of IT beyond traditional organizational boundaries—making location practically irrelevant.

IT agility, flexibility, and mobility are tightly interrelated and fully dependent on an organization’s IT infrastructure and architecture, which are discussed in Chapter 2.

With mobile devices, applications, platforms, and social media becoming inseparable parts of work life and corporate collaboration and with more employees working from home, the result is the rapid consumerization of IT. **IT consumerization** is the migration of consumer

Competitive advantage is an edge that enables a company to outperform its average competitor.

Agility means being able to respond quickly.

Responsiveness means that IT capacity can be easily scaled up or down as needed, which essentially requires cloud computing.

Flexibility means having the ability to quickly integrate new business functions or to easily reconfigure software or applications.

technology into enterprise IT environments. This shift has occurred because personally owned IT is as capable and cost-effective as its enterprise equivalents. **IT at Work 1.1** demonstrates how FitBit has maintained a competitive advantage with its fitness tracker.

As evidenced by the mergers of Grubhub/Seamless in food delivery and Handybook/Exec in home services, consolidation will accelerate as competition grows. This trend is underscored in the following examples:

- Collaboration of complementary, noncompetitive businesses will become commonplace as a means to collectively educate consumers about the benefits of on-demand services, increase awareness, and provide added value to core users. These range from cross-promotions similar to that of Instacart/Washio to close partnerships such as that of WunWun/Gett.
- Legacy providers in hospitality, transportation, and other Fortune 500s will “partner with” or “acquire” more innovative on-demand companies. Ken Chenault, Chairman and CEO of American Express, conceded in their annual report, “Our industry is being redefined by many forces, including the continued revolution in online and mobile technologies, which is transforming commerce and society.” Given the emerging influence of on-demand services, Amex launched a partnership with Uber earlier this year. American Express was able to get a foot in the door by allowing customers to earn 2× points for its spend on Uber with an American Express credit card.
- As on-demand businesses solve for the current technological and logistical challenges, design will increasingly become one of the most meaningful forms of competitive advantage. Creating a memorable, frictionless user interface is the next battleground for addressing consumers’ insatiable appetite for greater simplicity and convenience. Scott Belsky points out, “A new cohort of design-driven companies are adding a layer of convenience between us and the underlying services and utilities that improve our lives. This could change everything.”

IT at Work 1.1

FitBit: Smart, Connected Device Transforms Competition and Promotes Sustainability

In the first year of its existence, FitBit sold 100,000 devices. At the time, there were countless weight loss and exercise programs, plans, and gimmicks. But smart, connected wearable activity trackers were virtually nonexistent. Five years later, FitBit managed to take the title *biggest selling manufacturer of wearable tech* when it sold a whopping 21 million devices in 1 year. It still holds that title today.

Vision: Simple Approach Plus Smart Device

FitBit was launched in San Francisco, California, by Eric Friedman and James Park. These entrepreneurs took a basic approach to personal health and fitness—eating right and keeping active. Their vision was to develop a smart device that would motivate users to be more active, eat a well-rounded diet, and ultimately become healthier.

Throughout the day, FitBit logs data about the wearer’s activities, including the number of steps taken, distance travelled, calories burned, and what needs to be done to reach a personal daily goal, for example, walking 2 miles. FitBit’s internal memory can store at least a week of activity data.

One of FitBit’s competitive strengths is the app that is accessible from a smartphone. Users can sync FitBit devices and view their online profile, activity levels, and sleep patterns on **dashboards** that display on more than 150 mobile devices, including iOS, Android, and Windows Phone products. This compatibility maximizes the number

of friends and family in each user’s network to share performance stats. It also motivates and increases user retention.

First Class Fitness

A smart wearable product that fits effortlessly into users’ life styles launched an industry and made FitBit a market leader. In the second quarter of 2015 (2Q15), FitBit shipped 4.4 million units, up 159% from the same quarter a year ago (2Q14) and held 24.3% global market share. Second in line was Apple with 3.6 million units shipped in 2Q15 and 19.9% global market share. Thanks to the technology that enabled FitBit and the company’s growing reputation, Friedman and Park are likely to be in business for a long time.

IT at Work Questions

1. How did FitBit manage to take the title of biggest selling manufacturer of **wearable technology** tech and sustain it?
2. What could other companies who produce fitness trackers challenge FitBit in the marketplace?
3. What other features do you think consumers would like FitBit to incorporate into its fitness tracker to further improve it? How would consumers and FitBit benefit from these improvements?

Sources: Compiled from Ashcroft (2015), Nusca (2015), Grand View Research (2016), and Fitbit.com (2016).

Software Support for BPM

The purpose of **business process management (BPM)** is to help enterprises become more agile and effective by enabling them to better understand, manage, and adapt their business processes. Vendors, consulting and tech firms offer BPM expertise, services, software suites, and tools.

BPM software is used to map processes performed either by computers or manually—and to design new ones. The software includes built-in templates showing workflows and rules for various functions, such as rules for credit approval. These templates and rules provide consistency and high-quality outcomes. For example, Oracle's WebLogic Server Process Edition includes server software and process integration tools for automating complex business processes, such as handling an insurance claim.

But, BPM initiatives can be extremely challenging, and in order to be successful, BPM requires buy-in from a broad cross section of the business, the right technology selection, and highly effective change management processes.

Business process management consists of methods, tools, and technology to support and continuously improve business processes.

Questions

1. What is a business process? Give three examples.
2. What is the difference between business deliverables and objectives?
3. List and give examples of the three components of a business process.
4. Explain the differences between formal and informal processes.
5. What is an SOP?
6. What is the purpose of BPM?

1.3 IT Innovation and Disruption

Digital technology creates new markets, businesses, products, and careers. As digital changes the way consumers and retailers buy and sell products, companies must adapt and innovate to ensure their product offerings, platforms, technologies, and search options cater to these changing needs.

Social–Mobile–Analytics–Cloud (SMAC) Model

We are in the era of **social–mobile–analytics–cloud (SMAC)** computing that is reshaping business strategies and day-to-day operations (**Figure 1.6**).

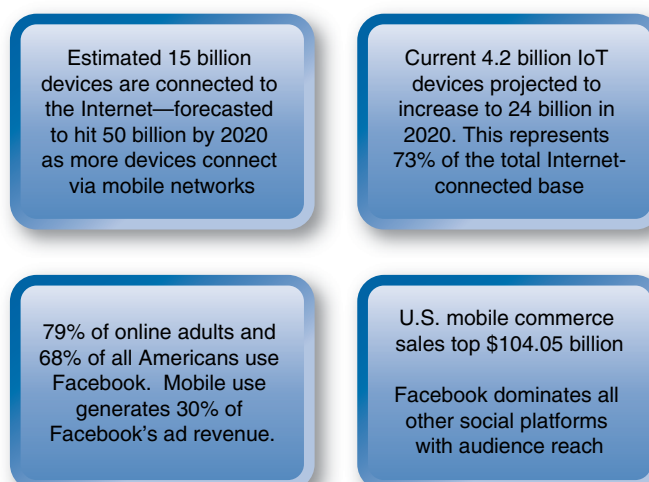


FIGURE 1.6 SMAC reshapes business strategies and day-to-day operations.

The cloud consists of huge data centers accessible via the Internet and forms the core by providing 24/7 access to storage, applications, and services. Handhelds and wearables, such as FitBit, Pebble, and the Apple Watch, and their users form the edge. Social channels connect the core and edge. The SMAC integration creates the technical and services infrastructure needed for digital business. This infrastructure makes it possible to meet the expectations of employees, customers, and business partners given that almost everyone is connected (social), everywhere they go (mobile), gets the information they need (analytics), and has 24/7 access to products and services (cloud).

Here are three examples of SMAC's influence:

1. **Powerful social influences impact advertising and marketing** Connections and feedback via social networks have changed the balance of influence. Consumers are more likely to trust tweets from ordinary people than recommendations made by celebrity endorsements. And, negative sentiments posted or tweeted can damage brands.
2. **Consumer devices go digital and offer new services** The Nike+ FuelBand wristband helps customers track their exercise activities and calories burned. The device links to a mobile app that lets users post their progress on Facebook.
3. **eBay's move to cloud technology improves sellers' and buyers' experiences** The world's largest online marketplace, eBay, moved its IT infrastructure to the cloud. With cloud computing, eBay is able to introduce new types of landing pages and customer experiences without the delay associated with having to buy additional computing resources.

The balance of power has shifted as business is increasingly driven by individuals for whom mobiles are an extension of their body and mind. They expect to use location-aware services, apps, alerts, social networks, and the latest digital capabilities at work and outside work. To a growing extent, customer loyalty and revenue growth depend on a business's ability to offer unique customer experiences that wow customers more than competitors can.

Technology Mega Trends

Mega trends are forces that shape or create the future of business, the economy, and society.

For 21st-century enterprises, connectivity, big data and analytics, and digitization are technology **mega trends** that cannot be ignored. Business breakthroughs and innovation would be impossible without them. They also mark the difference between outdated 20th-century business models and practices and those of today's on-demand economy.

The most influential IT mega trends driving digital transformation of companies in the on-demand economy are discussed next.

Connectivity Companies need to connect with consumers and business partners across multiple channels and devices using digital platforms that consist of hardware, software (mobile apps), networks (social media), (embedded sensors), and **cloud computing**.

For example, rather than run applications or programs from software stored on a computer or server owned by the company, cloud computing allows companies to access the same kinds of applications through the Internet. Major business cloud computing providers include Amazon Web Services (AWS), Cisco Powered, Dell Cloud Solutions, Google Cloud, IBM Cloud Solutions, and Teradata Cloud. One of the many benefits of cloud is that it provides the flexibility to acquire or expand connectivity and computing power as needed for operations, business transactions, and communication.

Expanded connectivity supports smart products, which have the ability to sense, process, report, and take corrective action, such as smart clothing, watches, phones, to smart buildings and smart cities. This IoT is becoming a driving force in the on-demand economy.

Connectivity pushes other sub trends, like big data, to create market opportunities for new products and services, such as social sentiment analysis, open innovation, new insurance business models, and micro personalized marketing and medicines. Big data is one of the many disruptive technologies that are impacting people, processes, and profits.

Big Data and Data Analytics There is no question that the increasing volume of data can be valuable, but only if they are processed and available when and where they are needed. The problem is that the amount, variety, structure, and speed of data being generated or collected by enterprises differ significantly from traditional data. **Big data** are what high-volume, mostly text data are called. Big data stream in from multiple channels and sources, including the following:

- Mobile devices and machine-to-machine sensors embedded in everything from airport runways to casino chips (Later in this chapter, you will read more about the IoT.)
- Social content from texts, tweets, posts, blogs
- Clickstream data from the Web and Internet searches
- Video data and photos from retail and user-generated content
- Financial, medical, research, customer, and **business-to-business** transactions.

Big data are 80% to 90% unstructured. **Unstructured data** do not have a predictable format like a credit card application form. Huge volumes of unstructured data flooding into an enterprise are too much for traditional technology to process and analyze quickly. Big data tend to be more time-sensitive than traditional (or small) data. Data collected from social, mobile, and other channels are analyzed to gain insights and make smart decisions that drive up the bottom line. Machine-generated data from sensors and social media texts are main sources of big data.

Big data has been one of the most disruptive forces businesses have seen in a long time. But when an enterprise harnesses its data and is able to act on analytic insights, it can turn the challenges into opportunities.

Digitization Across industries, companies are attempting to transform their disconnected or disjointed approaches to customers, products, services, and operating models to an always-on, real-time, and information-rich marketplace. Some leaders are redesigning their capabilities and operating models to take full advantage of digital technologies to keep step with the “connected” consumer and attract talent. Others are creating qualitatively new business models—and tremendous value—around disruptive digital opportunities. In doing so, these companies secure not only continued relevance but also superior returns.

Digitization often requires that old wisdom be combined with new skills, for example, by training a merchandising manager to program a pricing algorithm and creating new roles, such as user-experience designer. The benefits of digitizing processes, through business process reengineering, are huge. By digitizing information-intensive processes, costs can be cut by up to 90% and turnaround times improved by several orders of magnitude.

Examples span multiple industries. For example, one bank digitized its mortgage application and decision process, cutting the cost per new mortgage by 70% and slashing time to preliminary approval from several days to just one minute. A telecommunications company created a self-serve, prepaid service where customers could order and activate phones without back-office involvement. A shoe retailer built a system to manage its in-store inventory that enabled it to know immediately whether a shoe and size was in stock—saving time for customers and sales staff. An insurance company built a digital process to automatically adjudicate a large share of its simple claims.

In addition, replacing paper and manual processes with software allows businesses to automatically collect data that can be mined to better understand process performance, cost drivers, and causes of risk. Real-time reports and dashboards on digital-process performance enable managers to address problems before they get out of control. For example, quality issues in a company’s supply chain can be identified and remedied more rapidly by monitoring customer buying behavior and feedback in digital channels.

Digitization is the process of transforming any kind of activity or information into a digital format that can be collected, stored, searched, and analyzed electronically—and efficiently.

Machine-to-Machine Technology Sensors can be embedded in most products. Objects that connect themselves to the Internet include cars, heart monitors, stoplights, and appliances. Sensors are designed to detect and react, such as Ford’s rain-sensing front wipers that use an advanced optical sensor to detect the intensity of rain or snowfall and adjust wiper speed

Internet of Things (IoT) refers to a set of capabilities enabled when physical things are connected to the Internet via sensors.

accordingly. **Machine-to-machine (M2M) technology** enables sensor-embedded products to share reliable real-time data via radio signals. M2M and the **Internet of Things (IoT)** are widely used to automate business processes in industries ranging from transportation to health care. By adding sensors to trucks, turbines, roadways, utility meters, heart monitors, vending machines, and other equipment they sell, companies can track and manage their products remotely.

When devices or products are embedded with sensors, companies can track their movements or monitor interactions with them. Business models can be adjusted to take advantage of what is learned from this behavioral data. For example, an insurance company offers to install location sensors in customers' cars. By doing so, the company develops the ability to price the drivers' policies on how a car is driven and where it travels. Pricing is customized to match the actual risks of operating a vehicle rather than based on general proxies—driver's age, gender, or location of residence. **Table 1.2** lists a number of opportunities for improvement through the application of embedded physical things.

TABLE 1.2 Improvement Opportunities from Embedded Sensors

Industry Sector	Application	Payoff
Oil and gas	Exploration and development rely on extensive sensor networks placed in the earth's crust. Sensors can produce accurate readings of the location, structure, and dimensions of potential fields	Lower development costs and improved oil flows
Health care	Sensors and data links can monitor patients' behavior and symptoms in real time and at low cost allowing physicians to more precisely diagnose disease and prescribe treatment regimens	Reduce hospitalization and treatment costs by \$1 billion per year in the United States
Retail	Sensors can capture shoppers' profile data stored in their membership cards to help close purchases	Additional information and discounts at point of sale
Farming	Ground sensors can take into account crop and field conditions and adjust the amount of fertilizer that is spread on areas that need more nutrients	Reduction in time and cost
Advertising	Billboards can scan people passing by, assessing how they fit consumer profiles, and instantly change displayed messages based on those assessments	Better targeted marketing campaigns; flexibility; increased revenues
Automotive	Systems can detect imminent collisions and take evasive action, such as automatic braking systems	Potential accident reduction savings of more than \$100 billion annually

Lessons Learned from Companies Using Disruptive Technologies

Those companies who have adapted to change by exploiting digital technology and software are outperforming their peers. According to a survey conducted by CA Associates, companies who have turned the way they use technology from being a cost center and operational function to being a genuine competitive differentiator are reaping the benefits. Many reported doubling their revenue growth, experiencing a higher profit by a factor of 2.5 and increasing new business-based revenue by a factor of 1.5 (Vaughn-Brown, 2014). The five factors to which companies attribute these benefits can be summed up in the following Lessons Learned:

- 1. Exploit the power of software** Become “app-centric” and extend core business functions to include software development.
- 2. Develop, deliver, disrupt—quickly!** Embrace agile development techniques and broadly implement DevOps.
- 3. Boost speed and efficiency with automated programming interfaces (APIs)** Take a managed approach to use APIs for building full-function Web applications (particularly mobile apps) and for integrating back-office systems.

4. **Leverage third-party innovation** Take a more managed approach to use APIs for integrating third-party services into applications and enable external developer access to systems and data.
5. **Maximize returns with smarter IT investments** Get smarter at assessing and prioritizing IT investments to maximize return on investment and put portfolio management in place to prioritize and track IT programs.

Business opportunities and challenges presented by today's technology innovations are on an unprecedented scale. **Cloud services**, big data, mobility, digitization, and the IoT are likely to disrupt many industries and shake up competitive positions.

Innovation is necessary for any company that wants to remain relevant, retain customers, and increase profits. Increased competition, expanded global markets, and empowered customers define today's on-demand business environment.

Cloud service is any computing resource that is provided over the Internet on demand.

Questions

1. What are the benefits of cloud computing?
2. What is M2M technology? Give an example of a business process that could be automated with M2M.
3. Describe the relationships in the SMAC model.
4. What impacts is the SMAC model having on business?
5. Why have mobile devices given consumers more power in the marketplace?
6. Explain why connectivity is important in today's on-demand economy.
7. In what ways is IT disrupting business?

1.4 IT and You

Today, IT and information systems touch nearly all aspects of our lives. IT is a part of our social life, our work, and every business process, and it is no longer the sole responsibility of the IT department. Just think about much of your day you spend interacting with technology—your iPad, PC, and smartphone. It has been reported that the average American checks his/her phone 46 times every day! That's an increase of 35% over the 33 looks per day reported in a similar study just one year earlier. Aggregated across the 185 million American smartphone users, that's 8 billion "looks" per day (Eadicicco, 2015).

IT impacts the way you work, the way you learn, the way you communicate and socialize and the way you entertain yourself. Today, success in any field, be it health care, marketing, finance, accounting, law, education, sports, entertainment, etc. requires much more than a cursory knowledge of IT. IT is and will remain the foundation of the global economy and is especially important in the on-demand economy.

On-Demand Workers

A recent survey of on-demand economy (Chriss, 2016) in the United States and online talent recruiters reported an increase in people working in the on-demand economy who are enjoying a new way of working. Other facts and stats from the survey reveal the U.S. On-Demand workforce as a community of 45 million workers, the majority of whom are younger, optimistic, and urban-based (**Table 1.3**).

The survey also revealed that fewer and fewer people are looking for traditional employment. For example, 91% like the control over where, when, and how they work that the on-demand jobs offer them. The motivation for most is not to replace a traditional job, but to earn supplemental income (**Table 1.4**).

TABLE 1.3 Profile of U.S. On-Demand Workers

Characteristic	Number
Financial situation had improved over the past year	23 million
Expected their financial situation to improve over the coming year	28.8 million
Under 35 years of age	23 million
Live in urban areas	18.45 million

TABLE 1.4 Motivation to Work in the On-Demand Economy

Motivation	Percent
Earn supplemental income	63
Create and control their own schedule	46
Turned to on-demand work because they couldn't find another traditional job	11

The data also shows a strong entrepreneurial drive behind people choosing on-demand work. Just over one-third of respondents owned full- or part-time business and approximately one quarter reporting they are working in the on-demand economy to build a business. This entrepreneurial spirit is reflected in the ways that on-demand workers are compensated. While the 40-hour workweek is still alive and well, sources of income have changed. Instead of one paycheck, on-demand workers typically receive their income from three different sources:

1. On-demand work
2. Contracting and consulting
3. Running a business

Changes in Work Status While the on-demand economy provides positive opportunities, it can also offer limited benefits and inferior infrastructure. Take, for example, the “contractor” model that companies like Uber use. Initially, Uber set the standard for on-demand business by labeling its drivers “independent contractors” and essentially claiming that all of its 160,000 drivers were self-employed. This pushed many of the costs of doing business onto the independent contractors’ shoulders and deprived them of baseline labor protections such as worker’s compensation, social security contributions, minimum wage, and discrimination protections.

This business model also allowed companies using the Uber model to sidestep federal, state, and county taxes and insurance premiums and undercuts competitors that used a traditional W-2 hiring model. However, not all on-demand businesses use the Uber model.

Some companies treat their workers as employees from the start, while others have switched to the W-2 model and both approaches are reaping benefits. Shyp CEO Kevin Gibbon posted on LinkedIn that the move to employee status was “an investment in a longer-time relationship with our couriers, which we believe will ultimately create the best experience for our customers.” After moving to the W-2 model, Shyp had only 1 out of 245 employees quit and customer complaints decreased at the package delivery company. And Instacart, a food shopping and delivery service, offered its shoppers the option to convert to part-time employees so they could offer training to ensure a consistent customer experience and greater customer satisfaction (National Employment Law Project, 2016).

Regardless of their work status, it would appear that overall on-demand workers are highly satisfied with their work environment, perhaps because it fits a unique need. Intuit’s on-demand economy survey reported the following:

- 70% of on-demand workers are satisfied with their work.
- 81% plan to continue working with the same provider over the next year.
- 63% are happier to be working in the on-demand economy.

Overall, on-demand workers are forward-looking, eager to embrace new opportunities, and want to take charge of their careers.

IT Adds Value to Your Performance and Career

Whether you join the ranks of the on-demand workers, or choose to stay in a traditional job, IT can greatly enhance your performance at work and the ways you move through your career path.

Staying current in emerging technologies affecting markets is essential to the careers of knowledge workers, entrepreneurs, managers, and business leaders—not just IT and CIO.

In the current marketplace, organizations are finding it particularly difficult to find qualified IT Talent, as illustrated in [IT at Work 1.2](#).

IT at Work 1.2

Scott Zulpo Is Facing Stiff Competition

He's adding a senior project manager, a network analyst, and a help desk worker to his 55-member IT staff at BCU, a Vernon Hills, Illinois-based credit union where he is vice president of IT. And, Zulpo will need to add even more people to keep up with an increasing demand for tech-driven innovations.

"The challenge is twofold—first finding talent, and then determining if that talent has the skills, experience and personality to thrive in the position," says Zulpo, who's mindful that "the cost and impact of not hiring an 'A' player is huge."

Zulpo has his work cut out for him. He's hiring at time when very few IT professionals are out of work. Consequently, competition for tech talent is fierce. The unemployment rate for tech workers is about 2%, according to reports on recent data from the U.S. Bureau of Labor Statistics (Bureau of Labor Statistics, 2016).

And, Zulpo isn't the only one who's having a difficult time finding good IT talent. Many of his fellow IT leaders are seeking the same skills. Computerworld's Forecast 2017 survey of 196 IT professionals found that both project management and technical

support were among the top 10 most sought-after skills among companies that plan to recruit in the new year.

"The IT labor market is still very hot. The candidate is very much in the driver's seat," says Jason Hayman, market research manager for IT staffing firm TEKsystems.

Hayman cites a government report that estimates that 500,000 to 1 million IT jobs go unfilled every year, but notes that some analysts say the figure is closer to 2 million. He says there's a classic supply-and-demand scenario working here, with demand for talent far exceeding supply.

The takeaway is that there are not enough IT workers!

Compiled from: Bureau of Labor Statistics (2016), Pratt (2016), and Computerworld (2017).

IT at Work Questions

1. What are two reasons why Zulpo had trouble finding qualified IT talent?
2. What of position was Zulpo trying to fill?
3. What methods would you recommend to Zulpo help him in his efforts to recruit new IT personnel?

IT as a Career Fueled by corporate growth, systems expansion, need for competitive or unique services to increase business and security initiatives, companies are increasing their IT hires. Companies need new tech hires who have a background in both technology and business and who can articulate IT's value in meeting business goals. In particular, companies are seeking IT employees with skills in programming, application development, technical support, security, cloud, business intelligence, Web development, database administration, and project management.

According to the [U.S. Department of Labor \(2016\)](#), IT job growth is estimated at 12% from 2014 to 2024, faster than the average for all other occupations. This means about 488,500 new jobs. The median annual wage for computer and IT occupations was \$81,430 in May 2015, which was considerably higher than the median annual wage of \$36,200 for all other occupations. Here are some common IT jobs and their activities:

IT managers Play a vital role in the implementation and administration of digital technology. They plan, coordinate, and direct research on the computer-related activities of firms. In consultation with other managers, they help determine the goals of an organization and then implement technology to meet those goals.

Chief technology officers (CTOs) Evaluate the newest and most innovative technologies and determine how they can be applied for competitive advantage. CTOs develop technical standards, deploy technology, and supervise workers who deal with the daily IT issues of the firm. When innovative and useful new ITs are launched, the CTO determines implementation strategies, performs cost-benefit or **SWOT analysis**, and reports those strategies to top management, including the CIO.

IT project managers Develop requirements, budgets, and schedules for their firm's IT projects. They coordinate such projects from development through implementation, working with their organization's IT workers, as well as clients, vendors, and consultants. These managers are increasingly involved in projects that upgrade the information security of an organization.

Data scientists Manage and analyze massive sets of data for purposes such as target marketing, trend analysis, and the creation of individually tailored products and services. Enterprises that want to take advantage of big data use real-time data from tweets, sensors, and their big data sources to gain insights into their customers' interests and preference, to create new products and services, and to respond to changes in usage patterns as they occur. Big data analytics has increased the demand for data scientists, as described in [Career Insight 1.1](#).

Career Insight 1.1

Data Scientists Analyze Business Data for Actionable Business Intelligence

Online searches for data scientist are outpacing the number of job postings by more than 20% and the large business consulting firm, Price-Waterhouse-Cooper, recently announced they would be adding more than 1,000 data scientists during the next 2 years.

Big data, analytics tools, powerful networks, and greater processing power have contributed to growth of the field of **data science**. According to Glassdoor data (glassdoor.com, 2017), the median annual salary for data scientists in the United States is \$113,436 and experienced data scientists who manage teams of 5 to 10 people are earning more than \$250,000 per annum.

But, it's not just about the money—data scientists enjoy what they do. The job is interesting, spanning many different aspects of the organization and in some cases involves analyzing community outreach programs supported by organizations.

What Does a Data Scientist Do?

Enterprises need people who are capable of analyzing and finding insights in data captured from a range of sources, including customer transactions, click streams, sensors, social media, log files, and GPS plots. Their mission is to unlock valuable and predictive insights that will influence business decisions and spur a competitive advantage. According to Gregg Gordon, VP of the Big Data practice group at Kronos, provider of workforce management solutions in the cloud,

It's not sitting in a room all day – we take our work and apply it to customer problems. We're working and interacting with customers on a daily basis talking about real problems, then attempting to replicate, model and solve them.

An interesting example of what a data scientist can achieve can be found by studying Jonathan Goldman, the person who transformed LinkedIn. At the time Goldman joined, LinkedIn had less than 8 million members. Goldman noticed that existing members were inviting their friends and colleagues to join, but they were not making connections with other members at the rate executives had expected. A LinkedIn manager said, "It was like arriving at a conference reception and realizing you don't know anyone. So you just stand in the corner sipping your drink—and you probably leave

early." Goldman began analyzing the data from user profiles and looked for patterns to predict whose networks a given profile would land in. While most LinkedIn managers saw no value in Goldman's work, Reid Hoffman, LinkedIn's cofounder and CEO at the time, understood the power of analytics because of his experiences at PayPal. With Hoffman's approval, Goldman applied data analytics to test what would happen if a member were presented with names of other members they had not yet connected with, but seemed likely to know. He displayed the three best new matches for each member based on his or her LinkedIn profile. Within days, the click-through rate on those matches skyrocketed and things really took off. Thanks to this one feature, LinkedIn's growth increased dramatically.

Artist or Scientist?

The most successful—and sought-after—data scientists possess a combination of analytical skills, technical prowess and business acumen needed to effectively analyze massive data sets while thinking critically and shifting assumptions on the fly, ultimately transforming raw intelligence into concise and actionable insights.

The LinkedIn example shows that good data scientists do much more than simply try to solve obvious business problems. Creative and critical thinking are part of their job—that is, part analyst and part artist. They dig through incoming data with the goal of discovering previously hidden insights that could lead to a competitive advantage or detect a business crisis in enough time to prevent it. Data scientists often need to evaluate and select those opportunities and threats that would be of greatest value to the enterprise or brand.

Questions

1. What types of IT career have the most potential in the current hiring market?
2. What factors does Zulpo take into consideration when he's evaluating job applicants?
3. Why is IT a major enabler of business performance and success?
4. Explain why it is beneficial to be an informed user of IT.
5. Do you think IT job prospects are strong? Explain.

Sources: Darrow (2015), Marr (2016), U.S. Department of Labor (2016), and Bureau of Labor Statistics (2016).

IT Job Prospects In 2017, only 2% of all IT workers are unemployed. Workers with specialized technical knowledge and strong communications and business skills, as well as those with an MBA with a concentration in an IT area, will have the best prospects. Job openings will be the result of employment growth and the need to replace workers who transfer to other occupations or leave the labor force (Bureau of Labor Statistics, 2016)

Dow Chemical set up its own social network to help managers identify the talent they need to carry out projects across its diverse business units and functions. To expand its talent pool, Dow extended the network to include former employees and retirees.

Other companies are using networks to tap external talent pools. These networks include online labor markets such as Amazon Mechanical Turk and contest services such as InnoCentive that help solve business problems.

- Amazon Mechanical Turk is a marketplace for work that requires human intelligence. Its Web service enables companies to access a diverse, on-demand workforce.
- InnoCentive is an “open innovation” company that takes R&D problems in a broad range of areas such as engineering, computer science, and business and frames them as “challenge problems” for anyone to solve. It gives cash awards for the best solutions to solvers who meet the challenge criteria.

Becoming an Informed IT User

Knowing how best to use IT and how and when to interact with IT personnel, and they with you, will help you perform better at home and at work and enable you to become an **informed user** of technology.

The department or functional area that handles the collection, processing, storing, analysis and distribution of information using a computer-based tool can be referred to by many names—some companies refer to it as information technology (IT), while others refer to it as information systems (IS), management information systems (MIS), IT support, IT services or computer information systems (CIS). Whatever the name, its purpose is the same—to support a company’s information needs by developing, operating, securing, and maintaining one or more information systems.

To become an informed IT user, you will learn how the six components of an information system—hardware, software, procedures, people, networks, and data—interact to provide you with the information that you need, when you need it, and in the format you need. These components will be discussed in detail in Chapter 2.

By reading this book, you will become an informed user and you gain more value from IT to improve your performance and widen your career opportunities. For example, you will learn to

- Understand how using IT can improve organizational performance
- Understand how and why IT can benefit organizational growth
- Understand how business can use IT to enhance the customer experience
- Use how companies use IT to analyze business data and offer important insights
- Be able to offer input into the development and use of IT
- Be able to recommend and select IT applications at work
- Know how to find emerging technologies to make radical improvement in business processes
- Understand how IT can facilitate teamwork and improve individual productivity
- Appreciate the importance of ethical behavior when using IT and explain the associated risks and responsibilities
- Foster your entrepreneurial tendencies to start your own on-demand business.

Informed user is a person knowledgeable about information systems and IT.

Key Terms

agility 11	cross-functional business process 9	IT project managers 20
barriers to entry 23	customer experience 7	machine-to-machine (M2M) technology 16
big data 15	cycle time 22	mega trends 14
business model 3	dashboards 12	objectives 8
business process 9	data analytics 3	on-demand economy 4
business process management (BPM) 13	data science 20	productivity 7
business process reengineering (BPR) 10	deliverables 9	responsiveness 11
business-to-business 15	digital business model 7	social, mobile, analytics and cloud (SMAC) 13
chief technology officers (CTOs) 19	digitization 15	standard operating procedures (SOPs) 9
cloud computing 14	flexibility 11	SWOT analysis 19
cloud services 17	informed user 21	unstructured data 15
competitive advantage 11	Internet of Things (IoT) 16	wearable technology 12
critical success factor (CSF) 10	IT consumerization 11	

Assuring Your Learning

Discuss: Critical Thinking Questions

1. Why are businesses experiencing a digital transformation?
2. More data are collected in a day now than existed in the world 10 years ago. What factors have contributed to this volume of data?
3. Assume you had no smartphone, other mobile device, or mobile apps to use for 24 hours. How would that mobile blackout disrupt your ability to function?
4. Name three highly disruptive digital technologies. Give an example of one disruption for each technology.
5. Why are enterprises adopting cloud computing?
6. What is the value of M2M technology? Give two examples.
7. Starbucks monitors tweets and other sources of big data. How might the company increase revenue from big data analytics?
8. Select three companies in different industries, such as banking, retail store, supermarket, airlines, or package delivery, that you do business with. What digital technologies does each company use to engage you, keep you informed, or create a unique customer experience? How effective is each use of digital technology to keeping you a loyal customer?
9. Describe two examples of the influence of SMAC on the financial industry.
10. What is the potential impact of the IoT on the health-care industry?
11. Why does reducing the **cycle time** of a business process also help to reduce errors?
12. Research firm Gartner defines competitive advantage as a difference between a company and its competitors *that matters to customers*. Describe one use of M2M technology that could provide a manufacturer with a competitive advantage.
13. What IT careers are forecasted to be in high demand? Explain why.
14. Why or how would understanding the latest IT trends influence your career?

Explore: Online and Interactive Exercises

1. Research the growing importance of the IoT. Find two forecasts of its growth. What do they forecast?
2. Go to “9 Successful Digital Disruption Examples” on the IT Business Edge website. Close the pop-up to view the slideshow and read the descriptions of each of the ways in which technology is disrupting our lives. Answer the following questions:
 - a. Which of the following disruptions resonated best with you and your lifestyle? Explain.
 - b. Which of the disruptions was most surprising to you? Why?
 - c. Rank order the disruptions in their order of importance to you? Write a short report explaining your rankings.

Analyze & Decide: Apply IT Concepts to Business Decisions

1. A transportation company is considering investing in a truck tire with embedded sensors. Outline the benefits of this investment. Would this investment create a long-term competitive advantage for the transportation company?
2. Visit the website of UPS (ups.com), Federal Express (fedex.com), and one other logistics and delivery company.
 - a. At each site, describe what information is available to customers before and after they send a package?
 - b. Compare the three customer experiences. Which one do you prefer? Why?
3. Visit Dell.com and Apple.com to simulate buying a laptop computer. Compare and contrast the selection process, degree of customization, and other buying features. What are the **barriers to entry** into this market, based on what you learned from this exercise?

Case 1.2

Business Case: The Internet of Things Comes to the NFL

People love sports statistics and the more the better. Responding to this customer demand, the NFL increased the quality and quantity of statistics available to coaches and fans with radio frequency identification (RFID) chips.

Player RFID Project

When the 2015 National Football League played its first game in New England, each player was equipped with a set of RFID sensors. Each sensor, about the size of a quarter, is embedded in players' shoulder pads and emits a unique radio frequency. Every stadium used by the NFL is equipped with 20 receivers to pick up the RFID signals and pinpoint every player on the field. It also records speed, distance traveled, acceleration in real time, and the direction the player is facing.

The NFL plans to use the data it collects to power an Xbox One and Windows NFL apps to allow fans to call up stats for each player tied into the highlight clips posted on the app. The data will also be fed to broadcasters, leveraged for in-stadium displays, and provided to coaching staff and players.

"We've always had these traditional NFL stats," says Matt Swenson, senior director of Emerging Products and Technology at the NFL. "The league has been very interested in trying to broaden that and bring new statistics to the fans. Along the way, there's been more realization about how the data can be leveraged to make workflow more efficient around the game."

Zebra Technologies Software Vendor

The NFL's technology partner in its IoT push was Zebra Technologies of Lincolnshire, Illinois.

Zebra was well known for its manufacturing and selling marking, tracking and printing technologies such as thermal barcode label and receipt printers, RFID smart label printer/encoders, and card and kiosk

printers. As it moved into IoT and M2M applications, Zebra launched its MotionWorks Sports Solution, which powers the NFL IoT initiative. Zebra was able to develop RFID tags that blink up to 85 times per second to track motion of athletes in subseconds. Then it had to find a customer for the product—so it turned to the biggest fish in the pond—the NFL. Zebra trialed the tags by equipping more than 2,000 players, 18 NFL stadiums and officials, markers, and pylons. Over the course of the season, more than 1.7 billion sets of XY player coordinates were measured, transmitted, and stored during the games. Every stadium was connected to a command station in San Jose, California, that controls when the data are collected, where they are sent, and stores them in the cloud.

The Need for the Right People

An important lesson that Zebra learned is that generic data scientists weren't sufficient to gain insight into the data. Zebra needed football experts. "When you look at analytics in football, you really need people. We had to go out and hire football people. The analytics from manufacturing weren't the same as the analytics from football. We could see correlations in the data that seemed important and then found out they weren't. We had to bring in people that had the football expertise who could say 'Look, this is why it matters,'" said Jill Stelfox, Zebra Technologies Vice President and General Manager, Location Solutions.

The latest development in this IoT initiative is its integration with NFL's fantasy football offerings.

Questions

1. Why did NFL equip its players with RFID tags?
2. What factors contributed to the success of the IoT initiative at the NFL?
3. What other types of IoT applications can you think of that could be used in sports stadiums?

Case 1.3

Video Case: Knowing More and Doing More

Teradata is a leading provider of big data and data analytics solutions. In a video, Teradata explains that when you know the right thing to do, you can do more of what truly matters for your business and your customers. Visit [Teradata's website](#), search for and view the video entitled "Manufacturing: What Would You Do If You Knew?"™ (the video runs for 1:26 minutes).

Questions

1. What did you learn from the video?
2. What is the value of knowing more?

References

- Accenture. *Accenture Technology Vision* 2013.
- Ashcroft, S. "Fitbit sold 4.5 million trackers last quarter and smashed financial estimates." *Wearable.com*, August 6, 2015.
- Barry Libert, B., Y. Wind, and M.B. Fenley. "What Airbnb, Uber, and Alibaba Have in Common." *Harvard Business Review*, November 20, 2014.
- Basu, C. "The Six Important Business Objectives of Information Technology." *Chron*, 2017. Accessible from <http://smallbusiness.chron.com/six-important-business-objectives-information-technology-25220.html>
- Boland, M. "What's Driving the Local on Demand Economy?" *BIA/Kelsey blog*, May 5, 2015.
- Bureau of Labor Statistics. *Occupational Outlook Handbook*. U.S. Department of Labor, 2016–2017.
- Chandler, N. "How FitBit Works." *HowStuffWorks.com*, May 2, 2012.
- Chriss, A. "How the On-Demand Economy Is Reshaping the 40 Hour Work Week." 2016.
- Computerworld. "2017 Tech Forecast: IT Sharpens Its Focus." *Computerworld*, 2017.
- Darrow, B. "Data Science Is Still White Hot, but Nothing Lasts Forever." *Fortune*, 2015.
- Eadicicco, L. "American Check Their Phones 8 Billion Times a Day." *Time*, December 15, 2015.
- glassdoor.com. "Data Scientist Salaries." April 7, 2017.
- Grand View Research. "Wearable Technology Market Analysis By Product (Wrist-Wear, Eye-Wear, Foot-Wear, Neck-Wear, Body-Wear), by Application (Fitness & Wellness, Healthcare, Infotainment, Defense, Enterprise and Industrial) and Segment Forecasts to 2022." Grand View Research, 2016.
- Hammer, M. and J. Champy. "Re-engineering the Corporation: A Manifesto for Business Revolution." Updated and revised edition. Harper Business Essentials, 2006.
- Jaconi, M. "The 'On-Demand Economy' Is Revolutionizing Consumer Behavior—Here's How." *Business Insider*, July 13, 2014.
- Kappelman, L., E. McLean, V. Johnson, R. Torres, Q. Nguyen, C. Maurer, and M. Snyder. "The 2016 SIM IT Key Issues and Trends Study." *MIS Quarterly Executive*, March 2017.
- MacIver, K. "Digital Business in an Era of Disruptive Innovation." *I-CIO.com*, November 2015.
- Marr, B. "Is Being a Data Scientist Really the Best Job in America?" *Forbes*, February 25, 2016.
- National Employment Law Project. "Employers in the On-Demand Economy: Why Treating Workers as Employees is Good for Business." <http://www.nelp.org/content/uploads/Fact-Sheet-Employers-in-the-On-Demand-Economy.pdf>, 2016.
- Nusca, A. "The Numbers Are in: Apple Is No. 2 in Wearables." *Fortune*, August 27, 2015.
- Pratt, M. "10 Hottest Tech Skills for 2017." *Computerworld*, December 7, 2016.
- Primack, D. "GrubHub Makes Major Move in Restaurant Delivery Wars." *Fortune*, February 5, 2015.
- Schmidt-Subramanian, M., H. Manning, J. Knott, and Murphy. "The Business Impact of Customer Experience, 2013." *Forrester Research*, June 10, 2013.
- Storbaek, D. "The 5-Step Uber Playbook That Will Disrupt the On-Demand Economy." *TechCrunch.com*, October 15, 2015.
- U.S. Department of Labor (2016). "Computer and Information Technology Occupations". <https://www.bls.gov/ooh/computer-and-information-technology/home.htm>
- Vaughn-Brown, J. "The Digital Transformation Journey: Key Technology Considerations." *CA Technologies*, 2014.
- Winkler, R. and D. Macmillan. "The Secret Math of Airbnb's \$24 Billion Valuation." *The Wall Street Journal*, June 17, 2015.

Information Systems, IT Architecture, Data Governance, and Cloud Computing

CHAPTER OUTLINE

LEARNING OBJECTIVES

Case 2.1 Opening Case: Detoxing Location-Based Advertising Data at MEDIATA

2.1 IS Concepts and Classifications

2.1 Name the six components of an **information** system and match the various types of information systems to the type of support needed by business operations and decision-makers.

2.2 IT Infrastructure, IT Architecture, and Enterprise Architecture

2.2 Describe an IT infrastructure, an IT architecture, and an enterprisewide architecture (EA) and compare and contrast their roles in guiding IT growth and sustaining long-term performance.

2.3 Information Management and Data Governance

2.3 Explain the business benefits of information management and understand the importance of data governance and master data management in providing trusted data that is available when and where needed to support sustainability.

2.4 Data Centers and Cloud Computing

2.4 Understand the concepts of data centers and cloud computing and understand how they add value in an organization.

2.5 Cloud Services and Virtualization

2.5 Describe the different types of cloud services and the various forms of virtualization and understand how they add value in an organization.

Case 2.2 Business Case: Data Chaos Creates Risk

Case 2.3 Video Case: Cloud Computing at Coca-Cola Is Changing Everything

Introduction

One of the most popular business strategies for achieving success is the development of a competitive advantage. Competitive advantage exists when a company has superior resources and capabilities than its competitors that allow it to achieve either a lower cost structure or a differentiated product. For long-term business success, companies strive to develop *sustainable competitive advantages*, or competitive advantages that cannot be easily copied by the competition (Porter, 1998). To stay ahead, corporate leaders must constantly seek new ways to grow their business in the face of rapid technology changes, increasingly empowered consumers and employees, and ongoing changes in government regulation. Effective ways to thrive over the long term are to launch new business **models** and strategies or devise new ways to outperform competitors. Because these new business models, strategies, and performance capabilities will frequently be the result of advances in technology, the company's ability to leverage technological innovation over time will depend on its approach to enterprise IT architecture, information management, and data governance. The *enterprisewide IT architecture*, or simply the **enterprise architecture (EA)**, guides the evolution, expansion, and integration of information systems (ISs), digital technology, and business processes. This guidance enables companies to more effectively leverage their IT capability to achieve maximum competitive advantage and growth over the long term. **Information management** guides the acquisition, custodianship, and distribution of corporate data and involves the management of data systems, technology, processes, and corporate strategy. Data governance, or *information governance*, controls enterprise data through formal policies and procedures. One goal of data governance is to provide employees and business partners with high-quality data they can trust and access on demand.

Dirty data are data of such poor quality that they cannot be trusted or relied upon for decisions.

Bad decisions can result from the analysis of inaccurate data, which is widely referred to as **dirty data**, and lead to increased costs, decreased revenue, and legal, reputational, and performance-related consequences. For example, if data is collected and analyzed based on inaccurate information because advertising was conducted in the wrong location for the wrong audience, marketing campaigns can become highly skewed and ineffective. Companies must then begin costly repairs to their datasets to correct the problems caused by dirty data. This creates a drop in customer satisfaction and a misuse of resources in a firm. One example of an organization taking strides to clean the dirty data collected through inaccurate marketing is the data management platform, MEDIATA, which runs bidding systems and ad location services for firms looking to run ads on websites (see [Table 2.1](#)). Let's see how they did this.

Case 2.1 Opening Case



Courtesy of Billy Ray

MEDIATA

PLATFORM



Detoxing Location-Based Advertising Data at MEDIATA

TABLE 2.1 Opening Case Overview

Company	MEDIATA was launched as Valued Interactive Media (VIM) in 2009. Rebranded in 2013 as MEDIATA
Industry	Communications; Advertising
Product Lines	Wide range of programmatic solutions and products to provide practical solutions for digital marketing campaigns to deliver successful online advertising campaigns to organizations across Australia, Hong Kong, and New Zealand
Digital Technology	Information management and data governance to increase trust and accessibility of data to facilitate a company's vision
Business Vision	Shake up the online advertising industry. Improve transparency and foster greater cooperation between partners
Website	www.mediataplatform.com

Company Overview

MEDIATA uses its audience and media delivery platform to deliver thousands of successful online advertising campaigns across Australia, Hong Kong, and New Zealand. Known as a “programmatic solution specialist,” the MEDIATA platform is truly cutting-edge. It runs bidding systems and ad location services for companies that are looking to run ads on websites and provides its clients with high-impact, fully managed, 100% transparent advertising campaigns that produce results. MEDIATA is committed to shaking up the online advertising industry and is evolving into a fast-growing international business. MEDIATA clients include Qantas, LG, Virgin Money, Konica Minolta, Optus, Carlsberg, Honda, ACCOR Hotels, Air New Zealand, Heinz, Woolworths, Citibank, and JP Morgan.

The Problem

MEDIATA uses IP address data to locate customers and ad effectiveness. Unfortunately, *as much as 80% of ad inventories come with an incorrect location* and MEDIATA realized that this “dirty data” was adversely affecting their business. Location-based advertising provides organizations and companies alike with massive benefits. Target customers can be reached easily and effectively through marketing campaigns tailored specifically for them. For example, utility companies and internet service providers usually have certain areas or regions that they service. Using location-based targeting (see [Figure 2.1](#)), these companies can target television, newspaper, and online display ads to attract new customers. Another benefit includes the reduced waste of running marketing campaigns in unprofitable areas. Firms can choose precisely where their advertisements are displayed without wasting resources on customer segments that will not respond because of location or preference discrepancies.

Advanced data analytics in location-based advertising also allows companies like MEDIATA to reach customers where and when they are in decision-making mode using programmatic bidding algorithms and ad inventories. Browser-based ads use these algorithms to predict which customer segments will click on certain ads at certain times of the day. Automated bidding then ensues, with the ad spot on the page going to the highest bidder (Cailean, 2016). However, the data must be accurate to be useful and MEDIATA realized that their data could be much better than it was. Given the importance of this technology to advertisers and digital advertising agencies, there are overwhelming issues to overcome.

The issues stem from outdated methods of locating Internet users through IP addresses. These old systems do not pinpoint where exactly traffic is coming from, rather they give advertising agencies broad geographic regions to work with, and the ads go to random coordinates within the regions. Since the value of these activities comes from having accurate targeting, the inaccuracies of the antiquated systems severely impact profitability. As targeting regions shrink, information becomes more valuable and accurate, but even small inaccuracies dilute the value of demographic information applied to an audience.

The Solution

In 2016, MEDIATA established a data governance program in which it partnered with Skyhook, a U.S. global location software company to



FIGURE 2.1 Location-based advertising.

improve the effectiveness of MEDIATA's user profile data by more precisely locating IP addresses resolving MEDIATA's challenges related to dirty data. Skyhook's Context Accelerator Hyperlocal IP uses big data analytics to provide over 1 billion IP addresses to advertising platforms and cleaned MEDIATA's dirty data to pinpoint customers within 100 meters, thus increasing ad effectiveness for its clients. Hyperlocal IP achieves this by using big data analytics to provide over 1 billion IP addresses to advertising platforms.

Now, every time a device like a cell phone or laptop requests a location, the on-device software scans for Wi-Fi, GPS, or cell tower data. Combining all of these data points allows Skyhook to provide extremely accurate coordinates and pass this information along to MEDIATA to use.

While this approach still is not perfect, it allows MEDIATA's advertisements to become closer than ever to their target customers. A nine-month study conducted after implementing Skyhook showed that MEDIATA saw a 20% increase in marketing campaign effectiveness.

Creating and employing this data governance system allowed MEDIATA to clean its datasets and create new, effective methods to reach target audiences.

Questions

1. What business challenges did MEDIATA face because of its dirty data?
2. What is the function of location-based advertising?
3. Why is it important to maintain accurate location data?
4. How did Skyhook and data governance enable MEDIATA to achieve its vision?
5. What benefits did MEDIATA achieve as a result of implementing data governance?

Sources: Compiled from Cailean (2016), Schneider (2014), and Schneider (2015).

2.1 IS Concepts and Classification

Information systems (ISs) is a combination of information technology and people's activities using the technology to support business processes, operations, management, and decision-making at different levels of the organization.

IPOS is the cycle of inputting, processing, outputting, and storing information in an information system.

Before we begin to explore the value of **information systems (ISs)** to an organization, it's useful to understand what an IS is, what it does, and what types of ISs are typically found at different levels of an organization.

In addition to supporting decision-making, coordination, and control in an organization, ISs also help managers and workers analyze problems, visualize complex sets of data, and create new products. ISs collect (**input**) and manipulate data (**process**), and generate and distribute reports (**output**) and based on the data-specific IT services, such as processing customer orders and generating payroll, are delivered to the organization. Finally, the ISs save (**store**) the data for future use. In addition to the four functions of **IPOS**, an information needs **feedback** from its users and other stakeholders to help improve future systems as demonstrated in **Figure 2.2**.

The following example demonstrates how the components of the IPOS work together: To access a website, Amanda opens an Internet browser using the keyboard and enters a Web address into the browser (input). The system then uses that information to find the correct website (processing) and the content of the desired site is displayed in the Web browser (output). Next, Amanda bookmarks the desired website in the Web browser for future use (storage). The system then records the time that it took to produce the output to compare actual versus expected performance (feedback).

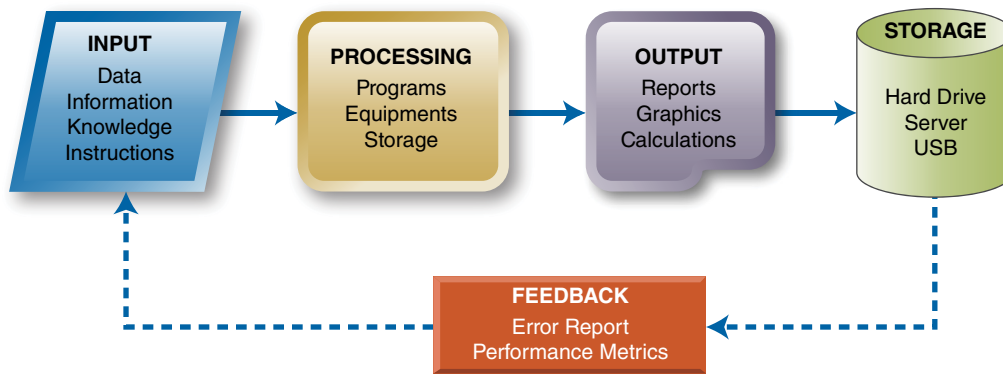


FIGURE 2.2 IPOS cycle.

Components of an IS

A computerized IS consists of six interacting components. Regardless of type and where and by whom they are used within an organization, the components of an IS must be carefully managed to provide maximum benefit to an organization (see [Figure 2.3](#)).

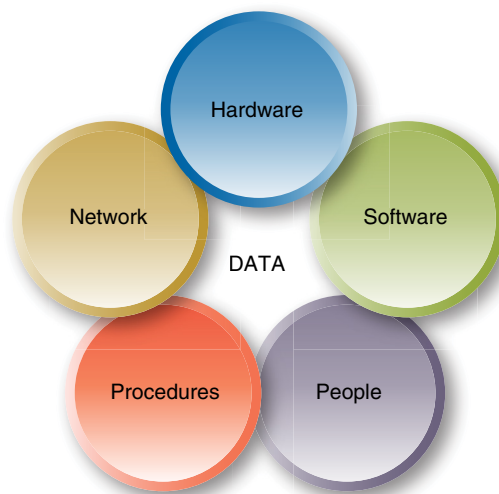


FIGURE 2.3 Components of an IS.

- 1. Hardware** Any physical device used in a computerized IS. Examples include central processing unit (CPU), sound card, video card, network card, hard drive, display, keyboard, motherboard, processor, power supply, modem, mouse, and printer.
- 2. Software** A set of machine-readable instructions (code) that makes up a computer application that directs a computer's processor to perform specific operations. Computer software is nontangible, contrasted with system hardware, which is the physical component of an IS. Examples include Internet browser, operating system (OS), Microsoft Office, Skype, and so on.
- 3. People** Any person involved in using an IS. Examples include programmers, operators help desk, and end-users.
- 4. Procedures** Documentation containing directions on how to use the other components of an IS. Examples include operational manual and user manual.
- 5. Network** A combination of lines, wires, and physical devices connected to each other to create a telecommunications network. In computer networks, networked computing

devices exchange data with each other using a data link. The connections between nodes are established using either cable media or wireless media. Networks can be internal or external. If they are available only internally within an organization, they are called “intranets.” If they are available externally, they are called “internets.” The best-known example of a computer network is the World Wide Web.

- 6. Data** Raw or unorganized facts and figures (such as invoices, orders, payments, customer details, product numbers, product prices) that describe conditions, ideas, or objects.

Data, Information, Knowledge, and Wisdom

Data describe products, customers, events, activities, and transactions that are recorded, classified, and stored.

Information is data that have been processed, organized, or put into context so that they have meaning and value to the person receiving them.

Knowledge adds understanding, experience, accumulated learning, and expertise as they apply to a current problem or activity, to information.

As you can see in Figure 2.3, **data** is the central component of any **information** system. Without data, an IS would have no purpose and companies would be unable to conduct business. Generally speaking, ISs process **data** into meaningful **information** that produces corporate **knowledge** and ultimately creates **wisdom** that fuels corporate strategy.

Data are the raw material from which information is produced; the quality, reliability, and integrity of the data must be maintained for the information to be useful. Data are the raw facts and figures that are not organized in any way. Examples are the number of hours an employee worked in a certain week or the number of new Ford vehicles sold from the first quarter (Q1) of 2015 through the second quarter (Q2) of 2017 (**Figure 2.4**).

Information is an organization’s most important asset, second only to people. Information provides the “who,” “what,” “where,” and “when” of data in a given context. For example,

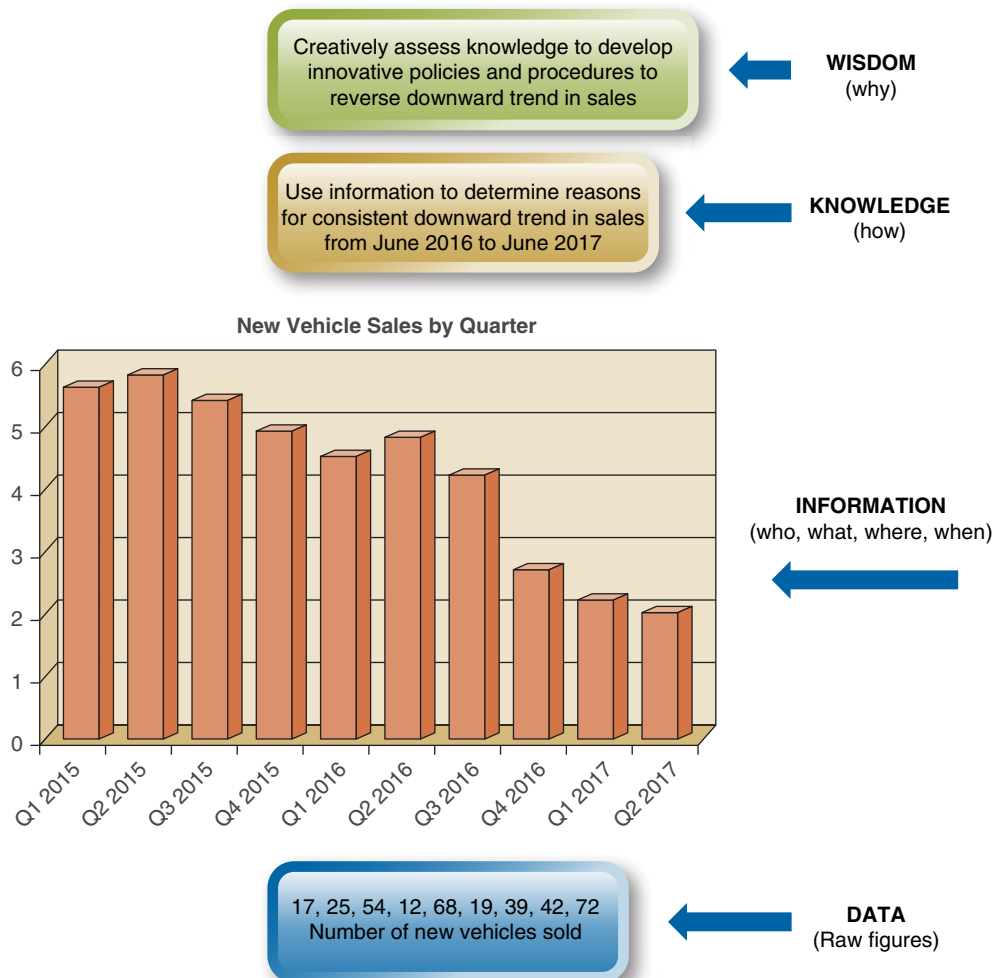


FIGURE 2.4 Examples of data, information, knowledge, and wisdom.

summarizing the quarterly sales of new Ford vehicles from Q1 2015 through Q2 2017 provides information that shows sales have steadily decreased from Q2 2016.

Knowledge is used to answer the question “how.” In our example, it would involve determining how the trend can be reversed, for example, customer satisfaction can be improved, new features can be added, and pricing can be adjusted.

Wisdom is more abstract than data and information (that can be harnessed) and knowledge (that can be shared). Wisdom adds value and increases effectiveness. It answers the “why” in a given situation. In the Ford example, wisdom would be corporate strategists evaluating the various reasons for the sales drop, creatively analyzing the situation as a whole, and developing innovative policies and procedures to reverse the recent downward trend in new vehicle sales.

ISs collect or input and process data to create and distribute reports or other outputs based on information gleaned from the raw data to support decision-making and business processes that, in turn, produce corporate knowledge that can be stored for future use. **Figure 2.5** shows the input-processing-output-storage (IPOS) cycle.

Wisdom is a collection of values, ethics, moral codes, and prior experiences that form an evaluated understanding or common-sense judgment.

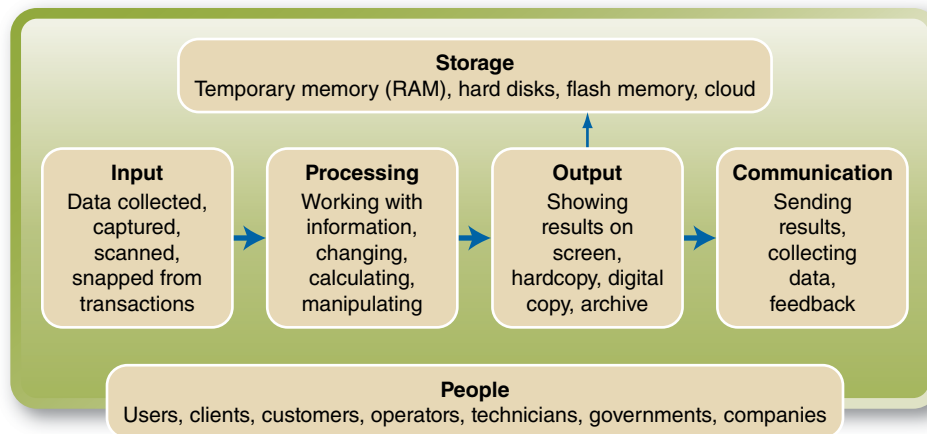


FIGURE 2.5 Input-processing-output-storage model.

Types of ISs

An IS may be as simple as a single computer and a printer used by one person, or as complex as several thousand computers of various types (tablets, desktops, laptops, mainframes) with hundreds of printers, scanners, and other devices connected through an elaborate network used by thousands of geographically dispersed employees. Functional ISs that support business analysts and other departmental employees range from simple to complex, depending on the type of employees supported. The following examples show the support that IT provides to major functional areas.

1. **Marketing** Utilizing IBM software, Bolsa de Comercio de Santiago, a large stock exchange in Chile, is able to process its ever-increasing, high-volume trading in microseconds. The Chilean stock exchange system can do the detective work of analyzing current and past transactions and market information, learning, and adapting to market trends and connecting its traders to business information in real time. Immediate throughput in combination with analytics allows traders to make more accurate decisions.
2. **Sales** According to the *New England Journal of Medicine*, one in five patients suffers from preventable readmissions, which cost taxpayers over \$17 billion a year. In the past, hospitals have been penalized for high readmission rates with cuts to the payments they receive from the government (Zuckerman et al., 2016). Using effective management information systems (MISs), the health-care industry can leverage unstructured information in ways not possible before, according to Matt McClelland, manager of information

governance for Blue Cross Blue Shield of North Carolina. “With proper support, information governance can bridge the gaps among the need to address regulation and litigation risk, the need to generate increased sales and revenue, and the need to cut costs and become more efficient. When done right, information governance positively impacts every facet of the business,” McClelland said in the Information Governance Initiative (Jarousse, 2016).

Figure 2.6 illustrates the classification of the different types of ISs used in organizations, the typical level of workers who use them and the types of input/output (I/O) produced by each of the ISs. At the operational level of the organization, line workers use **transaction processing systems (TPSs)** to capture raw *data* and pass it along (output) to middle managers. The raw data is then input into office automation (OA) and MISs by middle managers to produce *information* for use by senior managers. Next, information is input into **decision support systems (DSSs)** for processing into explicit *knowledge* that will be used by senior managers to direct current corporate strategy. Finally, corporate executives input the explicit knowledge provided by the DSSs into **executive information systems (EISs)** and apply their experience, expertise, and skills to create *wisdom* that will lead to new corporate strategies.

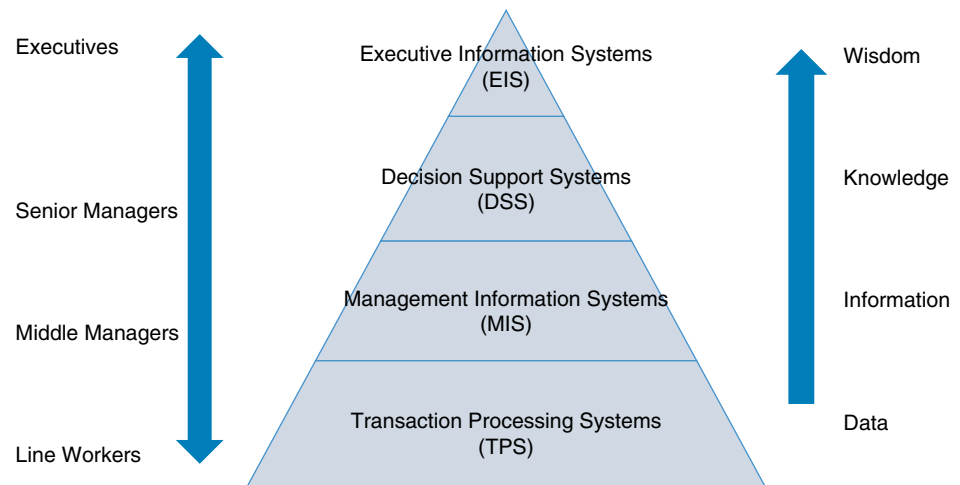


FIGURE 2.6 Hierarchy of ISs, input/output, and user levels.

Transaction Processing System (TPS)

A TPS is designed to process specific types of data input from ongoing transactions. TPSs can be manual, as when data are typed into a form on a screen, or automated by using scanners or sensors to capture barcodes or other data (**Figure 2.7**). TPSs are usually operated directly by frontline workers and provide the key data required to support the management of operations.

Organizational data are processed by a TPS, for example, sales orders, reservations, stock control, and payments by payroll, accounting, financial, marketing, purchasing, inventory control, and other functional departments. The data are usually obtained through the automated or semiautomated tracking of low-level activities and basic transactions. Transactions are either:

- **internal transactions** that originate within the organization or that occur within the organization, for example, payroll, purchases, budget transfers, and payments (in accounting terms, they are referred to as *accounts payable*); or
- **external transactions** that originate from outside the organization, for example, from customers, suppliers, regulators, distributors, and financing institutions.

TPSs are essential systems. Transactions that are not captured can result in lost sales, dissatisfied customers, unrecorded payments, and many other types of data errors with financial



FIGURE 2.7 Scanners automate the input of data into a transaction processing system (TPS).

© Jan_Neville/Stockphoto

impacts. For example, if the accounting department issued a check to pay an invoice (bill) and it was cashed by the recipient, but information about that transaction was not captured, then two things happen. First, the amount of cash listed on the company's financial statements is incorrect because no deduction was made for the amount of the check. Second, the accounts payable (A/P) system will continue to show the invoice as unpaid, so the accounting department might pay it a second time. Likewise, if services are provided, but the transactions are not recorded, the company will not bill for them and thus lose service revenue.

Batch versus Online Real-Time Processing Data captured by a TPS are processed and stored in a **database**; they then become available for use by other systems. Processing of transactions is done in one of two modes:

1. **Batch processing** A TPS in **batch processing** mode collects all transaction for a day, shift, or other time period, and then processes the data and updates the data stores. Pay-roll processing done weekly or bi-weekly is an example of batch mode.
2. **Online transaction processing (OLTP) or real-time processing** The TPS processes each transaction as it occurs, which is what is meant by the term *real-time processing*. In order for OLTP to occur, the input device or website must be directly linked via a network to the TPS. Airlines need to process flight reservations in real time to verify that seats are available.

Batch processing costs less than real-time processing. A disadvantage is that data are inaccurate because they are not updated immediately, in real time.

Processing Impacts Data Quality As data are collected or captured, they are validated to detect and correct obvious errors and omissions. For example, when a customer sets up an account with a financial services firm or retailer, the TPS validates that the address, city, and postal code provided are consistent with one another and also that they match the credit card holder's address, city, and postal code. If the form is not complete or errors are detected, the customer is required to make the corrections before the data are processed any further.

Data errors detected later may be time-consuming to correct or cause other problems. You can better understand the difficulty of detecting and correcting errors by considering identity theft. Victims of identity theft face enormous challenges and frustration trying to correct data about them.

Management Information System (MIS)

An MIS is built on the data provided by TPS. MISs are management-level systems that are used by middle managers to help ensure the smooth running of an organization in the short to medium term. The highly structured information provided by these systems allows managers

to evaluate an organization's performance by comparing current with previous outputs. Functional areas or departments—accounting, finance, production/operations, marketing and sales, human resources, and engineering and design—are supported by ISs designed for their particular reporting needs. General-purpose reporting systems are referred to as **management information systems (MISs)**. Their objective is to provide reports to managers for tracking operations, monitoring, and control.

Typically, a functional system provides reports about such topics as operational efficiency, effectiveness, and productivity by extracting information from databases and processing it according to the needs of the user. Types of reports include the following:

- **Periodic** These reports are created or run according to a pre-set schedule. Examples are daily, weekly, and quarterly. Reports are easily distributed via e-mail, blogs, internal web-sites (called *intranets*), or other electronic media. Periodic reports are also easily ignored if workers do not find them worth the time to review.
- **Exception** **Exception reports** are generated only when something is outside the norm, either higher or lower than expected. Sales in hardware stores prior to a hurricane may be much higher than the norm. Or sales of fresh produce may drop during a food contamination crisis. Exception reports are more likely to be read because workers know that some unusual event or deviation has occurred.
- **Ad hoc, or on demand** **Ad hoc reports** are unplanned reports. They are generated to a mobile device or computer on demand *as needed*. They are generated on request to learn more about a situation, problem, or opportunity.

Reports typically include interactive data visualizations, such as column and pie charts, as shown in **Figure 2.8**.



FIGURE 2.8 Sample report produced by an MIS.

Decision Support System (DSS)

A DSS is a knowledge-based system used by senior managers to facilitate the creation of knowledge and allow its integration into the organization. More specifically, a DSS is an interactive application that supports decision-making by manipulating and building upon the information from an MIS and/or a TPS to generate insights and new information.

Configurations of a DSS range from relatively simple applications that support a single user to complex enterprisewide systems. A DSS can support the analysis and solution of a specific problem, evaluate a strategic opportunity, or support ongoing operations. These systems support unstructured and semistructured decisions, such as make-or-buy-or-outsource decisions, or what products to develop and introduce into existing markets.

Degree of Structure of Decisions Decisions range from structured to unstructured. Structured decisions are those that have a well-defined method for solving and the

data necessary to reach a sound decision. An example of a structured decision is determining whether an applicant qualifies for an auto loan, or whether to extend credit to a new customer—and the terms of those financing options. **Structured decisions** are relatively straightforward and made on a regular basis, and an IS can ensure that they are done consistently.

At the other end of the continuum are **unstructured decisions** that depend on human intelligence, knowledge, and/or experience—as well as data and models to solve. Examples include deciding which new products to develop or which new markets to enter. Semistructured decisions fall in the middle of the continuum. DSSs are best suited to support these types of decisions, but they are also used to support unstructured ones. To provide such support, DSSs have certain characteristics to support the decision-maker and the overall decision-making process.

The main characteristic that distinguishes a DSS from an MIS is the inclusion of models. Decision-makers can manipulate models to conduct experiments and sensitivity analyses, for example, *what-if* and *goal seeking*. **What-if analysis** refers to changing assumptions or data in the model to observe the impacts of those changes on the outcome. For example, if sales forecasts are based on a 5% increase in customer demand, a what-if analysis would replace the 5% with higher and/or lower estimates to determine *what* would happen to sales *if* demand changed. With **goal seeking**, the decision-maker has a specific outcome in mind and needs to determine how that outcome could be achieved and whether it is feasible to achieve that desired outcome. A DSS can also estimate the risk of alternative strategies or actions.

California Pizza Kitchen (CPK) uses a DSS to support inventory decisions. CPK has over 200 locations in 32 U.S. states and 13 other countries, including 17 California Pizza Kitchen non-traditional, franchise concepts designed for airports, universities, and stadiums. Maintaining optimal inventory levels at all its restaurants was challenging and time-consuming. The original MIS was replaced by a DSS to make it easy for the chain's managers to maintain updated records, generate reports as and when needed, and make corporate- and restaurant-level decisions. Many CPK restaurants reported a 5% increase in sales after the DSS was implemented.

Executive Information System (EIS)

EISs are strategic-level information systems that help executives and senior managers analyze the environment in which the organization exists. They typically are used to identify long-term trends and to plan appropriate courses of action. The information in such systems is often weakly structured and comes from both internal and external sources. EISs are designed to be operated directly by executives without the need for intermediaries and easily tailored to the preferences of the individual using them. An EIS organizes and presents data and information from both external data sources and internal MIS or TPS in an easy-to-use dashboard format to support and extend the inherent capabilities of senior executives.

Initially, EISs were custom-made for an individual executive. However, a number of off-the-shelf EIS packages now exist and some enterprise-level systems offer a customizable EIS module.

The ways in which the different characteristics of the various types of ISs are classified is shown in [Table 2.2](#).

Here's an example of how these ISs are used together to add value in an organization. Day-to-day transaction data collected by the TPS are converted into prescheduled summarized reports by middle managers using an MIS. The findings in these reports are then analyzed by senior managers who use a DSS to support their semistructured or unstructured decision-making. DSSs contain models that consist of a set of formulas and functions, such as statistical, financial, optimization, and/or simulation models. Corporations, government agencies, the military, health care, medical research, major league sports, and nonprofits depend on their DSSs to answer what-if questions to help reduce waste in production operations, improve inventory management, support investment decisions, and predict demand and help sustain a competitive edge.

Customer data, sales, and other critical data produced by the DSS are then selected for further analysis, such as trend analysis or forecasting demand and are input into an EIS for

TABLE 2.2 Characteristics of Types of Information Systems

Type	Characteristics
TPS	Used by operations personnel Produce information for other ISs Use internal and external data Efficiency oriented
MIS	Used by lower and middle managers Based on internal information Support structured decisions Inflexible Lack analytical capabilities Focus on past and present data
DSS	Used by senior managers Support semistructured or unstructured decisions Contain models or formulas that enable sensitivity analysis, what-if analysis, goal seeking, and risk analysis Use internal and external data plus data added by the decision-maker who may have insights relevant to the decision situation Predict the future
EIS	Used by C-level managers Easy-to-use, customizable interface Support unstructured decisions Use internal and external data sources Focus on effectiveness of the organization Very flexible Focus on the future

use by top level management, who add their experience and expertise to make unstructured decisions that will affect the future of the business.

Figure 2.9 shows how the major types of ISs relate to one another and how data flow among them. In this example,

1. Data from online purchases are captured and processed by the TPS and then stored in the transactional database.
2. Data needed for reporting purposes are extracted from the database and used by the MIS to create periodic, ad hoc, or other types of reports.
3. Data are output to a DSS where they are analyzed using formulas, financial ratios, or models.

ISs Exist within Corporate Culture

It is important to remember that ISs do not exist in isolation. They have a purpose and a social (organizational) context. A common *purpose* is to provide a solution to a business problem. The *social context* of the system consists of the values and beliefs that determine what is admissible and possible within the culture of the organization and among the people involved. For example, a company may believe that superb customer service and on-time delivery are critical success factors. This belief system influences IT investments, among other factors.

The business value of IT is determined by the people who use them, the business processes they support, and the culture of the organization. That is, IS value is determined by the

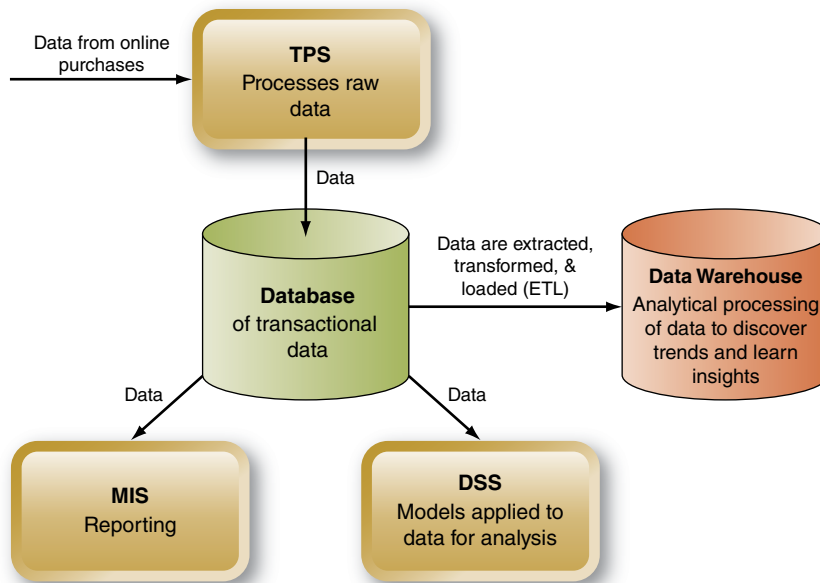


FIGURE 2.9 Flow of data from point of sale (POS) through processing, storage, reporting, decision support, and analysis. Also shows the relationships among different types of ISs.

relationships among ISs, people, and business processes—all of which are influenced strongly by organizational culture.

In an organization, there may be a culture of distrust between the technology and business employees. No enterprise IT architecture methodology or data governance can bridge this divide unless there is a genuine commitment to change. That commitment must come from the highest level of the organization—senior management. Methodologies cannot solve people problems; they can only provide a framework in which those problems can be solved.

Questions

1. Name the six components of an IS.
2. Describe the differences between data, information, knowledge, and wisdom.
3. Define TPS and give an example.
4. Explain why TPSs need to process incoming data before they are stored.
5. Define MIS and DSS and give an example of each.
6. What characteristic distinguishes a DSS from an MIS?
7. What level of personnel typically uses an EIS?
8. What factors determine IS value?

2.2 IT Infrastructure, IT Architecture, and Enterprise Architecture

Every enterprise has a core set of ISs and business processes that execute the transactions that keep it in business. Transactions include processing orders, order fulfillment and delivery, purchasing inventory and supplies, hiring and paying employees, and paying bills. To most effectively utilize its IT assets, an organization must create an IT infrastructure, IT architecture, and an enterprise architecture (EA) as shown in **Figure 2.10**.

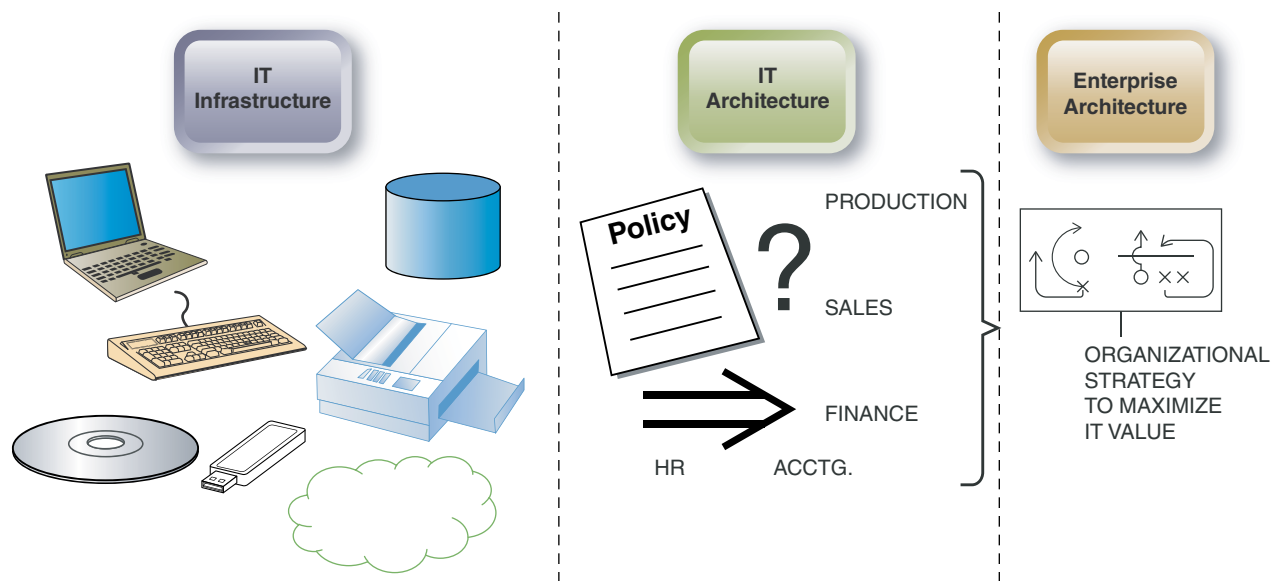


FIGURE 2.10 Comparing IT infrastructure, IT architecture, and enterprise architecture.

IT infrastructure is an inventory of the physical IT devices that an organization owns and operates. The IT infrastructure describes an organization's entire collection of hardware, software, networks, **data centers**, facilities, and other related equipment used to develop, test, operate, manage, and support IT services. It does **NOT** include the people or process components of an information system.

IT architecture guides the process of planning, acquiring, building, modifying, interfacing, and deploying IT resources in a single department within an organization. The IT architecture should offer a way to systematically identify technologies that work together to satisfy the needs of the departments' users. The IT architecture is a blueprint for how future technology acquisitions and deployment will take place. It consists of standards, investment decisions, and product selections for hardware, software, and communications. The IT architecture is developed first and foremost based on department direction and business requirements.

Enterprise architecture (EA) reviews all the information systems across all departments in an organization to develop a strategy to organize and integrate the organization's IT infrastructures to help it meet the current and future goals of the enterprise and maximize the value of technology to the organization. In this way, EA provides a holistic view of an organization with graphic and text descriptions of strategies, policies, information, ISs, and business processes and the relationships between them.

The EA adds value in an organization in that it can provide the basis for organizational change just as architectural plans guide a construction project. Since a poorly crafted **enterprise architecture (EA)** can also hinder day-to-day operations and efforts to execute business strategy, it is more important than ever before to carefully consider the EA within your organization when deciding on an approach to business, technology, and corporate strategy. Simply put, EA helps solve two critical challenges: **where** an organization is going, and **how** it will get there.

The success of EA is measured not only in financial terms, such as profitability and return on investment (ROI), but also in nonfinancial terms, for example, improved customer satisfaction, faster speed to market, and lower employee turnover as diagrammed in [Figure 2.11](#) and demonstrated in [IT at Work 2.1](#).

EA Helps to Maintain Sustainability

As you read in Chapter 1, the volume, variety, and speed of data being collected or generated have increased dramatically over the past decade. As enterprise ISs become more complex,

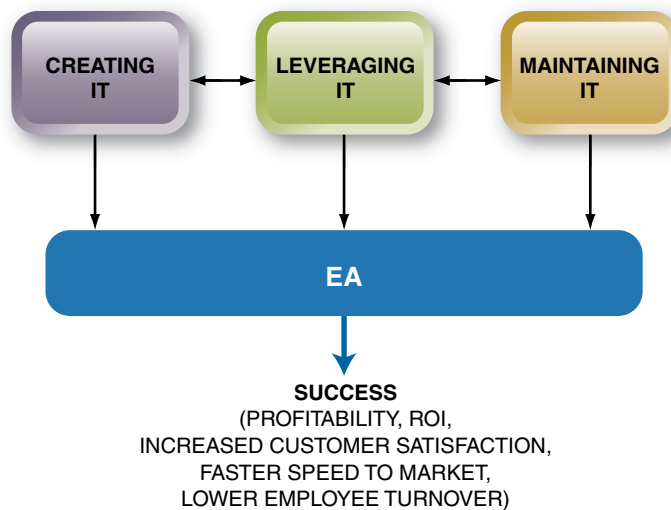


FIGURE 2.11 Enterprise architecture success.

IT at Work 2.1

A New Enterprise Architecture Improves Data Quality and EIS Use

Executives at a large chemical corporation were supported by an IS specifically designed for their needs—called an executive information system (EIS). The EIS was designed to provide senior managers with internal and external data and key performance indicators (KPIs) that were relevant to their specific needs. [Tech Note 2.1](#) describes KPIs. As with any system, the value of the EIS depends on the data quality.

Too Much Irrelevant Data

Unfortunately, the EIS was a failure. Executives soon discovered that only half of the data available through the EIS related to their level of analysis and decision-making at the corporate level. A worse problem was that the data they needed were not available when and how they wanted them. For example, executives needed to analyze current detailed sales revenue and cost data for every strategic business unit (SBU), product line, and operating business to compare performance. But, data were not in standardized format as needed, making analysis difficult or impossible. A large part of the problem was that SBUs reported sales revenues in different time frames (e.g., daily, weekly, monthly, or quarterly), and many of those reports were not available when needed. As a result, senior management could not get a *trusted* view of the company's current overall performance and did not know which products were profitable.

There were two reasons for the failure of the EIS:

1. IT architecture was not designed for customized reporting

The design of the IT architecture had been based on financial accounting rules. That is, the data were organized to make it easy to collect and consolidate the data needed to prepare financial statements and reports that had to be submitted to the SEC (Securities and Exchange Commission) and other regulatory agencies. These statements and reports have well-defined or standardized formats and only need to be prepared at specific times during the year, typically annually or quarterly. The organization of the data (for financial reporting)

did not have the flexibility needed for the customized ad hoc (unplanned) data needs of the executives. For example, it was nearly impossible to generate customized sales performance (nonfinancial) reports or do ad hoc analyses, such as comparing inventory turnover rates by product for each region for each sales quarter. Because of lags in reports from various SBUs, executives could not trust the underlying data.

2. **Complicated user interface** Executives could not easily review the KPIs. Instead, they had to sort through screens packed with too much data—some of interest and some irrelevant. To compensate for poor interface design, several IT analysts themselves had to do the data and KPI analyses for the executives—delaying response time and driving up the cost of reporting.

Solution: New Enterprise Architecture with Standardized Data Formats

The CIO worked with a task force to design and implement an entirely new EA. Data governance policies and procedures were implemented to standardize data formats companywide. Data governance eliminated data inconsistencies to provide reliable KPI reports on inventory turns, cycle times, and profit margins of all SBUs.

The new architecture was business-driven instead of financial reporting-driven. It was easy to modify reports—eliminating the costly and time-consuming ad hoc analyses. Fewer IT resources are needed to maintain the system. Because the underlying data are now relatively reliable, EIS use by executives increased significantly.

IT at Work Questions

1. Why was an EIS designed and implemented at the large chemical corporation?
2. What problems did the executives have with the EIS?
3. What were the two reasons for those EIS problems?
4. How did the CIO improve the EIS?
5. What are the benefits of the new IT enterprise architecture?
6. What are the benefits of data governance?

Tech Note 2.1

Key Performance Indicators (KPIs)

KPIs are a set of quantifiable measures used to evaluate factors that are crucial to the success of an organization. KPIs present data in easy-to-comprehend and comparison-ready formats to gauge or compare performance in terms of meeting an organization's operational and strategic goals. KPIs are used in four main areas: increasing revenue; reducing costs; improving process cycle-time; and improving customer satisfaction. Examples of key comparisons include actual versus budget, actual versus forecasted, and the ROI for this year versus prior years. KPIs help reduce the complex nature of organizational performance to a small number of understandable measures, including:

- Financial KPIs: accounts payable turnover; inventory turnover; net profit margin; sum of difference between planned and actual project budgets
- Social media KPIs: social traffic and conversions (number of visitors who are converted to customers); likes; new followers per week; social visits and leads
- Sales and marketing KPIs: cost per lead; how much revenue a marketing campaign generates; number of customer complaints; cycle time from customer request to delivery, percentage of correspondence replied to on time
- Operational and supply chain KPIs: units per transaction; carrying cost of inventory; order status; back order rate
- Environmental and carbon-footprint KPIs: energy, water, or other resource use; spend by utility; weight of landfill waste.

long-range IT planning is critical. Companies cannot simply add storage, new apps, or data analytics on an as-needed basis and expect those additional IT assets to work with existing systems.

The relationship between complexity and planning for the future is easier to see in physical things such as buildings and transportation systems. For example, if you are constructing a simple holiday cabin in a remote area, there is no need to create a detailed plan for future expansion. On the other hand, if you are building a large commercial development in a highly populated area, you're not likely to succeed without a detailed project plan. Relating this to the case of enterprise ISs, if you are building a simple, single-user, nondistributed system, you would not need to develop a well-thought-out growth plan. However, this approach would not be feasible to enable you to successfully manage big data, copious content from mobiles and social networks, and data in the cloud. Instead, you would need a well-designed set of plans, or blueprints, provided by an EA to align IT with business objectives by guiding and controlling hardware acquisition, software add-ons and upgrades, system changes, network upgrades, choice of cloud services, and other digital technology investments that you will need to make your business sustainable.

There are two specific strategic issues that the EA is designed to address:

- 1. IT systems' complexity** IT systems have become unmanageably complex and expensive to maintain.
- 2. Poor business alignment** Organizations find it difficult to keep their increasingly expensive IT systems aligned with business needs.

Business and IT Benefits of EA Having the right EA in place is important for the following reasons:

- EA cuts IT costs and increases productivity by giving decision-makers access to information, insights, and ideas where and when they need them.
- EA determines an organization's competitiveness, flexibility, and IT economics for the next decade and beyond. That is, it provides a long-term view of a company's processes, systems, and technologies so that IT investments do not simply fulfill immediate needs.
- EA helps align IT capabilities with business strategy—to grow, innovate, and respond to market demands, supported by an IT practice that is 100% in accord with business objectives.
- EA can reduce the risk of buying or building systems and enterprise applications that are incompatible or unnecessarily expensive to maintain and integrate.

Developing an Enterprise Architecture (EA)

Developing an EA starts with the organization's goals, for example, *where does it want to be in three years?* and identifies the strategic direction in which it is heading and the business drivers to which it is responding. The goal is to make sure that everyone understands and shares a single vision. As soon as managers have defined this single shared vision of the future, they then consider the impact this vision will have on the business, technical, information, and solutions architectures of the enterprise. This shared vision of the future will dictate changes in all these architectures, assign priorities to those changes, and keep those changes grounded in business value.

According to Microsoft, the EA should include the four different perspectives shown in [Table 2.3](#).

TABLE 2.3 Components of an Enterprise Architecture

Business architecture	How the business works. Includes broad business strategies and plans for moving the organization from where it is now to where it wants to be. Processes the business uses to meet its goals.
Application architecture	Portfolio of organization's applications. Includes descriptions of automated services that support business processes; descriptions of interactions and interdependencies between the organization's ISs.
Information architecture	What the organization needs to know to perform its business processes and operations. Includes standard data models; data management policies and descriptions of patterns of information production and use in an organization.
Technology architecture	Hardware and software that supports the organization. Examples include desktop and server software; OSs; network connectivity components; printers, modems.

It is important to recognize that the EA must be dynamic, not static. To sustain its effectiveness, it should be an ongoing process of aligning the creation, operation, and maintenance of IT across the organization with the ever-changing business objectives. As business needs change, so must the EA, as demonstrated in [IT at Work 2.2](#).

IT at Work 2.2

EA Must Be Dynamic and Evolving

In order to keep IT aligned with the business, the EA must be a dynamic plan. As shown in the model in [Figure 2.12](#), the EA evolves toward the target architecture, which represents the company's future IT needs. According to this model, EA defines the following:

1. The organization's mission, business functions, and future direction
2. Information and information flows needed to perform the mission
3. The current baseline architecture
4. The desired target architecture
5. The sequencing plan or strategy to progress from the baseline to the target architecture.

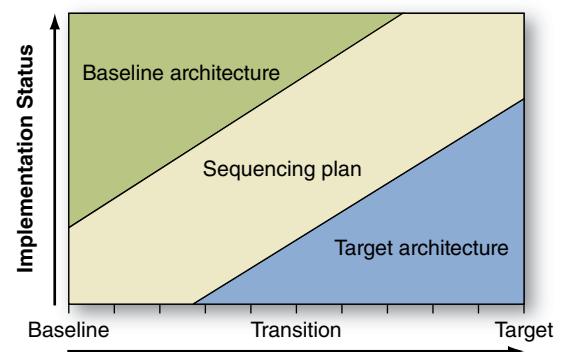


FIGURE 2.12 The importance of viewing EA as a dynamic and evolving plan The purpose of the EA is to maintain IT-business alignment. Changes in priorities and business are reflected in the target architecture to help keep IT aligned with them (Bloomberg, 2016).

Career Insight 2.1

Essential Skills of an Enterprise Architect (EA)

Enterprise architects need much more than technology skills. On a daily basis, an enterprise architect's activities can change quickly and significantly. Ideally, enterprise architects should come from a highly technical background. Even though enterprise architects deal with many other factors besides technology, it is still important to keep technical skills current. The job performance and success of such an architect—or anyone responsible for large-scale IT projects—depend on a broad range of skills.

- **Interpersonal or people skills** The job requires interacting with people and getting their cooperation.
- **Ability to influence and motivate** A large part of the job is motivating users to comply with new processes and practices.
- **Negotiating skills** The project needs resources—time, money, and personnel—that must be negotiated to get things accomplished.

- **Critical-thinking and problem-solving skills** Architects face complex and unique problems. Being able to expedite solutions prevents bottlenecks.
- **Business and industry expertise** Knowing the business and industry improves the outcomes and the architect's credibility.
- **Process orientation** Thinking in terms of process is essential for an enterprise architect. Building repeatable and reusable processes as artifacts from the work they do and how they work themselves.

The most common function an enterprise architect will perform is that of overseeing a large-scale program. Programs are a group of related projects and as such, managing EA implementations requires someone who is able to handle multiple aspects of a project at one time. Project management is covered in Chapter 13.

Questions

1. What is the purpose of the IT infrastructure?
2. How is the IT infrastructure different from the IT architecture?
3. What is the purpose of an EA?
4. What are the business benefits of EA?
5. Explain why it is necessary to ensure that an EA maintains alignment between IT and business strategy?
6. Explain KPIs and give an example.

2.3 Information Management and Data Governance

As shown in Figure 2.3, data is the heart of the business and the central component of an IS. Most business initiatives succeed or fail based on the quality of their data. Effective planning and decision-making depend on systems being able to make data available in usable formats on a timely basis. Almost everyone manages information. You manage your social and cloud accounts across multiple mobile devices and computers. You update or synchronize (“synch”) your calendars, appointments, contact lists, media files, documents, and reports. Your productivity depends on the compatibility of devices and applications and their ability to share data. Not being able to transfer and synch whenever you add a device or app is bothersome and wastes your time. For example, when you switch to the latest mobile device, you might need to reorganize content to make dealing with data and devices easier. To simplify add-ons, upgrades, sharing, and access, you might leverage cloud services such as iTunes, Instagram, Diigo, and Box.

This is just a glimpse at some of the **information management** situations that organizations face today and shows why a continuous plan is needed to guide, control, and govern IT growth. As with building construction (**Figure 2.13**), blueprints and models help guide and govern future IT and digital technology investments.

Information management is the use of IT tools and methods to collect, process, consolidate, store, and secure data from sources that are often fragmented and inconsistent.



© Martin Barraud/Alamy

FIGURE 2.13 Blueprints and models, like those used for building construction, are needed to guide and govern an enterprise's IT assets.

Information Management Harnesses Scattered Data

Business information is generally scattered throughout an enterprise, stored in separate systems dedicated to specific purposes, such as operations, supply chain management, or customer relationship management. Major organizations have over 100 data repositories (storage areas). In many companies, the integration of these disparate systems is limited—as is users' ability to access all the information they need. As a result, despite all the information flowing through companies, executives, managers, and workers often struggle to find the information they need to make sound decisions or do their jobs. The overall goal of information management is to eliminate that struggle through the design and implementation of a sound data governance program and a well-planned EA.

Providing easy access to large volumes of information is just one of the challenges facing organizations. The days of simply managing structured data are over. Now, organizations must manage semistructured and unstructured content from social and mobile sources even though that data may be of questionable quality.

Information management is critical to data security and compliance with continually evolving regulatory requirements, such as the Sarbanes-Oxley Act, Basel III, the Computer Fraud and Abuse Act (CFAA), the USA PATRIOT Act, and the Health Insurance Portability and Accountability Act (HIPAA).

Issues of information access, management, and security must also deal with information degradation and disorder—where people do not understand what data mean or how the data can be useful.

Reasons for Information Deficiencies

Organizational information and decision support technologies have developed over many decades. During that time management teams' priorities have changed along with their understanding of the role of IT within the organization; technology has advanced in unforeseeable ways, and IT investments have been increased or decreased based on competing demands on the budget. Other common reasons why information deficiencies are still a problem include:

1. **Data silos** Information can be *trapped* in departmental **data silos** (also called *information silos*), such as marketing or production databases. Data silos are illustrated in **Figure 2.14**. Since silos are unable to share or exchange data, they cannot consistently be updated. When data are inconsistent across multiple enterprise applications, data quality cannot (and should not) be trusted without extensive verification. Data silos exist when there is no overall IT architecture to guide IT investments, data coordination, and communication. Data silos support a single function and, as a result, do not support an organization's cross-functional needs.

Data silo are stand-alone data stores. Their data are not accessible by other ISs that need it or outside that department.

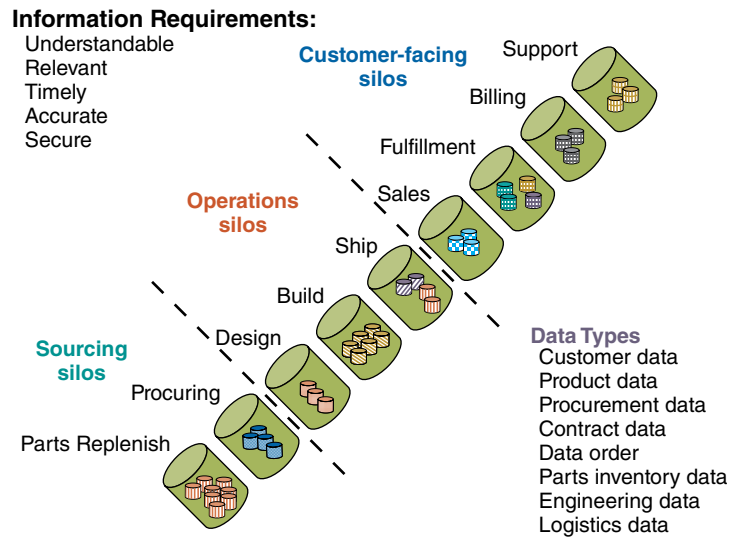


FIGURE 2.14 Data (or information) silos are ISs that do not have the capability to exchange data with other systems, making timely coordination and communication across functions or departments difficult.

For example, most health-care organizations are drowning in data, yet they cannot get reliable, actionable insights from these data. Physician notes, registration forms, discharge summaries, documents, and more are doubling every five years. Unlike structured machine-ready data, these are messy data that take too much time and effort for health-care providers to include in their business analysis. So, valuable messy data are routinely left out. Millions of insightful patient notes and records sit inaccessible or unavailable in separate clinical data silos because historically there has been no easy way to analyze the information they contain.

2. **Lost or bypassed data** Data can get lost in transit from one system to another. Or, data might never get captured because of inadequately tuned data collection systems, such as those that rely on sensors or scanners. Or, the data may not get captured in sufficient detail, as described in [Tech Note 2.2](#).
3. **Poorly designed interfaces** Despite all the talk about user-friendly interfaces, some ISs are horrible to deal with. Poorly designed interfaces or formats that require extra time and effort to figure out increase the risk of errors from misunderstanding the data or ignoring them.
4. **Nonstandardized data formats** When users are presented with data in inconsistent or nonstandardized formats, errors increase. Attempts to compare or analyze data are more difficult and take more time. For example, if the Northeast division reports weekly gross sales revenues per product line and the Southwest division reports monthly net sales per product, you cannot compare their performance without converting the data to a common format. Consider the extra effort needed to compare temperature-related sales, such as air conditioners, when some temperatures are expressed in degrees Fahrenheit and others in Centigrade.
5. **Difficult to hit moving targets** The information that decision-makers want keeps changing—and changes faster than ISs can respond to because of the first four reasons in this list. Tracking tweets, YouTube hits, and other unstructured content requires expensive investments—which managers find risky in an economic downturn.

These are the data challenges managers have to face when there is little or no information management. Companies undergoing fast growth or merger activity or those with decentralized systems (each division or business unit manages its own IT) will end up with a patchwork of reporting processes. As you would expect, patchwork systems are more complicated to modify, too rigid to support an agile business, and more expensive to maintain.

Tech Note 2.2

Need to Measure in Order to Manage

A residential home construction company had two divisions: standard homes and luxury homes. The company was not capturing material, labor, and other costs associated with each type of construction. Instead, these costs were pooled, making it impossible to allocate costs to each type of construction and then to calculate the profit margins of each division. They had no way of calculating profit margins on each type of home within the divisions. Without the ability to measure costs, they did not have any cost control.

After upgrading their ISs, they began to capture detailed data at the division level. They discovered a wide profit margin on standard homes, which was hiding the losses occurring in the luxury home division. Without cost control data, the profitable standard homes division had been subsidizing the luxury home division for many years. Based on the cost control data, the company decided to focus more on standard homes and adjust their pricing on luxury homes. This new cost control strategy increased the company's long-term performance.

Factors Driving the Shift from Silos to Sharing and Collaboration

Senior executives and managers are aware of the problems associated with their data silos and information management problems, but they also know about the huge cost and disruption associated with converting to newer IT architectures. The “silo effect” occurs when different departments of an organization do not share data and/or communicate effectively enough to maintain productivity. Surprisingly, 75% of employers believe team work and collaboration are essential, but only 18% of employees receive communication evaluations during performance critiques (Marchese, 2016). In the new age of efficiency of service, many companies like Formaspace, an industrial manufacturing and service corporation, must work toward complete cloud integration of old silos to increase customer service and generate more revenue. Enabling applications to interact with one another in an automated fashion to gain better access to data increases meaningful productivity and decreases time and effort spent in manual collaboration efforts. In an illustration of how silo integration is essential for a modern corporation, IT technician at Formaspace, Loddie Alspach, claims that in 2015, the company managed to increase revenues by 20% using Amazon-based cloud technology (Shore, 2015). However, companies are struggling to integrate thousands of siloed global applications, while aligning them to business operations. To remain competitive, they must be able to analyze and adapt their business processes quickly, efficiently, and without disruption.

Greater investments in collaboration technologies have been reported by the research firm Forrester (Keitt, 2014). A recent study identified four main factors that have influenced the increased use of cloud technologies, as shown in **Table 2.4** (Rai et al., 2015).

TABLE 2.4 Key Factors Leading to Increased Migration to the Cloud

Cost Savings
Efficient Use of Resources
Unlimited Scalability of Resources
Lower Maintenance

Business Benefits of Information Management

Based on the examples you have read, the obvious benefits of information management are:

- 1. Improves decision quality** Decision quality depends on accurate and complete data.
- 2. Improves the accuracy and reliability of management predictions** It is essential for managers to be able to predict sales, product demand, opportunities, and competitive

threats. Management predictions focus on “what is going to happen” as opposed to financial reporting on “what has happened.”

3. **Reduces the risk of noncompliance** Government regulations and compliance requirements have increased significantly in the past decade. Companies that fail to comply with laws on privacy, fraud, anti-money laundering, cybersecurity, occupational safety, and so on face harsh penalties.
4. **Reduces the time and cost** of locating and integrating relevant information.

Data Governance: Maintaining Data Quality and Cost Control

The success of every data-driven strategy or marketing effort depends on data governance. **Data governance** policies must address structured, semistructured, and unstructured data (discussed in Section 2.3) to ensure that insights can be trusted.

Data governance is the control of enterprise data through formal policies and procedures to help ensure data can be trusted and are accessible.

Enterprisewide Data Governance With an effective data governance program, managers can determine where their data are coming from, who owns them, and who is responsible for what—in order to know they can trust the available data when needed. Data governance is an enterprisewide project because data cross boundaries and are used by people throughout the enterprise. New regulations and pressure to reduce costs have increased the importance of effective data governance. Governance eliminates the cost of maintaining and archiving bad, unneeded, or inaccurate data. These costs grow as the volume of data grows. Governance also reduces the legal risks associated with unmanaged or inconsistently managed information.

Three industries that depend on data governance to comply with regulations or reporting requirements are the following:

- **Food industry** In the food industry, data governance is required to comply with food safety regulations. Food manufacturers and retailers have sophisticated control systems in place so that if a contaminated food product, such as spinach or peanut butter, is detected, they are able to trace the problem back to a particular processing plant or even the farm at the start of the food chain.
- **Financial services industry** In the financial services sector, strict reporting requirements of the Dodd–Frank Wall Street Reform and Consumer Protection Act of 2010 are leading to greater use of data governance. The Dodd–Frank Act regulates Wall Street practices by enforcing transparency and accountability in an effort to prevent another significant financial crisis like the one that occurred in 2008.
- **Health-care industry** Data are health care’s most valuable asset. Hospitals have mountains of electronic patient information. New health-care accountability and reporting obligations require data governance models for transparency to defend against fraud and to protect patients’ information.

Master Data and Master Data Management (MDM) **Master data** is the term used to describe business-critical information on customers, products and services, vendors, locations, employees, and other things needed for operations and business transactions. Master data are fundamentally different from the high volume, velocity, and variety of big data and traditional data. For example, when a customer applies for automobile insurance, data provided on the application become the master data for that customer. In contrast, if the customer’s vehicle has a device that sends data about his or her driving

behavior to the insurer, those machine-generated data are transactional or operational, but not master data.

Data are used in two ways—both depend on high-quality trustworthy data:

1. **For running the business** Transactional or operational use
2. **For improving the business** Analytic use

Master data are typically quite stable and typically stored in a number of different systems spread across the enterprise. **Master data management (MDM)** links and synchronizes all critical data from those disparate systems into one file called a **master file**, to provide a common point of reference. MDM solutions can be complex and expensive. Given their complexity and cost, most MDM solutions are out of reach for small and medium companies. Vendors have addressed this challenge by offering cloud-managed MDM services. For example, in 2013, Dell Software launched its next-generation Dell Boomi MDM. Dell Boomi provides MDM, data management, and data quality services (DQS)—and they are 100% cloud-based with near real-time synchronization.

Data governance and MDM manage the availability, usability, integrity, and security of data used throughout the enterprise. Strong data governance and MDM are needed ensure data are of sufficient quality to meet business needs. The characteristics and consequences of weak or nonexistent data governance are listed in **Table 2.5**.

Data governance and MDM are a powerful combination. As data sources and volumes continue to increase, so does the need to manage data as a strategic asset in order to extract its full value. Making business data consistent, trusted, and accessible across the enterprise is a critical first step in **customer-centric** business models. With data governance, companies are able to extract maximum value from their data, specifically by making better use of opportunities that are buried within behavioral data.

TABLE 2.5

Characteristics and Consequences of Weak or Nonexistent Data Governance and MDM

- Data duplication causes isolated data silos.
- Inconsistency exists in the meaning and level of detail of data elements.
- Users do not trust the data and waste time verifying the data rather than analyzing them for appropriate decision-making.
- Leads to inaccurate data analysis.
- Bad decisions are made on perception rather than reality, which can negatively affect the company and its customers.
- Results in increased workloads and processing time.

Questions

1. What is information management?
2. What is the “silo effect” and how does it affect business performance?
3. What three factors are driving collaboration and information sharing?
4. What are the business benefits of information management?
5. Explain why it is important to develop an effective data governance program?
6. Explain the purposes of master data management.
7. Why has interest in data governance and MDM increased?

2.4 Data Centers and Cloud Computing

Data centers and cloud computing are types of **IT infrastructures** or computing systems. Data center also refers to the building or facility that houses the servers and equipment. In the past, there were few IT infrastructure options. Companies owned their servers, storage, and network components to support their business applications and these computing resources were on their premises. Now, there are several choices for an IT infrastructure strategy—including cloud computing. As is common to IT investments, each infrastructure configuration has strengths, weaknesses, and cost considerations.

Data Centers

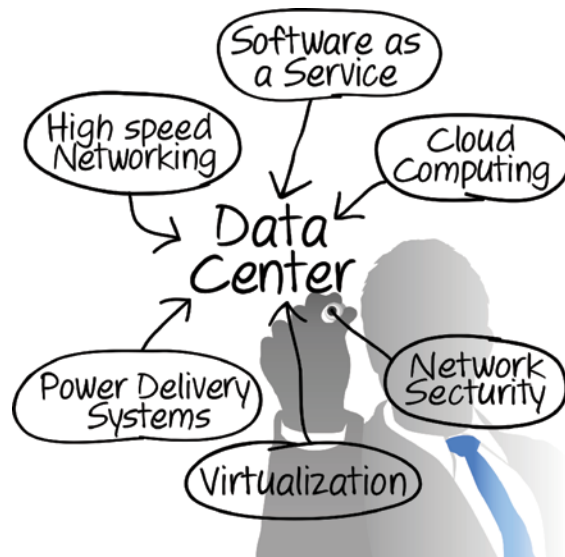
Traditionally, data and database technologies were kept in data centers that were typically run by an in-house IT department (**Figure 2.15**) and consisted of on-premises hardware and equipment that store data within an organization's local area network.

Today, companies may own and manage their own on-premises data centers or pay for the use of their vendors' data centers, such as in cloud computing, virtualization, and software-as-a-service arrangements (**Figure 2.16**).



© Oleksiy Mark/Shutterstock

FIGURE 2.15 A row of network servers in a data center.



© Michael D Brown/Shutterstock

FIGURE 2.16 Data centers are the infrastructure underlying cloud computing, virtualization, networking, security, delivery systems, and software-as-a-service.

In an on-premises data center connected to a local area network, it is easier to restrict access to applications and information to authorized, company-approved people and equipment. In the cloud, the management of updates, security, and ongoing maintenance are outsourced to a third-party cloud provider where data is accessible to anyone with the proper credentials and Internet connection. This arrangement can make a company more vulnerable since it increases exposure of company data at many more entry and exit points. Here are some examples of data centers.

- **National Climatic Data Center** The National Climatic Data Center is an example of a public data center that stores and manages the world's largest archive of weather data.
- **U.S. National Security Agency** The National Security Agency's (NSA) data center, shown in [Figure 2.17](#) is located in Bluffdale, UT. It is the largest spy data center for the NSA. People who think their correspondence and postings through sites like Google, Facebook, and Apple are safe from prying eyes should rethink that belief. You will read more about reports exposing government data collection programs in Chapter 5.
- **Apple** Apple has a 500,000-square-foot data center in Maiden, NC, that houses servers for various iCloud and iTunes services. The center plays a vital role in the company's back-end IT infrastructure. In 2014 Apple expanded this center with a new, smaller 14,250-square-foot tactical data center that also includes office space, meeting areas, and breakrooms.

Since only the company owns the infrastructure, a data center is more suitable for organizations that run many different types of applications and have complex workloads. A data center, like a factory, has limited capacity. Once it is built, the amount of storage and the workload the center can handle does not change without purchasing and installing more equipment.



FIGURE 2.17 The NSA data center in Bluffdale, UT.

When a Data Center Goes Down, so Does Business Data center failures disrupt all operations regardless of who owns the data center. Here are two examples.

- **Uber** The startup company Uber experienced an hour-long outage in February 2014 that brought its car-hailing service to a halt across the country. The problem was caused by an outage at its vendor's West Coast data center. Uber users flooded social media sites with complaints about problems kicking off Uber's app to summon a driver-for-hire.
- **WhatsApp** WhatsApp also experienced a server outage in early 2014 that took the service offline for 2.5 hours. WhatsApp is a smartphone text-messaging service that had been bought by Facebook for \$19 billion. "Sorry we currently experiencing server issues. We hope to be back up and recovered shortly," WhatsApp said in a message on Twitter that was retweeted more than 25,000 times in just a few hours. The company has grown rapidly to 450 million active users within five years, nearly twice as many as Twitter. More than two-thirds of these global users use the app daily. WhatsApp's server failure drove millions of users to a competitor. Line, a messaging app developed in Japan, added 2 million new registered users within 24 hours of WhatsApp's outage—the biggest increase in Line's user base within a 24-hour period.

These outages point to the risks of maintaining the complex and sophisticated technology needed to power digital services used by millions or hundreds of millions of people.

Integrating Data to Combat Data Chaos

An enterprise's data are stored in many different or remote locations—creating data chaos at times. And some data may be duplicated so that they are available in multiple locations that need a quick response. Therefore, the data needed for planning, decision-making, operations, queries, and reporting are scattered or duplicated across numerous servers, data centers, devices, and cloud services. Disparate data must be unified or integrated in order for the organization to function.

Data Virtualization As organizations have transitioned to a cloud-based infrastructure, data centers have become virtualized. For example, Cisco offers data virtualization, which gives greater IT flexibility. The process of data virtualization involves abstracting, transforming, merging, and delivering data from disparate sources. The main goal of data virtualization is to provide a single point of access to the data. By aggregating data from a wide range of sources users can access applications without knowing their exact location. Using data virtualization methods, enterprises can respond to change more quickly and make better decisions in real time without physically moving their data, which significantly cuts costs. Cisco Data Virtualization makes it possible to:

- Have instant access to data at any time and in any format.
- Respond faster to changing data analytics needs.
- Cut complexity and costs.

Compared to traditional (nonvirtual) data integration and replication methods, data virtualization accelerates time to value with:

- **Greater agility** Speeds 5–10 times faster than traditional data integration methods
- **Streamlined approach** 50–75% time savings over data replication and consolidation methods
- **Better insight** Instant access to data

Software-Defined Data Center Data virtualization has led to the latest development in data centers—the **software-defined data center (SDDC)**. An SDDC facilitates the integration of the various infrastructures of the SDDC silos within organizations and optimizes the use of resources, balances workloads, and maximizes operational efficiency by dynamically distributing workloads and provisioning networks. The goal of the SDDC is to decrease costs and increase agility, policy compliance, and security by deploying, operating, managing, and maintaining applications. In addition, by providing organizations with their own private cloud, SDDCs provide greater flexibility by allowing organizations to have on-demand access to their data instead of having to request permission from their cloud provider (see [Figure 2.18](#)).

The base resources for the SDDC are *computation, storage, networking, and security*. Typically, the SDDC includes limited functionality of service portals, applications, OSs, VM hardware, hypervisors, physical hardware, software-defined networking, software-defined storage, a security layer, automation and management layers, catalogs, a gateway interface module, and third-party plug-ins ([Figure 2.19](#)).

It is estimated that the market share for SDDCs will grow from the current level of \$22 billion to more than \$77 billion in the next five years. As the use of SDDCs grows at this extraordinary rate, data center managers will be called upon to scale their data centers exponentially at a moment's notice. Unfortunately, this is impossible to achieve using the traditional data center infrastructure. In the SDDC, software placement and optimization decisions are based on business logic, not technical provisioning directives. This requires changes in culture,

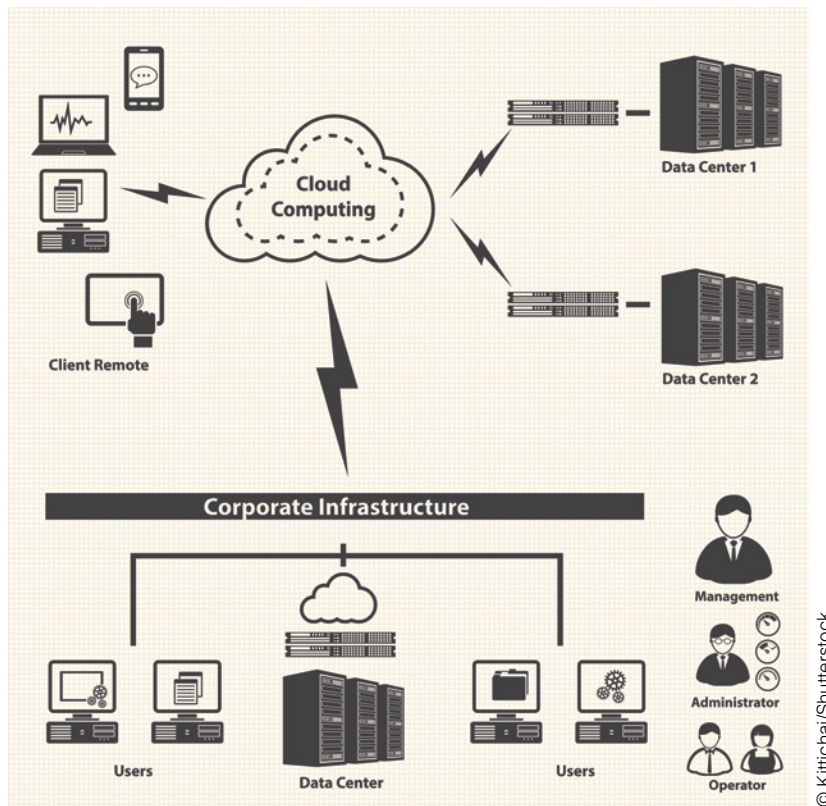


FIGURE 2.18 Corporate IT infrastructures can consist of an on-premises data center and off-premises cloud computing.

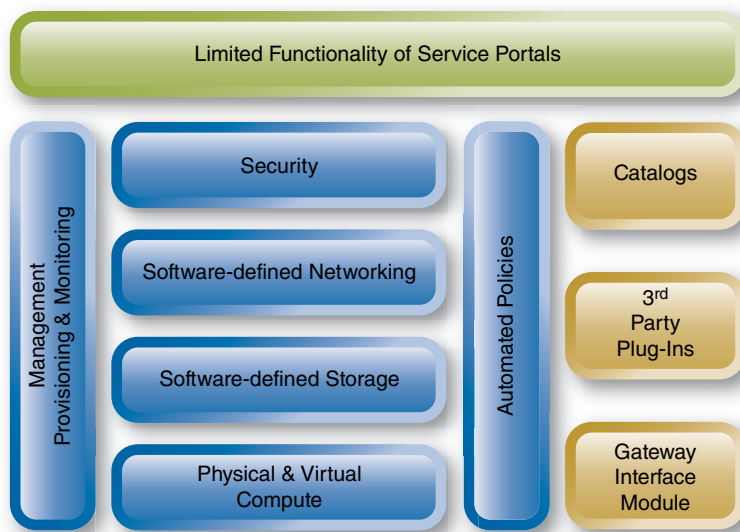


FIGURE 2.19 SDDC infrastructure (adapted from Sturm et al., 2017).

processes, structure, and technology. The SDDC isolates the application layer from the physical infrastructure layer to facilitate faster and more effective deployment, management, and monitoring of diverse applications. This is achieved by finding each enterprise application an optimal home in a public or private cloud environment or draw from a diverse collection of resources.

From a business perspective moving to a SDDC is motivated by the need to improve security, increase alignment of the IT infrastructure with business objectives and provision of applications more quickly.

Traditional data centers had dedicated, isolated hardware that results in poor utilization of resources and very limited flexibility. Second-generation virtualization data cases improved resource use by consolidating virtualized servers. By reducing the steps needed to decrease the time it takes to deploy workloads, facilitating the definition of applications and resource needs, the SDDC creates an even more flexible environment in which enterprise applications can be quickly reconfigured and supported to provide infrastructure-as-a service (IaaS). Transitioning to an SDDC enables an organization to optimize its resource usage, provide capacity on demand, improve business-IT alignment, improve agility and flexibility of operations, and save money (Figure 2.20).

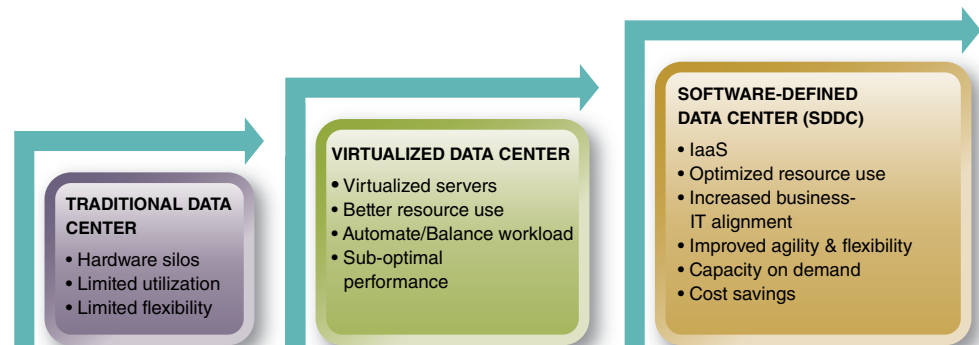


FIGURE 2.20 Evolution of data centers (adapted from Sturm et al., 2017).

Cloud Computing

In a business world where first movers gain the advantage, IT responsiveness and agility provide a competitive edge and lead to sustainable business practices. Yet, many IT infrastructures are extremely expensive to manage and too complex to easily adapt. A common solution is cloud computing. **Cloud computing** is the general term for infrastructures that use the Internet and private networks to access, share, and deliver computing resources. More specifically, IBM defines cloud computing as “the delivery of on-demand computing resources—everything from applications to data centers—over the Internet on a pay-for-use basis” (IBM, 2016).

Cloud computing is the delivery of computing and storage resources as a service to end-users over a network. Cloud systems are *scalable*. That is, they can be adjusted to meet changes in business needs. At the extreme, the cloud’s capacity is unlimited depending on the vendor’s offerings and service plans. A drawback of the cloud is control because a third party manages it. Unless the company uses a **private cloud** within its network, it shares computing and storage resources with other cloud users in the vendor’s **public cloud**. Public clouds allow multiple clients to access the same virtualized services and utilize the same pool of servers across a public network. In contrast, private clouds are single-tenant environments with stronger security and control for regulated industries and critical data. In effect, private clouds retain all the IT security and control provided by traditional IT infrastructures with the added advantages of cloud computing.

Selecting a Cloud Vendor

Because cloud is still a relatively new and evolving business model, the decision to select a cloud service provider should be approached with even greater diligence than other IT decisions. As cloud computing becomes an increasingly important part of the IT delivery model, assessing and selecting the right cloud provider also become the most strategic decisions that business leaders undertake. Providers are not created equally, so it is important to investigate each provider’s offerings prior to subscribing. When selecting and investing in cloud services, there are several service factors a vendor needs to address. These evaluation factors are listed in Table 2.6.

TABLE 2.6 Service Factors to Consider when Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Delays	What are the estimated server delays and network delays?
Workloads	What is the volume of data and processing that can be handled during a specific amount of time?
Costs	What are the costs associated with workloads across multiple cloud computing platforms?
Security	How are data and networks secured against attacks? Are data encrypted and how strong is the encryption? What are network security practices?
Disaster recovery and business continuity	How is service outage defined? What level of redundancy is in place to minimize outages, including backup services in different geographical regions? If a natural disaster or outage occurs, how will cloud services be continued?
Technical expertise and understanding	Does the vendor have expertise in your industry or business processes? Does the vendor understand what you need to do and have the technical expertise to fulfill those obligations?
Insurance in case of failure	Does the vendor provide cloud insurance to mitigate user losses in case of service failure or damage? This is a new and important concept.
Third-party audit or an unbiased assessment of the ability to rely on the service provided by the vendor	Can the vendor show objective proof with an audit that it can live up to the promises it is making?

Vendor Management and Cloud Service Agreements (CSAs) The move to the cloud is also a move to vendor-managed services and **cloud service agreements (CSAs)**. Also referred to as cloud service level agreements (SLAs), the CSA or SLA is a negotiated agreement between a company and service provider that can be a legally binding contract or an informal contract. You can review a sample CSA used by IBM by visiting http://www-05.ibm.com/support/operations/files/pdf/csa_us.pdf.

Staff experienced in managing outsourcing projects may have the necessary expertise for managing work in the cloud and policing SLAs with vendors. The goal is not building the best CSA terms, but negotiating the terms that align most closely with the business needs. For example, if a server becomes nonoperational and it does not support a critical business operation, it would not make sense to pay a high premium for reestablishing the server within one hour. On the other hand, if the data on the server support a business process that would effectively close down the business for the period of time that it was not accessible, it would be prudent to negotiate the fastest possible service in the CSA and pay a premium for that high level of service.

In April 2015, the Cloud Standards Customer Council (CSCC) published the *Practical Guide to Cloud Service Agreements, Version 2.0*, to reflect changes that have occurred since 2012 when it first published the *Practical Guide to Cloud Service Level Agreements*. The new guide provides a practical reference to help enterprise IT and business decision-makers analyze CSAs from different cloud service providers. The main purpose of a CSA is to set clear expectations for service between the cloud customer (buyer) and the cloud provider (seller), but CSAs should also exist between a customer and other cloud entities, such as the cloud carrier, the cloud broker, and even the cloud auditor. Although the various service delivery models, that is, IaaS, PaaS, SaaS, and so on, may have different requirements, the guide focuses on the requirements that are common across the various service models (Cloud Standards Customer Council, 2015, p. 4).

Implementing an effective management process is an important step in ensuring internal and external user satisfaction with cloud services. **Table 2.7** lists the 10 steps that should be taken by cloud customers to evaluate cloud providers' CSAs in order to compare CSAs across multiple providers or to negotiate terms with a selected provider.

TABLE 2.7 Ten Steps to Evaluate a CSA

1. Understand roles and responsibilities of the CSA customer and provider
2. Evaluate business-level policies and compliance requirements relevant to the CSA customer
3. Understand service and deployment model differences
4. Identify critical performance objectives such as availability, response time, and processing speed. Ensure they are measurable and auditable
5. Evaluate security and privacy requirements for customer information that has moved into the provider's cloud and applications, functions, and services being operated in the cloud to provide required service to the customer
6. Identify service management requirements such as auditing, monitoring and reporting, measurement, provisioning, change management, and upgrading/patching
7. Prepare for service failure management by explicitly documenting cloud service capabilities and performance expectations with remedies and limitations for each
8. Understand the disaster recovery plan
9. Develop a strong and detailed governance plan of the cloud services on the customer side
10. Understand the process to terminate the CSA

Cloud Infrastructure

The cloud has greatly expanded the options for enterprise IT infrastructures because any device that accesses the Internet can access, share, and deliver data. Cloud computing is a valuable infrastructure because:

1. It is dynamic, not static and provides a way to make applications and computing power available on demand. Applications and power are available on demand because they are provided *as a service*. For example, any software that is provided on demand is referred to as **software as a service (SaaS)**. Typical SaaS products are Google Apps and www.Salesforce.com. Section 2.5 discusses SaaS and other cloud services.
2. Helps companies become more agile and responsive while significantly reducing IT costs and complexity through improved workload optimization and service delivery.

Move to Enterprise Clouds A majority of large organizations have hundreds or thousands of software licenses that support business processes, such as licenses for Microsoft Office, Oracle database management, IBM CRM (customer relationship management), and various network security software. Managing software and their licenses involves deploying, provisioning, and updating them—all of which are time-consuming and expensive. Cloud computing overcomes these problems.

Issues in Moving Workloads from the Enterprise to the Cloud

Building a cloud strategy is a challenge, and moving existing applications to the cloud is stressful. Despite the business and technical benefits, the risk exists of disrupting operations or customers in the process. With the cloud, the network and WAN (wide area network) become an even more critical part of the IT infrastructure. Greater network bandwidth is needed to support the increase in network traffic. And, putting part of the IT architecture or workload into the cloud requires different management approaches, different IT skills, and knowing how to manage vendor relationships and contracts.

Infrastructure Issues There is a big difference because cloud computing runs on a shared infrastructure, so the arrangement is less customized to a specific company's requirements. A comparison to help understand the challenges is that outsourcing is like renting an apartment, while the cloud is like getting a room at a hotel.

With cloud computing, it may be more difficult to get to the root of performance problems, like the unplanned outages that occurred with Google's Gmail and Workday's human resources apps. The trade-off is cost versus control.

Increasing demand for faster and more powerful computers, and increases in the number and variety of applications are driving the need for more capable IT architectures.

Questions

1. What is a data center?
2. What is the difference between on-premises data centers and cloud computing?
3. What is an SDDC?
4. What are the advantages of using an SDDC?
5. How can cloud computing solve the problems of managing software licenses?
6. What factors should be considered when selecting a cloud vendor or provider?
7. When are private clouds used instead of public clouds?
8. Explain three issues that need to be addressed when moving to cloud computing or services.

2.5 Cloud Services and Virtualization

Managers want streamlined, real-time, data-driven enterprises, yet they may face budget cuts. Sustaining performance requires the development of new business applications and analytics capabilities, which comprise the *front end* and the data stores and digital infrastructure, or back end, to support them. The back end is where the data reside. The problem is that data may have to navigate through a congested IT infrastructure that was first designed decades ago. These network or database bottlenecks can quickly wipe out the competitive advantages from big data, mobility, and so on. Traditional approaches to increasing database performance—manually tuning databases, adding more disk space, and upgrading processors—are not enough when you are dealing with streaming data and real-time big data analytics. Cloud services help to overcome these limitations. Cloud services are outsourced to a third-party cloud provider who manages the updates, security, and ongoing maintenance.

At first glance, virtualization and cloud computing may appear to be quite similar. However, cloud computing and virtualization are inherently different. Unlike cloud computing that involves multiple computers or hardware devices sending data through vendor-provided networks, virtualization is the replacement of a tangible physical component with a virtual one. Each of these concepts are described and discussed in the following sections.

Anything as a Service (XAAS) Models

The cloud computing model for on-demand delivery of and access to various types of computing resources also extends to the development of business apps. **Figure 2.21** shows four “as a service” (XaaS) solutions based on the concept that the resource—software, platform, infrastructure, or data—can be provided on demand regardless of geolocation. As these as service solutions develop, the focus is changing from massive technology implementation costs to business-reengineering programs that enable XaaS platforms (Fresht, 2014).

Cloud services are services made available to users on demand via the Internet from a cloud computing provider's servers instead of being accessed through an organization's

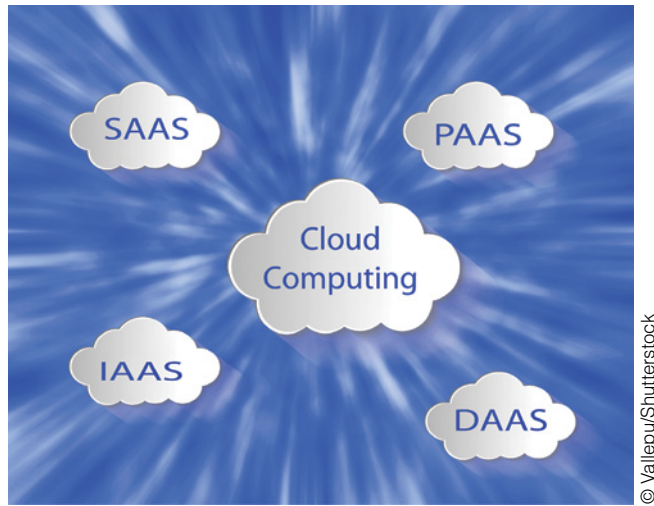


FIGURE 2.21 Four as a service solutions: software, platform, infrastructure, and data as a service.

on-premises servers. Cloud services are designed to provide easy, scalable access to applications, resources, and services, and are fully managed by a cloud services provider.

Cloud computing is often referred to as a “**stack**” or broad range of services built on top of each other under the name *cloud*. These cloud services can be defined as follows:

- **Software as a service (SaaS)** is a widely used model in which software is available to users from a service provider as needed. A provider licenses a SaaS application to customers as an on-demand service, through a subscription, a pay-as-you-go model, or free of charge (where revenue can be generated by other means, such as through sale of advertisements).
- **Platform as a service (PaaS)** is a computing platform that enables the quick and easy creation, testing, and deployment of web applications without the necessity of buying and maintaining the software and infrastructure underneath it. It is a set of tools and services that make coding and deploying these applications faster and more efficient.
- **Infrastructure as a service (IaaS)** is a way of delivering servers, storage, networks, work-load balancers, and OSs as an on-demand service.
- **Data as a service (DaaS)** is an information provision and distribution model in which data files (including text, images, sounds, and videos) are made available to customers over a network by a service provider.

Software as a Service (SaaS) SaaS is a rapidly growing method of delivering software and is particularly useful in applications in which there are considerable interactions between the organization and external entities that do not confer a competitive advantage, for example, e-mail and newsletters. It is also useful when an organization is going to be needing a particular type of software for a short period of time or for a specific project, and for software that is used periodically, for example, tax, payroll, or billing software. SaaS is not appropriate for accessing applications that require fast processing of real-time data or applications where regulation does not permit data being hosted externally.

Other terms for SaaS are *on-demand computing* and *hosted services*. The idea is basically the same: Instead of buying and installing expensive packaged enterprise applications, users can access software applications over a network, using an Internet browser. To use SaaS, a service provider hosts the application at its data center and customers access it via a standard Web browser.

The SaaS model was developed to overcome the common challenge to an enterprise of being able to meet fluctuating demands on IT resources efficiently. It is used in many business functions, primarily customer relationship management (CRM), accounting, human resources (HR), service desk management, communication, and collaboration.

There are thousands of SaaS vendors. www.Salesforce.com is one of the most widely known SaaS providers. Other examples are Google Docs and collaborative presentation software Prezi. For instance, instead of installing Microsoft Word on your own computer, and then loading Word to create a document, you use a browser to log into Google Docs. Only the browser uses your computer's resources.

Platform as a Service (PaaS) PaaS provides a standard unified platform for developing, testing, and deploying software over the Web. This computing platform allows the creation of Web applications quickly and easily without the complexity of buying and maintaining the underlying infrastructure. Without PaaS, the cost of developing some applications would be prohibitive. Examples of PaaS include databases, Web servers, development tools, and execution runtime. PaaS is particularly useful when multiple software developers are working on a software development project or when other external parties need to interact with the development process and for when developers want to automate testing and deployment services. It is less useful in those instances where application performance needs to be customized to the underlying hardware and software or an application needs to be highly portable in terms of where it is hosted. Some examples of PaaS include Microsoft Azure Service, www.Force.com, and Google App Engine.

Infrastructure as a Service (IaaS) Rather than purchasing all the components of its IT infrastructure, organizations buy their computing resources as a fully outsourced Infrastructure as a Service (IaaS) on demand. Generally, IaaS can be acquired as a Public or Private infrastructure or a combination of the two (Hybrid). A public IaaS is one that consists of shared resources deployed on a self-service basis over the Internet. On the other hand, a private IaaS is provided on a private network. And, a hybrid IaaS is a combination of both public and private. IaaS is useful where organizations experience significant highs and lows in terms of demand on the infrastructure, for new or existing organizations who have budgetary constraints on hardware investment and in situations where an organization has temporary infrastructure needs. Some IaaS providers you may be familiar with include Amazon Web Services (AWS) and Rackspace.

Data as a Service (DaaS)—The New Kid on the Block DaaS is the newest entrant into the XaaS arena. DaaS enables data to be shared among clouds, systems, apps, and so on regardless of the data source or where they are stored. Data files, including text, images, sound, and video, are made available to customers over a network, typically the Internet. DaaS makes it easier for data architects to select data from different pools, filter out sensitive data, and make the remaining data available on demand.

A key benefit of DaaS is that it transfers the risks and responsibilities associated with data management to a third-party cloud provider. Traditionally, organizations stored and managed their data within a self-contained storage system, however, as data become more complex, it is increasingly difficult and expensive to maintain using the traditional data model. Using DaaS, organizational data are readily accessible through a cloud-based platform and can be delivered to users despite organizational or geographical constraints. This model is growing in popularity as data become more complex, difficult, and expensive to maintain. Some of the most common business applications currently using DaaS are CRM and enterprise resource planning (ERP). For an example of DaaS, see [IT at Work 2.3](#).

IT at Work 2.3

Slack

Slack, the successful social chat app for companies and their executives and/or employees, has announced a “deep product partnership” with Salesforce (Lunden, 2016). The partnership includes a new data sharing platform for businesses to easily share

information about conversations they are having within the app. More specifically, businesses will be able to share details about client accounts in real time with automatic updates for new leads about the accounts. The new partnership will allow Slack and its users to be even more effective in collaboration and data sharing across many platforms and departments (Lunden, 2016).

As a Service Models Are Enterprisewide and Can Trigger Lawsuits The various As a Service models are used in various aspects of business. You will read how these specific services, such as CRM and HR management, are being used for operational and strategic purposes in later chapters. Companies are frequently adopting software, platform, infrastructure, data management, and starting to embrace *mobility as a service* and *big data as a service* because they typically no longer have to worry about the costs of buying, maintaining, or updating their own data servers. Both hardware and human resources expenses can be cut significantly. Service arrangements all require that managers understand the benefits and trade-offs—and how to negotiate effective SLAs and CSAs. Regulations mandate that confidential data be protected regardless of whether the data are on-premises or in the cloud. Therefore, a company's legal department needs to get involved in these IT decisions. Put simply, moving to cloud services is not simply an IT decision because the stakes around legal and compliance issues are very high.

Going Cloud

Cloud services can advance the core business of delivering superior services to optimize business performance. Cloud can cut costs and add flexibility to the performance of critical business apps. And, it can improve responsiveness to end-consumers, application developers, and business organizations. But to achieve these benefits, there must be IT, legal, and senior management oversight because a company still must meet its legal obligations and responsibilities to employees, customers, investors, business partners, and society.

Virtualization and Virtual Machines

There are many types of virtualization, such as virtual storage devices, virtual desktops, virtual OSs, and virtual servers for network virtualization. You can think of virtualization as a model for a physical component that is built into computer code, to create a software program that acts in the same way as the physical component it is modeling. For example, a virtual machine is a software representation of a computer, rather than an actual computer and a virtual server sends and receives signals just like a physical one, even though it doesn't have its own circuitry and other physical components.

You might ask why organizations want to virtualize their physical computing and networking devices. The answer is a gross underutilization of inefficient use of resources. Computer hardware had been designed to run a single OS and a single app, which leaves most computers vastly underutilized. Virtualization is a technique that creates a virtual (i.e., nonphysical) layer and multiple virtual machines (VMs) to run on a single physical machine. The virtual (or virtualization) layer makes it possible for each VM to share the resources of the hardware. **Figure 2.22** shows the relationship among the VMs and physical hardware.

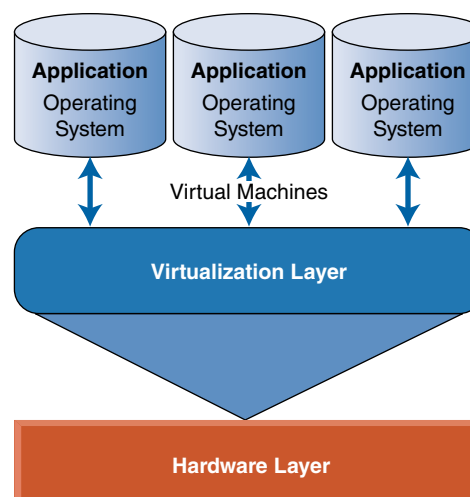


FIGURE 2.22 Virtual machines running on a simple computer hardware layer.

What Is a Virtual Machine? Just as *virtual reality* is not real, but a software-created world, a virtual machine is a software-created computer. Technically, a **virtual machine (VM)** is created by a software layer, called the *virtualization layer*, as shown in Figure 2.22. That layer has its own Windows or other OS and apps, such as Microsoft Office, as if it were an actual physical computer. A VM behaves exactly like a physical computer and contains its own virtual—that is, *software-based*—CPU, RAM (random access memory), hard drive, and network interface card (NIC). An OS cannot tell the difference between a VM and a physical machine, nor can applications or other computers on a network tell the difference. Even the VM thinks it is a “real” computer. Users can set up multiple real computers to function as a single PC through virtualization to pool resources to create a more powerful VM.

Virtualization is a concept that has several meanings in IT and therefore several definitions. The major type of virtualization is hardware virtualization, which remains popular and widely used. Virtualization is often a key part of an enterprise’s disaster recovery plan. In general, virtualization separates business applications and data from hardware resources. This separation allows companies to pool hardware resources—rather than dedicate servers to applications—and assign those resources to applications as needed.

Different types of virtualization include:

- *Storage virtualization* is the pooling of physical storage from multiple network storage devices into what appears to be a single storage device managed from a central console.
- *Server virtualization* consolidates multiple physical servers into virtual servers that run on a single physical server.
- *Desktop virtualization* is software technology that separates the desktop environment and associated application software from the physical machine that is used to access it.
- *Application virtualization* is the practice of running software from a remote server rather than on the user’s computer.
- *Network virtualization* combines the available resources in a network by splitting the network load into manageable parts, each of which can be assigned (or reassigned) to a particular server on the network.
- *Hardware virtualization* is the use of software to emulate hardware or a total computer environment other than the one the software is actually running in. It allows a piece of hardware to run multiple OS images at once. This kind of software is sometimes known as a virtual machine.

Virtualization Characteristics and Benefits Virtualization increases the flexibility of IT assets, allowing companies to consolidate IT infrastructure, reduce maintenance and administration costs, and prepare for strategic IT initiatives. Virtualization is not primarily about cost-cutting, which is a tactical reason. More importantly, for strategic reasons, virtualization is used because it enables flexible sourcing and cloud computing.

The characteristics and benefits of virtualization are as follows:

- 1. Memory-intensive** VMs need a huge amount of RAM (random access memory, or primary memory) because of their massive processing requirements.
- 2. Energy-efficient** VMs minimize energy consumed running and cooling servers in the data center—representing up to a 95% reduction in energy use per server.
- 3. Scalability and load balancing** When a big event happens, such as the Super Bowl, millions of people go to a website at the same time. Virtualization provides load balancing to handle the demand for requests to the site. The VMware infrastructure automatically distributes the load across a cluster of physical servers to ensure the maximum performance of all running VMs. Load balancing is key to solving many of today’s IT challenges.

Virtualization consolidates servers, which reduces the cost of servers, makes more efficient use of data center space, and reduces energy consumption. All of these factors reduce the total cost of ownership (TCO). Over a three-year life cycle, a VM costs approximately 75% less to operate than a physical server. **IT at Work 2.4** describes one example of how virtualization can help organizations provide higher levels of customer service and improve productivity.

IT at Work 2.4

Business Continuity with Virtualization

Liberty Wines supplies to restaurants, supermarkets, and independent retailers from its headquarters in central London. Recipient of multiple international wine awards—including the International Wine Challenge on Trade Supplier of the Year for two years running—Liberty Wines is one of the United Kingdom’s foremost wine importers and distributors.

IT Problems and Business Needs

As the business expanded, the existing servers did not have the capacity to handle increased data volumes, and maintenance of the system put a strain on the IT team of two employees. Existing systems were slow and could not provide the responsiveness that employees expected.

Liberty Wines had to speed up business processes to meet the needs of customers in the fast-paced world of fine dining. To provide the service their customers expect, employees at Liberty Wines needed quick and easy access to customer, order, and stock

information. In the past, the company relied on 10 physical servers for applications and services, such as order processing, reporting, and e-mail.

Virtualized Solution

Liberty Wines deployed a virtualized server solution incorporating Windows Server 2008 R2. The 10 servers were replaced with 3 physical servers, running 10 virtual servers. An additional server was used as part of a backup system, further improving resilience and stability.

By reducing the number of physical servers from 10 to 4, power use and air conditioning costs were cut by 60%. Not only was the bottom line improved, but the carbon footprint was also reduced, which was good for the environment.

The new IT infrastructure cut hardware replacement costs by £45,000 (U.S. \$69,500) while enhancing stability with the backup system. Applications now run faster, too, so employees can provide better customer service with improved productivity. When needed, virtual servers can be added quickly and easily to support business growth.

Questions

- 1. What is SaaS?
- 2. What is PaaS?
- 3. What is IaaS?
- 4. How might companies risk violating regulation or compliance requirements with cloud services?
- 5. In what ways is a virtualized information system different from a traditional information system?
- 6. Describe the different types of virtualization.
- 7. What is load balancing and why is it important?

Key Terms

ad hoc report 34	data silo 43	information management 42
batch processing 33	database 33	information systems (ISs) 28
cloud computing 52	decision support systems (DSS) 32	infrastructure as a service (IaaS) 56
cloud service agreements (CSAs) 53	dirty data 26	IT infrastructure 38
customer-centric 47	enterprise architecture (EA) 26	IPOS 28
data 30	exception report 34	knowledge 30
data as a service (DaaS) 56	executive information systems (EISs) 32	management information systems (MIS) 34
data center 38	goal seeking 35	master data 46
data governance 46	information 30	master data management (MDM) 47

master file 47	real-time processing 33	transaction processing systems (TPS) 32
model 26	service level agreement (SLA) 61	unstructured decisions 35
online transaction processing (OLTP) 33	software as a service (SaaS) 54	virtualization 59
platform as a service (PaaS) 56	software-defined data center (SDDC) 50	virtual machine (VM) 59
private cloud 52	stack 56	what-if analysis 35
public cloud 52	structured decisions 35	wisdom 31

Assuring Your Learning

Discuss: Critical Thinking Questions

- Why is a strong market position or good profit performance only temporary?
- Assume you had:
 - A tall ladder with a sticker that lists a weight allowance only five pounds more than you weigh. You know the manufacturer and model number.
 - Perishable food with an expiration date two days into the future.
 - A checking account balance that indicates you have sufficient funds to cover the balance due on an account.

In all three cases, trusting the data to be exactly correct could have negative consequences. Explain the consequences of trusting the data in each instance. How might you determine the correct data for each instance? Which data might not be possible to verify? How does dirty data impact your decision-making?
- If business data are scattered throughout the enterprise and not synched until the end of the month, how does that impact day-to-day decision-making and planning?
- Assume a bank's data are stored in silos based on financial product—checking accounts, saving accounts, mortgages, auto loans, and so on. What problems do these data silos create for the bank's managers?
- Why do managers and workers still struggle to find information that they need to make decisions or take action despite advances in digital technology? That is, what causes data deficiencies?
- According to a Tech CEO Council Report, Fortune 500 companies waste \$480 billion every year on inefficient business processes. What factors cause such huge waste? How can this waste be reduced?
- Explain why organizations need to implement EA and data governance.
- What two problems can EA solve?
- Name two industries that depend on data governance to comply with regulations or reporting requirements. Given an example of each.
- Why is it important for data to be standardized? Give an example of unstandardized data.
- Why are TPSs critical systems?
- Discuss why the cloud acts as the *great IT delivery frontier*.
- What are the functions of data centers?
- What factors need to be considered when selecting a cloud vendor?
- What protection does an effective SLA or CSA provide?
- Why is an SLA or a CSA a legal document?
- How can virtualization reduce IT costs while improving performance?

Explore: Online and Interactive Exercises

- When selecting a cloud vendor to host your enterprise data and apps, you need to evaluate the **service level agreement (SLA)**.
 - Research the SLAs of two cloud vendors, such as Rackspace, Amazon, or Google.
 - For the vendors you selected, what are the SLAs' uptime percentages? Expect them to be 99.9% or less.
 - Does each vendor count both scheduled downtime and planned downtime toward the SLA uptime percentage?
 - Compare the SLAs in terms of two other criteria.
 - Decide which SLA is better based on your comparisons.
 - Report your results and explain your decision.
- Many organizations initiate data governance programs because of pressing compliance issues that impact data usage. Organizations may need data governance to be in compliance with one or more regulations, such as the Gramm–Leach Bliley Act (GLB), HIPAA, Foreign Corrupt Practices Act (FCPA), Sarbanes–Oxley Act, and several state and federal privacy laws.
 - Research and select two U.S. regulations or privacy laws.
 - Describe how data governance would help an enterprise comply with these regulations or laws.
- Visit www.eWeek.com Cloud Computing Solutions Center for news and reviews at www.eweek.com/c/s/Cloud-Computing. Select one of the articles listed under Latest Cloud Computing News. Prepare an executive summary of the article.
- Visit Rackspace.com and review the company's three types of cloud products. Describe each of those cloud solutions.

5. Visit Oracle.com. Describe the types of virtualization services offered by Oracle.
6. Visit YouTube.com and search for two videos on virtualization. For

each video, report what you learned. Specify the complete URL, video title, who uploaded the video and the date, video length, and number of views.

Analyze & Decide: Apply IT Concepts to Business Decisions

1. Financial services firms experience large fluctuations in business volumes because of the cyclical nature of financial markets. These fluctuations are often caused by crises—such as the subprime mortgage problems, the discovery of major fraud, or a slowdown in the economy. These fluctuations require that executives and IT leaders have the ability to cut spending levels in market downturns and quickly scale up when business volumes rise again. Research SaaS solutions and vendors for the financial services sector. Would investment in SaaS help such firms align their IT capacity with their business needs and also cut IT costs? Explain your answer.
2. Despite multimillion-dollar investments, many IT organizations cannot respond quickly to evolving business needs. Also, they cannot

adapt to large-scale shifts like mergers, sudden drops in sales, or new product introductions. Can cloud computing help organizations improve their responsiveness and get better control of their IT costs? Explain your answer.

3. Describe the relationship between enterprise architecture and organizational performance.
4. Identify four **KPIs** for a major airline (e.g., American, United, Delta) or an automobile manufacturer (e.g., GM, Ford, BMW). Which KPI would be the easiest to present to managers on an online dashboard? Explain why.

Case 2.2

Business Case: Data Chaos Creates Risk

Data chaos often runs rampant in service organizations, such as health care and the government. For example, in many hospitals, each line of business, division, and department has implemented its own IT applications, often without a thorough analysis of its relationship with other departmental or divisional systems. This arrangement leads to the hospital having IT groups that specifically manage a particular type of application suite or data silo for a particular department or division.

Data Management

When applications are not well managed, they can generate terabytes of irrelevant data, causing hospitals to drown in such data. This data chaos could lead to medical errors. In the effort to manage excessive and massive amounts of data, there is increased risk of relevant information being lost (missing) or inaccurate—that is, faulty or dirty data. Another risk is data breaches.

- **Faulty data** By 2015, 96% of health-care organizations had adopted electronic health records, or EHRs (Office of the National Coordinator for HIT, 2016). It is well known that an unintended consequence of EHR is faulty data. According to a study published in the *Journal of the American Medical Association*, data in EHR systems may not be as accurate and complete as expected (Conn, 2016). Incorrect lab values, imaging results, or physician documentation lead to medical errors, harm patients, and damage the organization's accreditation and reputation.

- **Data breaches** More than 25 million people have been affected by health-care system data breaches since the Office for Civil Rights, a division of the U.S. Department of Health and Human Services, began reporting breaches in 2009. Most breaches involved lost or stolen data on laptops, removable drives, or other portable media. Breaches are extremely expensive and destroy trust.

Accountability in health-care demands compliance with strong data governance efforts. Data governance programs verify that data input into EHR, clinical, financial, and operational systems are accurate and complete—and that only authorized edits can be made and logged.

Vanderbilt University Medical Center Adopts EHR and Data Governance

Vanderbilt University Medical Center (VUMC) in Nashville, TN, was an early adopter of EHR and implemented data governance in 2009. VUMC's experience provides valuable lessons.

VUMC consists of three hospitals and the Vanderbilt Clinic, which have 918 beds, discharge 53,000 patients each year, and count 1.6 million clinic visits each year. On average, VUMC has an 83% occupancy rate and has achieved HIMSS Stage 6 hospital EHR adoption. HIMSS (Healthcare Information and Management Systems Society, himss.org) is a global, nonprofit organization dedicated to better health-care outcomes through IT. There are seven stages of EHR adoption, with Stage 7 being a fully paperless environment. That means all clinical data are part of an electronic medical record and, as a result, can be shared

across and outside the enterprise. At Stage 7, the health-care organization is getting full advantage of the *health information exchange (HIE)*. HIE provides interoperability so that information can flow back and forth among physicians, patients, and health networks (NextGen Healthcare, 2016).

VUMC began collecting data as part of its EHR efforts in 1997. By 2009, the center needed stronger, more disciplined data management. At that time, hospital leaders initiated a project to build a data governance infrastructure.

Data Governance Implementation

VUMC's leadership team had several concerns.

1. IT investments and tools were evolving rapidly, but they were not governed by HIM (Healthcare Information and Management) policies.
2. As medical records became electronic so they might be transmitted and shared easily, they became more vulnerable to hacking.
3. As new uses of electronic information were emerging, the medical center struggled to keep up.

Health Record Executive Committee

Initially, VUMC's leaders assigned data governance to their traditional medical records committee, but that approach failed. Next, they hired consultants to help develop a data governance structure and organized a *health record executive committee* to oversee the project. The committee reports to the medical board and an executive committee to ensure executive involvement and sponsorship. The committee is responsible for developing the strategy for standardizing health record practices, minimizing risk, and maintaining compliance. Members include the chief medical information officer (CMIO), CIO, legal counsel, medical staff, nursing informatics, HIM, administration, risk management, compliance, and accreditation. In addition, a legal medical records team was formed to support additions, corrections, and deletions to the EHR. This team defines procedures for removal of

duplicate medical record numbers and policies for data management and compliance.

Costs of Data Failure

Data failures incur the following costs:

- Rework
- Loss of business
- Patient safety errors
- Malpractice lawsuits
- Delays in receiving payments because billing or medical codes data are not available.

Benefits Achieved from Data Governance

As in other industries, in health care, data are the most valuable asset. The handling of data is the real risk. EHRs are effective only if the data are accurate and useful to support patient care. Effective ongoing data governance has achieved that goal at VUMC.

Questions

1. What might happen when each line of business, division, and department develops its own IT apps?
2. What are the consequences of poorly managed apps?
3. What two risks are posed by data chaos? Explain why.
4. What are the functions of data governance in the health-care sector?
5. Why is it important to have executives involved in data governance projects?
6. List and explain the costs of data failure.
7. Why are data the most valuable asset in health care?

Sources: Compiled from NextGen Healthcare (2016), Office of the National Coordinator for HIT (2016), and Conn (2016).

Case 2.3

Video Case: Cloud Computing at Coca-Cola Is Changing Everything

When organizations say they are “using the cloud,” they can mean a number of very different things. Using an IaaS service such as Amazon EC2 or Terremark is different from using Google Apps to outsource e-mail, which is different again from exposing an API in Facebook.

In this video Alan Boehme, CIO of the Coca-Cola Company discusses how Coca-Cola uses cloud computing to more effectively interact with its customers and describes the challenges Coca-Cola is facing in establishing SaaS partnerships with new start-ups.

Complete these three steps:

1. Visit <https://www.youtube.com/watch?v=hCxmsSED2DY>
2. View the 13-minute video.
3. Answer each of the three parts of the following question.

Question

1. Explain the value of Coca-Cola's cloud partnerships with start-up companies to:
 - a. Coca-Cola
 - b. The start-up companies
 - c. Coca-Cola's customers

IT Toolbox

Accurately Measuring the Value of Data Governance

When developing a data governance program, it's important to present a strong business case to get buy-in from top executives and stakeholders. A crucial part of the business case is an estimate of the data governance program's return on investment (ROI) to show how it will add value to the company. You will need to justify the ROI based on both business and IT strategy to ensure that available funds are used to best meet the business objectives.

To do this you will need to carefully analyze the IT infrastructure with regard to how different components of the IT infrastructure work together to support business processes, how data needed by one system can be received and used by another, how easily data can be communicated and/or repurposed. You will also need to factor in risks and adverse events such as costs associated with rework in data collection, costs associated with unreliable or unfit data, and delays associated with untimely or unavailable data. Now, all of these costs must be quantified and your level of confidence in the corporate data has to be calculated to ensure your business case accurately reflects the value of a data governance program.

One metric used to make this calculation is the *confidence in data-dependent assumptions* metric, or CIDDA (Reeves & Bowen, 2013). The CIDDA identifies specific areas of deficiency.

So, to sum up, when building a data governance model, it is necessary to:

1. Establish a leadership team
2. Define the program's scope
3. Calculate the ROI using the CIDDA.

CIDDA is computed by multiplying three confidence estimates using the following formula:

$$\text{CIDDA} = G \times M \times TS$$

where

G = Confidence that data are *good* enough for their intended purpose

M = Confidence that data *mean* what you think they do

TS = Confidence that you know where the data come from and *trust the source*.

CIDDA is a subjective metric for which there are no industry benchmarks, yet it can be evaluated over time to gauge improvements in data quality confidence.

To ensure your understanding of this IT Toolbox item, calculate the CIDDA of Company A over time, using the stated levels of confidence in the different aspects of its corporate data over Q1–Q4 2017:

Q1_2017: $G = 40\%$, $M = 50\%$, $TS = 20\%$

Q2_2017: $G = 50\%$, $M = 55\%$, $TS = 30\%$

Q3_2017: $G = 60\%$, $M = 60\%$, $TS = 40\%$

Q4_2017: $G = 60\%$, $M = 70\%$, $TS = 45\%$

References

- Bloomberg, J. "Change as Core Competency: Transforming the Role of the Enterprise Architect." *Forbes*, June 16, 2016.
- Cailean, I. "What Role Do Algorithms Play in Programmatic Advertising?" *Trade Mod*, January 6, 2016. <http://www.trademob.com/what-role-do-algorithms-play-in-programmatic-advertising>
- Cloud Standards Customer Council. *Practical Guide to Cloud Service Agreements, Version 2.0*. April 2015. <http://www.cloud-council.org/deliverables/CSCC-Practical-Guide-to-Cloud-Service-Agreements.pdf>
- Conn, J. "EHRs vs. Paper: A Split-decision on Accuracy." *Modern Healthcare*, July 8, 2016.
- Fresht, P. "The Ten Tenets Driving the As-a-service Economy." *Horses for Sources*, October 6, 2014. http://www.horsesforsources.com/as-a-service-economy_100614
- IBM. "What is Cloud Computing?" *IBM*, June 6, 2016. <https://www.ibm.com/cloud-computing/learn-more/what-is-cloud-computing>
- Jarousse, L. A. "Information Governance for Hospitals." *Hospitals & Health Networks*, February 18, 2016.
- Keitt, T. J. "Collaboration Technology Should Be Part of Your Customer Experience Tool Kit." *Forrester.com*, June 30, 2014.
- Lunden, I. "Enterprise Chat App Slack Ties up with Salesforce in a Deep Product Partnership." *Tech Crunch*, September 27, 2016. <https://techcrunch.com/2016/09/27/enterprise-chat-app-slack-ties-up-with-salesforce-in-a-deep-platform-partnership>
- Marchese, L. "How the 'Silo Effect' Is Hurting Cross Team Collaboration." *Trello*, May 10, 2016.
- NextGen Healthcare. "Health Information Exchange (HIE)." *NextGen Healthcare*, March 31, 2016.
- Office of the National Coordinator for Health Information Technology. "Percent of Hospitals, By Type, that Possess Certified Health IT." *Office of the National Coordinator for Health Information Technology*, May 31, 2016.
- Porter, M. *Competitive Advantage: Creating and Sustaining Superior Performance*. Free Press, 1998.
- Rai, R., G. Sahoo, and S. Mehruz. "Exploring the Factors Influencing the Cloud Computing Adoption: A Systematic Study on Cloud Migration." *Springerplus*, April 25, 2015, 4, 197.
- Reeves, M. G. and R. Bowen. "Developing a Data Governance Model in Health Care." *Healthcare Financial Management*, February 2013, 67(2): 82–86.
- Schneider, M. "Case Study: How MEDIATA Increased Campaign Performance with Hyperlocal Targeting." *Skyhook Wireless*, July 22, 2014.
- Schneider, M. "Solving the Dirty Data Problem in Location-Based Advertising." *Street Fight*, January 7, 2015.
- Shore, J. "Cloud-Based Integration Seeks to Tear Down Data Silos." *Tech Target*, August 19, 2015.
- Sturm, R., C. Pollard, and J. Craig. *Application Performance Management in the Digital Enterprise*. Elsevier, March 2017.
- Zuckerman, M.P.H., Sheingold, Ph.D., Orav, Ph.D., Ruhter, M.P.P., M.H.S.A., and Epstein, M.D. "Readmissions, Observation, and the Hospital Readmissions Reduction Program." *The New England Journal of Medicine*, April 21, 2016.

e-COMMERCE SYSTEMS



CHAPTER HIGHLIGHTS

e-Commerce Fundamentals

Introduction to e-Commerce
The Scope of e-Commerce

REAL WORLD CHALLENGE: Ticketmaster—
New Clients, New Outlets, New Needs

Essential e-Commerce Processes
Electronic Payment Processes

e-Commerce Applications and Issues

Business-to-Consumer e-Commerce
Web Store Requirements
Business-to-Business e-Commerce
e-Commerce Marketplaces
Clicks and Bricks in e-Commerce

REAL WORLD SOLUTION: Ticketmaster—Changing
the Way Tickets Are Sold

REAL WORLD CASE: Sony, 1-800-Flowers, Starbucks,
and Others: Social Networks, Mobile Phones, and
the Future of Shopping

REAL WORLD CASE: LinkedIn, Umbria, Mattel, and
Others: Driving the “Buzz” on the Web

LEARNING OBJECTIVES

- 9-1** Identify the major categories and trends of e-commerce applications.
- 9-2** Identify the essential processes of an e-commerce system, and give examples of how it is implemented in e-commerce applications.
- 9-3** Identify and give examples of several key factors and Web store requirements needed to succeed in e-commerce.
- 9-4** Identify and explain the business value of several types of e-commerce marketplaces.
- 9-5** Discuss the benefits and trade-offs of several e-commerce clicks-and-bricks alternatives.

e-Commerce Fundamentals

INTRODUCTION TO e-COMMERCE LO 9-1

E-commerce is changing the shape of competition, the speed of action, and the streamlining of interactions, products, and payments from customers to companies and from companies to suppliers.

For most companies today, **electronic commerce** is more than just buying and selling products online. Rather, it encompasses the entire online process of developing, marketing, selling, delivering, servicing, and paying for products and services transacted on inter-networked, global marketplaces of customers, with the support of a worldwide network of business partners. In fact, much like the term e-business, many consider the term *e-commerce* to be somewhat antiquated. Given that the incoming generation of businesspeople have grown up in a world in which online commerce has always been available, it may soon be time to eliminate the distinction between e-commerce and e-business and accept that it is all just “business as usual.” Until then, as before, we will retain the term *e-commerce* because it allows for a clearer picture of the differences between online and more traditional business transactions.

Because e-commerce involves more than just buying or selling something on the Internet, we must have a definition that is broad enough to allow for the vast array of activities defined by it. For our purposes, we will define e-commerce as *the online exchange of value*. Using this definition, all of the activities identified above, and more, can be defined as e-commerce.

As we will see in this chapter, e-commerce systems rely on the resources of the Internet and many other information technologies to support every step of this process. We will also see that most companies, large and small, are engaged in some form of e-commerce activities. Therefore, developing an e-commerce capability has become a competitive necessity for most businesses in today’s marketplace.

Read the Real World Challenge on the next page. We can learn a lot about the many issues that companies face as a result of changing customer preferences. See Figure 9.1.

THE SCOPE OF e-COMMERCE

Figure 9.2 illustrates the range of business processes involved in the marketing, buying, selling, and servicing of products and services in companies that engage in e-commerce. Companies involved in e-commerce as either buyers or sellers rely on Internet-based technologies and e-commerce applications and services to accomplish *marketing, discovery, transaction processing, and product and customer service processes*. For example, e-commerce can include *interactive marketing, ordering, payment, and customer support processes* at e-commerce catalog and auction sites on the World Wide Web. However, e-commerce also includes e-business processes such as extranet access of inventory databases by customers and suppliers (transaction processing), intranet access of customer relationship management systems by sales and customer service reps (service and support), and customer collaboration in product development via e-mail exchanges and Internet newsgroups (marketing/discovery).

The advantages of e-commerce allow a business of virtually any size that is located anywhere on the planet to conduct business with just about anyone, anywhere. Imagine a small olive oil manufacturer in a remote village in Italy selling its wares to major department stores and specialty food shops in New York, London, Tokyo, and other large metropolitan markets. The power of e-commerce allows geophysical barriers to disappear, making all consumers and businesses on earth potential customers and suppliers.

e-Commerce Technologies

Which technologies are necessary for e-commerce? The short answer is that most information technologies and Internet technologies that we discuss in this text are, in some form, involved in e-commerce systems. A more specific answer is illustrated in Figure 9.3,

Ticketmaster—New Clients, New Outlets, New Needs

It may seem hard to believe, but world leading entertainment ticketing company Ticketmaster started less than 40 years ago, in 1976, in a garage in Phoenix, Arizona, founded by two college students and a businessman. One year later, in 1977, it had signed its first ticketed concert, the Electric Light Orchestra appearing at the University of New Mexico. In 1978 the company signed its first international client in Oslo, Norway; its first major venue (the Louisiana Superdome); and its first major league team, the NBA New Orleans Jazz (now the Utah Jazz). Today, Ticketmaster has become one of the largest e-commerce sites in the world, operating more than 7,000 retail outlets in 20 markets internationally. In 2009, more than one million paperless tickets were sold.

Although most of us may think of Ticketmaster as the place—real or virtual—where we go to buy tickets to a concert, show, or game for our own use, a major portion of Ticketmaster's business relates to the provision of technology and services to more than 10,000 corporate clients such as arenas, stadiums, professional and college sports teams and leagues, museums, theaters, and other entertainment venues. Indeed, in 2008 alone Ticketmaster sold more than 140 million tickets worth more than \$9 billion on behalf of these clients. For example, Ticketmaster was

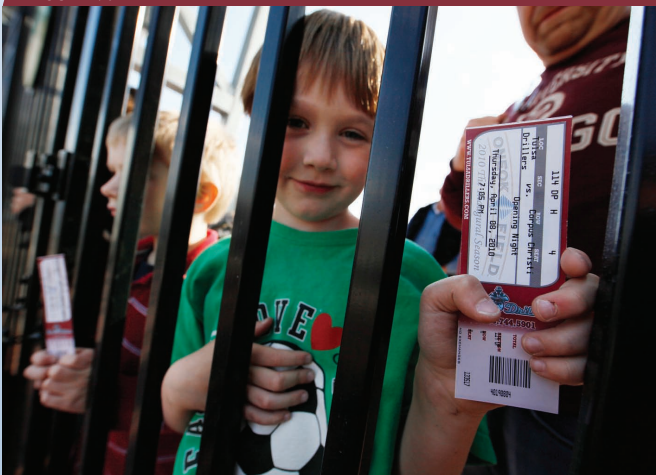
the official ticket provider for the 2008 Summer Olympic Games in Beijing and will do the same for the 2012 Summer Olympic Games in London. These arrangements, however, go beyond “merely” selling the tickets.

When that much money is at stake, it is important for Ticketmaster to provide its corporate clients with the tools necessary to run their businesses more efficiently; this in turn generates even more revenue for the company and more strongly ties its clients to the platform. The ultimate goal—which at the end of the day drives the entire industry—is that we, the final users of those tickets, have the best possible experience when purchasing them and later when attending our favorite events. For that to happen, however, many different things must go right, all behind the scenes.

Consider, for example, the Washington Capitals, an NFL franchise team in Washington, D.C., that makes the Verizon Center its home arena. The season ticket base for the team has about 4,500 ticket holders, with more than 50 percent of them renewing or upgrading their tickets online every year. On the one hand, online renewing and ticketing tremendously reduces the amount of manual labor involved in the production and delivery of those tickets, with obvious benefits for the team. On the other hand, that means that the software that supports the process must be very reliable with little, if any, downtime. In the current economic environment, going out for a game, concert, or the like is a major effort for most end customers—it may be the one big thing they do for the month. As a result, both Ticketmaster and its clients, like the Washington Capitals, want to do all they can to make the process as simple as possible. At the same time, consumers have become more demanding—the economic downturn has helped their attitude as well—and they are not content anymore with just buying a ticket. They want to do much more.

Think about how “buying a ticket” has changed in the last 15 or 20 years. Before, one would stand in line at a stadium or arena, sometimes for hours, to buy tickets to a game or concert. After purchasing the tickets, the next step was to show up at the specified date, time, and place, and enjoy the event. If the ticket holder got sick or had a job commitment, it was too bad, but tickets could not easily be sold or transferred. Unless the ticket holder knew somebody who wanted to go, it was difficult to resell the tickets. Although some places would buy them, the discounts were

FIGURE 9.1



The rise of the Internet and new customer practices are putting pressure on the ticketing business to adapt.

SOURCE: © AP Wide World Photos.

significant. If a game was sold out, and you were not one of the lucky ones to get a ticket, then it was also difficult to buy them in the secondary market. Scalpers charged major premiums, and buying these tickets is illegal in many states; a ticket holder would be uncertain whether the tickets were valid and whether he or she would be able to get into the game. We—the customers—do not want to do any of that today. This means that new tools and services must be made available for both Ticketmaster and its corporate clients so the end customers can be better served.

To meet the increasingly more complex needs of its corporate clients, Ticketmaster sought to develop a ticketing platform that provides granular-level control over seat inventory, customer relationship management, and full ownership of all customer data by the team, franchise, or facility. In addition, the solution would need to handle many different types of transactions quickly, easily, and securely. The goal is not only to provide these clients with a better way to market their events and thus sell more tickets—although that is certainly an important goal—but also to provide better service and support to their fans in a variety of ways. For example, beyond keeping accurate count of the number of available seats in the different pricing sections of the stadium, organizations also want to be able to have the option to offer clients an upgrade to a season or multi-ticket package (say, for the next five home games), or to keep waiting lists to be filled as new tickets become available. And these are just two examples. By providing more options to

ticket holders, any solution implemented would improve the ticket use rate: how much revenue is generated by each individual seat sold. More advanced features, such as gaining insight into patterns of consumption, behavior, and preferences of individual fans is not out of the question either.

Created less than 40 years ago in Arizona, Ticketmaster is a major player in the entertainment industry today, serving thousands of corporate clients, venues, and teams throughout the world. For most large events, Ticketmaster provides the underlying processes related to managing the sales of all tickets and admission, and those are largely based online today. A number of different steps and processes make up a great e-commerce experience, and Ticketmaster, as the leading provider of ticketing services in the world, is determined to enhance each and every one of those steps. At the same time, it is important for Ticketmaster—or any other company contemplating such a major development—to manage the project carefully, lest it fail to deliver superior value on every step because it is trying to achieve too much at once. As in any other major project that involves technology, there are major risks and there are major benefits.

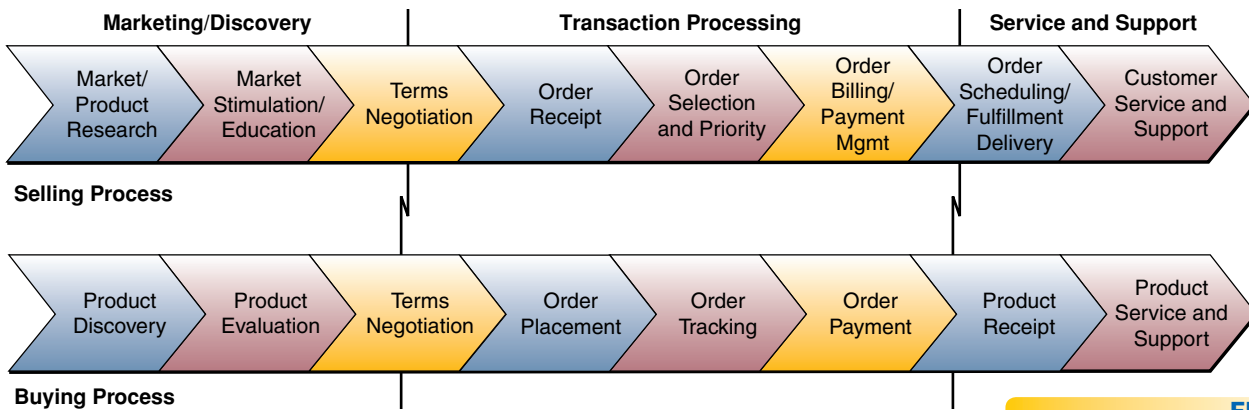
SOURCE: “ARCHTICS,” *Computerworld Honors Case Study*, 2010; “Ticketmaster,” *Sybase Customer Case Study*, 2010; “The Phoenix Suns Launch Ticketmaster’s Team Exchange, Online Ticket Forwarding Technology,” *Ticketmaster Press Release*, September 5, 2010; “Ticketmaster and the Golden State Warriors Sign Multi-Year Ticketing Agreement,” *Ticketmaster Press Release*, September 26, 2010; and www.ticketmaster.com, accessed May 29, 2011.

QUESTIONS TO CONSIDER

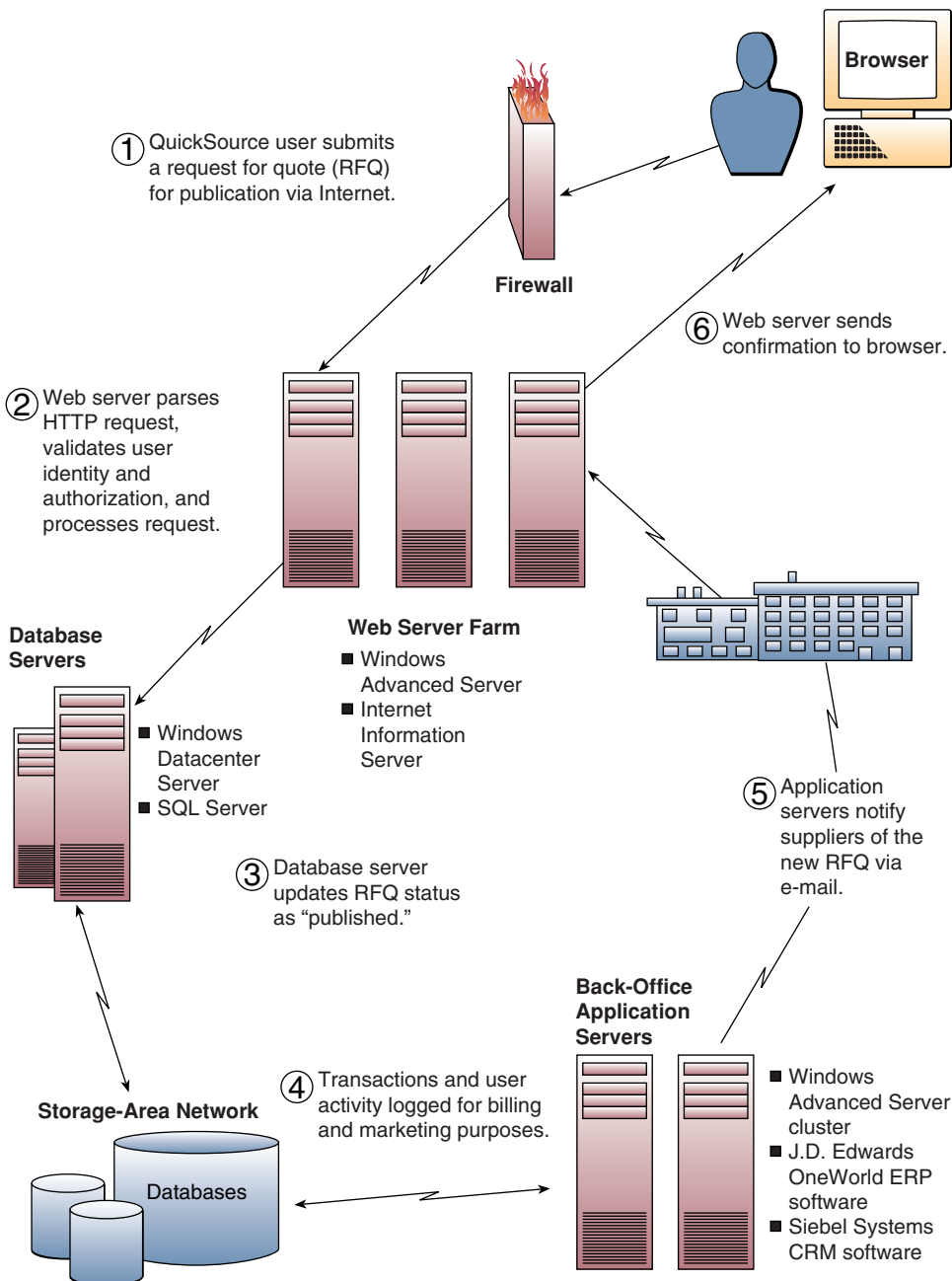
1. List all of the steps that purchasing a ticket entails. How can technology change the way in which each step is carried out? What alternatives could be implemented for each step?
2. What kind of information would be valuable for corporate clients, such as sport franchises, arenas, or the

like? What could those corporate clients do with that information?

3. Beyond those features discussed here, what other features should Ticketmaster consider incorporating into its future solution?

**FIGURE 9.2**

e-Commerce involves accomplishing a range of business processes to support the electronic buying and selling of goods and services.

**FIGURE 9.3**

The hardware, software, network, and database components and IT architecture of B2B online auctions provider FreeMarkets Inc. are illustrated in this example of its Internet-based QuickSource auction service.

which gives an example of the technology resources required by many e-commerce systems. The figure illustrates some of the hardware, software, data, and network components used by FreeMarkets Inc. to provide business-to-business (B2B) online auction e-commerce services.

Forrester: Web 2.0 Has a Bright Future

As a standard enterprise tool, Web 2.0 has a bright future, one for which companies are expected to spend \$4.6 billion by 2013 to integrate into their corporate computing environments, according to a Forrester Research report. Although still considered an upstart technology, Forrester believes that conventional Web 2.0 elements—social networking, RSS, blogs, wikis, mashups, podcasting, and widgets—are fast becoming the norm for communicating with employees and customers. The report highlights megacompanies such as General Motors, McDonald's, Northwestern Mutual Life Insurance, and Wells Fargo among those who have already jumped into the Web 2.0 pool with both feet. In addition, some 56 percent of North American and European enterprises consider Web 2.0 to be a priority.

"If I wanted to be anywhere in the Web 2.0 economy, I'd want to be on the enterprise side," says report author and Forrester Research analyst Oliver Young. "We're seeing enterprise-class software coming from startups, but we're seeing them through very low price points . . . so it [Web 2.0] will never be a mega market," says Young. "It will eventually disappear into the fabric of the enterprise, despite the major effects the technology will have on how businesses market their products and optimize their workforces."

The consumer-facing ad-funded Web 2.0 sites like Facebook, MySpace, and Delicious will also have difficulty as similar technologies are incorporated into the enterprise. "Even Google is having a hard time selling the advertising," Young said. Still, start-ups have much to gain in pursuing the Web 2.0 world, such as understanding how companies are adopting their technology. Small groups within a company are more likely to adopt blogs, wikis, mashups, and widgets. The key to adoption, he adds, is to show how there is a business value in using the Web 2.0 tools. "Web 2.0 is not a critical 'must have' for any company at this point, but it's more than likely that your competition is using it and is showing faster results because of it."

SOURCE: Michael Singer, "Web 2.0: Companies Will Spend \$4.6 Billion by 2013, Forrester Predicts," *InformationWeek*, April 21, 2008.

Categories of e-Commerce

Many companies today are participating in or sponsoring four basic categories of e-commerce applications: **business-to-consumer**, **business-to-business**, **consumer-to-consumer** and business-to-government e-commerce. Note: We do not explicitly cover business-to-government (B2G) and *e-government* applications because they are beyond the scope of this text, but many e-commerce concepts apply to such applications.

BUSINESS-TO-CONSUMER (B2C) e-COMMERCE. In this form of e-commerce, businesses must develop attractive electronic marketplaces to sell products and services to consumers. For example, many companies offer e-commerce Web sites that provide virtual storefronts and multimedia catalogs, interactive order processing, secure electronic payment systems, and online customer support. The B2C marketplace is growing like a wildfire but still remains the tip of the iceberg when compared with all online commerce.

CONSUMER-TO-CONSUMER (C2C) e-COMMERCE. The huge success of online auctions like eBay, where consumers (as well as businesses) can buy from and sell to one another in an auction process at an auction Web site, makes this e-commerce model an important e-commerce business strategy. Thus, participating in or sponsoring consumer or business auctions is an important e-commerce alternative for B2C, C2B (consumer-to-business), or B2B e-commerce. Electronic personal advertising of products or services to buy or sell by consumers at electronic newspaper sites, consumer e-commerce portals, or personal Web sites is also an important form of C2C e-commerce.

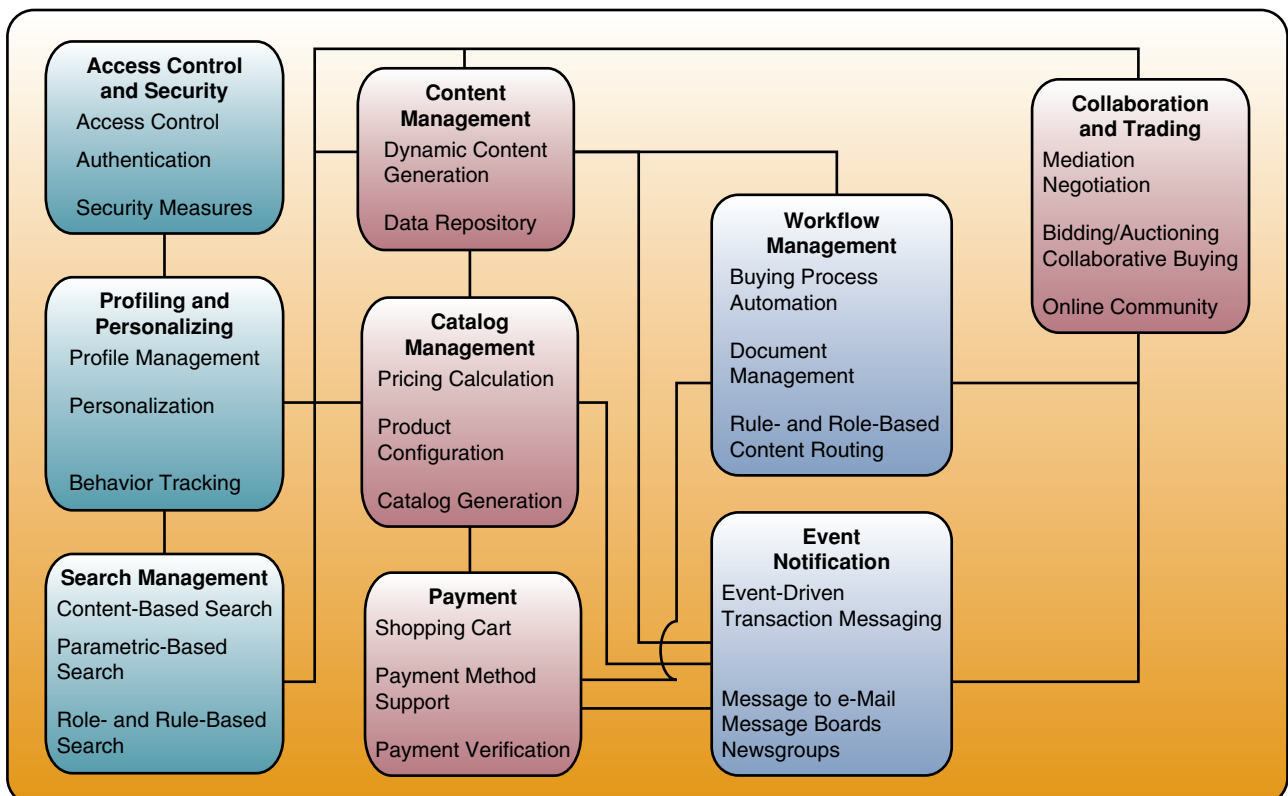
BUSINESS-TO-BUSINESS (B2B) e-COMMERCE. If B2C activities are the tip of the iceberg, B2B represents the part of the iceberg that is under the water—the biggest part. This category of e-commerce involves both e-business marketplaces and direct market links between businesses. For example, many companies offer secure Internet or extranet e-commerce catalog Web sites for their business customers and suppliers. Also very important are B2B e-commerce portals that provide auction and exchange marketplaces for businesses. Others may rely on electronic data interchange (EDI) via the Internet or extranets for computer-to-computer exchange of e-commerce documents with their larger business customers and suppliers.

The essential *e-commerce processes* required for the successful operation and management of e-commerce activities are illustrated in Figure 9.4. This figure outlines the nine key components of an *e-commerce process architecture* that is the foundation of the e-commerce initiatives of many companies today. We concentrate on the role these processes play in e-commerce systems, but you should recognize that many of these components may also be used in internal, noncommerce e-business applications. An example would be an intranet-based human resource system used by a company's employees, which might use all but the catalog management and product payment processes shown in Figure 9.4. Let's take a brief look at each essential process category.

ESSENTIAL e-COMMERCE PROCESSES LO 9-2

FIGURE 9.4

This e-commerce process architecture highlights nine essential categories of e-commerce processes.



Access Control and Security

E-commerce processes must establish mutual trust and secure access between the parties in an e-commerce transaction by authenticating users, authorizing access, and enforcing security features. For example, these processes establish that a customer and e-commerce site are who they say they are through user names and passwords, encryption keys, or digital certificates and signatures. The e-commerce site must then authorize access to only those parts of the site that an individual user needs to accomplish his or her particular transactions. Thus, you usually will be given access to all resources of an e-commerce site except for other people's accounts, restricted company data, and Web master administration areas. Companies engaged in B2B e-commerce may rely on secure industry exchanges for procuring goods and services or Web trading portals that allow only registered customers to access trading information and applications. Other security processes protect the resources of e-commerce sites from threats such as hacker attacks, theft of passwords or credit card numbers, and system failures. We discuss many of these security threats and features in Chapter 13.

Profiling and Personalizing

Once you have gained access to an e-commerce site, profiling processes can occur that gather data on you and your Web site behavior and choices, as well as build electronic profiles of your characteristics and preferences. User profiles are developed using profiling tools such as user registration, cookie files, Web site behavior tracking software, and user feedback. These profiles are then used to recognize you as an individual user and provide you with a personalized view of the contents of the site, as well as product recommendations and personalized Web advertising as part of a *one-to-one marketing* strategy. Profiling processes are also used to help authenticate your identity for account management and payment purposes and gather data for customer relationship management, marketing planning, and Web site management. Some of the ethical issues in user profiling are discussed in Chapter 13.

Search Management

Efficient and effective search processes provide a top e-commerce Web site capability that helps customers find the specific product or service they want to evaluate or buy. E-commerce software packages can include a Web site search engine component, or a company may acquire a customized e-commerce search engine from search technology companies like Google and Requisite. Search engines may use a combination of search techniques, including searches based on content (e.g., a product description) or parameters (e.g., above, below, or between a range of values for multiple properties of a product). While we tend to think of Google, Yahoo, or Bing when we think of search engines, currently there are more than 30 search engines on the Internet that perform functions that are unique to a particular search need. Try checking out *ChaCha*, *Faroo*, *Blekko*, *Dogpile* or *DuckDuckGo*. You can even search at *Gigablast*, which has more than 200 billion pages and is powered entirely via wind energy!

Content and Catalog Management

Content management software helps e-commerce companies develop, generate, deliver, update, and archive text data and multimedia information at e-commerce Web sites. For example, German media giant Bertelsmann, part owner of BarnesandNoble.com, uses StoryServer content manager software to generate Web page templates that enable online editors from six international offices to easily publish and update book reviews and other product information, which are sold (syndicated) to other e-commerce sites.

E-commerce content frequently takes the form of multimedia catalogs of product information. As such, generating and managing catalog content is a major subset of content management, or catalog management. For example, W.W. Grainger & Co., a multibillion-dollar industrial parts distributor, uses the CenterStage catalog management software suite to retrieve data from more than 2,000 supplier databases, standardize the data, translate it into HTML or XML for Web use, and organize and enhance the data for speedy delivery as multimedia Web pages at its www.grainger.com Web site. At K and G Cycles, a large online motorcycle parts and accessories retailer, more than 250,000 products organized into more than 5,000 categories are managed via X-Cart software such that customers can find whatever they are looking for in four clicks or less (www.kandgcycles.com)!

Content and catalog management software works with the profiling tools we mentioned previously to personalize the content of Web pages seen by individual users. For example, Travelocity.com uses OnDisplay content manager software to push personalized promotional information about other travel opportunities to users while they are involved in an online travel-related transaction.

Finally, content and catalog management may be expanded to include *product configuration* processes that support Web-based customer self-service and the *mass customization* of a company's products. Configuration software helps online customers select the optimum feasible set of product features that can be included in a finished product. For example, both Dell Computer and Cisco Systems use configuration software to sell built-to-order computers and network processors to their online customers.

Nothing is as heart-wrenching to an e-tailer as watching a customer abandon a full cart just seconds before consummating the deal. To be so close yet so cashless is more than frustrating; it's harmful to an e-tailer's health. A virtual armory of tools are in use to woo, cajole, prompt, and push consumers to make the buy—but are they working, or are they turning even more customers away?

"Most fall woefully short," says Matthew Brown, senior director of e-commerce and interactive marketing at MarketNet. "Instead of focusing on using tools and technologies to help the customer, much more thought and time needs to go into Web site architecture in the first place."

Many theories are being tossed about as to why consumers turn fickle a hair short of the finish line. For each theory, there are a multitude of technological solutions. "Retailers continue to launch and test technologies and features aimed at reducing abandonment or increasing online conversion," says Jessica Ried, a director of retail strategy at Resource Interactive. "In our experience, it is difficult to know for sure if any particular one is going to be effective for a given retailer without testing it with that retailer's customer base, or at least having a solid understanding of existing customer behaviors on the site through site analytics and surveys."

Once an e-tailer understands the true obstacles to closing the deal, there are a range of tools available to clear the way to bigger profits. The most commonly deployed are live chat, pop-up discounts, and follow-up e-mail programs; some are achieved through the standard use of cookies, others via pixel-based triggers. Third-person endorsements are also frequently used. "Hosting consumer-generated content such as ratings and reviews has typically allowed retailers to improve conversions," explains Ried, "as customers are more confident with their selections. That's because they have access to an 'unbiased' opinion, building trust rather than having to rely solely on the marketing copy on the retailer's site."

e-Commerce Tools to Close the Deal

“We use Liveperson chat extensively. It has been an incredible tool for answering any last-minute doubts during the last few states of the transaction,” notes Adrian Salamunovic, cofounder of DNA 11, a multimillion-dollar e-commerce art retailer. “Our average transaction is over US\$500, so this is very important to us.”

“It pays for itself many times over each month,” he adds. “For us, interrupting the client with pop-ups or invitations to chat really doesn’t work—in fact, it does the opposite. We’ve watched customers bounce (exit) quite quickly after being interrupted with pop-ups.”

Therein lies the conundrum. No two customers are identical. At least some personalized customization is essential. There is a point, however, at which actions considered helpful by the retailer are perceived as intrusive by the consumer. “Some customers welcome the help; others are unnerved by the Big Brother effect it can suggest,” says Resource Interactive’s Ried. “Start by considering what is known about consumer behavior in evaluating which technologies, features, and functionalities to explore first.”

SOURCE: Pam Baker, “Rescuing the e-Commerce Deal When the Customer’s Walking Way,” *E-Commerce Times*, April 24, 2009.

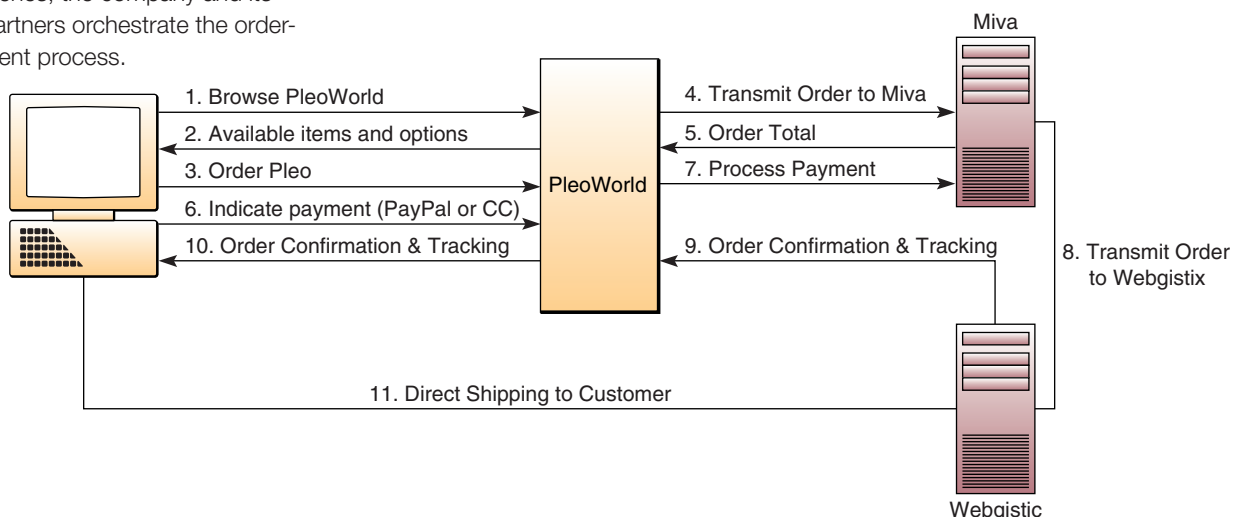
Workflow Management

Many of the business processes in e-commerce applications can be managed and partially automated with the help of **workflow management** software. E-business workflow systems for enterprise collaboration help employees electronically collaborate to accomplish structured work tasks within knowledge-based business processes. Workflow management in both e-business and e-commerce depends on a *workflow software engine* containing software models of the business processes to be accomplished. The workflow models express the predefined sets of business rules, roles of stakeholders, authorization requirements, routing alternatives, databases used, and sequence of tasks required for each e-commerce process. Thus, workflow systems ensure that the proper transactions, decisions, and work activities are performed, and the correct data and documents are routed to the right employees, customers, suppliers, and other business stakeholders.

As many of you begin your business careers, you will be charged with the responsibility of driving cost out of existing business processes while maintaining or improving

FIGURE 9.5

The PleoWorld site serves as the only interface with which customers work, whereas, behind the scenes, the company and its two partners orchestrate the order-fulfillment process.



the effectiveness of those processes. As you continue to acquire a greater appreciation for, and understanding of, how technology can benefit business, you will explore workflow management as the key to this optimization of cost and effectiveness throughout the business.

For example, Figure 9.5 illustrates the e-commerce processes of the PleoWorld site by Innvo Labs. Customers use this site to browse, choose, and order the animatronic dinosaur toy Pleo, while partners Miva and Webgistic take care of order validation, billing, and payments—including those in foreign currency—and direct shipping to the customer, all happening behind the interface of the PleoWorld site. Making all these processes transparent to the customer was a major design consideration.

Pleo is an animatronic dinosaur toy designed to emulate the appearance and imagined behavior of a week-old baby *Camarasaurus*. Created and designed by the inventor of the Furby, this particular species was chosen because its relatively large cranium provided the ideal place to hide the various sensors and motors necessary for its operation. Each individual unit would learn from its unique experiences and would, over time, develop its own personality. It was manufactured and distributed by a company called Ugobe, which went bankrupt in 2008. In 2009, Innvo Labs Corporation was formed and acquired the rights to the toy.

Under Ugobe, orders were placed through a Web site, with order fulfillment performed manually at a central location on an order-by-order basis, which significantly slowed down deliveries. Rather, the new owner wanted a process that would be as much “hands-off” as possible, from order to delivery. In addition, the new Web site would need to support international deliveries, order customization, and membership in the site, which allowed registered users to receive news, software updates, and other information. To address fraud concerns associated with international orders, buyer authentication and foreign currency processing were also implemented behind the scenes.

“The most successful e-commerce stores follow a conservative approach to design. The PleoWorld site does that, and allows for shoppers to have a seamless, simple and secure shopping process that promotes a high conversion rate,” says Rick Wilson, executive vice president at Miva Merchant, the platform vendor selected to run the new site. Another company involved in the project was Webgistic, which provided automated order fulfillment. One key aspect of the site is that as the transaction is processed, customers always stay within the PleoWorld environment; they are not bounced from interface to interface to complete all the steps, which don’t require user input. Figure 9.5 shows the complete sequence of steps in more detail.

Since the site went live in 2009, more than 20,000 member registrations have been received, including 7,500 existing Pleo owners who reregistered their existing units. In 2010, Innvo Labs launched a second-generation version, the Pleo rb, which, among other things, snores while it sleeps.

Innvo Labs: Automated e-Commerce Processes

SOURCE: “Innvo Labs—How PleoWorld’s Ecommerce Runs Itself,” Planetmagpie Case Study, October 2010.

Event Notification

Most e-commerce applications are *event-driven* systems that respond to a multitude of events—from a new customer’s first Web site access, to payment and delivery processes, to innumerable customer relationship and supply chain management activities. That is why event notification processes play an important role in e-commerce systems;

customers, suppliers, employees, and other stakeholders must be notified of all events that might affect their status in a transaction. **Event notification** software works with workflow management software to monitor all e-commerce processes and record all relevant events, including unexpected changes or problem situations. Then it works with user-profiling software to notify all involved stakeholders automatically of important transaction events using appropriate user-preferred methods of electronic messaging, such as e-mail, newsgroup, pager, and fax communications. This notification includes a company's management, who then can monitor their employees' responsiveness to e-commerce events and customer and supplier feedback.

For example, when you purchase a product at a retail e-commerce Web site like Amazon.com, you automatically receive an e-mail record of your order. Then you may receive e-mail notifications of any change in product availability or shipment status and, finally, an e-mail message notifying you that your order has been shipped and is complete. This type of event notification uses a software application referred to as an *autoresponder*. As events occur, the autoresponder application is triggered to send a particular email or notice to the customer. Autoresponder events can be timed such that a single event can trigger a series of messages set to be sent at specific intervals in the future.

Collaboration and Trading

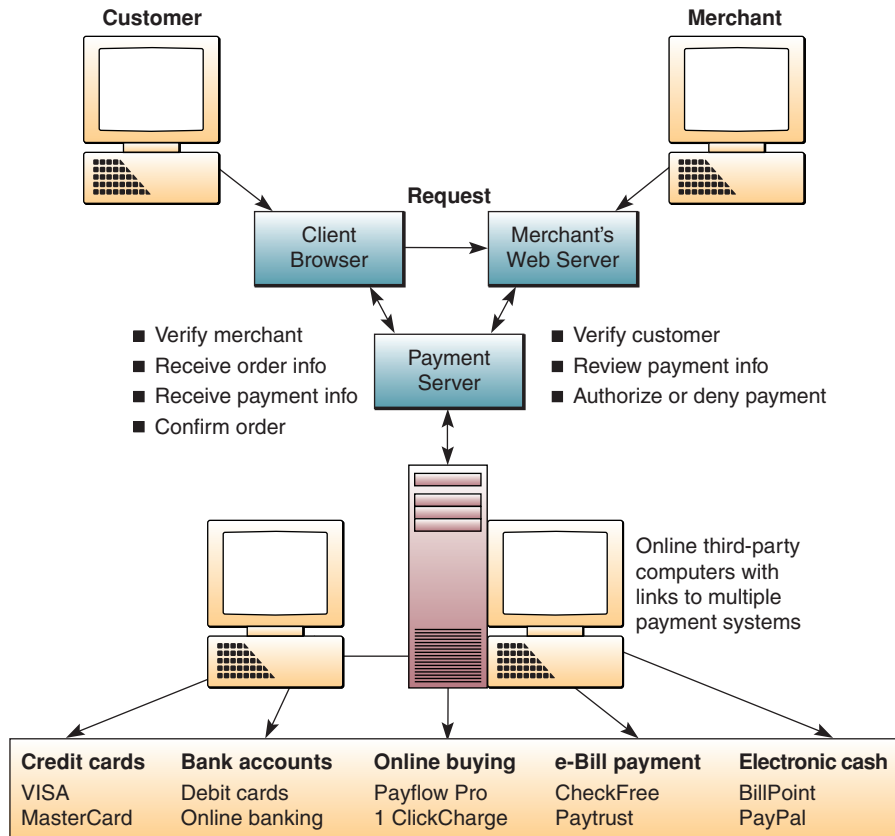
This major category of e-commerce processes consists of those that support the vital collaboration arrangements and trading services needed by customers, suppliers, and other stakeholders to accomplish e-commerce transactions. Thus, in Chapter 2, we discussed how a customer-focused e-business uses tools such as e-mail, chat systems, and discussion groups to nurture online *communities of interest* among employees and customers to enhance customer service and build customer loyalty in e-commerce. The essential collaboration among business trading partners in e-commerce may also be provided by Internet-based trading services. For example, B2B e-commerce Web portals provided by companies like Ariba and Commerce One support matchmaking, negotiation, and mediation processes among business buyers and sellers. In addition, B2B e-commerce is heavily dependent on Internet-based trading platforms and portals that provide online exchange and auctions for e-business enterprises. Therefore, the online auctions and exchanges developed by companies like FreeMarkets are revolutionizing the procurement processes of many major corporations. We will discuss these and other e-commerce applications later in the chapter.

ELECTRONIC PAYMENT PROCESSES LO 9-2

Payment for the products and services purchased is an obvious and vital set of processes in e-commerce transactions. Payment processes, however, are not simple because of the nearly anonymous electronic nature of transactions taking place between the networked computer systems of buyers and sellers and the many security issues involved. E-commerce payment processes are also complex because of the wide variety of debit and credit alternatives, as well as the financial institutions and intermediaries that may be part of the process. Therefore, a variety of **electronic payment systems** have evolved over time. In addition, new payment systems are being developed and tested to meet the security and technical challenges of e-commerce over the Internet.

Web Payment Processes

Most e-commerce systems on the Web involving businesses and consumers (B2C) depend on credit card payment processes, but many B2B e-commerce systems rely on more complex payment processes based on the use of purchase orders, as was illustrated in Figure 9.5. However, both types of e-commerce typically use an electronic

**FIGURE 9.6**

An example of a secure electronic payment system with many payment alternatives.

shopping cart process, which enables customers to select products from Web site catalog displays and temporarily put them in a virtual shopping basket for later checkout and processing. Figure 9.6 illustrates and summarizes a B2C electronic payment system with several payment alternatives.

Electronic Funds Transfer

Electronic funds transfer (EFT) systems are a major form of electronic payment systems in banking and retailing industries. EFT systems use a variety of information technologies to capture and process money and credit transfers between banks and businesses and their customers. For example, banking networks support teller terminals at all bank offices and automated teller machines (ATMs) at locations throughout the world. Banks, credit card companies, and other businesses may support pay-by-phone services. Very popular also are Web-based payment services, such as PayPal and BillPoint for cash transfers, and CheckFree and Paytrust for automatic bill payment, that enable the customers of banks and other bill payment services to use the Internet to pay bills electronically. In addition, most point-of-sale terminals in retail stores are networked to bank EFT systems, which makes it possible for you to use a credit card or debit card to pay instantly for gas, groceries, or other purchases at participating retail outlets.

Secure Electronic Payments

When you make an online purchase on the Internet, your credit card information is vulnerable to interception by *network sniffers*, software that easily recognizes credit card number formats. Several basic security measures are being used to solve this security problem: (1) encrypt (code and scramble) the data passing between the

FIGURE 9.7

VeriSign provides electronic payment, security, and many other e-commerce services.



SOURCE: Courtesy of VeriSign Inc.

customer and merchant, (2) encrypt the data passing between the customer and the company authorizing the credit card transaction, or (3) take sensitive information off-line. Note: Because encryption and other security issues are discussed in Chapter 13, we will not explain how they work in this section.

For example, many companies use the Secure Socket Layer (SSL) security method originally developed by Netscape Communications that automatically encrypts data passing between your Web browser and a merchant's server. However, sensitive information is still vulnerable to misuse once it's decrypted (decoded and unscrambled) and stored on a merchant's server, so a digital wallet payment system was developed. In this method, you add security software add-on modules to your Web browser. That enables your browser to encrypt your credit card data in such a way that only the bank that authorizes credit card transactions for the merchant gets to see it. All the merchant is told is whether your credit card transaction is approved or not.

The Secure Electronic Transaction (SET) standard for electronic payment security extends this digital wallet approach. In this method, software encrypts a digital envelope of digital certificates specifying the payment details for each transaction. VISA, MasterCard, IBM, Microsoft, Netscape, and most other industry players have agreed to SET. Therefore, a system like SET may become the standard for secure electronic payments on the Internet. See Figure 9.7.

e-Commerce Applications and Issues

LO 9-1

E-commerce is here to stay. The Web and e-commerce are key industry drivers. It's changed how many companies do business. It's created new channels for our customers. Companies are at the e-commerce crossroads, and there are many ways to go.

Thus, e-commerce is now defining how companies do business both internally and externally with their customers, suppliers, and other business partners. As managers confront a variety of e-commerce alternatives, the way companies apply e-commerce to their businesses is also subject to change. The applications of e-commerce by many companies have gone through several major stages as e-commerce matures in the world of business. For example, e-commerce between businesses and consumers (B2C) moved from merely offering multimedia company information at corporate Web sites (*brochureware*) to offering products and services at Web storefront sites via electronic catalogs and online sales transactions. B2B e-commerce, in contrast, started with Web site support to help business customers serve themselves, and then moved toward automating intranet and extranet procurement systems. One of the most important things to understand about e-commerce is that by converting a business model from bricks and mortar to an e-commerce approach, the transaction costs (i.e., the costs of doing business with a customer or supplier) drop dramatically. Thus, anything that can be digital will be digital. See Figure 9.8.

e-Commerce Trends

Figure 9.9 illustrates some of the trends taking place in the e-commerce applications that we introduced earlier in the chapter. Notice how B2C e-commerce moves from simple Web storefronts to interactive marketing capabilities that provide a personalized shopping experience for customers, and then toward a totally integrated Web store that supports a variety of customer shopping experiences. B2C e-commerce is also moving toward a self-service model in which customers configure and customize the products and services they wish to buy, aided by configuration software and online customer support as needed.

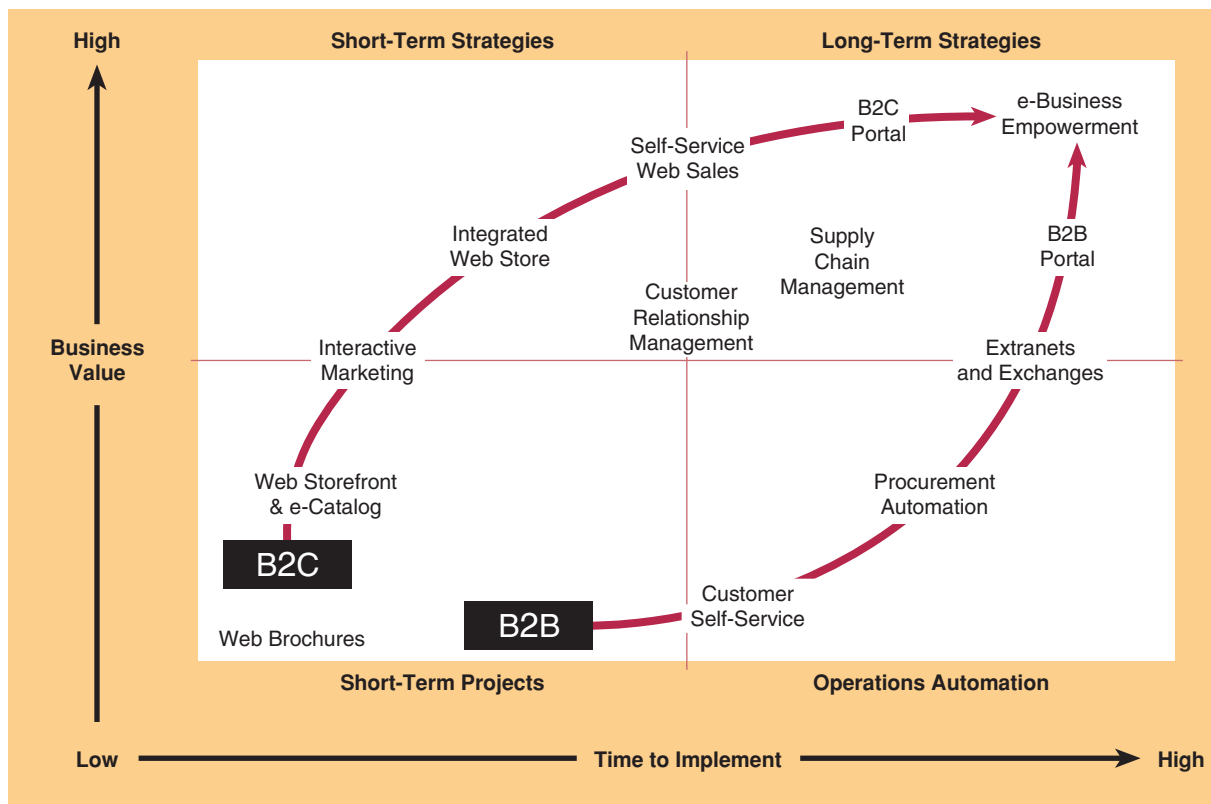
B2B e-commerce participants moved quickly from self-service on the Web to configuration and customization capabilities and extranets connecting trading partners. As B2C e-commerce moves toward full-service and wide-selection retail Web portals, the trend for B2B is toward the use of e-commerce portals that provide catalog, exchange,



SOURCE: © Digital Vision/PunchStock.

FIGURE 9.8

Online opinion leaders may be tapping into underlying trends that are critical to marketers.

**FIGURE 9.9**

Trends in B2C and B2B e-commerce, and the business strategies and value driving these trends.

SOURCE: Jonathan Rosenoer, Douglas Armstrong, and J. Russell Gates, *The Clickable Corporation: Successful Strategies for Capturing the Internet Advantage* (New York: The Free Press, 1999), p. 24.

and auction markets for business customers within or across industries. Of course, both of these trends are enabled by e-business capabilities like customer relationship management and supply chain management, which are the hallmarks of the customer-focused and inter-networked supply chains of a fully e-business-enabled company.

BUSINESS-TO-CONSUMER e-COMMERCE

LO 9-3

LO 9-4

E-commerce applications that focus on the consumer share an important goal: to attract potential buyers, transact goods and services, and build customer loyalty through individual courteous treatment and engaging community features.

What does it take to create a successful B2C e-commerce business venture? That's the question that many are asking in the wake of the failures of many pure B2C *dot-com* companies. One obvious answer would be to create a Web business initiative that offers attractive products or services of great customer value, with a business plan based on realistic forecasts of profitability within the first year or two of operation—a condition that was lacking in many failed dot-coms. Such failures, however, have not stemmed the tide of millions of businesses, both large and small, that are moving at least part of their business to the Web. So let's take a look at some essential success factors and Web site capabilities for companies engaged in either B2C or B2B e-commerce. Figure 9.10 provides examples of a few top-rated retail Web companies.

e-Commerce Success Factors

On the Internet, the barriers of time, distance, and form are broken down, and businesses are able to transact the sale of goods and services 24 hours a day, 7 days a week, 365 days a year with consumers all over the world. In certain cases, it is even possible to convert a physical good (CDs, packaged software, a newspaper) to a virtual good (MP3 audio, downloadable software, information in HTML format).

Top Retail Web Sites		
• Amazon.com	\$34.20B 2010 Web sales volume	www.amazon.com Amazon.com is the exception to the rule that consumers prefer to shop “real world” retailers online. The mother of all shopping sites, Amazon features a vast selection of books, videos, DVDs, CDs, toys, kitchen items, electronics, and even home and garden goods sold to millions of loyal customers.
• Staples, Inc.	\$9.8B 2010 Web sales volume	www.staples.com Staples tops the “Big 3” office supply giants in terms of Internet sales, although Office Depot and OfficeMax are also members of the top 10 retail Web sites list. Consumers can access the entire catalog online and can have their purchases delivered to their home or office within 24 hours and often within the same business day.
• Dell, Inc.	\$4.5B 2010 Web sales volume	www.dell.com Dell has created an online shopping experience for their customers that makes buying and configuring a computer system to meet a specific need almost effortless.
• Apple, Inc.	\$4.25B 2010 Web sales volume	www.apple.com Apple has captured the world with the iPod, iPhone, and iPad. 2011 is expected to be a great year for Apple with an estimated 50 million iPhones being sold.

FIGURE 9.10

Examples of a few top-rated retail Web sites.

A basic fact of Internet retailing is that all retail Web sites are created equal as far as the “location, location, location” imperative of success in retailing is concerned. No site is any closer to its Web customers, and competitors offering similar goods and services may be only a mouse click away. This scenario makes it vital that businesses find ways to build customer satisfaction, loyalty, and relationships so that customers keep coming back to their Web stores. Thus, the key to e-tail (retail business conducted online) success is to optimize several key factors, such as *selection and value*, *performance and service efficiency*, *the look and feel of the site*, *advertising and incentives to purchase*, *personal attention*, *community relationships*, and *security and reliability*. Let’s briefly examine each of these factors that are essential to the success of a B2C Web business. See Figure 9.11.

SELECTION AND VALUE. Obviously, a business must offer Web shoppers a good selection of attractive products and services at competitive prices, or the shoppers will quickly click away from a Web store. However, a company’s prices don’t have to be

e-Commerce Success Factors
• Selection and Value. Attractive product selections, competitive prices, satisfaction guarantees, and customer support after the sale.
• Performance and Service. Fast and easy navigation, shopping, and purchasing, and prompt shipping and delivery.
• Look and Feel. Attractive Web storefront, Web site shopping areas, multimedia product catalog pages, and shopping features.
• Advertising and Incentives. Targeted Web page advertising and e-mail promotions, discounts, and special offers, including advertising at affiliate sites.
• Personal Attention. Personal Web pages, personalized product recommendations, Web advertising and e-mail notices, and interactive support (online chat, Skype, and so forth) for all customers.
• Community Relationships. Virtual communities of customers, suppliers, company representatives, and others via newsgroups, chat rooms, and links to related sites.
• Security and Reliability. Security of customer information and Web site transactions, trustworthy product information, and reliable order fulfillment.
• Great Customer Communication. Easy-to-find contact information, online order status, product support specialists.

FIGURE 9.11

Some of the key factors for success in e-commerce.

the lowest on the Web if it builds a reputation for high-quality, guaranteed satisfaction and top customer support while shopping and after the sale. For example, top-rated e-tailer REI.com helps you select quality outdoor gear for hiking and other activities with a “How to Choose” section and gives a money-back guarantee on your purchases.

PERFORMANCE AND SERVICE. People don’t want to be kept waiting when browsing, selecting, or paying in a Web store. A site must be efficiently designed for ease of access, shopping, and buying, with sufficient server power and network capacity to support Web site traffic. Web shopping and customer service must also be friendly and helpful, as well as quick and easy. In addition, products offered should be available in inventory for prompt shipment to the customer.

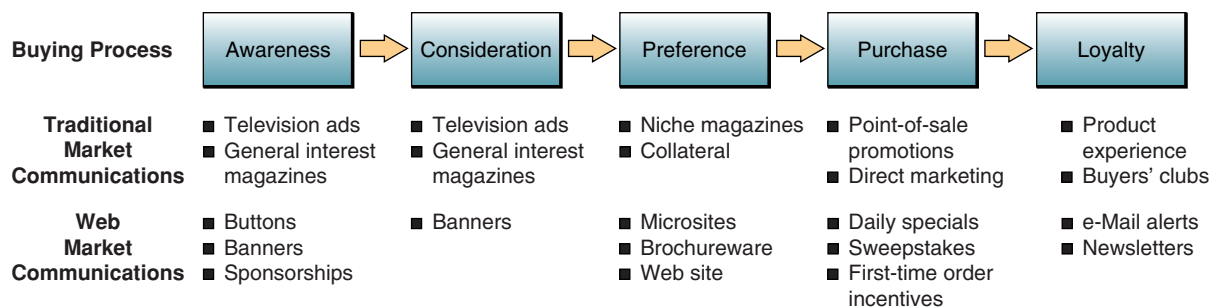
LOOK AND FEEL. B2C sites can offer customers an attractive Web storefront, shopping areas, and multimedia product catalogs. These could range from an exciting shopping experience with audio, video, and moving graphics to a more simple and comfortable look and feel. Thus, most retail e-commerce sites let customers browse product sections, select products, drop them into a virtual shopping cart, and go to a virtual check-out station when they are ready to pay for their order.

ADVERTISING AND INCENTIVES. Some Web stores may advertise in traditional media, but most advertise on the Web with targeted and personalized banner ads and other Web page and e-mail promotions. Most B2C sites also offer shoppers incentives to buy and return. Typically, these incentives mean coupons, discounts, special offers, and vouchers for other Web services, sometimes with other e-tailers at cross-linked Web sites. Many Web stores also increase their market reach by being part of Web banner advertising exchange programs with thousands of other Web retailers. Figure 9.12 compares major marketing communications choices in traditional and e-commerce marketing to support each step of the buying process.

PERSONAL ATTENTION. Personalizing your shopping experience encourages you to buy and make return visits. Thus, e-commerce software can automatically record details of your visits and build user profiles for you and other Web shoppers. Many sites also encourage you to register with them and fill out a personal interest profile. Then, whenever you return, you are welcomed by name or with a personal Web page, greeted with special offers, and guided to those parts of the site in which you are most interested. This *one-to-one marketing* and relationship-building power is one of the major advantages of personalized Web retailing.

FIGURE 9.12

How traditional and Web marketing communications differ in supporting each step of the buying process.



COMMUNITY RELATIONSHIPS. Giving online customers with special interests a feeling of belonging to a unique group of like-minded individuals helps build customer loyalty and value. Thus, Web site relationship and affinity marketing programs build and promote virtual communities of customers, suppliers, company representatives, and others via a variety of Web-based collaboration tools. Examples include discussion forums or newsgroups, chat rooms, message board systems, and cross-links to related Web site communities.

SECURITY AND RELIABILITY. As a customer of a successful Web store, you must feel confident that your credit card, personal information, and details of your transactions are secure from unauthorized use. You must also feel that you are dealing with a trustworthy business whose products and other Web site information you can trust to be as advertised. Having your orders filled and shipped as you requested, in the time frame promised, and with good customer support are other measures of an e-tailer's reliability.

GREAT CUSTOMER COMMUNICATIONS. As more consumers shift their habits from the traditional brick-and-mortar approach to an online shopping experience, one thing becomes even more important than ever: the need for constant and informative communication channels with the customer. Despite the conveniences associated with online shopping, consumers still have questions that need to be answered by a human being. Issues ranging from product information to order status or modification are often still handled the "old fashioned way." K and G Cycles, the large online motorcycle parts and accessories retailer, provides telephone, live chat space, and Skype access to customer representatives who will even help you pick out your purchases in real time.

Amazon.com has just launched an application on Facebook that enables members of the social network to buy gifts for each other based on wish lists registered with the online retailer. *Amazon Giver* also provides Facebook members with the option of viewing suggested items for friends based on interests listed on their profile pages. A second Facebook application, *Amazon Grapevine*, provides a news feed of friends' activity on Amazon, such as when they update their wish lists, write reviews, or tag products. Both applications share information only with Facebook members who have opted in to the service.

"By combining Amazon's vast selection of products with Facebook's millions of users, we are able to make activities like gift-giving more efficient and rewarding for Facebook users," says Eva Manolis, vice president of Amazon.

By adding the *Amazon Giver* application to their profile, Facebook members get the option of clicking directly to a secure Amazon checkout page. If the recipient has a wish list, then Amazon can ship the item without the buyer entering a shipping address, which would already be on file. In order for people to view a wish list, it would have to be set as "public." With *Amazon Grapevine*, people have the option to choose what type of activity they would be willing to share with friends through the news feed. Activity updates are entirely opt-in.

Amazon.com has also introduced a new way for online merchants to leverage Amazon's infrastructure to ship physical products. "The *Amazon Fulfillment Web Service* (Amazon FWS) allows merchants to tap in to Amazon's network of fulfillment centers and our expertise in logistics," says Amazon Web

Amazon.com: Partnering and Leveraging Infrastructure

Services evangelist Jeff Barr. “Merchants can store their own products to our fulfillment centers and then, using a simple Web service interface, fulfill orders for the products.”

Amazon FWS is designed to complement *Fulfillment By Amazon* (FBA), the fulfillment service Amazon has offered since 2006, by making the fulfillment process accessible programmatically. Amazon also maintains a separate fulfillment program called *Amazon Advantage*, which allows content publishers to send Amazon music, books, and videos for sale on consignment, with a 55 percent fee.

The idea, Barr explains, is to be able to ship a product with a simple Web service call. By making it possible for merchants to further automate their e-commerce and fulfillment efforts, Amazon is demonstrating its commitment to selling “muck,” as CEO Jeff Bezos has referred to his company’s e-commerce infrastructure.

SOURCE: Antone Gonsalves, “Amazon.com Launches Shopping Apps on Facebook,” *InformationWeek*, March 13, 2008; and Thomas Claburn, “Amazon Introduces Fulfillment Web Service,” *InformationWeek*, March 20, 2008.

WEB STORE REQUIREMENTS LO 9-3

FIGURE 9.13

To develop a successful e-commerce business, these Web store requirements must be implemented by a company or its Web site hosting service.

Most business-to-consumer e-commerce ventures take the form of retail business sites on the World Wide Web. Whether a huge retail Web portal like Amazon.com or a small specialty Web retailer, the primary focus of such e-tailers is to develop, operate, and manage their Web sites so they become high-priority destinations for consumers who will repeatedly choose to go there to buy products and services. Thus, these Web sites must be able to demonstrate the key factors for e-commerce success that we have just covered. In this section, let’s discuss the essential Web store requirements that you would have to implement to support a successful retail business on the Web, as summarized and illustrated in Figure 9.13.

Developing a Web Store		
Build - Web site design tools - Site design templates - Custom design services - Web site hosting	Market - Web page advertising - E-mail promotions - Web advertising exchanges with affiliate sites - Search engine registrations and optimization	
Serving Your Customers		
Serve - Personalized Web pages - Dynamic multimedia catalog - Catalog search engine - Integrated shopping cart	Transact - Flexible order process - Credit card processing - Shipping and tax calculations - E-mail order notifications	Support - Web site online help - Customer service e-mail - Discussion groups and chat rooms - Links to related sites
Managing a Web Store		
Manage - Web site usage statistics - Sales and inventory reports - Customer account management - Links to accounting system	Operate 24×7 Web site hosting - Online tech support - Scalable network capacity - Redundant servers and power	Protect - User password protection - Encrypted order processing - Encrypted Web site administration - Network firewalls and security monitors

Stork Craft Manufacturing Inc. (“Stork Craft”) is an industry leader in the design, manufacture, and distribution of juvenile products. Stork Craft is one of the world’s leading juvenile furniture manufacturers. Founded more than 60 years ago in Vancouver, Canada, the company started with its founder selling hand-crafted furniture out of his basement. Over the years it has racked up a number of awards, including the Ernst & Young Entrepreneur of the Year award in 2006, the Sears Supplier of the Year in 2007, and a finalist to Canada’s Best Managed Companies in 2008.

Stork Craft has had a retail and distribution partnership with Wal-Mart for the United States and Canada since 2005, with both companies jointly developing a Web site to purchase Stork Craft products in the United States in 2006. The initiative proved very profitable to the company, which was able to acquire its three main competitors in a span of 18 months. In 2008, the goal was to develop a similar partnership with Wal-Mart Canada, with similar results, Stork Craft hoped.

The driving force behind the project was Irene Jeremic, then corporate executive advisor and chief information officer for Stork Craft. “It is not often that you have a business-to-business partner trying to cater to the general public,” says Jeremic. “The fact that Wal-Mart was open to this was perhaps the biggest factor in getting the site to work.”

Up to then, interested customers could only browse, but not purchase, Stork Craft products through the Web site of Wal-Mart Canada. After assembling a cross-functional team with executives from both companies, Stork Craft’s IT group built the entire technology platform, which included a Web site in which Wal-Mart Canada customers could shop, as well as management and administrative functionality. In addition, instead of shipping to Wal-Mart Canada, which in turn would ship to the end customers, Stork Craft agreed to ship its products directly to the final consumers, as well as provide direct-sale support such as handling returns and warranty and service inquiries. As an alternative to in-store sales of Stork Craft products at Wal-Mart stores, the online sale plus direct shipping alternative allowed Stork Craft to cut delivery times by up to 85 percent, as well as keep better control over the delivery process.

The co-branded site, where Wal-Mart Canada gets a portion of every sale made, went live in 2008 when shoppers could purchase more than 200 different cribs, tables, and toddler beds. Since then, more than 350,000 customers have visited the site, which is not an insignificant fraction of the total adult population of the country. Jeremic says the Web site has provided “great value” to both partners.

Stork Craft Manufacturing Helps Wal-Mart Canada



SOURCE: Matt Villano, “Ringling Up Baby,” *CIO Magazine*, August 1, 2009, and www.storkcraftdirect.com, accessed May 30, 2011.

Getting Customers to Find You

Just because your Web store has been launched does not mean customers will come flocking to your cyber front door. Your Web store needs to be discovered by your customers, and this means getting listed in the popular search engines.

You can submit your Web site to search engines such as Yahoo, Google, Live, and others, and each will begin looking at your Web pages and listing you when appropriate search terms are entered. Waiting for your site to show up competitively ranked with all the other similar sites could take weeks and even months. There is a science to search engine ranking and it is an essential element in Web store success.

Search engine optimization (SEO) is considered a subset of search engine marketing, and it focuses on improving the number and/or quality of visitors to a Web site

over “natural” (also called “organic” or “algorithmic” search engine) listings. The term SEO can also refer to search engine optimizers, an industry of consultants who carry out optimization projects on behalf of clients.

Search engines display different kinds of listings on a results page, including paid advertising in the form of pay-per-click (PPC) advertisements and paid inclusion listings, as well as unpaid organic search results and keyword-specific listings, such as news stories, definitions, map locations, and images. As an Internet marketing strategy, SEO considers how search engines work and what people search for.

Optimizing a Web site primarily involves editing its content and HTML coding, both to increase its relevance to specific keywords and to remove barriers to the indexing activities of search engines. Because SEO requires making changes to the source code of a site, it is often most effective when incorporated into the initial development and design of a site, leading to the use of the term *search engine friendly* to describe designs, menus, content management systems, and shopping carts that can be optimized easily and effectively.

A range of strategies and techniques are employed in SEO, including changes to a site’s code (referred to as *inside SEO*) and getting links from other sites (referred to as *outside SEO*). These techniques include two broad categories: techniques that search engines recommend as part of good design, and those techniques that search engines do not approve of and attempt to minimize the effect of, referred to as spamdexing. Methods such as *link farms*, where a group of Web sites is set up so that all hyperlink to every other Web site in the group, and *keyword stuffing*, where a Web page is loaded with keywords in the meta tags or in content, are examples of techniques considered “black hat” SEO. Such techniques serve only to degrade both the relevance of search results and the user experience of search engines.

SEO, as a marketing strategy, can often generate a good return. However, as the search engines are not paid for the traffic they send from *organic search*, the algorithms used can and do change, and there are many factors that can cause search engine problems when crawling or ranking a site’s pages. In contrast to black hat SEO techniques, *white hat SEO* requires a series of tools that constantly look at keywords and variants of search terms for specific items and then assist in the optimization of various pages on a website such that it shows up closer to the front of the pack when a specific set of search terms are entered. This activity can be very demanding when it comes to time and often small businesses must pay large sums of money to outside vendors specializing in SEO services. There are no guarantees of success, either in the short or long term. Because of the lack of guarantees and certainty, SEO is often compared to traditional public relations (PR), with PPC advertising closer to traditional advertising.

Serving Your Customers

Once your retail store is on the Web and receiving visitors, the Web site must help you welcome and serve them personally and efficiently so that they become loyal customers. So most e-tailers use several Web site tools to create user profiles, customer files, and personal Web pages and promotions that help them develop a one-to-one relationship with their customers. This effort includes creating incentives to encourage visitors to register, developing *Web cookie files* to identify returning visitors automatically, or contracting with Web site tracking companies like DoubleClick and others for software to record and analyze the details of the Web site behavior and preferences of Web shoppers automatically.

Of course, your Web site should have the look and feel of an attractive, friendly, and efficient Web store. That means having e-commerce features like a dynamically changing and updated multimedia catalog, a fast catalog search engine, and a convenient shopping cart system that is integrated with Web shopping, promotions, payment, shipping, and customer account information. Your e-commerce order processing software should be fast and able to adjust to personalized promotions and customer options like gift handling, special discounts, credit card or other payments, and shipping and tax alternatives.

Also, automatically sending your customers e-mail notices to document when orders are processed and shipped is a top customer service feature of e-tail transaction processing.

Providing customer support for your Web store is an essential Web site capability. Thus, many e-tail sites offer help menus, tutorials, and lists of FAQs (frequently asked questions) to provide self-help features for Web shoppers. Of course, e-mail correspondence with customer service representatives of your Web store offers more personal assistance to customers. Establishing Web site discussion groups and chat rooms where your customers and store personnel can interact helps create a more personal community that can provide invaluable support to customers, as well as build customer loyalty. Providing links to related Web sites from your Web store can help customers find additional information and resources, as well as earning commission income from the affiliate marketing programs of other Web retailers. For example, the Amazon.com affiliate program pays commissions of up to 15 percent for purchases made by Web shoppers clicking to its Web store from your site.

Managing a Web Store

A Web retail store must be managed as both a business and a Web site, and most e-commerce hosting companies offer software and services to help you do just that. For example, companies like FreeMerchant, Prodigy Biz, and Verio provide their hosting clients with a variety of management reports that record and analyze Web store traffic, inventory, and sales results. Other services build customer lists for e-mail and Web page promotions or provide customer relationship management features to help retain Web customers. Also, some e-commerce software includes links to download inventory and sales data into accounting packages like QuickBooks for bookkeeping and preparation of financial statements and reports.

Of course, Web-hosting companies must enable their Web store clients to be available online 24 hours a day and seven days a week all year. This availability requires them to build or contract for sufficient network capacity to handle peak Web traffic loads and redundant network servers and power sources to respond to system or power failures. Most hosting companies provide e-commerce software that uses passwords and encryption to protect Web store transactions and customer records, as well as to employ network firewalls and security monitors to repel hacker attacks and other security threats. Many hosting services also offer their clients 24-hour tech support to help them with any technical problems that arise. We will discuss these and other e-commerce security management issues in Chapter 13.

Historically, luxury brands have been slow to embrace e-commerce. But in recent years, high-end retail sites like Net-a-Porter and Yoox, as well as discount luxury flash sales like those on Gilt Groupe and Rue La La, are forcing executives to rethink the benefits of online sales. Bain & Co. estimates that the \$4.9 billion online luxury market grew by 20 percent in 2009.

Richemont, which owns luxury names like Cartier, Van Cleef & Arpels, Montblanc, and Jaeger-LeCoultre, has a 33 percent stake in Net-a-Porter and will buy the remaining 66 percent of the company, with founder Natalie Massenet remaining as the executive chairman. Richemont made the offer, valuing Net-a-Porter at \$534 million.

Net-a-Porter, founded in 2000 by former fashion journalist Natalie Massenet, has been a forerunner in selling expensive designer women's clothes and accessories online. That is a space that was long overlooked by big luxury-goods houses like Richemont, Burberry PLC, and LVMH Moët Hennessy Louis Vuitton SA, which jumped on the online sales bandwagon far later than their lower-priced counterparts did.

Luxury Goes Digital: Fashion House Embraces Online Shopping

With the acquisition of a successful luxury e-tailer—Net-a-Porter saw sales of \$183 million in 2009—Richemont is clearly making a commitment to boosting its presence in the online luxury space. Just one month earlier, Cartier had launched its U.S. transactional site.

At the time, Cartier North America CEO Emmanuel Perrin acknowledged the importance of selling on the Web. “The Internet has been a medium taking an increasing part in our client’s lifestyle and means of interaction,” he says.

Being available online is no longer a stigma to luxury brands, and things like holograms allow them to help consumers identify authorized resellers online. High-end designers like Narciso Rodriguez and Norma Kamali have even created exclusive collections for eBay.

The big luxury brands have made digital retailing a higher priority, having recognized that shoppers are increasingly willing to buy very expensive products on the Web. But selling \$1,000 dresses online is different from hawking groceries or secondhand books: Customers want an editorial element, a guiding hand to replace the in-store salesperson and signal what’s in style, which is where Net-a-Porter has carved out its niche.

“It’s just as much a magazine as it is a store,” says Massenet. “That really has served us well, because when you’re online you lose the offline experience of walking into a store.”

Says Massenet, “Richemont has completely embraced our vision and strategy since they came on board as a shareholder and together we are going to continue to build the 21st-century model for luxury fashion retailing.”

That model would be online shopping.

SOURCE: Anne C. Lee, “Luxury Goes Digital: Fashion House Richemont Embraces E-Commerce,” *Fast Company*, April 1, 2010; and Paul Sonne, “Richemont to Buy Net-a-Porter,” *The Wall Street Journal*, April 2, 2010.

BUSINESS-TO-BUSINESS e-COMMERCE

LO 9-4

Business-to-business e-commerce is the wholesale and supply side of the commercial process, where businesses buy, sell, or trade with other businesses. B2B e-commerce relies on many different information technologies, most of which are implemented at e-commerce Web sites on the World Wide Web and corporate intranets and extranets. B2B applications include electronic catalog systems, electronic trading systems such as exchange and auction portals, electronic data interchange, electronic funds transfers, and so on. All of the factors for building a successful retail Web site that we discussed previously also apply to wholesale Web sites for business-to-business e-commerce.

In addition, many businesses are integrating their Web-based e-commerce systems with their e-business systems for supply chain management, customer relationship management, and online transaction processing, as well as with their traditional, or legacy, computer-based accounting and business information systems. This integration ensures that all e-commerce activities are integrated with e-business processes and supported by up-to-date corporate inventory and other databases, which in turn are automatically updated by Web sales activities.

Avnet Tears Up the B2B e-Commerce Playbook

When does a global distributor of electronic components need to start operating more like Amazon.com and other consumer-focused companies? When the market demands that it move in that direction.

At Avnet Inc., they came to just that realization a few years ago as they saw a shift taking place within their electronic components market. While large manufacturers continued to buy large quantities of components for their designs, a growing segment of engineers and smaller companies wanted to buy

low volumes (including product samples) online, instead of by phone or face to face. Many of their customers had to either be really patient or just stubborn to make a successful purchase on the e-commerce site they offered at the time.

Avnet realized the need to shift their B2B e-commerce approach to incorporate a B2C perspective. While they were dealing with business customers, their online purchasing expectations were shaped by their experiences on consumer-friendly Web sites such as Amazon.com and HomeDepot.com. The problem was that the experience and functionality that those kinds of sites provide users isn't easily replicated in a B2B environment, especially within the components industry. For example, Avnet deals with millions of parts, and each part has dozens of technical attributes that must be precisely specified for engineers to determine whether it's the part they need. Additionally, legal and country-specific regulations determine which companies and individuals Avnet can ship certain parts to around the world.

So Avnet went ahead and made a few changes. First, they eliminated the need to register. Previously, all customers had to register to get on the commerce site. To make matters even more complicated, all customers had to qualify for credit before they could search for parts, even if they were purchasing with a credit card. Now anyone can search for part information without having to register and share personal details. Only when customers reach a purchase point does the site then ask them for their information. And if the customer is paying with a credit card, credit checks are out the window.

Previously, customers could search for parts only by entering precise supplier part numbers, which may be up to 50 characters—the equivalent of making a reader search for a book by the unique ISBN number. Furthermore, the search result displayed information on only the part corresponding to the number entered, not on alternative parts that may also meet the customer's requirements. Customers can now search by part number, product name, description, and technical attributes. Returned search results now include similar products that match the engineer's requirements, so the engineer can make informed decisions about alternative parts based on factors such as availability, cost, and manufacturer.

The new e-commerce site, featuring more than 3.5 million electronic components, took two years to develop and deploy, and Avnet keeps on adding functionality based on customer feedback. So far, however, results tell them that customers already like what they see: There has been a 75 percent annual increase in e-commerce revenue and a 50 percent annual increase in site visitors.

SOURCE: Steve Phillips and Beth Ely, "Global CIO: Avnet Tears Up the B2B E-Commerce Playbook," *InformationWeek*, June 15, 2009.

The latest e-commerce transaction systems are scaled and customized to allow buyers and sellers to meet in a variety of high-speed trading platforms: auctions, catalogs, and exchanges.

Businesses of any size can now buy everything from chemicals to electronic components, excess electrical energy, construction materials, or paper products at business-to-business **e-commerce marketplaces**. Figure 9.14 outlines five major types of e-commerce marketplaces used by businesses today. However, many B2B **portals** provide several types of marketplaces. Thus, they may offer an electronic **catalog** shopping and ordering site for products from many suppliers in an industry. Or they may serve as an **exchange** for buying and selling via a bid-ask process or at negotiated prices. Very popular are electronic **auction** Web sites for B2B auctions

e-COMMERCE MARKETPLACES LO 9-4

FIGURE 9.14

Types of e-commerce marketplaces.

e-Commerce Marketplaces
• One to Many. Sell-side marketplaces. Host one major supplier, who dictates product catalog offerings and prices. Examples: Cisco.com and Dell.com.
• Many to One. Buy-side marketplaces. Attract many suppliers that flock to these exchanges to bid on the business of a major buyer like GE or AT&T.
• Some to Many. Distribution marketplaces. Unite major suppliers who combine their product catalogs to attract a larger audience of buyers. Examples: VerticalNet and Works.com.
• Many to Some. Procurement marketplaces. Unite major buyers who combine their purchasing catalogs to attract more suppliers and thus more competition and lower prices. Examples: the auto industry.
• Many to Many. Auction marketplaces used by many buyers and sellers that can create a variety of buyers. Examples: eBay and FreeMarkets.

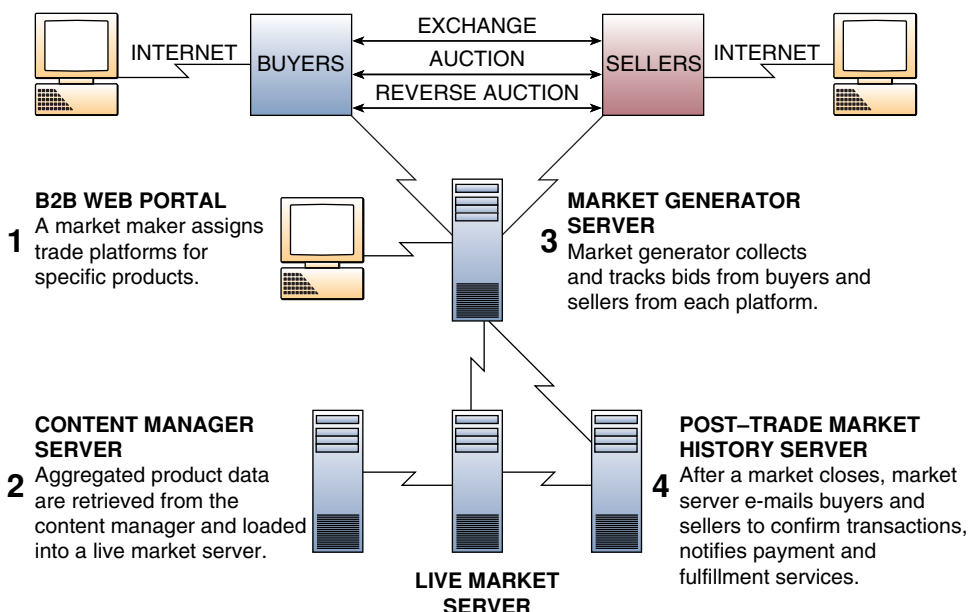
of products and services. Figure 9.15 illustrates a B2B trading system that offers exchange, auction, and reverse auction (where sellers bid for the business of a buyer) electronic markets.

Many of these B2B e-commerce portals are developed and hosted by third-party *market-maker* companies who serve as **infomediaries** that bring buyers and sellers together in catalog, exchange, and auction markets. Infomediaries are companies that serve as intermediaries in e-business and e-commerce transactions. Examples are Ariba, Commerce One, and VerticalNet, to name a few successful companies. All provide e-commerce marketplace software products and services to power business Web portals for e-commerce transactions.

These B2B e-commerce sites make business purchasing decisions faster, simpler, and more cost effective because companies can use Web systems to research and transact with many vendors. Business buyers get one-stop shopping and accurate purchasing information. They also get impartial advice from infomediaries that they can't get from the sites hosted by suppliers and distributors. Thus, companies can negotiate or bid for better prices from a larger pool of vendors. Of course, suppliers benefit from easy access to customers from all over the globe. Now, let's look at a real-world example.

FIGURE 9.15

An example of a B2B e-commerce Web portal that offers exchange, auction, and reverse auction electronic markets.



Online marketplaces like Craigslist and Freecycle allow consumers to make low-cost sales—or even exchange goods for free—through sophisticated technological systems that make such transactions efficient.

Some companies are attempting to apply a similar model to online business-to-business marketplaces.

The FCC holds auctions to grant licenses for radio spectrums, and most of these are used by cell phone carriers, or for first responders and their communication gear. But some of these spectrums aren't being used for a variety of reasons.

Spectrum Bridge's Web site, SpecEx.com, aims to create a secondary market for these unused spectrums. The company says the site can provide an easy and effective way to connect buyers and sellers. The market could potentially be large, as public safety agencies and major wireless carriers like Verizon Wireless and AT&T routinely purchase spectrums on the secondary market. The cable companies could also become potential buyers, especially as some are eyeing the wireless voice space. Spectrum Bridge makes money by taking a percentage of the transaction.

All transfers of spectrum would have to be approved by the FCC, but the agency has been supportive of spectrum trading in the past.

The idea of organizing the secondary spectrum market isn't a new one, but previous attempts have not been successful because they couldn't get enough buyers and sellers. "The spectrum world is almost tribal," says Peter Stanforth, chief technology officer for Spectrum Bridge. "It consists of small groups of people who know each other—and do everything manually." That is not an efficient system for smaller parcels—SpecEx's sweet spot. "By automating a lot of functions and bringing in a wider audience of buyers and sellers, we are making these smaller pieces more liquid and valuable," explains Stanforth.

Rick Rotondo, chief marketing officer of Spectrum Bridge, compares the SpecEx service to Craigslist, a favorite site for consumer bargains. With its launch several years ago, Craigslist made the sale of small consumer items efficient, which is what SpecEx aims to do with respect to the sale of wireless spectrum parcels. "Let's say you had used sunglasses you wanted to sell, for maybe \$25. Before online classifieds were introduced, it would not have been cost-efficient to try to sell them to a huge audience in a paper, because the ad probably would have cost you \$20." Same thing with wireless spectrum, he says. "Transaction costs are eating up most of the value for small buyers and sellers."

E-commerce technology can standardize much of the process, notes Stanforth. "What we are trying to do is be the eBay of the wireless spectrum world—a one-stop shop where companies can go to monetize excess or idle spectrum, and spectrum seekers can go to find reasonably priced unused spectrums."

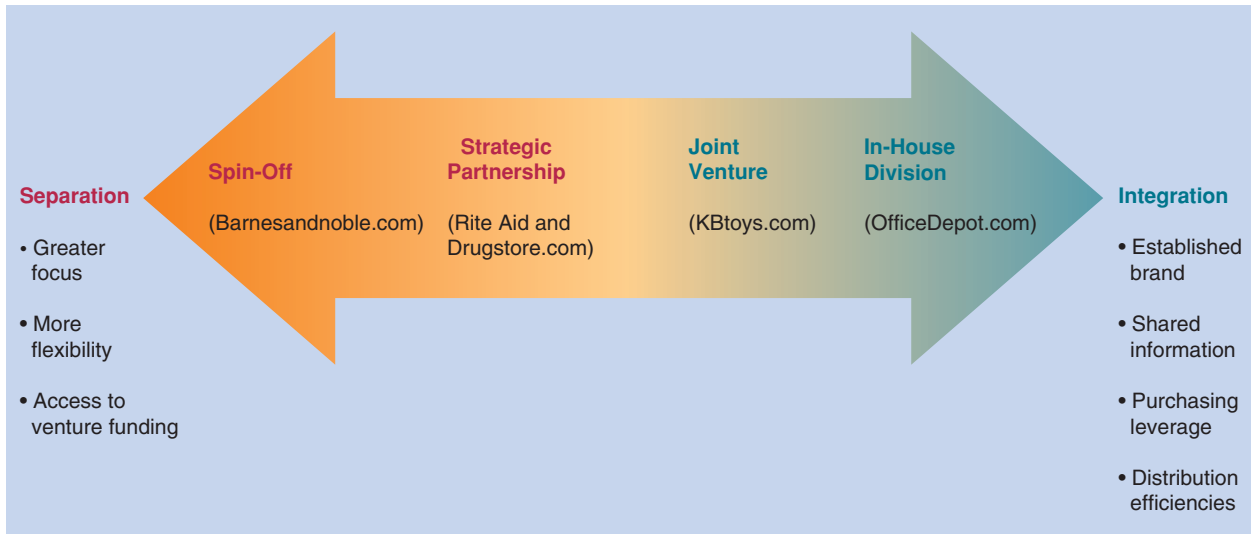
SpecEx.com: B2B Trading of Wireless Spectrum

SOURCE: Erika Morphy, "The Corporate Bargain Hunters' Quest for a Business Model," *E-Commerce Times*, January 20, 2009; and Marin Perez, "Spectrum Bridge Launches Online Secondary Market," *InformationWeek*, September 5, 2008.

Companies are recognizing that success will go to those who can execute clicks-and-mortar strategies that bridge the physical and virtual worlds. Different companies will need to follow very different paths when deciding how closely—or loosely—to integrate their Internet initiatives with their traditional operations.

Figure 9.16 illustrates the spectrum of alternatives and benefit trade-offs that e-business enterprises face when choosing an e-commerce **clicks-and-bricks strategy**. E-business managers must answer this question: Should we integrate our e-commerce virtual business operations with our traditional physical business operations or keep them separate? In today's networked world, a bricks-and-mortar business that does not

CLICKS AND BRICKS IN
e-COMMERCE
LO 9-5

**FIGURE 9.16**

Companies have a spectrum of alternatives and benefit trade-offs when deciding on an integrated or separate e-commerce business.

adopt a clicks-and-bricks strategy is sometimes referred to as a brink-and-mortar business, suggesting it is on the brink of extinction. While that may be somewhat of an exaggeration, businesses of any size that have not yet adopted an Internet presence, at minimum, are very likely feeling the effects in terms of lost sales and market share.

In February 2011, bricks-and-mortar mainstay Borders Inc. filed for Chapter 11 bankruptcy protection. As competition for consumers intensified amid the growth of the book superstore in the 1990s, Borders made a number of crucial gaffes, including transferring its Internet operations to Amazon in 2001 and embarking on an overseas expansion that swelled its debt. In recent years, Borders had started to shift the focus away from its physical presence by halting expansion plans and identifying unproductive stores for closure, but their recognition of the need for a strategic clicks-and-bricks approach came too late to prevent the bankruptcy. Borders claimed to be focusing on a restructuring plan that would more closely target the Internet sales channel than was done previously. Later in 2011, Borders closed its doors for good. As Figure 9.16 shows, successful companies have implemented a range of integration/separation strategies and made key benefit trade-offs in answering the integration question. Let's take a look at several alternatives.

e-Commerce Integration

The Internet is just another channel that gets plugged into the business architecture.

So says Bill Seltzer, former chief information officer of the office supply retailer Office Depot, which fully integrates its OfficeDepot.com e-commerce sales channel into its traditional business operations. Thus, Office Depot is a prime example of why many companies have chosen integrated clicks-and-bricks strategies, where their e-commerce business is integrated in some major ways into the traditional business operations of a company. The business case for such strategies rests on:

- Capitalizing on any unique strategic capabilities that may exist in a company's traditional business operations that could be used to support an e-commerce business.
- Gaining several strategic benefits of integrating e-commerce into a company's traditional business, such as sharing established brands and key business information, joint buying power, and distribution efficiencies.

For example, Office Depot already had a successful catalog sales business with a professional call center and a fleet of more than 2,000 delivery trucks. Its 1,825 stores and 30 warehouses were networked by a sophisticated information system that

provided complete customer, vendor, order, and product inventory data in real time. These business resources made an invaluable foundation for coordinating Office Depot's e-commerce activities and customer services with its catalog business and physical stores. In 2010, Office Depot boasted 9 public Web sites in North America, more than 40 public Web sites outside North America, and more than 15 corporate contract customized Web sites. Thus, customers can shop at OfficeDepot.com at their home, business, or in-store kiosks. Then they can choose to pick up their purchases at the store or have them delivered. In addition, the integration of Web-enabled e-commerce applications within Office Depot's traditional store and catalog operations has helped increase the traffic at their physical stores and improved the catalog operation's productivity and average order size.

Borders Group Inc. is an international book and music retailer based in Ann Arbor, Michigan. Before filing for bankruptcy protection in 2011 and later going out of business, the company operated close to 1,000 stores in the United States under the Borders and Waldenbooks brands. Computerworld columnist Frank Hayes believes there are a number of IT lessons to be learned from this debacle:

First, you cannot run a company without IT-savvy management. When founded by brothers Tom and Louis Borders, the company had what was then a state-of-the-art punch card system that allowed them to track and adjust inventory in a near real-time basis. It was reportedly one of the main reasons Kmart bought Borders in 1992: to use the system to run its Waldenbooks chain. The brothers, however, cashed out and all managers who came after them never had a strong grasp of IT. Sometimes even the best technology does not scale well; the system worked fine for a few stores, but not for hundreds.

Companies need to keep getting better all of the time. In 2000, according to Forrester, Borders overtook Amazon.com as the best online bookseller. Amazon kept improving, while Borders turned its attention elsewhere. Borders also missed the chance to emulate, borrow, or steal the best ideas that were introduced in the Web site of its main competitor, Amazon. At some point, Borders fell too far behind.

Right now most analysts decry the deal that, in 2001, gave Amazon.com control of the Borders online operations. Then, however, Gartner called it "a step in the right direction for both companies," which would allow Borders a stronger presence online and Amazon a foothold in the brick-and-mortar retail market. It turns out that all media—including books and music—was heading online. After the original deal, Borders tried to revive its inventory management system, but it failed twice (badly the second time).

Finally, Borders never created its own competitor to the Kindle and Nook e-book offerings. Although it sold e-readers from Sony and Kobo in its stores, it never developed the tight integration between reader and bookstore that makes Amazon the preferred place for buying Kindle books, whereas those that have Nooks go to Barnes & Noble.

"Sure, you can save a little money by using exactly the same technology that anyone can buy. But you only get a real business advantage from IT when you are different," says Hayes. "And if you think a little money is better than a real business advantage, consider this: Right now, 'a little money' is all Borders has left."

IT Lessons from the Demise of Borders

SOURCE: Frank Hayes, "Seven IT Lessons from the Failure of Borders," *Computerworld*, March 7, 2011.

Other Clicks-and-Bricks Strategies

As Figure 9.16 illustrates, other clicks-and-bricks strategies range from partial e-commerce integration using joint ventures and strategic partnerships to complete separation via the spin-off of an independent e-commerce company.

The strategic partnership of the Rite Aid retail drugstore chain and Drugstore.com is a good example of a less integrated e-commerce venture. Rite Aid only owns about 25 percent of Drugstore.com, which has an independent management team and a separate business brand. However, both companies share the decreased costs and increased revenue benefits of joint buying power, an integrated distribution center, co-branded pharmacy products, and joint prescription fulfillment at Rite Aid stores.

Finally, let's look at an example of the benefits and challenges of a completely separate clicks-and-bricks strategy. Barnesandnoble.com was created as an independent e-commerce company that was spun off by the Barnes & Noble book retail chain. This status enabled it to gain several hundred million dollars in venture capital funding, create an entrepreneurial culture, attract quality management, maintain a high degree of business flexibility, and accelerate decision making. However, the book e-retailer has done poorly since its founding and failed to gain market share from Amazon.com, its leading competitor. Many business analysts say that the failure of Barnes & Noble to integrate some of the marketing and operations of Barnesandnoble.com within their thousands of bookstores meant it forfeited a key strategic business opportunity.

e-Commerce Channel Choices

Some of the key questions that the management of companies must answer in making a clicks-and-bricks decision and developing the resulting e-commerce channel are outlined in Figure 9.17. An **e-commerce channel** is the marketing or sales channel created by a company to conduct and manage its chosen e-commerce activities. How this e-commerce channel is integrated with a company's traditional sales channels (e.g., retail/wholesale outlets, catalog sales, and direct sales) is a major consideration in developing its e-commerce strategy.

Thus, the examples in this section emphasize that there is no universal clicks-and-bricks e-commerce strategy or e-commerce channel choice for every company, industry, or type of business. Both e-commerce integration and separation have major business benefits and shortcomings. Deciding on a clicks-and-bricks strategy and e-commerce channel depends heavily on whether a company's unique business operations provide strategic capabilities and resources to support a profitable business model successfully for its e-commerce channel. As these examples show, most companies are implementing some measure of clicks-and-bricks integration because "the benefits of integration are almost always too great to abandon entirely."

FIGURE 9.17

Key questions for developing an e-commerce channel strategy.

A Checklist for Channel Development

1. What audiences are we attempting to reach?
2. What action do we want those audiences to take? To learn about us, to give us information about themselves, to make an inquiry, to buy something from our site, to buy something through another channel?
3. Who owns the e-commerce channel within the organization?
4. Is the e-commerce channel planned alongside other channels?
5. Do we have a process for generating, approving, releasing, and withdrawing content?
6. Will our brands translate to the new channel or will they require modification?
7. How will we market the channel itself?

When outdoor equipment retailer REI wanted to boost in-store sales, the company looked to its Web site. In June 2003, REI.com launched free in-store pickup for customers who ordered online. The logic behind that thinking: People who visit stores to collect their online purchases might be swayed to spend more money upon seeing the colorful displays of clothing, climbing gear, bikes, and camping equipment.

REI's hunch paid off. "One out of every three people who buy something online will spend an additional \$90 in the store when they come to pick something up," says Joan Broughton, REI's vice president of multichannel programs. That tendency translates into a healthy 1 percent increase in store sales.

As Broughton sees it, the mantra for any multichannel retailer should be "a sale is a sale is a sale, whether online, in stores or through catalogs." The Web is not simply an isolated channel with its own operational metrics or exclusive group of customers.

As the Web has matured as a retail channel, consumers have turned to online shopping as an additional place to interact with a retailer, rather than as a replacement for existing channels such as stores or catalogs.

And to make that strategy as cost-efficient as possible, the company uses the same trucks that restock its stores to fulfill online orders slated for in-store pickup. To make this work, REI had to integrate order information from the Web site and replenishment orders from stores at its distribution warehouse in Washington state.

In and of itself, integrating the two types of order information wasn't complex, says Brad Brown, REI's vice president of information services. What was difficult, however, was coordinating fulfillment of both online and replenishment orders because "orders placed on the Web by customers are nothing like replenishment orders that stores place," he says. Online orders are picked from the warehouse at the time of the order and then put in a queue until the appropriate truck is loaded, whereas store orders are picked by an automated replenishment system that typically picks orders at one time based on either a weekly or biweekly replenishment schedule.

To make in-store pickup a reality, Brown's group wrote a "promise algorithm" that informs customers of a delivery date when they place an online order. Timing can get tricky when orders are placed the day before a truck is scheduled to depart the warehouse with a store-replenishment delivery. For example, if an online order is placed on a Monday night and a truck is scheduled to depart Tuesday morning, the system promises the customer a pickup date of a week later, as if the order would be placed on the following week's truck. However, REI will shoot for fulfilling the order that night; if it can do it, REI (and, ultimately, the customer) is happy because the order arrives sooner than was promised.

Creating effective business-to-consumer retail Web sites entails more than just calculating sales figures. It's about delivering the functionality that users expect and using the site to drive sales through other channels. And only IT integration can make this happen.

REI: Scaling e-Commerce Mountain

SOURCE: Megan Santosus, "Channel Integration—How REI Scaled e-Commerce Mountain," *CIO Magazine*, May 15, 2004.



Ticketmaster—Changing the Way Tickets Are Sold

Ticketmaster is the undisputed global leader in the provision of ticketing and related products and services to the live entertainment industry. Since it was founded in 1976, the way in which tickets to events are issued, distributed, purchased, exchanged, and used has changed dramatically, in no small part thanks to the rise of the Internet. Because these operations are so different from when the company started its business, Ticketmaster requires a new approach to its operations.

In order to answer these challenges, Ticketmaster deployed its Advanced Product Suite, at the center of which is a package called ARCHTICS, which handles ticketing. Although it may sound straightforward—you pay for a ticket and a ticket is issued—the package encompasses all steps: creating ticket offers and packages, selling tickets to the fans or customers, managing inventory control, collecting personal and demographic information, and only then taking credit card payments. The package runs on top of a Sybase SQL Anywhere database engine, which features self-tuning and self-managing capabilities that make it ideal for deployment in environments where the clients operating the application (sport teams, stadiums, arenas, theaters, and so forth) may have very limited information technology capabilities. As a result, the application must be as easy to deploy, run, and support as possible.

“Sybase SQL Anywhere does all the complex, heavy lifting of the ARCHTICS platform,” says Steve Davidson, vice president of Product Development at Ticketmaster. “Inventory control is absolutely essential to the system—there are only a certain number of seats and each must be accurately assigned. All that control is done through the Sybase database engine.”

“The ARCHTICS solution is a phenomenal asset for the Washington Capitals. The reliability and stability of the solution is second to none. The reporting features and the amount of data we can house and pull at any given time is tremendous,” says Christopher Sheap, director of Ticket Operations with the Washington Capitals.

The first instantiation of the ARCHTICS platforms was not originally a Ticketmaster development. It was developed by a small company that sold it to a number of clients. Ticketmaster recognized the value of the solution, acquired it, and expanded its capabilities beyond managing season ticket holders, which was the original focus. Par-

ticularly for sport franchises, however, season ticket holders form the core of their annual revenue, and they continue to be the central focus of attention. “Our ARCHTICS installation provides the technology to help us really get to know our fans. We track season ticket holder information in the CRM (customer relationship management) console in ARCHTICS so that we know who our VIP fans are and continuously develop programs to improve their event experience. Our fans are critical to the success of the Boston Celtics and it is imperative that we take good care of them. You need an organized solution to help you do that, and ARCHTICS does that for us,” says Paul Cacciatore, senior director of ticketing at the Boston Celtics NBA franchise.

While ARCHTICS is at the center of the Advanced Product Suite solution, Ticketmaster offers more to its customers than ticketing. For example, it also includes MailManager, a module used to manage e-mail campaigns and promotions to fans. The application sends out the e-mails, tracks whether each was opened and, if so, which links in each e-mail were clicked on. It also tracks the activities that users conducted while visiting the sites to which they were sent after clicking on those links—most importantly, whether the promotional e-mail generated a sale after all. All this allows marketing managers at the various sport franchises or venues to accurately pinpoint response rates to their campaigns, as well as to provide insights into what may work better in some scenarios, to fine-tune future efforts.

Selling a ticket, however, is only half—or maybe a third—of the story. From the moment a fan purchases the ticket until the date of the game, a number of different things may prevent the ticket holder from attending the game. What then? In the world before ARCHTICS, the ticket holder would almost certainly lose money and be frustrated by the complicated process. To help simplify the process for the ticket holder, Ticketmaster introduced Ticket Exchange, which also runs on ARCHTICS. The Ticket Exchange platform allows fans to post their tickets for sale, and those who are looking for tickets can search for them in a number of ways: by team, game, price, seating, and so forth. By purchasing tickets through Ticket Exchange, fans know they are getting real tickets posted directly by season ticket holders and other fans. A ticket holder who sells through

Ticket Exchange typically gets a better price than one who sells to a discount ticket broker.

“Ultimately, ARCHTICS helps us run our entire business better and give our valued fans a better purchasing experience, says Sheap. “One feature I particularly like is the online ticket management. It allows our season ticket holders to transfer tickets they are not going to use to someone else or sell them in a safe and secure manner on the open market. Additionally, fans who purchase tickets through Ticket Exchange know that they are getting valid tickets. This goes a long way in helping sell season tickets, and build fan loyalty and retention in the long run.”

After a ticket has been originally issued, and maybe changed hands along the way, the last third of the experience is actually coming into the arena to enjoy the game or event. This is where AccessManager, also part of the Advanced Product Suite centered around ARCHTICS, comes into play. At that time, security and fraud are major concerns, as team owners and arena operators want to ensure that only customers with valid tickets are granted access to the venue. In order to achieve this, each ticket features a bar code that is scanned by a handheld device as patrons enter the building. The device connects wirelessly to the local ARCHTICS instance, which validates the ticket on the spot. If it does not check out, you cannot come in. The system also allows managers to deactivate bar codes for tickets that are reported lost or stolen, and then reissue them to the original owners. As a result, the actual patrons can still come into the game, and those who stole their tickets cannot, which is a desirable state of affairs. Although this is certainly useful, a lot of data are collected in the process of security and access control that can be used for something else. Good data are too valuable to let go to waste.

The AccessManager module also identifies when a ticketed patron came into the venue (i.e., date and time information), through which door or gate the patron entered, and how many of the purchased tickets were used that night. All this information feeds back into ARCHTICS to provide team owners with a complete picture of attendance for each game, which in turn provides their marketing managers with the necessary information to boost attendance. After all, a fan who does not come to the game, even one who has paid for a ticket, is a fan who does not purchase something to eat or take a gift or souvenir home; revenues go beyond the price of the ticket alone. Managers can then alert season ticket holders who show a pattern of not regularly coming to the games that they can give their tickets to a friend or family member on a game-by-game basis, with no ongoing commitment; post them for resale for specific games; or use other means to help them come to the games—anything that makes a ticket sold mean that an actual customer comes through the door.

These modules, part of ARCHTICS and the Advanced Product Suite, started out as turnkey systems that Ticketmaster sold to its clients, who were responsible for operating them. The company is now looking to expand into offering remote hosting of the suite for smaller clients, who may not have the necessary in-house IT resources but still want to be able to access all of these features. Today, more than 400 corporate clients—teams, arenas, theaters, and so forth—are using ARCHTICS.

SOURCE: “ARCHTICS,” *Computerworld Honors Case Study*, 2010; “Ticketmaster,” *Sybase Customer Case Study*, 2010; “The Phoenix Suns Launch Ticketmaster’s Team Exchange, Online Ticket Forwarding Technology,” *Ticketmaster Press Release*, September 5, 2010; “Ticketmaster and the Golden State Warriors Sign Multi-Year Ticketing Agreement,” *Ticketmaster Press Release*, September 26, 2010; and www.ticketmaster.com, accessed May 29, 2011.

QUESTIONS TO CONSIDER

1. Since ARCHTICS and the Advanced Product Suite manage so much of the operational aspects of ticketing today, where does the value added by the independent clients—teams, franchises, and so forth—come from?
2. How would the relationship between Ticketmaster and its corporate clients change if Ticketmaster were to host and operate the applications for them outright? How would its corporate clients benefit? Should the corporate clients have any concerns?

- **e-Commerce.** E-commerce encompasses the entire online process of developing, marketing, selling, delivering, servicing, and paying for products and services. The Internet and related technologies, e-commerce Web sites on the World Wide Web, and corporate intranets and extranets serve as the business and technology platforms for e-commerce marketplaces for consumers and businesses in the basic categories of business-to-consumer (B2C), business-to-business (B2B), and consumer-to-consumer (C2C) e-commerce. The essential processes that should be implemented in all e-commerce applications—access control and security, personalizing and profiling, search management, content management, catalog management, payment systems, workflow management, event notification, and collaboration and trading—are summarized in Figure 9.4.
- **e-Commerce Issues.** Many e-business enterprises are moving toward offering full-service B2C and B2B e-commerce portals supported by integrated customer-focused processes and inter-networked supply chains, as illustrated in Figure 9.9. In addition, companies must evaluate a variety of e-commerce integration or separation alternatives and benefit trade-offs when choosing a clicks-and-bricks strategy and e-commerce channel, as summarized in Figures 9.16 and 9.17.
- **B2C e-Commerce.** Businesses typically sell products and services to consumers at e-commerce Web sites that provide attractive Web pages, multimedia catalogs, interactive order processing, secure electronic payment systems, and online customer support. However, successful e-tailers build customer satisfaction and loyalty by optimizing factors outlined in Figure 9.11, such as selection and value, performance and service efficiency, the look and feel of the site, advertising and incentives to purchase, personal attention, community relationships, and security and reliability. In addition, a Web store has several key business requirements, including building and marketing a Web business, serving and supporting customers, and managing a Web store, as summarized in Figure 9.13.
- **B2B e-Commerce.** Business-to-business applications of e-commerce involve electronic catalog, exchange, and auction marketplaces that use Internet, intranet, and extranet Web sites and portals to unite buyers and sellers, as summarized in Figure 9.14 and illustrated in Figure 9.15. Many B2B e-commerce portals are developed and operated for a variety of industries by third-party market-maker companies called infomediaries, which may represent consortiums of major corporations.

These are the key terms and concepts of this chapter. The page number of their first reference appears in parentheses.

- | | | |
|---|--|--|
| 1. Clicks-and-bricks strategy (393) | <i>b.</i> Collaboration and trading (378) | 5. Electronic commerce (368) |
| 2. E-commerce channel (396) | <i>c.</i> Content and catalog management (375) | <i>a.</i> Business-to-business (B2B) (372) |
| 3. E-commerce marketplaces (391) | <i>d.</i> Electronic payment systems (378) | <i>b.</i> Business-to-consumer (B2C) (372) |
| <i>a.</i> Auction (391) | <i>e.</i> Event notification (378) | <i>c.</i> Consumer-to-consumer (C2C) (372) |
| <i>b.</i> Catalog (391) | <i>f.</i> Profiling and personalizing (374) | 6. Electronic funds transfer (EFT) (379) |
| <i>c.</i> Exchange (391) | <i>g.</i> Search management (374) | 7. Infomediaries (392) |
| <i>d.</i> Portal (391) | <i>h.</i> Workflow management (376) | 8. Search engine optimization (387) |
| 4. E-commerce processes (373) | | |
| <i>a.</i> Access control and security (374) | | |

Match one of the key terms and concepts listed previously with each of the brief examples or definitions that follow. Try to find the best fit for the answers that seem to fit more than one term or concept. Defend your choices.

- | | |
|--|--|
| _____ 1. The online process of developing, marketing, selling, delivering, servicing, and paying for products and services. | _____ 12. Companies that serve as intermediaries in e-commerce transactions. |
| _____ 2. Business selling to consumers at retail Web stores is an example. | _____ 13. A process aimed at improving the volume and/or quality of traffic to a Web site. |
| _____ 3. Using an e-commerce portal for auctions by business customers and their suppliers is an example. | _____ 14. An e-commerce marketplace that may provide catalog, exchange, or auction service for businesses or consumers. |
| _____ 4. Using an e-commerce Web site for auctions among consumers is an example. | _____ 15. Buyers bidding for the business of a seller. |
| _____ 5. E-commerce applications must implement several major categories of interrelated processes, such as search and catalog management, in order to be effective. | _____ 16. Marketplace for bid (buy) and ask (sell) transactions. |
| _____ 6. Helps to establish mutual trust between you and an e-tailer at an e-commerce site. | _____ 17. The most widely used type of marketplace in B2C e-commerce. |
| _____ 7. Tracks your Web site behavior to provide you with an individualized Web store experience. | _____ 18. The marketing or sales channel created by a company to conduct and manage its e-commerce activities. |
| _____ 8. Develops, generates, delivers, and updates information to you at a Web site. | _____ 19. The processing of money and credit transfers between businesses and financial institutions. |
| _____ 9. Ensures that proper e-commerce transactions, decisions, and activities are performed to serve you more efficiently. | _____ 20. Ways to provide efficient, convenient, and secure payments in e-commerce. |
| _____ 10. Sends you an e-mail when your e-commerce order has been shipped. | _____ 21. Companies can evaluate and choose from several e-commerce integration alternatives. |
| _____ 11. Includes matchmaking, negotiation, and mediation processes among buyers and sellers. | _____ 22. Web sites and portals hosted by individual companies, consortiums, or intermediaries that bring together buyers and sellers to accomplish e-commerce transactions. |
| | _____ 23. A component of e-commerce sites that helps customers find what they are looking for. |

- Most businesses should engage in e-commerce on the Internet. Do you agree or disagree with this statement? Explain your position.
- Are you interested in investing in, owning, managing, or working for a business that is primarily engaged in e-commerce on the Internet? Explain your position.
- Consider the Real World Challenge at the beginning of the chapter. The e-commerce practices of today would not be possible without the Internet. Did customers place new demands for products and services on companies like Ticketmaster before the Internet appeared, and the technology allowed the companies to satisfy them? Or are the demands a result of the technology being there in the first place? Which one comes first?
- What are the critical success factors for retail e-business start-ups?
- Do the e-commerce success factors listed in Figure 9.11 guarantee success for an e-commerce business venture? Give a few examples of what else could go wrong and how you would confront such challenges.
- If personalizing a customer's Web site experience is a key success factor, then electronic profiling processes to track visitor Web site behavior are necessary. Do you agree or disagree with this statement? Explain your position.
- All corporate procurement should be accomplished in e-commerce auction marketplaces, instead of using B2B Web sites that feature fixed-price catalogs or negotiated prices. Explain your position on this proposal.
- Refer to the Real World Solution in the chapter. Given that corporate clients largely cannot operate without relying on Ticketmaster

today, do you think sport teams and franchises should be concerned about being edged out of the ticketing business by Ticketmaster in the future? What can they do to prevent this from happening?

9. If you were starting an e-commerce Web store, which of the business requirements summarized in Figure 9.13 would you primarily do yourself, and

which would you outsource to a Web development or hosting company? Why?

10. Refer to the Real World Case on social networks, mobile commerce, and online shopping in the chapter. Do you think that mobile devices (not just phones anymore) are becoming the major platform for shopping, communication, everything? What are the implications for companies?

1. Small Business e-Commerce Portals

On the Internet, small businesses have become big business, and a really big business, Microsoft, wants a piece of the action. The company's Small Business Center (www.microsoft.com/smallbusiness) is one of many sites offering advice and services for small businesses moving online. Most features, whether free or paid, are what you'd expect: lots of links and information along the lines established by Google Small Business Network (<http://www.google.com/smallbusinessnetwork>) or Entrabase.com.

Small Business Center, however, stands out for its affordable advertising and marketing services. Go check it out and see what you think.

One program helps businesses create banner ads and places them on a collection of Web sites that it claims are visited by 60 percent of the Web surfing community. With its "Banner Network Ads" program, buyers don't pay a huge fee up front, and they don't run the risk that a huge number of visitors will unexpectedly drive up click-through commissions. Instead, this program allows small business to pay a small, fixed fee for a guaranteed number of click-throughs (people who click on your banner ad to visit your Web site). Small Business Center rotates these banner ads around a network of participating Web sites and removes the ad as soon as it has received the guaranteed number of click-through visitors. This action eliminates the guesswork regarding both traffic and fees. The three packages—100, 250, and 1,000 visitors—break down to 50 cents per visitor.

- a. Check out Small Business Center and the other e-commerce portals mentioned. Identify several benefits and limitations for a business using these Web sites.
- b. Which Web site is your favorite? Why?
- c. Which site would you recommend or use to help a small business wanting to get into e-commerce? Why?

2. Car Buying

e-Commerce Web Sites for Car Buying

Nowadays new car buyers can configure the car of their dreams on Microsoft's MSN Autos Web site, as well as those of Ford, GM, and other auto giants. Many independent online car purchase and research companies offer similar services. See Figure 9.18. Car-buying information provided by manufacturers, brokerage sites, car dealers, financial institutions, and consumer advocate Web sites has exploded in the past few years. Yet in the age of the Internet, the auto industry remains a steadfast holdout to innovations that might threaten the well-established and well-connected supply chain, the car dealership. American new car buyers simply cannot skip the middleperson and purchase an automobile directly from the manufacturer. That's not just a business decision by the manufacturers; that's the law. Even so, many car buyers use the Internet as a place to research their purchases. Instead of selling new cars directly, Web sites such as Autobytel.com of Irvine, California, just put consumers in touch with a local dealer where they test-drive a vehicle and negotiate a price. Autobytel.com has been referring buyers to new and used car dealers since 1995. It also offers online financing and insurance. Online car-buying sites on the Web make consumers less dependent on what cars a dealer has on the lot. At online sites, buyers can customize a car—or van, truck, or sport utility vehicle—by selecting trim, paint, color, and other options before purchase. They can also use Web sites such as CarBuyingTips.com to help prepare for the final negotiating process.

- a. Check out several of the Web sites shown in Figure 9.18. Evaluate them based on ease of use, relevance of information provided, and other criteria you feel are important. Don't forget the classic: "Did they make you want to buy?"
- b. Which sites would you use or recommend if you or a friend actually wanted to buy a car? Why?

Manufacturer's Sites	
Ford Motor Company	(www.ford.com) Product specifications, pricing, build-your-own feature, finance calculators, and dealer finder.
General Motors Corporation	(www.gm.com) With access to nearly 6,000 GM dealerships, car shoppers can get a price quote, schedule a test drive, and buy.
Pro-Consumer Sites	
Cars.com	(www.cars.com) Research tools include automotive reviews, model reports, dealer locators, and financial information.
Auto Web	(www.autoweb.com) Buy new and used automobiles online, or sell your old one. Go to the new car research center to investigate autos. Find out what rebates and incentives are currently available, link to Kelly Blue Book, or chat with others in the forums.
Edmunds	(www.edmunds.com) For an objective opinion, Edmunds.com provides reviews, safety updates, and rebate news for car buyers.
Kelly Blue Book	(www.kbb.com) New and used car information and valuation.
CarSmart	(www.carsmart.com) Find new car price quotes, dealer locators, and used auto sales at the site "where America shops for cars." Report library allows visitors to download vehicle information and car-buying tips.
Cars.com	(www.cars.com) Research tools include automotive reviews, model reports, dealer locators, and financial information.
CarMax	(www.carmax.com) More than 20,000 new and used vehicles available to purchase online. Check out their online research tools. If the exact model you want isn't at a nearby store, it can be transferred to your local CarMax for a fee.
J.D. Power	(www.jdpower.com/auto) These folks award the coveted J.D. Powers quality awards. Hundreds of vehicles rated by thousands of people. Consistently accurate reviews on cars, boats, insurance, and finance. Automotive Ratings highly respected. Also see powerful buyers guides.
Cars Direct	(www.carsdirect.com) "America's #1 way to buy cars online" is now partnered with Amazon.com. Get prices on new cars. Search more than 50,000 used car listings. Anti-theft products, auto accessories, and vehicle history reports too.
Microsoft MSN Autos	(www.autos.msn.com) Auto reviews, detailed vehicle specifications, safety ratings, and buying services for new and used cars, including customizing your very own Ford.

FIGURE 9.18

Top car-buying Web sites.

- c. Check out the Consumer Federation of America's study on anticompetitive new car-buying state laws or similar studies online. How much does it estimate consumers would save if they could purchase cars directly from manufacturers online?

3. Which e-Commerce Site Is Best?

Comparing e-Commerce Sites

In this exercise, you will experiment with electronic shopping and compare alternative e-commerce sites. First, select a category of product widely available on the Web, such as books, CDs, or toys. Second, select five specific products to

price on the Internet, for example, five specific CDs you might be interested in buying. Third, search three prominent e-commerce sites selling this type of product and record the price charged for each product by each site.

- a. Using a spreadsheet, record a set of information similar to that shown for each product. (Categories describing the product will vary depending on the type of product you select—CDs might require the title of the CD and the performer[s], whereas toys or similar products would require the name of the product and its description.) See Figure 9.19.

FIGURE 9.19

Table for Problem 3.

Title of Book	Author	Price			Rating		
		Site A	Site B	Site C	A	B	C
The Return of Little Big Man	Berger, T.	15.00	16.95	14.50	2	3	1
Learning Perl/Tk	Walsh, N. & Mui, L.	26.36	25.95	25.95	3	1.5	1.5
Business at the Speed of Thought	Gates, W.	21.00	22.95	21.00	1.5	3	1.5
Murders for the Holidays	Smith, G.		8.25	7.95	4	2	1
Designs for Dullards	Jones	17.95	18.50	18.50	1	2.5	2.5
Sum of ratings (low score represents most favorable rating)					11.5	12	7.5

- For each product, rank each company on the basis of the price charged. Give a rating of 1 for the lowest price and 3 for the highest, and split the ratings for ties—two sites tying for the lowest price would each receive a 1.5. If a site does not have one of the products available for sale, give that site a rating of 4 for that product. Add the ratings across your products to produce an overall price/availability rating for each site.
- Based on your experience with these sites, rate them on their ease of use, completeness of information, and order-filling and shipping options. As in Part (b), give a rating of 1 to the site you feel is best in each category, a 2 to the second best, and a 3 to the poorest site.
- Prepare a set of PowerPoint slides or similar presentation materials summarizing the key results and including an overall assessment of the sites you compared.
- Search using the term “third-party escrow.” What legitimate function does this serve? Provide an example of a legitimate third-party escrow service for Internet transactions. How has the third-party escrow system been used to commit fraud on the Internet?

5. Ecosystems and e-Commerce

Does Google Have a Leg Up on Groupon?

If you use Facebook, you’ve heard of Groupon. Groupon advertises heavily on Facebook where the social networking medium serves not only as a targeted advertising platform but also as a highly effective viral marketing platform.

Groupon uses self-reported demographics, purchase history, and location to match users with money-saving deals. Users get great deals, and vendors attract more customers during slow periods. Groupon also has a mobile app with GPS-enabled searching and paperless coupons that enables on-the-spot savings.

Before buying stock in Groupon just yet, please note that the Google gorilla has entered the room. Google, with its own mobile operating system, Web browser, massive amounts of geographic and user data, and a well-established highly successful, online advertising system may pose a massive threat.

Google’s beta application “Google Offers” competes directly with Groupon. Google is now rolling out “Google Wallet”: a mobile application that allows users to make secure retail purchases in person directly via any Android-enabled mobile device.

Read Practical Ecommerce’s article, “*Google Wallet to Impact Ecommerce Merchants?*” (<http://bit.ly/iJZuEf>), view the embedded video, and answer the following questions.

- What are the pros and cons of PCI certification for brick-and-mortar stores?
- According to the video, what is “fun” about using Google Wallet?
- Google has had numerous successes and failures. Should Groupon be worried?

4. e-Commerce: The Dark Side Don’t Get Scammed

Anonymous transactions on the Internet may have a dark side. Scams abound. Research each of the terms below on the Web. Prepare a one-page report for each term researched. Your paper should describe the problem and provide real-world examples where possible. Conclude each paper with recommendations on how to guard against each type of fraud.

- Search using the terms “Ponzi scheme” or “Pyramid scheme.” You may find current examples by searching for “cash matrix,” or “gifting party.”
- Search using the terms “phishing.” Report your results. If possible, include a printout of a real-world example that you or an acquaintance may have received via e-mail.
- Search using the term “Nigerian scam.” Report your results. If possible, include a printout of a real-world example that you or an acquaintance may have received via e-mail.



Sony, 1-800-Flowers, Starbucks, and Others: Social Networks, Mobile Phones, and the Future of Shopping

A number of major retailers have been driven into bankruptcy protection during this recession, including RedEnvelope and Eddie Bauer, or gone out of business altogether, like Circuit City. Blockbuster, Virgin Megastores and many more have closed stores. Survivors, suffering deflated profits and slow sales, warn of a bleak future.

But smart retailers are going where it's warm: the hot little hands of cellphone- and laptop-toting consumers who want to shop right now, wherever they happen to be sipping their lattes or watching their kids playing soccer. Technology-backed projects to increase revenue include mobile e-commerce, coupons by text message, even storefronts on social networks. As enablers of these projects, chief information officers are moving ever closer to the customer.

"Out of recession develops one picture—finally—of what true business-IT alignment looks like," says Drew Martin, chief information officer of Sony Electronics, "IT is becoming part of the product offerings." Whether that's hotel kiosks, mobile banking, hospital patient portals or retail, chief information officers are getting their IT groups to the front line in the competition for consumer dollars. When a customer logs on to his new Sony e-book reader, for example, the device automatically connects him to his existing customer profile, from which he can start buying e-books. This feature is thanks to Martin's efforts to connect product development with Sony's internal customer relationship management system.

As exciting as it is to live on the progressive edge of the CIO profession, though, it's a new world to navigate at a time when wrong moves can severely hurt a company. "The challenge is that now you're entering into the revenue space," Martin says. "You need to commit to delivering your part of what needs to be delivered."

"Websites and e-mail—that's just too many steps now," says Brett Michalak, chief information officer with Tickets.com, which sells tickets to games, concerts, and other events, as well as its own ticketing technology.

Social media such as Twitter, Facebook, and YouTube take e-mail out of the equation, putting offers in front of customers on sites they already visit. Dell, JetBlue, Whole Foods and other big brands have pounced on Twitter as a marketing and promotion tool, tweeting special deals to followers. Dell, for example, attributes more than \$2 mil-

lion in sales to its 14 Twitter accounts that promote offers to 1.4 million followers. ("15% off any Dell Outlet Inspiron laptop. Enter code at checkout . . .")

Sony is using Twitter, among other social networking sites, to hype the SonyReader. A recent tweet included a link to a page at Sony's site comparing the product favorably to Amazon's Kindle. "You can't build a site and expect people to come. We are on YouTube, Facebook, and Twitter to go out and get them," Martin says.

1-800-Flowers intends to find out whether social networkers are also social shoppers. In July 2009, the \$714 million flower delivery company launched the first Facebook storefront. Collectively, Facebook's 300 million active members spend 8 billion minutes per day on the site, according to the company. An Experian survey found that dwell time for an adult visiting a social network is 19 minutes and 32 seconds. Meanwhile, 35 percent of adults who had been on a social network in the past month had also bought something online in that time period, the survey found—a ripe demographic.

"Still, there's a lot to do on Facebook, so any shopping has to be fast," says Vibhav Prasad, vice president of Web marketing and merchandising at 1-800-Flowers. The company's Facebook store, therefore, offers only 10 percent to 15 percent of the several hundred bouquets available from the main 1-800-Flowers Web site, and the checkout process has been pared down. No suggestions to buy related products pop up, for example, and four special occasions tabs span the top of the page, instead of the eight on the main site.

"It's a fairly impulsive purchase in this channel," Prasad says. "As simple and as quick as we can make it, the more effective we'll be." Impulsiveness is key. Every time Facebook members log in, they see updates about who among their friends is having a birthday. Prasad wants those regular reminders to spark flower buys. Going social was "a logical extension" for 1-800-Flowers, which was one of the first retailers to put up an e-commerce site, in the early 1990s, notes Kevin Ranford, director of Web marketing. "It comes from listening to customers and responding to the channels in which they're interacting," Ranford says.

Facebook users spend most of their time looking at their own home pages. They read their news feed—a

display of their friends' status updates, quizzes taken, notes posted, and games played. So, 1-800-Flowers is planning a way into the news feed. When a fan fills out a wish list to indicate which flowers she'd like to receive, notification would go into the feeds of her friends. Carol logs on to Facebook, sees Alice has a birthday on Thursday and wishes for the "Pleasantly Pink" bouquet. Ding! Carol clicks over to the 1-800-Flowers store and \$29.99-plus-shipping later, takes care of that gift without ever leaving Facebook. "We think people will do it because social networking is all about you expressing your interests and your friends responding," says Wade Gerten, chief executive officer of Alvenda, the Minneapolis software developer that built the Facebook store for 1-800-Flowers. "Shopping online can be social again, as it was in person."

People lose their credit cards and forget their wallets. But cell phones? There is perhaps no combination of vices so bursting with commercial promise than that of cell phone-plus-caffeine. Starbucks is there. In September 2009, the \$9.8 billion coffee chain began testing a system to let customers pay using their iPhones or iPod touch devices. They download the Starbucks Card Mobile App and type in the number of their Starbucks loyalty card, preloaded with spending money. A two-dimensional bar code appears that cashiers can scan.

Royal Oak Music Theatre, a Michigan music and comedy venue where acts such as Train and Bob Saget have played, started mobile ticketing three years ago and has adjusted its marketing to cover for finicky technology.

Anyone who's done self-check out at the supermarket knows that scanning takes a special, knowing touch. Still, scanning bar codes on the screens of mobile devices often requires extra wiggling of the phone and slanting it at different angles. It's slower than scanning paper tickets. To avoid ticking off patrons lined up to run in and grab general admission floor spots, Royal Oak created a separate VIP entrance for the mobile customers. There, staff use the newer model scanners required for reading mobile bar codes, and it's not so apparent that the scanning takes longer, says Diana Williams, box office manager.

Mobile customers are also allowed to get into the theater a few minutes before traditional customers, which encourages more people to buy their tickets by cell phone, she says. That's cheaper for the theater than handling

paper tickets—saving money and hassle time is Williams's goal. But it also positions the theater well for collecting future revenue.

"Mobile ticketing skews young," Williams observes. The theater does shows for all ages, and for a typical adult event, 16 percent of tickets sold are through the mobile channel. But for a recent show by the boy-band Hansen, popular with tween girls, mobile accounted for nearly 40 percent of tickets. "There's an age—around 22 or younger—where it would never occur to patrons that you couldn't buy a ticket from your phone," Williams says.

Mobile and social commerce projects will change the business of any company that invests in it, says Russ Stanley, managing vice president of ticket services and client relations for the San Francisco Giants. For example, instead of being a long-planned activity, a Major League Baseball game can become an impulse buy, Stanley says, bringing in more sales for the organization.

Every game day, the Giants have 40,000 seats to sell. If they've sold only 30,000, 10,000 spoil every bit as badly as old pears. Last year, the team changed prices daily on about 2,000 seats. Stanley imagines the day when he'll have a database of fans who, say, live within a mile of the ballpark to whom he can text last-minute offers. "Hey, the Giants have \$5 tickets left for tonight. For \$5, I'll walk down there," he says. "As they're walking up to the entrance, they're buying on the mobile."

The Giants started to offer mobile tickets midway through the 2008 season, when they sold about 100 tickets that way per game. In 2009, it was about 200 and Stanley expects to do about 400 per game in the coming years. "Fans who use it love it. It's getting the people to use it," he says.

Like hot dogs and cold beer, holding a ticket is part of the rite of baseball, he says. Plus, there's the souvenir value. When pitcher Jonathan Sanchez threw a no-hitter against the San Diego Padres in July 2009, about 50 mobile fans, as well as people who had bought tickets online and printed them on plain paper at home, later requested the team print "real" tickets for them to commemorate the event. "We did that for them. It's good relations," says Stanley. And, he adds, it could turn into a money-making service in the future.

SOURCE: Kim S. Nash, "Facebook, Mobile Phones, and the Future of Shopping," *CIO.com*, November 24, 2009.

CASE STUDY QUESTIONS

1. How do the companies involved benefit from the innovations discussed in the case? Is it about more efficient transaction processing, better reaching out to customers, or both? Use examples from the case to illustrate your answer.
2. "Shopping online can be social again, as it was in person," says Wade Gerten, chief executive officer of Alvenda. Do you think this is a stretch, or are we in the midst of a turning point in online shopping? Explain your answer.

3. Many of the applications discussed in the case are mostly used by the younger demographic, who grew up around technology. How do online behavior patterns change as

people become older, with more responsibilities, and more challenging jobs? Do applications like those discussed in the case become less important? More important?

REAL WORLD ACTIVITIES

1. Consider the examples discussed in the case. Go online and research what other companies or industries are doing in terms of the use of social networking sites and mobile commerce. What other examples can you find? Prepare a report that compares those in your research with the ones described here, highlighting similarities and differences. Can you spot any new trends?
2. How often, if ever, do you shop with your mobile phone? What do you think are some of the roadblocks that prevent the widespread adoption of mobile shopping? What would you suggest companies do to overcome those? Break into small groups with your classmates to develop a few recommendations.



LinkedIn, Umbria, Mattel, and Others: Driving the “Buzz” on the Web

David Hahn has spotted a trend. As director of advertising for the popular online business networking site LinkedIn, he’s being asked pointed questions by large advertisers about his ability to help them find “influentials”: those people within the LinkedIn community who are the most likely to go out and spread the word about a particular product or experience. “Some of them are requesting it specifically, while others are more implying it, but it comes down to the same thing,” Hahn says. “Marketers are very interested in the value of online social networks, and how leaders in those networks can be used to drive proactive behaviors in the population.”

Hahn isn’t alone in his observations. “The notion of the online influencer is quite the thing today in the marketing world,” says Janet Edan-Harris, chief executive officer of Umbria, which monitors chatter in cyberspace communities for corporations wanting to know what’s being discussed online about their brands and products. “Companies are incredibly eager to get to those people. Do that—or so the conventional wisdom says—and you’ll be in marketing heaven.”

But new research, as well as growing business experience, suggests that such thinking may be overly simplistic. The effectiveness of using online word-of-mouth campaigns—or using individuals rather than traditional media advertising to spread the word about products or services—is increasingly viewed as an effective way to reach consumers. But the popular notion that frequently accompanies this, that there are special individuals who hold the key to the hearts of entire online communities, is coming under fire.

Dave Balter certainly thinks so. As chief executive officer of BzzAgent, a word-of-mouth marketing firm, Balter had a revelation three years ago: The so-called influentials, or opinion leaders, in online communities can’t be influenced in a way that accelerates the success of a word-of-mouth campaign. “We actually believed in the idea that influentials drove market trends at that point,” says Balter. “But upon closer look, we found out it didn’t add up. The sales data of our campaigns didn’t match the profiles of the opinion leaders we had targeted, and it really caused us to re-evaluate some of our core assumptions.” Today, when a client comes in with the goal of influencing the influentials, “we tell them that’s fools’ gold,” says Balter. “It

sounds really great, it sounds really sexy, but the results simply don’t fly.”

This indeed is what Edan-Harris has concluded from her experiences working with online communities. “We said, ‘Wait a minute, is this really a correct assumption, that there are individuals on the Internet that have that much influence?’” she says. Her conclusion: “Not nearly as much as everyone seems to think.”

Despite this, companies are putting significant dollars into efforts to find these online opinion leaders, whether they’re bloggers, contributors to discussion boards, or members of online social networks. Indeed, a whole cottage industry has sprung up based upon the notion that all marketers need to kick off a successful marketing strategy with a list of Internet opinion leaders. And with the expanding universe of blogs, online communities, and social networks such as MySpace, FaceBook, and LinkedIn, the appeal of this idea has become even more entrenched. There’s a growing perception that the increasingly ubiquitous availability of broadband, coupled with the rise in popularity of blogs and online communities, make “influentials” even more influential.

It’s critical to understand, however, that all these proponents of opinion leaders as drivers of social and commercial trends aren’t talking about media stars or personalities, but about otherwise seemingly ordinary members of a community who, through accumulation of knowledge or number of connections with others, act as catalysts for change. Not surprisingly, marketers of all stripes began trying to take advantage of this almost at once—at first offline, and now increasingly within the online social networks rising in popularity.

“The largest companies had already established influence-based programs and are now extending that model into the online social networking space,” says Matthew Hurst, a scientist at Microsoft LiveLabs who follows online marketing trends. “It’s not the notion of influence that’s new, it’s the technology that is now enabling it to a greater degree.” Not surprisingly, a rapidly increasing number of companies have leaped into the fray to help firms identify the influentials in cyberspace. Buzzlogic is one of them. Launched in 2007, Buzzlogic is dedicated to this idea that opinion leaders in online social networks can be identified, and their influence measured.

An early Buzzlogic beta customer is Protuo.com, a Web-based career management portfolio service that provides matchmaking between employers and potential employees. Not having the funds to buy expensive marketing spots in television, radio, or mainstream print media, Jennifer Gerlach, vice president of marketing, hired Buzzlogic to find the people who are the most influential in the human resource/employee professional space, contact them, and get them to buzz about the product. “We noticed that once one blogger wrote about our service, then suddenly a bunch of other people were writing about it. All at once, there were reviewers everywhere,” says Gerlach, who just snagged a major feature in Inc. that she attributes to the online “influentials” campaign. She says she can map increases in site traffic precisely to blog mentions, and she views the campaign as a huge success.

But despite this apparent triumph, a steadily growing number of online marketing experts would argue that, rather than being responsible for the deluge of publicity that Protuo.com is experiencing, the bloggers targeted by Buzzlogic were simply tapping into a sort of zeitgeist waiting to happen—in this case, intense interest in how the Internet could be used to bring employers and candidates together more efficiently than traditional job boards are capable of doing.

Indeed, a growing school of thought is that “influentials” aren’t so much leading trends as acting as mouthpieces for underlying social movements that are either already in progress or lying fallow waiting to be triggered. Thus, successful marketing doesn’t depend so much on finding influential people and seeding them with ideas so much as doing the kind of research that exposes embryo trends, and then helping “influentials” discover them.

This in fact is what Umbria does by focusing on tracking online conversations taking place in discussion boards and social networks, as well as blogs. “It’s much more important to identify those themes that are gaining momentum than try to find opinion leaders,” says Edan-Harris. “You want to ride the wave rather than trying to start one on your own.” By listening first to the conversations and being nimble enough to use the Internet to craft campaigns that jump on an existing trend, “you get much better results than attempting to generate your own little epicenter,” she says.

Protuo.com’s Gerlach agreed with some aspects of that. “There has to be a story around your product, and that story has to resonate in the world for the opinion leader strategy to work,” she says.

Herein lies the problem with swallowing the “influentials” theory whole cloth. Much of the so-called evidence of how the process works is a matter of reverse engineering. Once something happens—if there’s a best-selling book coming out of nowhere, or a surprise political upset—you can always go back to the beginning and find the event or person that seems to have triggered it. You can always tell a causal story in retrospect.

Michael Shore, vice president of worldwide consumer insights for Mattel, directs an organization that increasingly monitors blogs, social networks, discussion boards, and forums to figure out what the market might want from toys in general and Mattel products in particular. But unlike many other global consumer-brands companies, Mattel isn’t interested in simply smoking out those individuals who are inordinately influential in their online communities and pushing top-down marketing messages onto them. Despite the fact that this has become the strategy du jour in the online world, Shore’s philosophy is a more holistic one.

“We’re not just interested in opinion leaders. We’d consider that too narrow a focus,” says Shore, who hired MarketTools.com to help him develop and get involved with online communities. Instead, he uses the online universe to do what he calls “cultural assessments” that involve analyzing language, behavioral patterns, and values. Armed with that information, Shore says, Mattel gets valuable information from the Internet that it uses to shape future product development, as well as marketing campaigns.

If there’s one thing that everyone agrees on, it’s that marketers need to invest a great deal more effort into how online social networks and Internet communities actually work with respect to selling products and services at the grass-roots level.

“It’s an emerging medium, and the rules haven’t yet been established,” says Umbria’s Edan-Harris. “We’re still learning what does and doesn’t work.”

SOURCE: Alice LaPlante, “Online Influencers: How the New Opinion Leaders Drive Buzz on the Web,” *InformationWeek*, May 5, 2007.

CASE STUDY QUESTIONS

1. How can companies benefit from the “cultural assessments” regularly performed by Mattel? How could the information obtained be used to create business value for those organizations? Provide multiple examples.
2. The case notes that, in spite of disconfirming evidence as to the effectiveness of targeting online opinion leaders,

companies are nonetheless increasing their efforts to identify and contact them. Why do you think this is the case?

3. One of the participants in the case states that “you want to ride the wave rather than trying to start one of your own.” What does she mean by that? If companies are not starting these “waves,” where are they coming from?

REAL WORLD ACTIVITIES

1. A number of technological and cultural developments in recent years has resulted in the emergence of extensive social networks and a large number of avidly followed blogs. Go online to research how companies are tapping into these trends and what new marketing practices have arisen as a result. Prepare a report to summarize your findings.
2. Reflect on your own purchasing behavior. How much do you rely on blogs, feedback, and recommendations from past customers to make your own purchasing decisions? Why do you (or do you not) rely on these sources of information? Do you believe they are largely unbiased? Break into small groups to discuss these issues with your classmates and compare perspectives on them.

Data Management, Data Analytics, and Business Intelligence

CHAPTER OUTLINE

LEARNING OBJECTIVES

Case 3.1 Opening Case: Coca-Cola Strategically Manages Data to Retain Customers and Reduce Costs

3.1 Data Management and Database Technologies

3.1 Describe the purpose and benefits of data management and how database technologies help support business processes.

3.2 Centralized and Distributed Database Architectures

3.2 Describe the differences between centralized and distributed database architectures and the importance of creating and maintaining data that can be trusted.

3.3 Data Warehouses

3.3 Understand the concepts of data analytics and data warehousing and evaluate their tactical and strategic benefits.

3.4 Data Analytics and Data Discovery

3.4 Explain benefits of data and text mining and business intelligence and how they benefit an organization.

3.5 Business Intelligence and Electronic Records Management

3.5 Describe electronic records management and how it helps companies to meet their compliance, regulatory, and legal obligations.

Case 3.2 Business Case: Big Data Analytics is the “Secret Sauce” for Revitalizing McDonald’s

Case 3.3 Video Case: Verizon Improves its Customer Experience with Data-Driven Decision-Making

Introduction

As discussed in Chapter 2, collecting and maintaining trusted data is a critical aspect of any business. Knowing how and where to find data, store it efficiently, analyze it in new ways to increase the organization's competitive advantage, and enable the right people to access it at the right time are all fundamental components of managing the ever-increasing amounts of corporate data. Indeed, data analytics is the primary differentiator when doing business in the 21st century. Transactional, social, mobile, cloud, Web, and sensor data offer enormous potential. But without tools to analyze these data types and volumes, there would not be much difference between business in the 20th century and business today—except for mobile access. High-quality data and human expertise are essential to the value of analytics.

Human expertise is necessary because analytics alone cannot explain the reasons for trends or relationships; know what action to take; or provide sufficient context to determine what the numbers represent and how to interpret them.

Database, data warehouse, data analytics, and business intelligence (BI) technologies interact to create a new biz-tech ecosystem. Data analytics and BI discover insights or relationships of interest that otherwise might not have been recognized. They make it possible for managers to make decisions and act with clarity, speed, and confidence. Data analytics is not just about managing more or varied data. Rather, it is about asking new questions, formulating new hypotheses, exploration and discovery, and making data-driven decisions. Ultimately, a big part of data analysis efforts is the use of new analytics techniques.

Mining data or text taken from day-to-day business operations reveals valuable information, such as customers' desires, products that are most important, or processes that can be made more efficient. These insights expand the ability to take advantage of opportunities, minimize risks, and control costs.

While you might think that physical pieces of paper are a relic of the past, in most offices the opposite is true. Aberdeen Group's survey of 176 organizations worldwide found that the volume of physical documents is growing by up to 30% per year. Document management technology archives digital and physical data to meet business needs, as well as regulatory and legal requirements (Eisenhauer, 2015).

Case 3.1 Opening Case



guyenwood/Getty Images



© Katherine Welles/Shutterstock



Fernando Madeira/Shutterstock

Coca-Cola Strategically Manages Data to Retain Customers and Reduce Costs

Coca-Cola's Data Management Challenges

The Coca-Cola Company is a Fortune 100 company with over \$43.7 billion in sales revenue and \$7.35 billion in profit (**Figure 3.1**). The market leader manages and analyzes several **petabytes** (Pb) of data generated or collected from more than 500 brands and consumers in 206 countries. To understand the size of one petabyte of data, it would take 223,000 DVDs (4.7 Gb each) to hold 1 Pb of data!

Coca-Cola's bottling partners provide sales and shipment data, while retail customers transmit transaction and merchandising data. Other data sources are listed in **Table 3.1**. Before the introduction of its newest BI system, Coca-Cola knew there were BI opportunities in the mountains of data its bottlers were storing, but finding and accessing all of that data for analytics proved to be nearly impossible. The disparate data sources caused long delays in getting analytics reports from IT to sales teams. The company decided to replace the legacy software at each bottling facility and standardize them on a new BI system—a combination of MicroStrategy and Microsoft BI products.

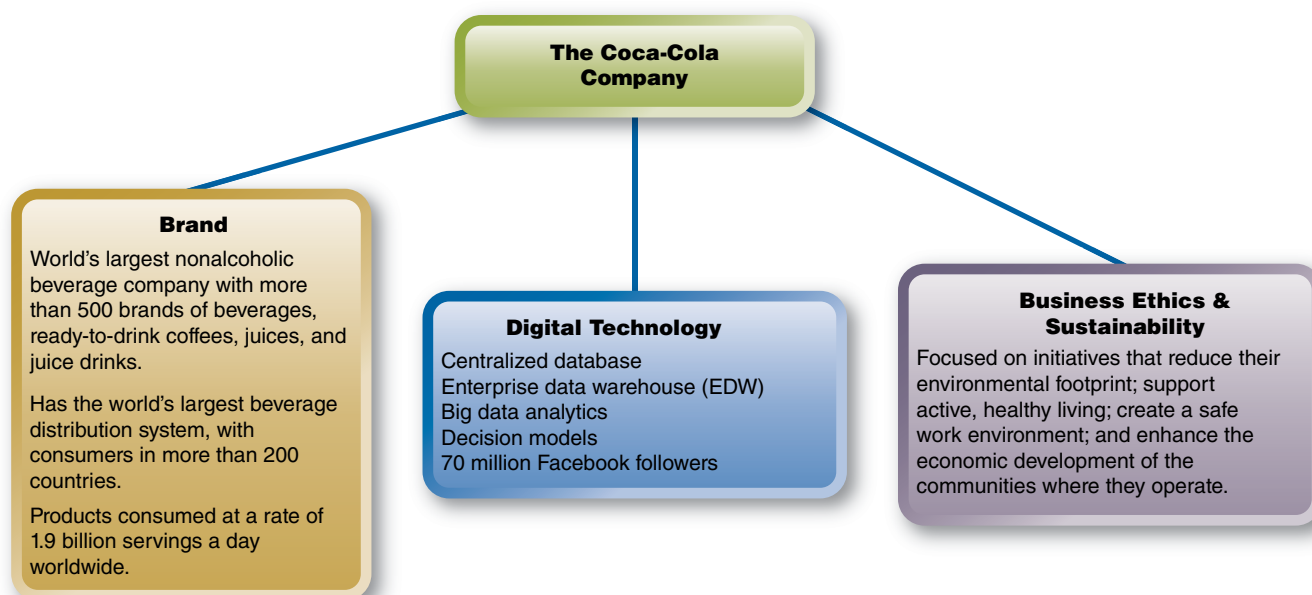


FIGURE 3.1 The Coca-Cola Company overview.

TABLE 3.1 Opening Case Overview

Company	- The Coca-Cola Company, www.coca-cola.com - Sustainability: www.coca-colacompany.com/sustainability \$43.7 billion in sales revenue and profits of \$7.35 billion, 2016
Industry	The global company manufactures, sells, and distributes nonalcoholic beverages
Product lines	More than 500 brands of still and sparkling beverages, ready-to-drink coffees, juices, and juice drinks
Digital technology	- Enterprise data warehouse (EDW) - Big data and analytics - Business intelligence - In 2014, moved from a decentralized approach to a centralized approach, where the data are combined centrally and available via the shared platforms across the organization
Business challenges	Coca-Cola had 74 unique databases, many of them used different software to store and analyze data. Dealing with incompatible databases and reporting systems was a major problem. Coca-Cola had to take a strategic approach instead of a tactical approach with big data
Global data sources	- Transaction and merchandising data - Data from nationwide network of more than 900 bottlers and manufacturing facilities - Multichannel retail data - Customer profile data from loyalty programs - Social media data - Supply chain data - Competitor data - Sales and shipment data from bottling partners
Taglines	"Taste the feeling!"
Website	www.coca-cola.com

Enterprise Data Management

Like most global companies, Coca-Cola relies on sophisticated enterprise data management, BI, and analytic technologies to sustain its performance in fiercely competitive markets. Data are managed in a **centralized database**. They use **data warehousing**, data analytics,

data modeling, and social media to respond to competitors' activity, market changes, and consumer preferences.

To support its business strategy and operations, Coca-Cola changed from a decentralized database approach to a centralized database approach (**Figure 3.2**). Now its data are combined centrally



FIGURE 3.2 Data from online and offline transactions are stored in databases. Data about entities such as customers, products, orders, and employees are stored in an organized way.

and accessible via shared platforms across the organization to help its major retail customers such as Walmart sell more Coca-Cola products and to improve the consumer experience and implemented a data governance program to ensure that cultural data sensitivities are respected.

Sustaining Business Performance

All data are standardized through a series of master data management (MDM) processes. An enterprise data warehouse (EDW) generates a single view of all multichannel retail data and creates a trusted view of customers, sales, and transactions. This enables Coca-Cola to respond quickly and accurately to changes in market conditions.

Throughout Coca-Cola huge volumes of data are analyzed to make more and better time-sensitive, critical decisions about products, shopper marketing, the supply chain, and production. Point-of-sale (POS) data are captured from retail channels and communicated via a centralized iPad reporting system to create customer profiles. POS data are analyzed to support collaborative planning, forecasting, and replenishment processes within its supply chain.

Coca-Cola's Approach to Big Data and Decision Models Coca-Cola takes a strategic approach instead of a tactical approach to big data. The company is far advanced in the use of big data to manage its products, sales revenue, and customer experiences in near real time and reduce costs. For example, it cut overtime costs almost in half by analyzing service center data. Big data help Coca-Cola relate to its millions of Facebook followers—many of whom bolster the Coke brand.

Big data play a key role in ensuring that its orange juice tastes the same year-round and is readily available anywhere in the world. Oranges used by Coca-Cola have a peak growing season of only three months. Producing orange juice with a consistent taste year-round despite the

inconsistent quality of the orange supply is complex. To deal with this complexity, an orange juice decision model was developed, the *Black Book model*. A **decision model** quantifies the relationship between variables to reduce uncertainty. Black Book combines detailed data on the 6001 flavors that make up an orange, weather, customer preferences, expected crop yields, cost pressures, regional consumer preferences, and acidity or sweetness rate. The model specifies how to blend the orange juice to create a consistent taste. Coke's Black Book juice model is considered one of the most complex **business analytics** apps. It requires analyzing up to 1 quintillion (10E18) decision variables to consistently deliver the optimal blend.

With the power of big data and decision models, Coca-Cola is prepared for disruptions in supply far in advance. According to Doug Bippert, Coca-Cola's vice president of business acceleration, "If we have a hurricane or a freeze, we can quickly re-plan the business in 5 or 10 minutes just because we've mathematically modeled it" (www.BusinessIntelligence.com, 2013b).

Questions

1. Why does the Coca-Cola Company have petabytes of data?
2. Why is it important for Coca-Cola to be able to process POS data in near real time?
3. How does Coca-Cola attempt to create favorable customer experiences?
4. What is the importance of having a trusted view of the data?
5. What is the benefit of a decision model?
6. What is the Black Book model?
7. Explain the strategic benefit of the Black Book model.

Sources: Compiled from Burns (2013), BusinessIntelligence.com (2013), CNNMoney (2014), HBS (2015), Liyakas (2015), and Ransbothom (2015).

3.1 Data Management and Database Technologies

Due to the incredible volume of data that the typical organization creates, effective data management is vital to prevent storage costs from spiraling out of control and controlling data growth while supporting greater performance. **Data management** oversees the end-to-end lifecycle of data from creation and initial storage to the time when it becomes obsolete and is deleted.

The objectives of data management include the following:

1. Mitigating the risks and costs of complying with regulations.
2. Ensuring legal requirements are met.
3. Safeguarding data security.
4. Maintaining accuracy of data and availability.
5. Certifying consistency in data that come from or go to multiple locations.
6. Ensuring that data conform to organizational best practices for access, storage, backup, and disposal.

Typically, newer data, and data that is accessed more frequently, is stored on faster, but more expensive storage media while less critical data is stored on cheaper, slower media.

The main benefits of data management include greater compliance, higher security, less legal liability, improved sales and marketing strategies, better product classification, and improved data governance to reduce risk. The following data management technologies keep users informed and support the various business demands:

- **Databases** store data generated by business apps, sensors, operations, and transaction-processing systems (TPS). Data in some databases can be extremely **volatile**. Medium and large enterprises typically have many databases of various types—centralized and distributed.
- **Data warehouses** integrate data from multiple databases and data silos across the organization, and organize them for complex analysis, knowledge discovery, and to support decision-making. For example, data are extracted from a database, processed to standardize their format, and then loaded into data warehouses at specific times, such as weekly. As such, data in data warehouses are nonvolatile—and are ready for analysis.
- **Data marts** are small-scale data warehouses that support a single function or one department. Enterprises that cannot afford to invest in data warehousing may start with one or more data marts.
- **Business intelligence (BI)**—tools and techniques process data and do statistical analysis for insight and discovery—that is, to discover meaningful relationships in the data, keep informed in real time, detect trends, and identify opportunities and risks.

Each of these database management technologies will be discussed in greater detail later in this chapter.

Data management is the management of the flow of data from creation and initial storage to the time when the data become obsolete and are deleted.

Databases are collections of data sets or records stored in a systematic way.

Database Management Systems and SQL

Data-processing techniques, processing power, and enterprise performance management capabilities have undergone revolutionary advances in recent years for reasons you are already familiar with—big data, mobility, and cloud computing. The last decade, however, has seen the emergence of new approaches, first in data warehousing and, more recently, for transaction processing. Given the huge number of transactions that occur daily in an organization, the data in databases are constantly in use or being updated. The volatility of databases makes it impossible to use them for complex decision-making and problem-solving tasks. For this reason, data are extracted from the database, transformed (processed to standardize the data), and then loaded into a data warehouse.

Database management systems (DBMSs) are software used to manage the additions, updates, and deletions of data as transactions occur, and to support data queries and reporting. They are online transaction-processing (OLTP) systems.

Database management systems (DBMSs) integrate with data collection systems such as TPS and business applications; store the data in an organized way; and provide facilities for accessing and managing that data. Factors to consider when evaluating the performance of a database management system are listed in **Tech Note 3.1**. Over the past 25 years, the **relational database** has been the standard database model adopted by most enterprises. Relational databases store data in tables consisting of columns and rows, similar to the format of a spreadsheet, as shown in **Figure 3.3**.

Tech Note 3.1

Factors That Determine the Performance of a DBMS

Factors to consider when evaluating the performance of a database management system include:

- Data latency** **Latency** is the elapsed time (or delay) between when data are created and when they are available for a **query** or report. Applications have different tolerances for latency. Database systems tend to have shorter latency than data warehouses. Short latency imposes more restrictions on a system.
- Ability to handle the volatility of the data** The database has the processing power to handle the volatility of the data. The rates at which data are added, updated, or deleted determine the workload that the database must be able to control to prevent problems with the response rate to queries.
- Query response time** The volume of data impacts response times to queries and data explorations. Many databases

pre-stage data—that is, summarize or precalculate results—so queries have faster response rates.

- Data consistency** **Immediate consistency** means that as soon as data are updated, responses to any new query will return the updated value. With **eventual consistency**, not all query responses will reflect data changes uniformly. Inconsistent query results could cause serious problems for analyses that depend on accurate data.
- Query predictability** The greater the number of ad hoc or unpredictable queries, the more flexible the database needs to be. Database or query performance management is more difficult when the workloads are so unpredictable that they cannot be prepared for in advance. The ability to handle the workload is the most important criterion when choosing a database.
- Query processing capabilities** Database queries are processed in real time and results are transmitted via wired or wireless networks to computer screen or handheld devices.

Query are ad hoc (unplanned) user requests for specific data.



FIGURE 3.3 Illustration of structured data format. Numeric and alphanumeric data are arranged into rows and predefined columns similar to those in an Excel spreadsheet.

Structured query language (SQL) is a standardized query language for accessing databases.

Relational management systems (RDBMSs) provide access to data using a declarative language—**structured query language (SQL)**. **Declarative languages** simplify data access by requiring that users only specify what data they want to access without defining how access will be achieved. The format of a basic SQL statement is

```
SELECT column_name(s)
FROM table_name
WHERE condition
```

An instance of SQL is shown in **Figure 3.4**.

```

SELECT      EmployeeID, FirstName, LastName, HireDate, Ci
FROM        Employees
WHERE       HireDate BETWEEN '1-june-2012' AND '15-decemb

SELECT EmployeeID, FirstName, LastName, HireDate, City
WHERE City IN ('Seattle', 'Tacoma', 'Redmond')

SELECT      EmployeeID, FirstName, LastName, HireDate, Ci
FROM        Employees
WHERE       HireDate NOT BETWEEN '1-june-2012' AND '15-de

```

© Piotr Adamowicz/Shutterstock

FIGURE 3.4 An instance of SQL to access employee information based on date of hire.

DBMS Functions An accurate and consistent view of data throughout the enterprise is needed so one can make informed, actionable decisions that support the business strategy. Functions performed by a DBMS to help create such a view are shown in **Figure 3.5**.

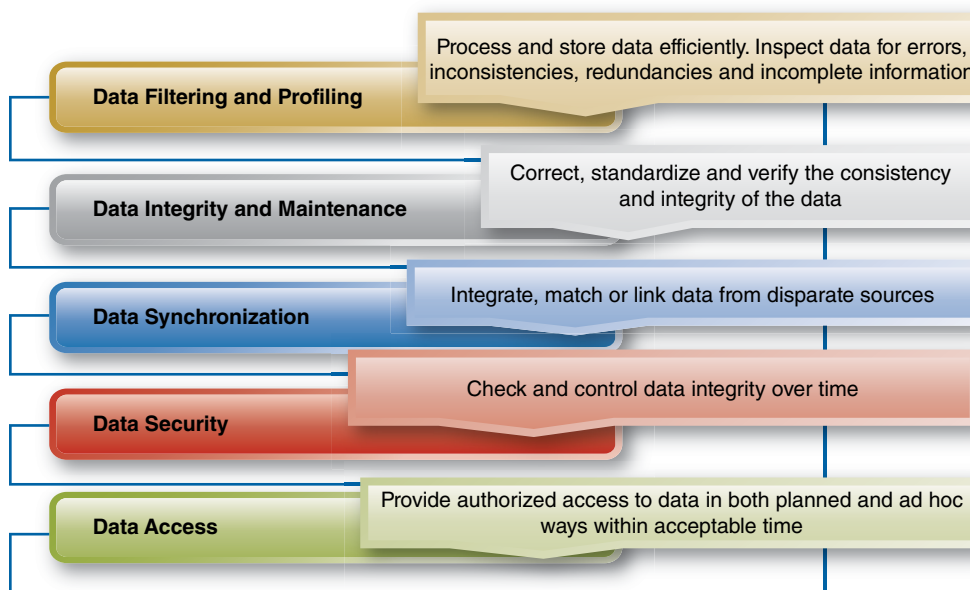


FIGURE 3.5 DBMS functions.

Online Transaction Processing and Online Analytics Processing When most business transactions occur—for instance, an item is sold or returned, an order is sent or cancelled, a payment or deposit is made—changes are made immediately to the database. These online changes are additions, updates, or deletions. DBMSs record and process transactions in the database, and support queries and reporting. Given their functions, DBMSs are referred to as **online transaction processing (OLTP) systems**. OLTP is a database design that

Online transaction processing (OLTP) systems are designed to manage transaction data, which are volatile.

breaks down complex information into simpler data tables to strike a balance between transaction-processing efficiency and query efficiency. OLTP databases process millions of transactions per second. However, databases cannot be optimized for data mining, complex **online analytics processing (OLAP) systems**, and decision support. These limitations led to the introduction of data warehouse technology. Data warehouses and data marts are optimized for OLAP, data mining, BI, and decision support. OLAP is a term used to describe the analysis of complex data from the data warehouse. In summary, databases are optimized for extremely fast transaction processing and query processing. Data warehouses are optimized for analysis.

DBMS and Data Warehousing Vendors Respond to Latest Data Demands

One of the major drivers of change in the data management market is the increased amount of data to be managed. Enterprises need powerful DBMSs and data warehousing solutions, analytics, and reporting. The four vendors that dominate this market—Oracle, IBM, Microsoft, and Teradata—continue to respond to evolving data management needs with more intelligent and advanced software and hardware. Advanced hardware technology enables scaling to much higher data volumes and workloads than previously possible, or it can handle specific workloads. Older general-purpose relational databases DBMSs lack the scalability or flexibility for specialized or very large workloads, but are very good at what they do.

Trend Toward NoSQL Systems RDBMSs are still the dominant database engines, but the trend toward **NoSQL** (short for “not only SQL”) systems is clear. NoSQL systems increased in popularity by 96% from 2014 to 2016. Although NoSQL have existed for as long as relational DBMS, the term itself was not introduced until 2009. That was when many new systems were developed in order to cope with the unfolding requirements for DBMS—namely, handling big data, scalability, and fault tolerance for large Web applications. **Scalability** means the system can increase in size to handle data growth or the load of an increasing number of concurrent users. To put it differently, scalable systems efficiently meet the demands of high-performance computing. **Fault tolerance** means that no single failure results in any loss of service.

NoSQL systems are such a heterogeneous group of database systems that attempts to classify them are not very helpful. However, their general advantages are the following:

- higher performance
- easy distribution of data on different nodes, which enables scalability and fault tolerance
- greater flexibility
- simpler administration

Starting in 2010 and continuing through 2016, Microsoft has been working on the first rewrite of SQL Server’s query execution since Version 7 was released in 1998. The goal is to offer NoSQL-like speeds without sacrificing the capabilities of a relational database.

With most NoSQL offerings, the bulk of the cost does not lie in acquiring the database, but rather in implementing it. Data need to be selected and migrated (moved) to the new database. Microsoft hopes to reduce these costs by offering migration solutions.

DBMS Vendor Rankings The top five enterprise database systems of 2016 are Oracle’s 12c Database, Microsoft SQL Server, IBM DB2, SAP Sybase ASE, and PostgreSQL:

1. **Oracle 12c Database** consolidates and manages databases as cloud services via Oracle’s multitenant architecture and in-memory data processing capabilities and can be rapidly provisioned.

2. **Microsoft SQL Server** ease of use, availability, and Windows operating system integration make it an easy choice for firms that choose Microsoft products for their enterprises.
3. **IBM DB2** is widely used in large data centers and runs on Linux, UNIX, Windows, IBM iSeries, and mainframes.
4. **SAP Sybase ASE** is a major force after 25 years of success and improvements. Supports partition locking, relaxed query limits, query plan optimization, and dynamic thread assignment.
5. **PostgreSQL** is the most advanced open source database, often used by online gaming applications and Skype, Yahoo!, and MySpace. This database runs on a wide variety of operating systems including Linux, Windows, FreeBSD, and Solaris.

Questions

1. Describe a database and a database management system (DBMS).
2. Explain what an online transaction-processing (OLAP) system does.
3. Why are data in databases volatile?
4. Describe the functions of a DBMS.
5. Describe the purpose and benefits of data management.
6. What is a relational database management system?

3.2 Centralized and Distributed Database Architectures

Databases can be **centralized** or **distributed**, as shown in **Figure 3.6**. Both types of databases need one or more backups and should be archived on- and offsite in case of a crash or security incident.

For decades the main database platform consisted of centralized database files on massive mainframe computers. Benefits of centralized database configurations include the following:

1. **Better control of data quality** Data consistency is easier when data are kept in one physical location because data additions, updates, and deletions can be made in a supervised and orderly fashion.
2. **Better IT security** Data are accessed via the centralized host computer, where they can be protected more easily from unauthorized access or modification.

A major disadvantage of centralized databases, like all centralized systems, is transmission delay when users are geographically dispersed. More powerful hardware and networks compensate for this disadvantage.

In contrast, distributed databases use client/server architecture to process information requests. The databases are stored on servers that reside in the company's data centers, a private cloud, or a public cloud (**Figure 3.7**). Advantages of a distributed database include reliability—if one site crashes, the system will keep running—and speed—it's faster to search a part of a database than the whole. However, if there's a problem with the network that the distributed database is using, it can cause availability issues and the appropriate hardware and software can be expensive to purchase.

Centralized database stores all data in a single central compute such as a mainframe or server.

Distributed database stores portions of the database on multiple computers within a network.

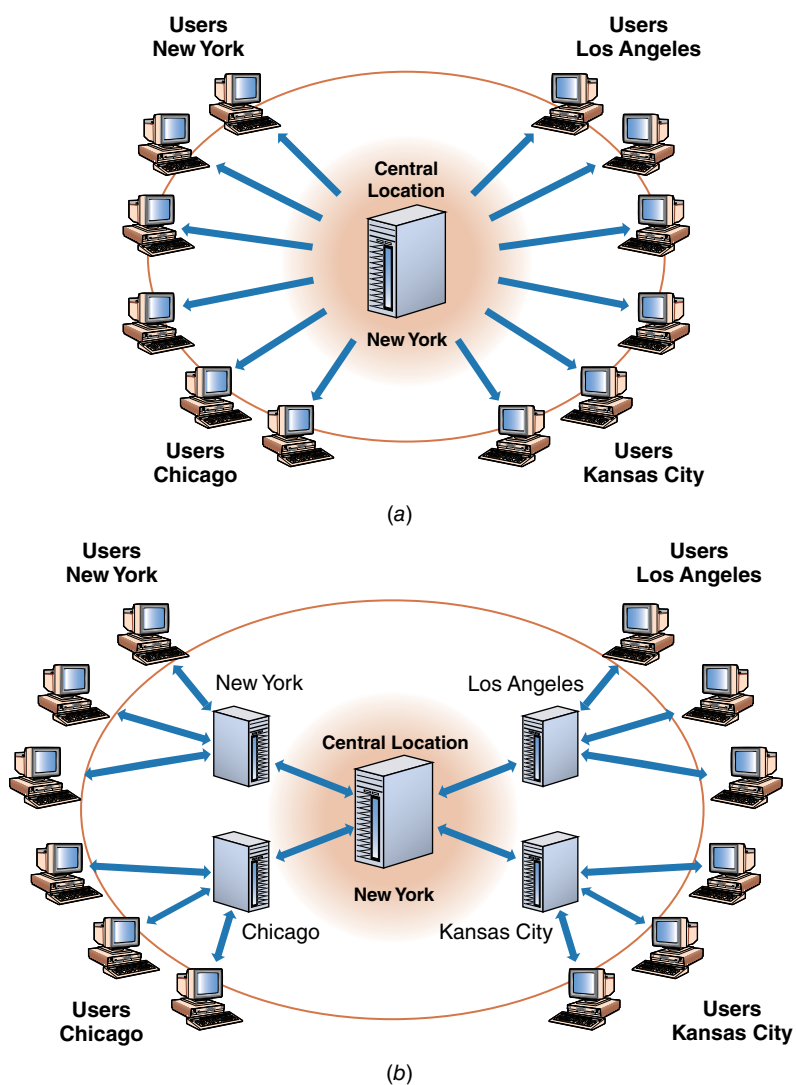


FIGURE 3.6 Comparison of (a) centralized and (b) distributed databases.

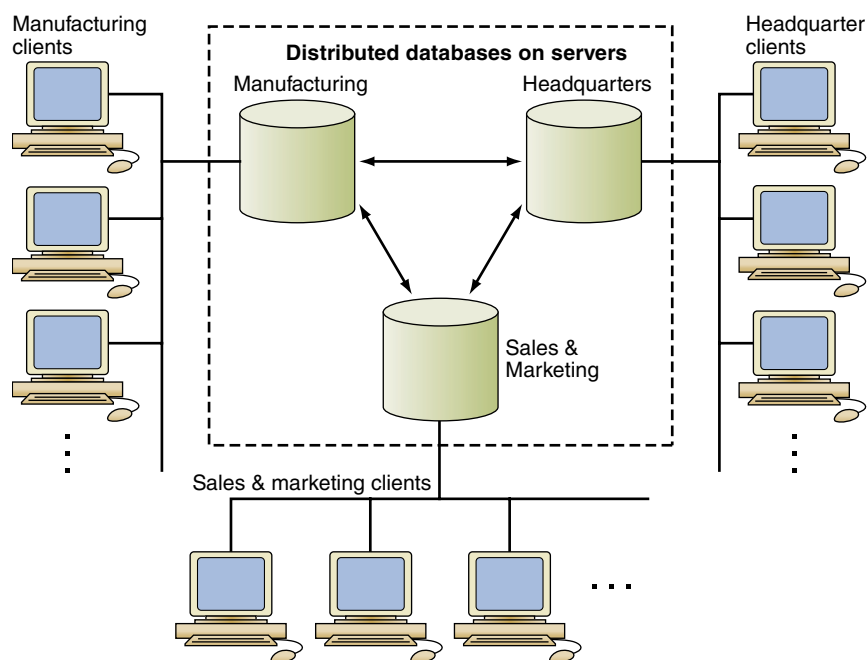


FIGURE 3.7 Distributed database architecture for headquarters, manufacturing, and sales and marketing.

Garbage In, Garbage Out

Data collection is a highly complex process that can create problems concerning the quality of the data being collected. Therefore, regardless of how the data are collected, they need to be validated so users know they can trust them. Classic expressions that sum up the situation are “garbage in, garbage out” (GIGO) and the potentially riskier “garbage in, gospel out.” In the latter case, poor-quality data are trusted and used as the basis for planning. For example, you have probably encountered data safeguards, such as integrity checks, to help improve data quality when you fill in an online form, such as when the form will not accept an e-mail address or a credit card number that is not formatted correctly.

Table 3.2 lists the characteristics typically associated with dirty or poor-quality data.

TABLE 3.2 Characteristics of Poor-Quality or Dirty Data

Characteristic of Dirty Data	Description
Incomplete	Missing data
Outdated or invalid	Too old to be valid or useful
Incorrect	Too many errors
Duplicated or in conflict	Too many copies or versions of the same data—and the versions are inconsistent or in conflict with each other
Nonstandardized	Data are stored in incompatible formats—and cannot be compared or summarized
Unusable	Data are not in context to be understood or interpreted correctly at the time of access

Dirty Data Costs and Consequences As discussed in Chapter 2, too often managers and information workers are actually constrained by data that cannot be trusted because they are incomplete, out of context, outdated, inaccurate, inaccessible, or so overwhelming that they require weeks to analyze. In such situations, the decision-maker is facing too much uncertainty to make intelligent business decisions.

On average, an organization experiences 40% data growth annually, and 20% of that data is found to be dirty. Each **dirty data** point, or record, costs \$100 if not resolved (RingLead, 2015). The costs of poor-quality data spread throughout a company, affecting systems from shipping and receiving to accounting and customer service. Data errors typically arise from the functions or departments that generate or create the data—and not within the IT department. When all costs are considered, the value of finding and fixing the causes of data errors becomes clear. In a time of decreased budgets, some organizations may not have the resources for such projects and may not even be aware of the problem. Others may be spending most of their time fixing problems, thus leaving them with no time to work on preventing them. However, the benefits of acting preventatively against dirty data are astronomical. It costs \$1 to prevent and \$10 to correct dirty data. While the short-run cost of cleaning and preventing dirty data is unrealistic for some companies, the long-term conclusion is far more expensive (Kramer, 2015).

Bad data are costing U.S. businesses hundreds of billions of dollars a year and affecting their ability to ride out the tough economic climate. Incorrect and outdated values, missing data, and inconsistent data formats can cause lost customers, sales, and revenue; misallocation of resources; and flawed pricing strategies.

Consider a corporation that follows the cost structure associated with clean/dirty data explained above with 100,000 data points. Over a three-year span, by cleaning the 20% of dirty data during the first year and using prevention methods for the following years, the corporation will save \$8,495,000. Purely based on the quality of its data, a corporation with a large amount of data can hypothetically increase its revenue by 70% (RingLead, 2015).

Dirty data is poor-quality data that lacks integrity and cannot be trusted.

The cost of poor-quality data may be expressed as a formula:

Cost of Poor-Quality Data = Lost Business + Cost to Prevent Errors + Cost to Correct Errors

Examples of these costs include the following:

- **Lost business** Business is lost when sales opportunities are missed, orders are returned because wrong items were delivered, or errors frustrate and drive away customers.
- **Time spent preventing errors** If data cannot be trusted, then employees need to spend more time and effort trying to verify information in order to avoid mistakes.
- **Time spent correcting errors** Database staff need to process corrections to the database. For example, the costs of correcting errors at U-rent Corporation are estimated as follows:
 - a. Two database staff members spend 25% of their workday processing and verifying data corrections each day:
 $2 \text{ people} * 25\% \text{ of } 8 \text{ hours / day} = 4 \text{ hours / day correcting errors}$
 - b. Hourly salaries are \$50 per hour based on pay rate and benefits:
 $\$50 / \text{hour} * 4 \text{ hours / day} = \$200 / \text{day correcting errors}$
 - c. 250 workdays per year:
 $\$200 / \text{day} * 250 \text{ days} = \$50,000 / \text{year to correct errors}$

For a particular company, it is difficult to calculate the full cost of poor-quality data and its long-term effects. Part of the difficulty is the time delay between the mistake and when it is detected. Errors can be very difficult to correct, especially when systems extend across the enterprise. Another concern is that the impacts of errors can be unpredictable, far-reaching, and serious.

Data Ownership and Organizational Politics

Compliance with numerous federal and state regulations relies on rock-solid data and trusted metrics used for regulatory reporting. Data ownership, data quality, and formally managed data are high priorities on the agenda of CFOs and CEOs who are held personally accountable if their company is found to be in violation of regulations.

Despite the need for high-quality data, organizational politics and technical issues make that difficult to achieve. The source of the problem is data ownership—that is, who owns or is responsible for the data. Data ownership problems exist when there are no policies defining responsibility and accountability for managing data. Inconsistent data formats of various departments create an additional set of problems as organizations try to combine individual applications into integrated enterprise systems.

The tendency to delegate data-quality responsibilities to the technical teams who have no control over data quality, as opposed to business users who do have such control, is another common pitfall that stands in the way of accumulating high-quality data.

Those who manage a business or part of a business are tasked with trying to improve business performance and retain customers. Compensation is tied to improving profitability, driving revenue growth, and improving the quality of customer service. These key performance indicators (KPIs) are monitored closely by senior managers who want to find and eliminate defects that harm performance. It is strange then that so few managers take the time to understand how performance is impacted by poor-quality data. Two examples make a strong case for investment in high-quality data.

Retail banking: For retail bank executives, risk management is the number one issue. Disregard for risk contributed to the 2008 financial services meltdown. Despite risk management strategies, many banks still incur huge losses. Part of the problem in many banks is that their ISs enable them to monitor risk only at the product level—mortgages, loans, or credit cards. Product-level risk management ISs monitor a customer's risk exposure for mortgages, or for loans, or for credit cards, and so forth—but not for a customer for all products. With product-level ISs, a bank cannot see the full risk exposure of a customer. The limitations of these siloed product-level risks have serious implications for business performance because bad-risk customers cannot be identified easily,

and customer data in the various ISs may differ. However, banks are beginning to use big data to analyze risk more effectively. Although they are still very limited to credit card, loan, and mortgage risk data, cheaper and faster computing power allows them to keep better and more inclusive records of customer data. Portfolio monitoring offers earlier detection and predictive analytics for potential customers, and more advanced risk models show intricate patterns unseen by the naked eye in large data sets. Also, more fact-based inputs and standardized organizational methods are being implemented to reduce loan and credit officer bias to take risks on undesirable customers.

Marketing: Consider what happens when each product-level risk management IS feeds data to marketing ISs. Marketing may offer bad-risk customers incentives to take out another credit card or loan that they cannot repay. And since the bank cannot identify its best customers either, they may be ignored and enticed away by better deals offered by competitors. This scenario illustrates how data ownership and data-quality management are critical to risk management. Data defects and incomplete data can quickly trigger inaccurate marketing and mounting losses. Banks' increasing dependence on business modeling requires that risk managers understand and manage model risk better. Although losses often go unreported, the consequences of errors in the model can be extreme. For instance, a large Asia-Pacific bank lost \$4 billion when it applied interest-rate models that contained incorrect assumptions and data-entry errors. Risk mitigation will entail rigorous guidelines and processes for developing and validating models, as well as the constant monitoring and improvement of them (Harle et al., 2016).

Manufacturing: Many manufacturers are at the mercy of a powerful customer base—large retailers. Manufacturers want to align their processes with those of large retail customers to keep them happy. This alignment makes it possible for a retailer to order centrally for all stores or to order locally from a specific manufacturer. Supporting both central and local ordering makes it difficult to plan production runs. For example, each manufacturing site has to collect order data from central ordering and local ordering systems to get a complete picture of what to manufacture at each site. Without accurate, up-to-date data, orders may go unfilled, or manufacturers may have excess inventory. One manufacturer who tried to keep its key retailer happy by implementing central and local ordering could not process orders correctly at each manufacturing site. No data ownership and lack of control over how order data flowed throughout business operations had negative impacts. Conflicting and duplicate business processes at each manufacturing site caused data errors, leading to mistakes in manufacturing, packing, and shipments. Customers were very dissatisfied.

These examples demonstrate the consequences of a lack of data ownership and data quality. Understanding the impact mismanaged data can have on business performance highlights the need to make data ownership and data accuracy a high priority.

Data Life Cycle and Data Principles

The data life cycle is a model that illustrates the way data travel through an organization as shown in [Figure 3.8](#). The data life cycle begins with storage in a database, to being loaded into a data warehouse for analysis, then reported to knowledge workers or used in business

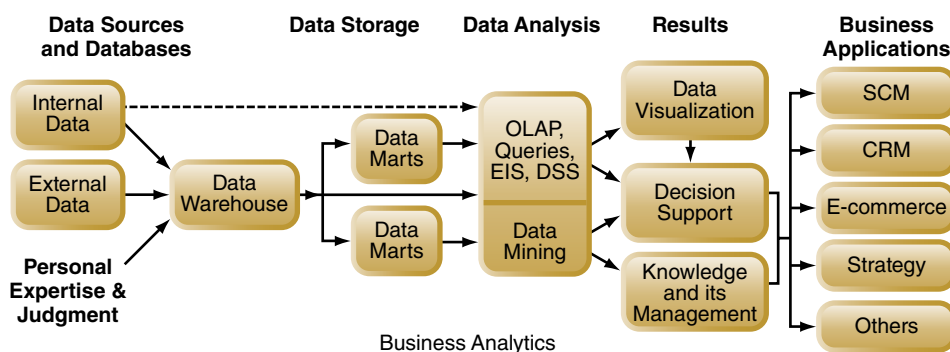


FIGURE 3.8 Data life cycle.

apps. Supply chain management (SCM), customer relationship management (CRM), and e-commerce are enterprise applications that require up-to-date, readily accessible data to function properly.

Three general data principles relate to the data life cycle perspective and help to guide IT investment decisions:

1. **Principle of diminishing data value** The value of data diminishes as they age. This is a simple, yet powerful principle. Most organizations cannot operate at peak performance with blind spots (lack of data availability) of 30 days or longer. Global financial services institutions rely on near real-time data for peak performance.
2. **Principle of 90/90 data use** According to the 90/90 data-use principle, a majority of stored data, as high as 90%, is seldom accessed after 90 days (except for auditing purposes). That is, roughly 90% of data lose most of their value after three months.
3. **Principle of data in context** The capability to capture, process, format, and distribute data in near real time or faster requires a huge investment in data architecture (Chapter 2) and infrastructure to link remote POS systems to data storage, data analysis systems, and reporting apps. The investment can be justified on the principle that data must be integrated, processed, analyzed, and formatted into “actionable information.”

Master Data and Master Data Management

As data become more complex and their volumes explode, database performance degrades. One solution is the use of master data and **master data management (MDM)** as introduced in Chapter 2. MDM processes integrate data from various sources or enterprise applications to create a more complete (unified) view of a customer, product, or other entity. **Figure 3.9** shows how master data serve as a layer between transactional data in a database and analytical data in a data warehouse. Although vendors may claim that their MDM solution creates “a single version of the truth,” this claim is probably not true. In reality, MDM cannot create a single unified version of the data because constructing a completely unified view of all master data is simply not possible.

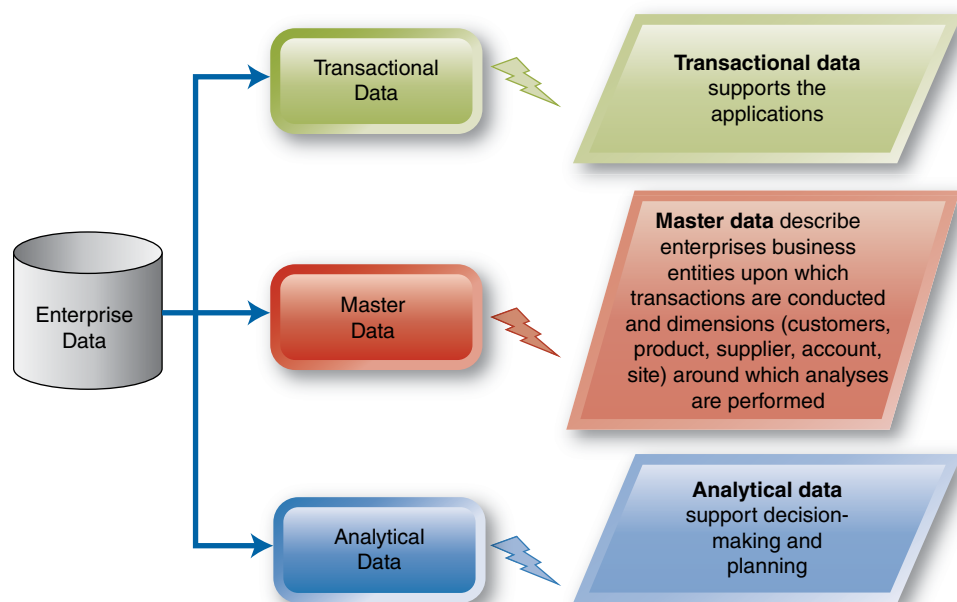


FIGURE 3.9 An enterprise has transactional, master, and analytical data.

Master Reference File and Data Entities Realistically, MDM consolidates data from various data sources into a master reference file, which then feeds data back to the applications, thereby creating accurate and consistent data across the enterprise. In **IT at Work 3.1**, participants in the health-care supply chain essentially developed a master reference file of its key data entities. A **data entity** is anything real or abstract about which a company wants to collect and store data. Master data entities are the main entities of a company, such as customers, products, suppliers, employees, and assets.

IT at Work 3.1

Data Errors Increase Costs Downstream

At an insurance company, the cost of processing each claim is \$1, but the average downstream cost due to errors in a claim is \$300. The \$300 average downstream costs included manual handling of exceptions, customer support calls initiated due to errors in claims, and reissuing corrected documents for any claims processed incorrectly the first time. In addition, the company faced significant soft costs from regulatory risk, lost revenues due to customer dissatisfaction, and overpayment on claims due to claims-processing errors. These soft costs are not included in the hard cost of \$300.

Every day health-care administrators and others throughout the health-care supply chain waste 24–30% of their time correcting data errors. Each transaction error costs \$60 to \$80 to correct.

In addition, about 60% of all invoices among supply chain partners contain errors, and each invoice error costs \$40 to \$400 to reconcile. Altogether, errors and conflicting data increase supply costs by 3–5%. In other words, each year billions of dollars are wasted in the health-care supply chain because of supply chain data disconnects, which refer to one organization's IS not understanding data from another's IS.

IT at Work Questions

1. Why are the downstream costs of data errors so high?
2. What are soft costs?
3. Explain how soft costs might exceed hard costs. Give an example.

Each department has distinct master data needs. Marketing, for example, is concerned with product pricing, brand, and product packaging, whereas production is concerned with product costs and schedules. A customer master reference file can feed data to all enterprise systems that have a customer relationship component, thereby providing a more unified picture of customers. Similarly, a product master reference file can feed data to all the production systems within the enterprise.

An MDM includes tools for cleaning and auditing the master data elements as well as tools for integrating and synchronizing data to make them more accessible. MDM offers a solution for managers who are frustrated with how fragmented and dispersed their data sources are.

Questions

1. Describe the data life cycle.
2. What is the function of master data management (MDM)?
3. What are the consequences of not cleaning “dirty data”?
4. Describe the differences between centralized and distributed databases.
5. Discuss how data ownership and organizational politics affect the quality of an organization's data.

3.3 Data Warehouses

Data warehouses are the primary source of cleansed data for analysis, reporting, and business intelligence (BI). Often the data are summarized in ways that enable quick responses to queries. For instance, query results can reveal changes in customer behavior and drive the decision to redevelop the advertising strategy.

Enterprise data warehouses (EDWs) is a data warehouse that integrates data from databases across an entire enterprise.

Data warehouses that pull together data from disparate sources and databases across an entire enterprise are called **enterprise data warehouses (EDWs)**.

Data warehouses store data from various source systems and databases across an enterprise in order to run analytical queries against huge datasets collected over long time periods.

The high cost of data warehouses can make them too expensive for a company to implement. **Data marts** are lower-cost, scaled-down versions of a data warehouse that can be implemented in a much shorter time, for example, in less than 90 days. Data marts serve a specific department or function, such as finance, marketing, or operations. Since they store smaller amounts of data, they are faster and easier to use, and navigate.

Procedures to Prepare EDW Data for Analytics

Consider a bank's database. Every deposit, withdrawal, loan payment, or other transaction adds or changes data. The volatility caused by constant transaction processing makes data analysis difficult—and the demands to process millions of transactions per second consume the database's processing power. In contrast, data in warehouses are relatively stable, as needed for analysis. Therefore, select data are moved from databases to a warehouse. Specifically, data are as follows:

1. *Extracted* from designated databases.
2. *Transformed* by standardizing formats, cleaning the data, integrating them.
3. *Loaded* into a data warehouse.

These three procedures—**extract, transform, and load**—are referred to by their initials **ETL** (Figure 3.10). In a warehouse, data are *read-only*; that is, they do not change until the next ETL.



FIGURE 3.10 Data enter databases from transaction systems. Data of interest are extracted from databases, transformed to clean and standardize them, and then loaded into a data warehouse. These three processes are called ETL.

Three technologies involved in preparing raw data for analytics include ETL, **change data capture (CDC)**, and data deduplication (“deduping the data”). CDC processes capture the changes made at data sources and then apply those changes throughout enterprise data stores to keep data synchronized. CDC minimizes the resources required for ETL processes by only dealing with data changes. Deduping processes remove duplicates and standardize data formats, which helps to minimize storage and data synch.

Building a Data Warehouse

Figure 3.11 diagrams the process of building and using a data warehouse. The organization's data from operational transaction processes systems are stored in operational databases

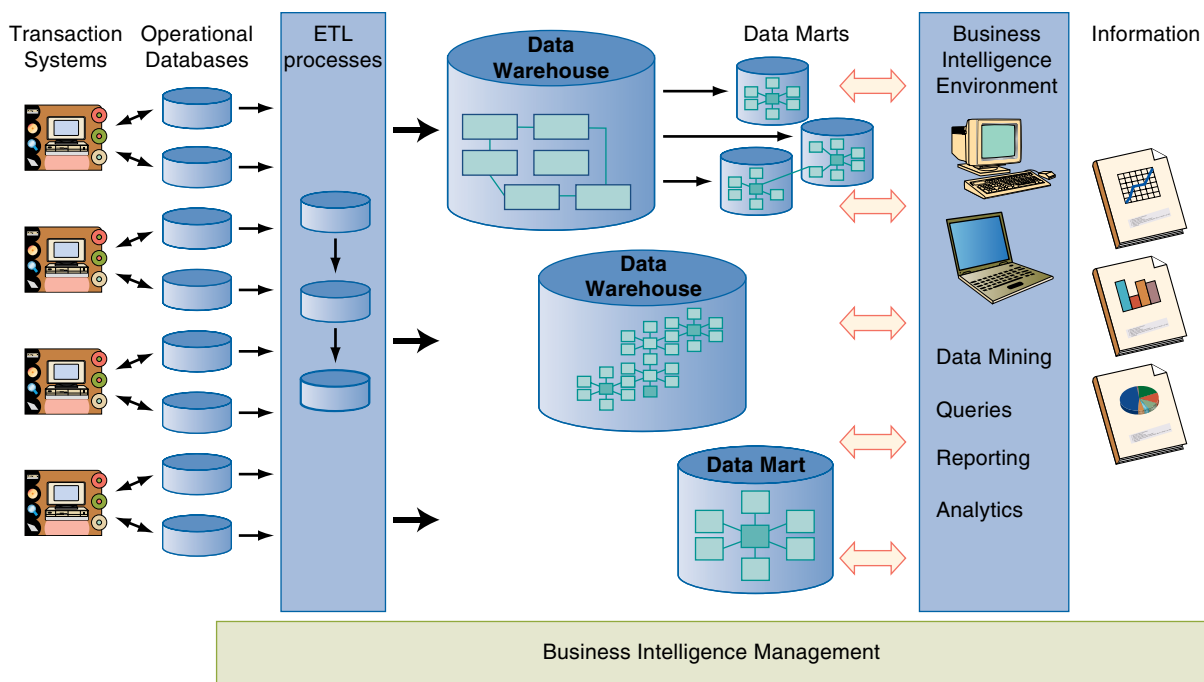


FIGURE 3.11 Database, data warehouse and marts, and BI architecture.

(left side of the figure). Not all data are transferred to the data warehouse. Frequently, only summary data are transferred. The warehouse organizes the data in multiple ways—by subject, functional area, vendor, and product. As shown, the data warehouse architecture defines the flow of data that starts when data are captured by transaction systems; the source data are stored in transactional (operational) databases; ETL processes move data from databases into data warehouses or data marts, where the data are available for access, reports, and analysis.

Real-Time Support from an Active Data Warehouse

Early data warehouse technology primarily supported strategic applications that did not require instant response time, direct customer interaction, or integration with operational systems. ETL might have been done once per week or once per month. But, demand for information to support real time customer interaction and operations leads to real-time data warehousing and analytics—known as an **active data warehouse (ADW)**. Massive increases in computing power, processing speeds, and memory made ADW possible. ADW are not designed to support executives' strategic decision-making, but rather to support operations. For example, shipping companies like DHL use huge fleets of trucks to move millions of packages. Every day and all day, operational managers make thousands of decisions that affect the bottom line, such as: “Do we need four trucks for this run?” “With two drivers delayed by bad weather, do we need to bring in extra help?” Traditional data warehousing is not suited for immediate operational support, but active data warehousing is. For example, companies with an ADW are able to:

- Interact with a customer to provide superior customer service.
- Respond to business events in near real time.
- Share up-to-date status data among merchants, vendors, customers, and associates.

Here are some examples of how two companies use ADW.

Capital One. Capital One uses its ADW to track each customer’s “profitability score” to determine the level of customer service to provide for that person. Higher-cost personalized service is only given to those with high scores. For instance, when a customer calls Capital One,

he or she is asked to enter a credit card number, which is linked to a profitability score. Low-profit customers get a voice response unit only; high-profit customers are connected to a live customer service representative (CSR) because the company wants to minimize the risk of losing those customers.

Travelocity. If you use Travelocity, an ADW is finding the best travel deals especially for you. The goal is to use “today’s data today” instead of “yesterday’s data today.” The online travel agency’s ADW analyzes your search history and destinations of interest; then predicts travel offers that you would most likely purchase. Offers are both relevant and timely to enhance your experience, which helps close the sale in a very competitive market. For example, when a customer is searching flights and hotels in Las Vegas, Travelocity recognizes the interest—the customer wants to go to Vegas. The ADW searches for the best-priced flights from all carriers, builds a few package deals, and presents them in real time to the customer. When customers see a personalized offer they are already interested in, the ADW helps generate a better customer experience. The real-time data-driven experience increases the conversion rate and sales.

Data warehouse content can be delivered to decision-makers throughout the enterprise via the cloud or company-owned intranets. Users can view, query, and analyze the data and produce reports using Web browsers. These are extremely economical and effective data delivery methods.

Data Warehousing Supports Action as well as Decisions Many organizations built data warehouses because they were frustrated with inconsistent data that could not support decisions or actions. Viewed from this perspective, data warehouses are infrastructure investments that companies make to support ongoing and future operations, including the following:

- **Marketing** Keeps people informed of the status of products, marketing program effectiveness, and product line profitability; and allows them to take intelligent action to maximize per-customer profitability.
- **Pricing and contracts** Calculates costs accurately in order to optimize pricing of a contract. Without accurate cost data, prices may be below or too near to cost; or prices may be uncompetitive because they are too high.
- **Forecasting** Estimates customer demand for products and services.
- **Sales** Calculates sales profitability and productivity for all territories and regions; analyzes results by geography, product, sales group, or individual.
- **Financial** Provides real-time data for optimal credit terms, portfolio analysis, and actions that reduce risk or bad debt expense.

Table 3.3 summarizes several successful applications of data warehouses.

TABLE 3.3 Data Warehouse Applications by Industry	
Industry	Applications
Airline	Crew assignment, aircraft deployment, analysis of route profitability, and customer loyalty promotions
Banking and financial	Customer service, trend analysis, product and service services promotions, and reduction of IS expenses
Credit card	Customer service, new information service for a fee, fraud detection
Defense contracts	Technology transfer, production of military applications
E-business	Data warehouses with personalization capabilities, marketing/shopping preferences allowing for up-selling and cross-selling
Government	Reporting on crime areas, homeland security
Health care	Reduction of operational expenses
Investment and insurance	Risk management, market movements analysis, customer tendencies analysis, and portfolio management
Retail chain	Trend analysis, buying pattern analysis, pricing policy, inventory control, sales promotions, and optimal distribution channel decision

Questions

1. What are the differences between databases and data warehouses?
2. What are the differences between data warehouses and data marts?
3. Explain ETL.
4. Explain CDC.
5. What is an advantage of an enterprise data warehouse (EDW)?
6. Why might a company invest in a data mart instead of a data warehouse?
7. What types of decisions can benefit from a data warehouse?

3.4 Big Data Analytics and Data Discovery

Like mobile and cloud, **big data** and advanced **data analytics** are reshaping organizations and business processes to increase efficiency and improve performance. Research firm IDC forecasts that big data and analytics spending will reach \$187 billion in 2019 (Ovalsrud, 2016).

Data analytics is an important tool across organizations, which helps users discover meaningful real-time insights to meet customer expectations, achieve better results and stay competitive. These deeper insights combined with human expertise enable people to recognize meaningful relationships more quickly or easily; and furthermore, realize the strategic implications of these situations. Imagine trying to make sense of the fast and vast data generated by social media campaigns on Facebook or by sensors attached to machines or objects. Low-cost sensors make it possible to monitor all types of physical things—while analytics makes it possible to understand those data in order to take action in real time. For example, sensors data can be analyzed in real time:

- To monitor and regulate the temperature and climate conditions of perishable foods as they are transported from farm to supermarket.
- To sniff for signs of spoilage of fruits and raw vegetables and detect the risk of *E. coli* contamination.
- To track the condition of operating machinery and predict the probability of failure.
- To track the wear of engines and determine when preventive maintenance is needed.

In this section, you will learn about the value, challenges, and technologies involved in putting data and analytics to use to support decisions and action, together with examples of skill sets currently in high demand by organizations expanding their efforts to train, hire and retain competent data professionals (**Career Insight 3.1**).

Big data is an extremely large data set that is too large or complex to be analyzed using traditional data processing techniques.

Data analytics is a technique of qualitatively or quantitatively analyzing a data set to reveal patterns, trends, and associations that often relate to human behavior and interactions, to enhance productivity and business gain.

Career Insight 3.1

Managing and Interpreting Big Data are High Demand Skills

Concerns about the analytics skills gap have existed for years. It is increasingly clear that the shortage isn't just in data scientists, but also data engineers, data analysts, and even the executives required to manage data initiatives. As a result, organizations and institutions are expanding their efforts to train, hire, and retain data professionals. Here are two of those skill sets that are in high demand.

Big data specialists manage and package big data collections, analyze, and interpret trends and present their findings in easy to understand ways to "C"-level executives. Those who can present the data through user-friendly data visualizations will be

particularly sought after. Skills required of these big data professionals include big data visualization, statistical analysis, Big Data reporting and presentation, Apache Hadoop, NoSQL Database Skills, and machine learning.

Business intelligence (BI) analysts use tools and techniques to go beyond the numbers of big data and take action based on the findings of the big data analyses. Successful BI professionals use self-service BI platforms, like Tableau, SAP, Oracle BI, Microsoft BI, and IBM Cognos, to create BI reports and visualizations to streamline the process and reduce reliance on additional staff. Additional skills of critical thinking, creative problem solving, effective communication, and presentations further enhance their attractiveness to employers (Hammond, 2015).

Big data analytics process of examining large and varied data sets to identify hidden patterns and correlations, market trends, customer preferences and other useful information to enable better business decisions.

When the data set is too large or complex to be analyzed using traditional data processing applications, **big data analytics** tools are used. One of the biggest sectors of customer relations relative to big data is customer value analytics (CVA). CVA studies the recent phenomenon that customers are more willing to use and purchase innovative products, services, and customer service channels while demanding an increasing amount of high-quality, personalized products. Companies and producers use big data analytics to capture this combination to transform the information into usable data to track and predict trends. If companies know what customers like, what makes them spend more, and when they are happy, they can leverage the information to keep them happy and provide better products and services.

Companies can also use big data analytics to store and use their data across the supply chain. To maximize the effectiveness of data analytics, companies usually complete these objectives throughout their input transformation process:

- Invest heavily in IT to collect, integrate, and analyze data from each store and sales unit.
- Link these data to suppliers' databases, making it possible to adjust prices in real time, to reorder hot-selling items automatically, and to shift items from store to store easily.
- Constantly test, integrate, and report information instantly available across the organization—from the store floor to the CFO's office.

These big data programs enable them to pinpoint improvement opportunities across the supply chain—from purchasing to in-store availability management. Specifically, the companies are able to predict how customers *will behave* and use that knowledge to be prepared to respond quickly. According to Louis Columbus at Forbes, the market demand for big data analytics is about to hit its largest increase in history. Software for business analytics will increase by more than 50% by 2019. Prescriptive analytics software will be worth \$1.1B in 2019, compared to its value of \$415M in 2014. Since increasing the focus on customer demand trends, effectively entering new markets and producing better business models, and enhancing organizational performance are the most important goals for 21st-century companies, business analytics will be needed in almost every instance. Taking advantage of the benefits of business intelligence is allowing sectors like health care to compete in areas they would have not been able to enter before (Columbus, 2016).

To be effective in using data analysis, organization must pay attention to the four Vs of analytics—variety, volume, velocity, and veracity—shown in **Figure 3.12**.

Big data can have a dramatic impact on the success of any enterprise, or they can be a low-contributing major expense. However, success is not achieved with technology alone. Many companies are collecting and capturing huge amounts of data, but spending very little effort to ensure the veracity and value of data captured at the transactional stage or point of origin. Emphasis in this direction will not only increase confidence in the datasets, but also significantly reduce the efforts for analytics and enhance the quality of decision-making. Success depends also on ensuring that you avoid invalid assumptions, which can be done by testing the assumptions during analysis.

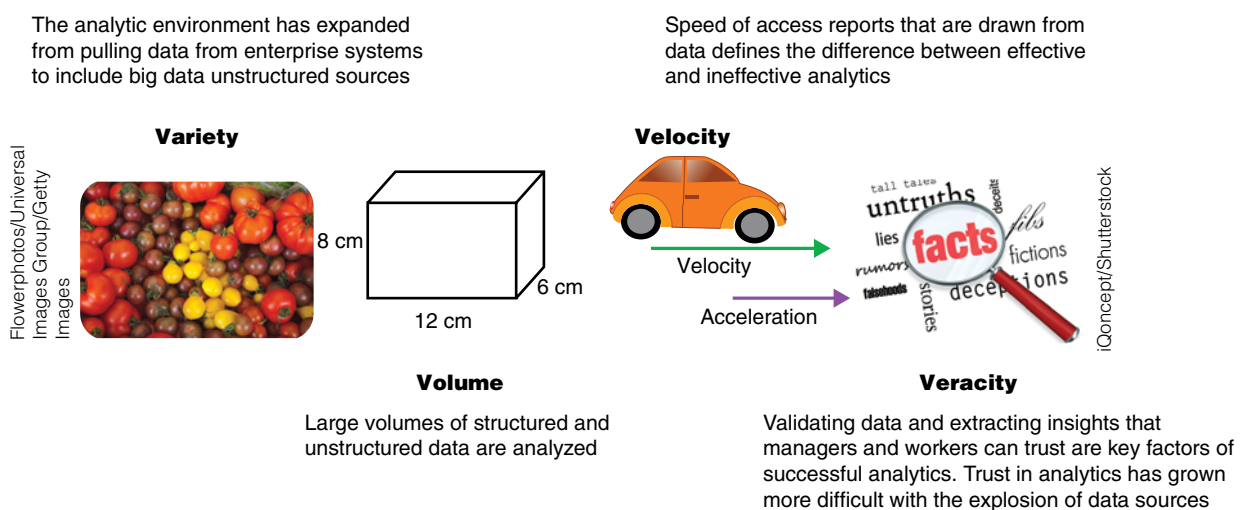


FIGURE 3.12 The four Vs of data analytics.

Human Expertise and Judgment are Needed

Human expertise and judgment are needed to interpret the output of analytics (refer to [Figure 3.13](#)). Data are worthless if you cannot analyze, interpret, understand, and apply the results in context. This brings up several challenges:

- **Data need to be prepared for analysis** For example, data that are incomplete or duplicated need to be fixed.
- **Dirty data degrade the value of analytics** The “cleanliness” of data is very important to data mining and analysis projects. Analysts have complained that data analytics is like janitorial work because they spend so much time on manual, error-prone processes to clean the data. Large data volumes and variety mean more data that are dirty and harder to handle.
- **Data must be put into meaningful context** If the wrong analysis or datasets are used, the output would be nonsense, as in the example of the Super Bowl winners and stock market performance. Stated in reverse, managers need context in order to understand how to interpret traditional and big data.

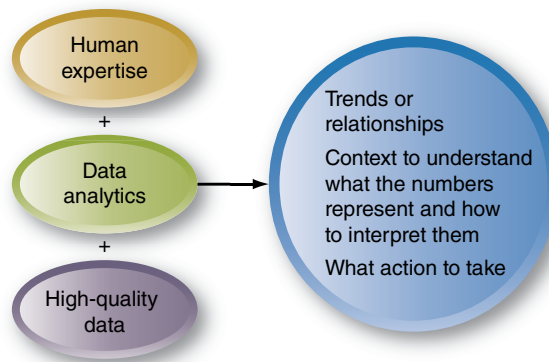


FIGURE 3.13 Data analytics, human expertise, and high-quality data are needed to obtain actionable information.

IT at Work 3.2 describes how big data analytics, collaboration, and human expertise have transformed the new drug development process.

Machine-generated sensor data are becoming a larger proportion of big data ([Figure 3.14](#)), according to a research report by IDC (2015). It is predicted that these data will increase to two-thirds of all data by 2020, representing a significant increase from the 11% level of 2005. In addition to its growth as a portion of analyzed data, the market for sensor data will increase to \$1.7 trillion in 2020.

On the consumer side, a significant factor in this market is the boom in wearable technology—products like FitBit and the Apple Watch. Users no longer even have to input data to these devices as it is automatically gathered and tracked in real time. On the public sector and enterprise side, sensor data and the Internet of Things (IoT) are being used in the advancement of IT-enabled business processes like automated factories and distribution centers and IT-enabled products like the wearable tech (IDC, 2015). Federal health reform efforts have pushed health-care organizations toward big data and analytics. These organizations are planning to use big data analytics to support revenue cycle management, resource utilization, fraud prevention, health management, and quality improvement.

Hadoop and MapReduce Big data volumes exceed the processing capacity of conventional database infrastructures. A widely used processing platform is [Apache Hadoop](#).

IT at Work 3.2

Researchers Use Genomics and Big Data in Drug Discovery

Drug development is a high-risk business. Almost 90% of new drugs ultimately fail to reach the market. One of the challenges has been the amount, variety, and complexity of the data that need to be systematically analyzed. Big data technologies and private–public partnerships have made biomedical analytics feasible.

New Drug Development Had Been Slow and Expensive

Biotechnology advances have produced massive data on the biological causes of disease. However, analyzing these data and converting discoveries into treatments are much more difficult. Not all biomedical insights lead to effective drug targets, and choosing the wrong target leads to failures late in the drug development process, costing time, money, and lives. Developing a new drug—from early discovery through Food and Drug Administration (FDA) approval—takes over a decade. As a consequence, each success ends up costing more than \$1 billion. Sometimes much more! For example, by the time Pfizer Inc., Johnson & Johnson, and Eli Lilly & Co. announced their new drugs had only limited benefit for Alzheimer’s patients in late-stage testing, the industry had spent more than \$30 billion researching amyloid plaque in the brain.

Reducing Risk of Failure

Drug makers, governments, and academic researchers have partnered to improve the odds of drug success and after years of decline, the pharmaceutical industry is beginning to experience a greater rate of success with its clinical trials. Partnerships bring together the expertise of scientists from biology, chemistry, bioinformatics, genomics, and big data. They are using big data to identify biological targets for drugs and eliminate failures before they reach the human testing stage and many anticipate that big data and the analytics that go with it could be a key element in further increasing the success rates in pharmaceutical R&D (Cattell et al., 2016).

GlaxoSmithKline, the European Bioinformatics Institute (EBI), and the Wellcome Trust Sanger Institute established the Centre for Therapeutic Target Validation (CTTV) near Cambridge, England. CTTV partners combine cutting-edge genomics with the ability to collect and analyze massive amounts of biological data. By not

developing drugs that target the wrong biological pathways, they avoid wasting billions of research dollars.

With biology now a data-driven discipline, collaborations such as CTTV are needed to improve efficiencies, cut costs, and provide the best opportunities for success. Other private–public partnerships that had formed to harness drug research and big data include the following:

- **Accelerating Medicines Partnership and U.S. National Institutes of Health (NIH)** In February 2014 the NIH announced that the agency, 10 pharmaceutical companies, and nonprofit organizations were investing \$230 million in the Accelerating Medicines Partnership.
- **Target Discovery Institute and Oxford University** Oxford University opened the Target Discovery Institute in 2013. Target Discovery helps to identify drug targets and molecular interactions at a critical point in a disease-causing pathway—that is, when those diseases will respond to drug therapy. Researchers try to understand complex biological processes by analyzing image data that have been acquired at the microscopic scale.

“The big data opportunity is especially compelling in complex business environments experiencing an explosion in the types and volumes of available data. In the health-care and pharmaceutical industries, data growth is generated from several sources, including the R&D process itself, retailers, patients and caregivers. Effectively utilizing these data will help pharmaceutical companies better identify new potential drug candidates and develop them into effective, approved and reimbursed medicines more quickly” (Cattell et al., 2016).

IT at Work Questions

1. What are the consequences of new drug development failures?
2. What factors have made biomedical analytics feasible? Why?
3. Large-scale big data analytics are expensive. How can the drug makers justify investments in big data?
4. Why would drug makers such as Glaxo and Pfizer be willing to share data given the fierce competition in their industry?

Sources: Compiled from Cattell et al. (2016), *HealthCanal* (2014), Kitamura (2014), and NIH (2014).

It places no conditions on the structure of the data it can process. Hadoop distributes computing problems across a number of servers. Hadoop implements **MapReduce** in two stages:

1. **Map stage** MapReduce breaks up the huge dataset into smaller subsets; then distributes the subsets among multiple servers where they are partially processed.
2. **Reduce stage** The partial results from the map stage are then recombined and made available for analytic tools.



© Kittichais/Stockphoto

FIGURE 3.14 Machine-generated data from physical objects are becoming a much larger portion of big data and analytics.

To store data, Hadoop has its own distributed file system, *Hadoop File System* (HDFS), which functions in three stages:

- Loads data into HDFS.
- Performs the MapReduce operations.
- Retrieves results from HDFS.

Figure 3.15 diagrams how Facebook uses database technology and Hadoop. **IT at Work 3.3** describes how First Wind has applied big data analytics to improve the operations of its wind farms and to support sustainability of the planet by reducing environmentally damaging carbon emissions.

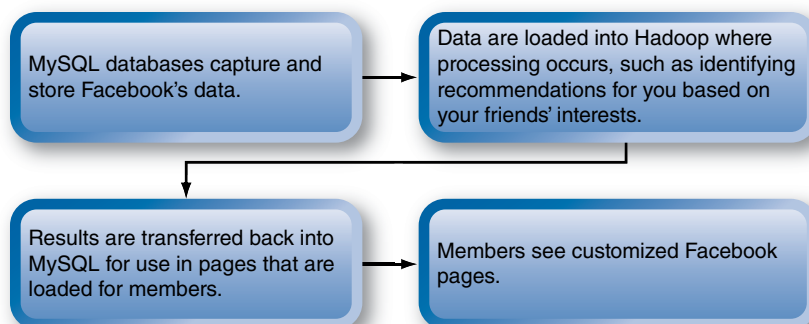


FIGURE 3.15 Facebook's MySQL database and Hadoop technology provide customized pages for its members.

IT at Work 3.3

Industrial Project Relies on Big Data Analytics

Wind power can play a major role in meeting America's rising demand for electricity—as much as 20% by 2030. Using more domestic wind power would reduce the nation's dependence on foreign sources of natural gas and also decrease carbon dioxide (CO₂) emissions that contribute to adverse climate change.

First Wind is an independent North American renewable energy company focused on the development, financing, construction, ownership, and operation of utility-scale power projects in the United States. Based in Boston, First Wind has developed and operates 980 megawatts (MW) of generating capacity at 16 wind energy projects in Maine, New York, Vermont, Utah, Washington, and Hawaii. First Wind has a large network of sensors embedded in the wind turbines, which generate huge volumes of data continuously. The data are transmitted in real time and analyzed on a 24/7 real time basis to understand the performance of each wind turbine.

Sensors collect massive amounts of data on the temperature, wind speeds, location, and pitch of the blades. The data are analyzed to study the operation of each turbine in order to adjust them to maximum efficiency. By analyzing sensor data, highly refined

measurements of wind speeds are possible. In wintry conditions, turbines can detect when they are icing up, and speed up or change pitch to knock off the ice. In the past, when it was extremely windy, turbines in the entire farm had been turned off to prevent damage from rotating too fast. Now First Wind can identify the specific portion of turbines that need to be shut down. Based on certain alerts, decisions often need to be taken within a few seconds.

Upgrades on 123 turbines on two wind farms have improved energy output by 3%, or about 120 megawatt hours per turbine per year. That improvement translates to \$1.2 million in additional revenue a year from these two farms.

IT at Work Questions

1. What are the benefits of big data analytics to First Wind?
2. What are the benefits of big data analytics to the environment and the nation?
3. How do big data analytics impact the performance of wind farms?

Sources: Compiled from www.FirstWind.com (2014) and U.S. Department of Energy (2015).

Data and Text Mining

Data and text mining are different from DBMS and data analytics. As you have read earlier in this chapter, a DBMS supports queries to extract data or get answers from huge databases. But, in order to perform queries in a DBMS you must first know the question you want to be answered. You also have read that Data Analytics describes the entire function of applying technologies, algorithms, human expertise, and judgment. **Data and text mining** are specific analytic techniques that allow users to discover knowledge that they didn't know existed in the databases.

Data mining software enables users to analyze data from various dimensions or angles, categorize them, and find correlations or patterns among fields in the data warehouse. Up to 75% of an organization's data are nonstructured word-processing documents, social media, text messages, audio, video, images and diagrams, faxes and memos, call center or claims notes, and so on.

IT at Work 3.4 describes one example of how the U.S. government is using data mining software to continuously improve its detection and deterrence systems.

Text mining is a broad category that involves interpreting words and concepts in context. Any customer becomes a brand advocate or adversary by freely expressing opinions and attitudes that reach millions of other current or prospective customers on social media. Text mining helps companies tap into the explosion of customer opinions expressed online. Social commentary and social media are being mined for **sentiment analysis** or to understand consumer intent. Innovative companies know they could be more successful in meeting their customers' needs, if they just understood them better. Tools and techniques for analyzing text, documents, and other nonstructured content are available from several vendors.

Combining data and text mining can create even greater value. Burns (2016) pointed out that mining text or nonstructural data enables organizations to forecast the future instead of merely reporting the past. He also noted that forecasting methods using existing structured data and nonstructured text from both internal and external sources provide the best view of what lies ahead.

Creating Business Value

Enterprises invest in data mining tools to add business value. Business value falls into three categories, as shown in **Figure 3.16**.

IT at Work 3.4

DoD and Homeland Security Use Data Mining Spy Machine for Threat Intelligence

Digital Reasoning, a large player in the field of big data analytics has upgraded its software that is currently contracted by the Department of Defense and Homeland Security. Synthesys 4, the name for the brand new software, allows the agencies to monitor threats in the homeland and gather data about potential attacks. Ironically, one of the main tactics employed with this software is to track and deter potential employees or contractors who have access to it. Vice President of Federal Programs Eric Hansen says that the software excels at monitoring behavioral patterns, language, and data to act and respond like a human detective would to a potential threat.

While Digital Reasoning also contracts out to other organizations like Goldman Sachs, the US government is probably its most interesting and important client. Using automatic computer software

to analyze data is much more effective at hindering attacks and quicker for analyzing large amounts of data about potential threats domestically and abroad. For instance, the software knows exactly what to look for without being bogged down and distracted by superfluous data. As available data and analytical capabilities increase, the US government is continuously aiming to improve its detection and deterrence systems using software like **Synthesys 4** (Bing, 2016).

IT at Work Questions

1. What is Synthesys 4?
2. How does data mining help the DoD achieve its mission?
3. What are the main threats to the government's data sources?
4. Why does the government see Synthesys 4 as essential to its threat deterrence measures?

Sources: Compiled from Bing (2016) and synthesys.net (2017).

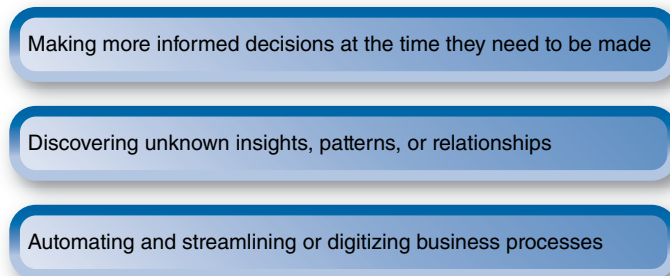


FIGURE 3.16 Business value falls into three buckets.

Here are some brief cases illustrating the types of business value created by data and text mining.

1. Using pattern analysis, **Argo Corporation**, an agricultural equipment manufacturer based in Georgia, was able to optimize product configuration options for farm machinery and real-time customer demand to determine the optimal base configurations for its machines. As a result, Argo reduced product variety by 61% and cut days of inventory by 81% while still maintaining its service levels.
2. The mega-retailer **Walmart** wanted its online shoppers to find what they were looking for faster. Walmart analyzed clickstream data from its 45 million monthly online shoppers; then combined that data with product- and category-related popularity scores. The popularity scores had been generated by text mining the retailer's social media streams. Lessons learned from the analysis were integrated into the Polaris search engine used by customers on the company's website. Polaris has yielded a 10% to 15% increase in online shoppers completing a purchase, which equals roughly \$1 billion in incremental online sales.
3. **McDonald's** bakery operation replaced manual equipment with high-speed photo analyses to inspect thousands of buns per minute for color, size, and sesame seed distribution. Automatically, ovens and baking processes adjust instantly to create uniform buns and reduce thousands of pounds of waste each year. Another food products company also uses photo analyses to sort every french fry produced in order to optimize quality.
4. **Infinity Insurance** discovered new insights that it applied to improve the performance of its fraud operation. The insurance company text mined years of adjuster reports to look for key drivers of fraudulent claims. As a result, the company reduced fraud by 75%, and eliminated marketing to customers with a high likelihood of fraudulent claims.

Text Analytics Procedure

With text analytics, information is extracted from large quantities of various types of textual information. The basic steps involved in text analytics include the following:

1. **Exploring** First, documents are explored. This might occur in the form of simple word counts in a document collection, or by manually creating topic areas to categorize documents after reading a sample of them. For example, what are the major types of issues (brake or engine failure) that have been identified in recent automobile warranty claims? A challenge of the exploration effort is misspelled or abbreviated words, acronyms, or slang.
2. **Preprocessing** Before analysis or the automated categorization of content, the text may need to be preprocessed to standardize it to the extent possible. As in traditional analysis, up to 80% of preprocessing time can be spent preparing and standardizing the data. Misspelled words, abbreviations, and slang may need to be transformed into consistent terms. For instance, BTW would be standardized to “by the way” and “left voice message” could be tagged as “lvm.”
3. **Categorizing and modeling** Content is then ready to be categorized. Categorizing messages or documents from information contained within them can be achieved using statistical models and business rules. As with traditional model development, sample documents are examined to train the models. Additional documents are then processed to validate the accuracy and precision of the model, and finally new documents are evaluated using the final model (scored). Models can then be put into production for the automated processing of new documents as they arrive.

Analytics Vendor Rankings

Analytics applications cover business intelligence functions sold as a standalone application for decision support or embedded in an integrated solution. The introduction of intuitive decision support tools, dashboards, and data visualization (discussed in detail in Chapter 11) have

TABLE 3.4 Top Analytics Vendors

Rank	Vendor	Focus	Products
1	SAP	Market lines of analytics products that cover BI and reporting, predictive analysis, performance management and governance, risk and compliance applications	SAP Business Objects Predictive Analytics SAP Business Objects BI SAP Business Objects Planning and Consolidation
2	SAS	Offer simple desktop solutions to high performance distributed processing solutions	SAS Analytics Pro SAS Enterprise Miner SAS Visual Analytics SAS Customer Intelligence 360
3	IBM	Allow users to quickly discover patterns and meanings in data with guided data discovery, automated predictive analytics, one-click analysis, self-service dashboards, and a natural language dialogue	Watson Analytics
4	Oracle	Offer a complete solution for connecting and collaborating with analytics in the cloud. Products allow users to aggregate, experiment, manage, and analyze/act	Oracle Data Integrator Oracle Big Data Cloud Service Oracle R Advanced Analytics for Hadoop BI Cloud Service Oracle Stream Explore
5	Microsoft	Provide a broad range of products from standalone solutions to integrated tools that provide data preparation, data discovery, and interactive dashboard capabilities in a single tool	Excel HDInsight Machine Learning Stream Analytics Power BI Embedded

added some interesting interactive components to big data analytics to bring the data to life and enable nonexperts to use it.

Organizations invest in analytics, BI, and data/text mining applications based on new features and capabilities beyond those offered by their legacy systems. Analytics vendors offer everything from simple-to-use reporting tools to highly sophisticated software for tackling the most complex data analysis problems. A list of the top five analytics and BI application vendors are shown in [Table 3.4](#).

Questions

1. Why are human expertise and judgment important to data analytics? Give an example.
2. What is the relationship between data quality and the value of analytics?
3. Why do data need to be put into a meaningful context?
4. How can manufacturers and health care benefit from data analytics?
5. How does data mining provide value? Give an example.
6. What is text mining?
7. What are the basic steps involved in text analytics?

3.5 Business Intelligence and Electronic Records Management

Continuing developments in data analytics and **business intelligence (BI)** make it increasingly necessary for organizations to be aware of the differences between these terms and the different ways in which they add value in an organization. The field of BI started in the late 1980s and has been a key to competitive advantage across industries and in enterprises of all sizes. Unlike data analytics that has predictive capabilities, BI is a comprehensive term that refers to analytics and reporting tools that were traditionally used to determine trends in historical data.

The key distinction between data analytics and BI is that analytics uses algorithms to statistically determine the relationships between data whereas BI presents data insights established by data analytics in reports, easy-to-use dashboards, and interactive visualizations. BI can also make it easier for users to ask data-related questions and obtain results that are presented in a way that they can easily understand.

What started as a tool to support sales, marketing, and customer service departments has widely evolved into an enterprise wide strategic platform. While BI software is used in the operational management of divisions and business processes, they are also used to support strategic corporate decision-making. The dramatic change that has taken effect over the last few years is the growth in demand for operational intelligence across multiple systems and businesses—increasing the number of people who need access to increasing amounts of data. Complex and competitive business conditions do not leave much slack for mistakes.

Unfortunately, some companies are not able to use their data efficiently, creating a higher cost to gather information than the benefits it provides. Luckily, BI software brings decision-making information to businesses in as little as two clicks. Small businesses have a shared interest with large corporations to enlist BI to help with decision-making, but they are usually unequipped to build data centers and use funds to hire analysts and IT consultants. However, small business BI software is rapidly growing in the analytics field, and it is increasingly cheaper to implement it as a decision-making tool. Small businesses do not always have workers specialized in certain areas, but BI software makes it easy for all employees to analyze the data and make decisions (King, 2016).

Business intelligence (BI) is a set of tools and techniques for acquiring and transforming raw data into meaningful and useful information for business analysis purposes in the forms of reports, dashboards, or interactive visualizations.

Business Benefits of BI

BI provides data at the *moment of value* to a decision-maker—enabling it to extract crucial facts from enterprise data in real time or near real time. A BI solution with a well-designed dashboard, for example, provides retailers with better visibility into inventory to make better decisions about what to order, how much, and when in order to prevent stock-outs or minimize inventory that sits on warehouse shelves.

Companies use BI solutions to determine what questions to ask and find answers to them. BI tools integrate and consolidate data from various internal and external sources and then process them into information to make smart decisions. BI answers questions such as these: Which products have the highest repeat sales rate in the last six months? Do customer likes on Facebook relate to product purchase? How does the sales trend break down by product group over the last five years? What do daily sales look like in each of my sales regions?

According to The Data Warehousing Institute, BI “unites data, technology, analytics, and human knowledge to optimize business decisions and ultimately drive an enterprise’s success. BI programs usually combine an enterprise data warehouse and a BI platform or tool set to transform data into usable, actionable business information” (The Data Warehousing Institute, 2014). For many years, managers have relied on business analytics to make better-informed decisions. Multiple surveys and studies agree on BI’s growing importance in analyzing past performance and identifying opportunities to improve future performance.

Common Challenges: Data Selection and Quality

Companies cannot analyze all of their data—and much of them would not add value. Therefore, an unending challenge is how to determine which data to use for BI from what seems like unlimited options (Oliphant, 2016). One purpose of a BI strategy is to provide a framework for selecting the most relevant data without limiting options to integrate new data sources. **Information overload** is a major problem for executives and for employees. Another common challenge is data quality, particularly with regard to online information, because the source and accuracy might not be verifiable.

Aligning BI Strategy with Business Strategy

Reports and dashboards are delivery tools, but they may not be delivering business intelligence. To get the greatest value out of BI, the CIO needs to work with the CFO and other business leaders to create a BI governance program whose mission is to achieve the following (Ladley, 2016):

1. Clearly articulate business strategies.
2. Deconstruct the business strategies into a set of specific goals and objectives—the *targets*.
3. Identify the key performance indicators (KPIs) that will be used to measure progress toward each target.
4. Prioritize the list of KPIs.
5. Create a plan to achieve goals and objectives based on the priorities.
6. Estimate the costs needed to implement the BI plan.
7. Assess and update the priorities based on business results and changes in business strategy.

After completing these activities, BI analysts can identify the data to use in BI and the source systems. This is a **business-driven development approach** that starts with a business strategy and work backward to identify data sources and the data that need to be acquired and analyzed.

Businesses want KPIs that can be utilized by both departmental users and management. In addition, users want real-time access to these data so that they can monitor processes with the smallest possible latency and take corrective action whenever KPIs deviate from their target

values. To link strategic and operational perspectives, users must be able to drill down from highly consolidated or summarized figures into the detailed numbers from which they were derived to perform in-depth analyses.

BI Architecture and Analytics

BI architecture is undergoing technological advances in response to big data and the performance demands of end-users (Wise, 2016). BI vendors are facing the challenges of social, sensor, and other newer data types that must be managed and analyzed. One technology advance that can help handle big data is BI in the cloud. **Figure 3.17** lists the key factors contributing to the increased use of BI. It can be hosted on a public or private cloud. Although cloud services come with more upkeep, optimizing the service and customizing it for one's company brings undeniable benefits in data security. With a public cloud, a service provider hosts the data and/or software that are accessed via an Internet connection. For private clouds, the company hosts its own data and software, but uses cloud-based technologies.

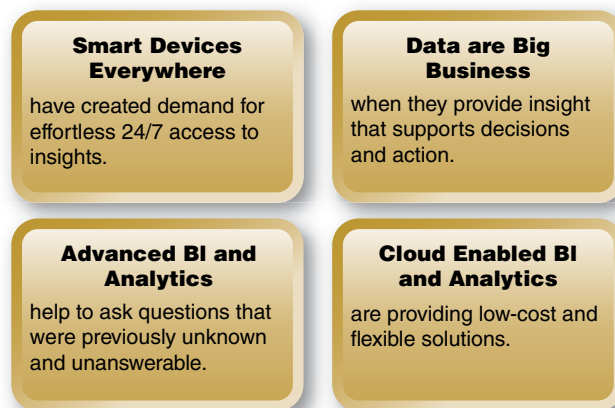


FIGURE 3.17 Four factors contributing to increased use of BI.

For cloud-based BI, a popular option offered by a growing number of BI tool vendors is software as a service (SaaS). MicroStrategy offers MicroStrategy Cloud, which provides fast deployment with reduced project risks and costs. This cloud approach appeals to small and mid-sized companies that have limited IT staff and want to carefully control costs. The potential downsides include slower response times, security risks, and backup risks.

Competitive Analytics in Practice: CarMax CarMax, Inc. is the nation's largest retailer of used cars and for a decade has remained one of *FORTUNE* Magazine's "100 Best Companies to Work For." CarMax was the fastest retailer in U.S. history to reach \$1 billion in revenues. In 2016 the company had over \$15 billion in net sales and operating revenues, representing a 6.2% increase over the prior year's results. The company grew rapidly because of its compelling customer offer—no-haggle prices and quality guarantees backed by a 125-point inspection that became an industry benchmark—and auto financing. As of November 30, 2016, CarMax operated in 169 locations across 39 U.S. states and had more than 22,000 full- and part-time employees.

CarMax continues to enhance and refine its information systems, which it believes to be a core competitive advantage. CarMax's IT includes the following:

- A proprietary IS that captures, analyzes, interprets, and distributes data about the cars CarMax sells and buys.
- Data analytics applications that track every purchase; number of test drives and credit applications per car; color preferences in every demographic and region.

- Proprietary store technology that provides management with real-time data about every aspect of store operations, such as inventory management, pricing, vehicle transfers, wholesale auctions, and sales consultant productivity.
- An advanced inventory management system that helps management anticipate future inventory needs and manage pricing.

Throughout CarMax, analytics are used as a strategic asset and insights gained from analytics are available to everyone who needs them.

Electronic Records Management

All organizations create and retain **business records**. A record is documentation of a business event, action, decision, or transaction. Examples are contracts, research and development, accounting source documents, memos, customer/client communications, hiring and promotion decisions, meeting minutes, social posts, texts, e-mails, website content, database records, and paper and electronic files. Business documents such as spreadsheets, e-mail messages, and word-processing documents are a type of record. Most records are kept in electronic format and maintained throughout their life cycle—from creation to final archiving or destruction by an **electronic records management system (ERMS)**.

Electronic records management system (ERMS) consists of hardware and software that manage and archive electronic documents and image paper documents; then index and store them according to company policy.

One application of an ERMS would be in a company that is required by law to retain financial documents for at least seven years, product designs for many decades, and e-mail messages about marketing promotions for a year. The major ERM tools are workflow software, authoring tools, scanners, and databases. ERM systems have query and search capabilities so documents can be identified and accessed like data in a database. These systems range from those designed to support a small workgroup to full-featured, Web-enabled enterprise-wide systems.

Legal Duty to Retain Business Records

Companies need to be prepared to respond to an audit, federal investigation, lawsuit, or any other legal action against them. Types of lawsuits against companies include patent violations, product safety negligence, theft of intellectual property, breach of contract, wrongful termination, harassment, discrimination, and many more.

Because senior management must ensure that their companies comply with legal and regulatory duties, managing electronic records (e-records) is a strategic issue for organizations in both the public and private sectors. The success of ERM depends greatly on a partnership of many key players, namely, senior management, users, records managers, archivists, administrators, and most importantly, IT personnel. Properly managed, records are strategic assets. Improperly managed or destroyed, they become liabilities.

ERM Best Practices

Effective ERM systems capture all business data and documents at their first touchpoint—data centers, laptops, the mailroom, at customer sites, or remote offices. Records enter the enterprise in multiple ways—from online forms, bar codes, sensors, websites, social sites, copiers, e-mails, and more. In addition to capturing the entire document as a whole, important data from within a document can be captured and stored in a central, searchable repository. In this way, the data are accessible to support informed and timely business decisions.

In recent years, organizations such as the Association for Information and Image Management (**AIIM**), National Archives and Records Administration (NARA), and **ARMA International** (formerly the Association of Records Managers and Administrators) have created and published industry standards for document and records management. Numerous best

practices articles, and links to valuable sources of information about document and records management, are available on their websites. The IT Toolbox describes ARMA's eight generally accepted recordkeeping principles framework.

ERM Benefits

Departments or companies whose employees spend most of their day filing or retrieving documents or warehousing paper records can reduce costs significantly with ERM. These systems minimize the inefficiencies and frustration associated with managing paper documents and workflows. However, they do not create a paperless office as had been predicted.

An ERM can help a business to become more efficient and productive by the following:

- Enabling the company to access and use the content contained in documents.
- Cutting labor costs by automating business processes.
- Reducing the time and effort required to locate information the business needs to support decision-making.
- Improving the security of content, thereby reducing the risk of intellectual property theft.
- Minimizing the costs associated with printing, storing, and searching for content.

When workflows are digital, productivity increases, costs decrease, compliance obligations are easier to verify, and green computing becomes possible. Green computing is an initiative to conserve our valuable natural resources by reducing the effects of our computer usage on the environment. You can read about green computing and the related topics of reducing an organization's carbon footprint, sustainability, and ethical and social responsibilities in Chapter 14.

ERM for Disaster Recovery, Business Continuity, and Compliance

Businesses also rely on their ERM system for disaster recovery and business continuity, security, knowledge sharing and collaboration, and remote and controlled access to documents. Because ERM systems have multilayered access capabilities, employees can access and change only the documents they are authorized to handle.

When companies select an ERM to meet compliance requirements, they should ask the following questions:

1. Does the software meet the organization's needs? For example, can the DMS be installed on the existing network? Can it be purchased as a service?
2. Is the software easy to use and accessible from Web browsers, office applications, and e-mail applications? If not, people will not use it.
3. Does the software have lightweight, modern Web and graphical user interfaces that effectively support remote users?
4. Before selecting a vendor, it is important to examine workflows and how data, documents, and communications flow throughout the company. For example, know which information on documents is used in business decisions. Once those needs and requirements are identified, they guide the selection of technology that can support the input types—that is, capture and index them so they can be archived consistently and retrieved on-demand.

IT at Work 3.5 describes how several companies currently use ERM. Simply creating backups of records is not sufficient because the content would not be organized and indexed to retrieve them accurately and easily. The requirement to manage records—regardless of whether they are physical or digital—is not new.

IT at Work 3.5

ERM Applications

Here are some examples of how companies use ERM in the health-care, finance, and education sectors:

- The Surgery Center of Baltimore stores all medical records electronically, providing instant patient information to doctors and nurses anywhere and at any time. The system also routes charts to the billing department, which can then scan and e-mail any relevant information to insurance providers and patients. The ERM system helps maintain the required audit trail, including the provision of records when they are needed for legal purposes. How valuable has ERM been to the center? Since it was implemented, business processes have been expedited by more than 50%, the costs of these processes have been significantly reduced, and the morale of office employees in the center has improved noticeably.
- American Express (AMEX) uses TELEform, developed by Alchemy and Cardiff Software, to collect and process more than 1 million customer satisfaction surveys every year. The data are collected in templates that consist of more than 600 different survey forms in 12 languages and 11 countries. AMEX integrated TELEform with AMEX's legacy system, which enables it to distribute processed results to many managers. Because the survey forms are now readily accessible, AMEX has reduced the number of staff who process these forms from 17 to 1, thereby saving the company more than \$500,000 a year.

- The University of Cincinnati provides authorized access to the personnel files of 12,000 active employees and tens of thousands of retirees. The university receives more than 75,000 queries about personnel records every year and then must search more than 3 million records to answer these queries. Using a microfilm system to find answers took days. The solution was an ERM that digitized all paper and microfilm documents, without help from the IT department, making them available via the Internet and the university's intranet. Authorized employees access files using a browser.

IT at Work Questions

1. What are the business benefits of BI?
2. What are two related challenges that must be resolved for BI to produce meaningful insights?
3. What are the steps in a BI governance program?
4. What does it mean to drill down into data, and why is it important?
5. What four factors are contributing to increased use of BI?
6. Why is ERM a strategic issue rather than simply an IT issue?
7. Why might a company have a legal duty to retain records? Give an example.
8. Why is creating backups an insufficient way to manage an organization's documents?

Key Terms

active data warehouse (ADW) 81
big data 83
big data analytics 84
business analytics 68
business intelligence (BI) 91
business record 94
business-driven development approach 92
centralized database 73
change data capture (CDC) 80
data analytics 83
data entity 79
data management 69
data marts 80
data mining 88
data warehouse 67
database 69

database management system (DBMS) 70
decision model 68
declarative language 70
dirty data 75
distributed database 73
electronic records management system (ERMS) 94
extract, transform and load (ETL) 80
enterprise data warehouses (EDWs) 80
eventual consistency 70
fault tolerance 72
Hadoop 85
information overload 92
immediate consistency 70
latency 70
MapReduce 86

master data management (MDM) 78
NoSQL 72
online transaction processing (OLTP) systems 71
online analytical processing (OLAP) systems 72
petabyte 66
query 70
relational database 70
relational management systems (RDBMSs) 70
sentiment analysis 88
scalability 72
structured query language (SQL) 70
text mining 88

Assuring Your Learning

Discuss: Critical Thinking Questions

1. What are the functions of databases and data warehouses?
2. How does data quality impact business performance?
3. List three types of waste or damages that data errors can cause.
4. What is the role of a master reference file?
5. Give three examples of business processes or operations that would benefit significantly from having detailed real-time or near real-time data and identify the benefits.
6. What are the tactical and strategic benefits of big data analytics?
7. Explain the four Vs of data analytics.
8. Select an industry. Explain how an organization in that industry could improve consumer satisfaction through the use of data warehousing.
9. Explain the principle of 90/90 data use.
10. Why is master data management (MDM) important in companies with multiple data sources?
11. Why would a company invest in a data mart instead of a data warehouse?
12. Why is data mining important?
13. What are the operational benefits and competitive advantages of business intelligence?
14. How can ERM decrease operating costs?

Explore: Online and Interactive Exercises

1. Visit www.YouTube.com and search for SAS Enterprise Miner Software Demo in order to assess the features and benefits of SAS Enterprise Miner. The URL is <https://www.youtube.com/watch?v=Nj4L5RFvkMg>.
 - a. View the SAS Enterprise Miner Software demo, which is about seven minutes long.
 - b. Based on what you learn in the demo, what skills or expertise are needed to build a predictive model?
 - c. At the end of the demo, you hear the presenter say that “SAS Enterprise Miner allows end-users to easily develop predictive models and to generate scoring to make better decisions about future business events.” Do you agree that SAS Enterprise Miner makes it easy to develop such models? Explain.
 - d. Do you agree that if an expert develops predictive models, it will help managers make better decisions about future business events? Explain.
 - e. Based on your answers to (c), (d), and (e), under what conditions would you recommend SAS Enterprise Miner?
2. Research two electronic records management vendors, such as Iron Mountain.
 - a. What are the retention recommendations made by the vendors? Why?
 - b. What services or solutions does each vendor offer?
3. View the “Edgenet Gain Real time Access to Retail Product Data with In-Memory Technology” video on YouTube. Explain the benefit of in-memory technology.

Analyze & Decide: Apply IT Concepts to Business Decisions

1. Visit www.Oracle.com. Click the Solutions tab to open the menu; then click Data Warehousing under Technology Solutions.
 - a. Scroll down to view “Procter & Gamble Drives 30X Performance Gains with Oracle Exadata.”
 - b. Describe the Procter & Gamble’s challenges, why it selected Oracle Exadata, and how that solution met their challenge.
2. Visit www.Teradata.com. Click Resources and open “Videos.” Select one of the videos related to data analytics. Explain the benefits of the solution chosen.
3. Spring Street Company (SSC) wanted to reduce the “hidden costs” associated with its paper-intensive processes. Employees jokingly predicted that if the windows were open on a very windy day, total chaos

would ensue as thousands of papers started to fly. If a flood, fire, or windy day occurred, the business would literally grind to a halt. The company's accountant, Sam Spring, decided to calculate the costs of its paper-driven processes to identify their impact on the bottom line. He recognized that several employees spent most of their day filing or retrieving documents. In addition, there were the monthly costs to warehouse old paper records. Sam measured the activities related to the handling of printed reports and paper files. His average estimates were as follows:

- a. Dealing with a file: It takes an employee 12 minutes to walk to the records room, locate a file, act on it, refile it, and return to his or her desk. Employees do this 4 times per day (five days per week).
- b. Number of employees: 10 full-time employees perform the functions.
- c. Lost document replacement: Once per day, a document gets "lost" (destroyed, misplaced, or covered with massive coffee stains) and must be recreated. The total cost of replacing each lost document is \$200.
- d. Warehousing costs: Currently, document storage costs are \$75 per month.

Sam would prefer a system that lets employees find and work with business documents without leaving their desks. He's most concerned about the human resources and accounting departments. These personnel are traditional heavy users of paper files and would greatly benefit

from a modern document management system. At the same time, however, Sam is also risk averse. He would rather invest in solutions that would reduce the risk of higher costs in the future. He recognizes that the U.S. PATRIOT Act's requirements that organizations provide immediate government access to records apply to SSC. He has read that manufacturing and government organizations rely on efficient document management to meet these broader regulatory imperatives. Finally, Sam wants to implement a disaster recovery system.

Prepare a report that provides Sam with the data he needs to evaluate the company's costly paper-intensive approach to managing documents. You will need to conduct research to provide data to prepare this report. Your report should include the following information:

1. How should SSC prepare for an ERM if it decides to implement one?
2. Using the data collected by Sam, create a spreadsheet that calculates the costs of handling paper at SSC based on average hourly rates per employee of \$28. Add the cost of lost documents to this. Then, add the costs of warehousing the paper, which increases by 10% every month due to increases in volume. Present the results showing both monthly totals and a yearly total. Prepare graphs so that Sam can easily identify the projected growth in warehousing costs over the next three years.
3. How can ERM also serve as a disaster recovery system in case of fire, flood, or break-in?
4. Submit your recommendation for an ERM solution. Identify two vendors in your recommendation.

Case 3.2

Business Case: Big Data Analytics is the "Secret Sauce" for Revitalizing McDonald's

With 62 million daily customers and an annual revenue of \$27 billion, McDonald's has a virtually unrivaled amount of data at its disposal to analyze. In order to dominate the market, retain its loyal customers, and attract new customers who are skeptical of McDonald's practices and quality, it lends itself to its data, becoming an "information centric organization." What does it mean to be information centric? Instead of using a fixed process of production, service, etc. as a business plan that is product driven, McDonald's uses customer data to dictate its next move as a customer-driven corporation. During the inception of McDonald's in 1940, the McDonald brothers derived a product-driven business centered around fast service and tasty food. While that method was successful before other restaurants entered the fast food market, growth was stunted due to a lack of innovation and change. So, the organization began to collect customer data as a means to monitor successful products, customer demands, and the results of marketing campaigns.

This venture led to McDonald's becoming the premier fast food chain across the United States in the 1980s. Soon after becoming a customer-driven corporation, McDonald's introduced the Happy Meal so families with small children could reduce costs and waste at dinner time, released the Egg McMuffin as the most successful breakfast item of all time, equipped professionals and teenagers with free Wi-Fi to expand its customer segmentation, and provided nutrition details to become the most transparent fast food chain at the time. All of these improvements derived from McDonald's using its immense amount of data to set its chain apart from the rest.

In 2008, to further improve its ability to leverage big data, McDonald's made the transition from average-based metrics to trend analytics. The issue with average-based metrics is that it is hard to compare regions and stores. A store could be growing in its sales and productivity but have the same average metrics as a store that is declining. Using trend analytics allowed McDonald's to combine multiple datasets from multiple data sources to visualize and understand cause-and-effect relationships in individual stores and regions. The correlations it found enabled its analysts to prescribe solutions to problems in sales, production, turnover, and supply chain management to reduce costs and save time. The variables it studies allows McDonald's to create a standardized experience across the world. However, analyzing local data in each store produces minor changes around the organization. For example, most McDonald's locations look the same, but each restaurant is slightly different and optimized for the local market.

A great example of McDonald's big data analysis in action is its updated drive-thru system. All fast food chains have bottlenecks in their drive-thru lanes, but McDonald's average customer wait time is about 3 minutes, which is close to the industry's longest wait time of 214 seconds. One of the most prominent issues in its drive-thru was that customers going through the line for dinner, ordering large meals and searching over the menu for an extended period of time, created a negative experience for each car in line behind them. In response, McDonald's optimized the drive-thru across three components: design, information, and people. Design focused on the improvements to the drive-thru, including better speaker quality and higher resolution, digital menu boards. Information centered around what was on the menu board. In order to decrease order times, McDonald's removed about 40% of the drive-thru menu board. In

its third aspect, people, the fast food chain attempted to reduce the negative experiences for those in line by creating a second drive-thru line with a designated order taker for each line, a third drive-thru window, and two production lines.

Another example showing McDonald's commitment to being a customer-driven corporation is its introduction of all day breakfast, which was the highest priority for customers across the United States. Being the corporation with by far the largest share of the fast food market, McDonald's will continue to use its growing data sets to provide the best experience and food to its customers (van Rijmenam, 2016).

Questions

1. Explain McDonald's mission and responsibilities.
2. What limitation did McDonald's face in gaining data that was meaningful to decision-making?
3. Describe trend analytics.
4. Is McDonald's product oriented or customer oriented?
5. Why is the ability to identify patterns and relationships critical to McDonald's operations?

Sources: Compiled from Van Rijmenam (2016) and McDonald's (2017).

Case 3.3

Video Case: Verizon Improves Its Customer Experience with Data Driven Decision-Making

Verizon leverages Teradata's data analytics platform to shift its operations from qualitative decision-making to evidence-based and data-driven decision-making to improve the customer experience. Visit www.Teradata.com and search for the video "Verizon: Using Advanced Analytics to Deliver on Their Digital Promise to Help Customers Innovate Their Lifestyle."

1. How does Verizon use Teradata to make decisions?
2. How do the three sectors of Verizon work together to create value for the customer?
3. How does Verizon use data analytics to "penetrate the market"?
4. What impact does customer behavior data have on Verizon's marketing strategy?

"IT Matters" Discussion Board

Research the concept of Big Data. Find at least one company that maintains a "Big Data" database.

1. What is Big Data? Give at least three examples of organizations that have Big Data sets. What are the applications?
2. Discuss one of the organizations that you identified that uses Big Data and briefly describe the use of Big Data in the organization.
3. Discuss as much as you can find about the amount of data they have and how they process it.
4. What did you learn or what lessons did you take away from this research?

Provide at least two hyperlinked references to back up your findings (one for the organization you chose to discuss and one for the concept of big data in general). Post your findings and respond to at least two comments posted by your fellow students.

IT Toolbox

Framework for Generally Accepted Recordkeeping Principles

The Framework for generally accepted recordkeeping principles is a useful tool for managing business records to ensure that they support an enterprise's current and future regulatory, legal, risk mitigation, environmental, and operational requirements.

The framework consists of eight principles or best practices, which also support data governance. These principles were created by ARMA International and legal and IT professionals.

1. **Principle of accountability** Assign a senior executive to oversee a recordkeeping program; adopt policies and procedures to guide personnel; and ensure program audit ability.
2. **Principle of transparency** Document processes and activities of an organization's recordkeeping program in an understandable manner and available to all personnel and appropriate parties.
3. **Principle of integrity** Ensure recordkeeping program is able to reasonably guarantee the authenticity and reliability of records and data.

4. **Principle of protection** Construct the recordkeeping program to ensure a reasonable level of protection to records and information that are private, confidential, privileged, secret, or essential to business continuity.
5. **Principle of compliance** Ensure recordkeeping program complies with applicable laws, authorities, and the organization's policies.
6. **Principle of availability** Maintain records in a manner that ensures timely, efficient, and accurate retrieval of needed information.
7. **Principle of retention** Maintain records and data for an appropriate time based on legal, regulatory, fiscal, operational, and historical requirements.
8. **Principle of disposition** Securely disposed of records when they are no longer required to be maintained by laws or organizational policies.

References

- Bing, C. "Data Mining Software Used by Spy Agencies just got more Powerful." *FedScoop*, June 21, 2016.
- Burns, E. "Coca-Cola Overcomes Challenges to Seize BI Opportunities." *TechTarget.com*. August 2013.
- Burns, E. "Text Analysis Tool Helps Lenovo Zero in on the Customer." *BusinessAnalytics*, April 8, 2016.
- BusinessIntelligence.com. "Coca-Cola's Juicy Approach to Big Data." July 29, 2013b. <http://businessintelligence.com/bi-insights/coca-colas-juicy-approach-to-big-data>
- Cattell, J., S. Chilukuri, and M. Levy. "How Big Data Can Revolutionize Pharmaceutical R&D." 2016. <http://www.mckinsey.com/industries/pharmaceuticals-and-medical-products/our-insights/how-big-data-can-revolutionize-pharmaceutical-r-and-d>
- CNNMoney, The Coca-Cola Co (NYSE:KO) 2014.
- Columbus, L. "Ten Ways Big Data Is Revolutionizing Marketing and Sales." *Forbes*, May 9, 2016.
- Eisenhauer, T. "The Undeniable Benefits of Having a Well-Designed Document Management System." *Axero Solutions*, August 5, 2015.
- FirstWind website www.firstwind.com, 2017.
- Forbes*. "Betting on Big Data." 2015.
- Hammond, T. "Top IT Job Skills for 2014: Big Data, Mobile, Cloud, Security." *TechRepublic.com*, January 31, 2014.
- Harle, P., A. Havas, and H. Samandari. "The Future of Bank Risk Management." McKinsey & Company, July 2016.
- Harvard Business School*. "How Coca-Cola Controls Nature's Oranges." November 22, 2015.
- HealthCanal*. "Where Do You Start When Developing a New Medicine?" March 27, 2014.
- IDC. "Explosive Internet of Things Spending to Reach \$1.7 Trillion in 2020, According to IDC." June 02, 2015.
- King, L. "How Business Intelligence Helps Small Businesses Make Better Decisions." *Huffington Post*, July 28, 2016.
- Kitamura, M. "Big Data Partnerships Tackle Drug Development Failures." *Bloomberg News*, March 26, 2014.
- Kramer, S. "The High Costs of Dirty Data." *Digitalist*, May 1, 2015.
- Ladley, J. "Business Alignment Techniques for Successful and Sustainable Analytics." *CIO*, May 13, 2016.
- Liyakasa, K. "Coke Opens Data-Driven Happiness, Builds Out Marketing Decision Engine." *Ad Exchanger*, October 14, 2015.
- McDonald's website 2017. <https://www.mcdonalds.com/us/en-us/about-us/our-history.html>
- NIH (National Institute of Health). "Accelerating Medicines Partnership." February 2014. <http://www.nih.gov/science/amp/index.htm>
- Oliphant, T. "How to Make Big Data Insights Work for You." *Business Intelligence*, February 24, 2016.
- Ovalsrud, T. "Big Data and Analytics Spending to Hit \$1.87 billion" *CIO*, May 24, 2016.
- Ransbothom, S. "Coca-Cola's Unique Challenge: Turning 250 Datasets into One." *MIT Sloan Management Review*, May 27, 2015.
- RingLead, Inc. "The True Cost of Bad (And Clean) Data." July 17, 2015.
- syntheses.net 2017.
- The Data Warehousing Institute (TDWI). tdwi.org/portals/business-intelligence.asp. 2014
- U.S. Department of Energy. "Wind Vision: A New Era for Wind Power in the United States." <http://energy.gov/eere/wind/maps/wind-vision>, March 12, 2015.
- Van Rijmenam, M. "From Big Data to Big Mac; how McDonalds leverages Big Data." *DataFloq.com*, August 15, 2016.
- Van Rijmenam, M. "How Coca-Cola Takes a Refreshing Approach on Big Data." *DataFloq*, July 18, 2016.
- Wise, L. "Evaluating Business Intelligence in the Cloud." *CIO*, March 9, 2016.

Introduction: Data-Analytic Thinking

*Dream no small dreams for they have no power to
move the hearts of men.*

—Johann Wolfgang von Goethe

The past fifteen years have seen extensive investments in business infrastructure, which have improved the ability to collect data throughout the enterprise. Virtually every aspect of business is now open to data collection and often even instrumented for data collection: operations, manufacturing, supply-chain management, customer behavior, marketing campaign performance, workflow procedures, and so on. At the same time, information is now widely available on external events such as market trends, industry news, and competitors' movements. This broad availability of data has led to increasing interest in methods for extracting useful information and knowledge from data—the realm of data science.

The Ubiquity of Data Opportunities

With vast amounts of data now available, companies in almost every industry are focused on exploiting data for competitive advantage. In the past, firms could employ teams of statisticians, modelers, and analysts to explore datasets manually, but the volume and variety of data have far outstripped the capacity of manual analysis. At the same time, computers have become far more powerful, networking has become ubiquitous, and algorithms have been developed that can connect datasets to enable broader and deeper analyses than previously possible. The convergence of these phenomena has given rise to the increasingly widespread business application of data science principles and data-mining techniques.

Probably the widest applications of data-mining techniques are in marketing for tasks such as targeted marketing, online advertising, and recommendations for cross-selling.

Data mining is used for general customer relationship management to analyze customer behavior in order to manage attrition and maximize expected customer value. The finance industry uses data mining for credit scoring and trading, and in operations via fraud detection and workforce management. Major retailers from Walmart to Amazon apply data mining throughout their businesses, from marketing to supply-chain management. Many firms have differentiated themselves strategically with data science, sometimes to the point of evolving into data mining companies.

The primary goals of this book are to help you view business problems from a data perspective and understand principles of extracting useful knowledge from data. There is a fundamental structure to data-analytic thinking, and basic principles that should be understood. There are also particular areas where intuition, creativity, common sense, and domain knowledge must be brought to bear. A data perspective will provide you with structure and principles, and this will give you a framework to systematically analyze such problems. As you get better at data-analytic thinking you will develop intuition as to how and where to apply creativity and domain knowledge.

Throughout the first two chapters of this book, we will discuss in detail various topics and techniques related to data science and data mining. The terms “data science” and “data mining” often are used interchangeably, and the former has taken a life of its own as various individuals and organizations try to capitalize on the current hype surrounding it. At a high level, *data science* is a set of fundamental principles that guide the extraction of knowledge from data. Data mining is the extraction of knowledge from data, via technologies that incorporate these principles. As a term, “data science” often is applied more broadly than the traditional use of “data mining,” but data mining techniques provide some of the clearest illustrations of the principles of data science.



It is important to understand data science even if you never intend to apply it yourself. Data-analytic thinking enables you to evaluate proposals for data mining projects. For example, if an employee, a consultant, or a potential investment target proposes to improve a particular business application by extracting knowledge from data, you should be able to assess the proposal systematically and decide whether it is sound or flawed. This does not mean that you will be able to tell whether it will actually succeed—for data mining projects, that often requires trying—but you should be able to spot obvious flaws, unrealistic assumptions, and missing pieces.

Throughout the book we will describe a number of fundamental data science principles, and will illustrate each with at least one data mining technique that embodies the principle. For each principle there are usually many specific techniques that embody it, so in this book we have chosen to emphasize the basic principles in preference to specific techniques. That said, we will not make a big deal about the difference between data

science and data mining, except where it will have a substantial effect on understanding the actual concepts.

Let's examine two brief case studies of analyzing data to extract predictive patterns.

Example: Hurricane Frances

Consider an example from a *New York Times* story from 2004:

Hurricane Frances was on its way, barreling across the Caribbean, threatening a direct hit on Florida's Atlantic coast. Residents made for higher ground, but far away, in Bentonville, Ark., executives at Wal-Mart Stores decided that the situation offered a great opportunity for one of their newest data-driven weapons ... predictive technology.

A week ahead of the storm's landfall, Linda M. Dillman, Wal-Mart's chief information officer, pressed her staff to come up with forecasts based on what had happened when Hurricane Charley struck several weeks earlier. Backed by the trillions of bytes' worth of shopper history that is stored in Wal-Mart's data warehouse, she felt that the company could 'start predicting what's going to happen, instead of waiting for it to happen,' as she put it. (Hays, 2004)

Consider *why* data-driven prediction might be useful in this scenario. It might be useful to predict that people in the path of the hurricane would buy more bottled water. Maybe, but this point seems a bit obvious, and why would we need data science to discover it? It might be useful to project the *amount of increase* in sales due to the hurricane, to ensure that local Wal-Marts are properly stocked. Perhaps mining the data could reveal that a particular DVD sold out in the hurricane's path—but maybe it sold out that week at Wal-Marts across the country, not just where the hurricane landing was imminent. The prediction could be somewhat useful, but is probably more general than Ms. Dillman was intending.

It would be more valuable to discover patterns due to the hurricane that were not obvious. To do this, analysts might examine the huge volume of Wal-Mart data from prior, similar situations (such as Hurricane Charley) to identify *unusual* local demand for products. From such patterns, the company might be able to anticipate unusual demand for products and rush stock to the stores ahead of the hurricane's landfall.

Indeed, that is what happened. *The New York Times* (Hays, 2004) reported that: "... the experts mined the data and found that the stores would indeed need certain products—and not just the usual flashlights. 'We didn't know in the past that strawberry Pop-Tarts increase in sales, like seven times their normal sales rate, ahead of a hurricane,' Ms. Dillman said in a recent interview. 'And the pre-hurricane top-selling item was beer.'"¹

1. Of course! What goes better with strawberry Pop-Tarts than a nice cold beer?

Example: Predicting Customer Churn

How are such data analyses performed? Consider a second, more typical business scenario and how it might be treated from a data perspective. This problem will serve as a running example that will illuminate many of the issues raised in this book and provide a common frame of reference.

Assume you just landed a great analytical job with MegaTelCo, one of the largest telecommunication firms in the United States. They are having a major problem with customer retention in their wireless business. In the mid-Atlantic region, 20% of cell phone customers leave when their contracts expire, and it is getting increasingly difficult to acquire new customers. Since the cell phone market is now saturated, the huge growth in the wireless market has tapered off. Communications companies are now engaged in battles to attract each other's customers while retaining their own. Customers switching from one company to another is called *churn*, and it is expensive all around: one company must spend on incentives to attract a customer while another company loses revenue when the customer departs.

You have been called in to help understand the problem and to devise a solution. Attracting new customers is much more expensive than retaining existing ones, so a good deal of marketing budget is allocated to prevent churn. Marketing has already designed a special retention offer. Your task is to devise a precise, step-by-step plan for how the data science team should use MegaTelCo's vast data resources to decide which customers should be offered the special retention deal prior to the expiration of their contracts.

Think carefully about what data you might use and how they would be used. Specifically, how should MegaTelCo choose a set of customers to receive their offer in order to best reduce churn for a particular incentive budget? Answering this question is much more complicated than it may seem initially. We will return to this problem repeatedly through the book, adding sophistication to our solution as we develop an understanding of the fundamental data science concepts.



In reality, customer retention has been a major use of data mining technologies—especially in telecommunications and finance businesses. These more generally were some of the earliest and widest adopters of data mining technologies, for reasons discussed later.

Data Science, Engineering, and Data-Driven Decision Making

Data science involves principles, processes, and techniques for understanding phenomena via the (automated) analysis of data. In this book, we will view the ultimate goal

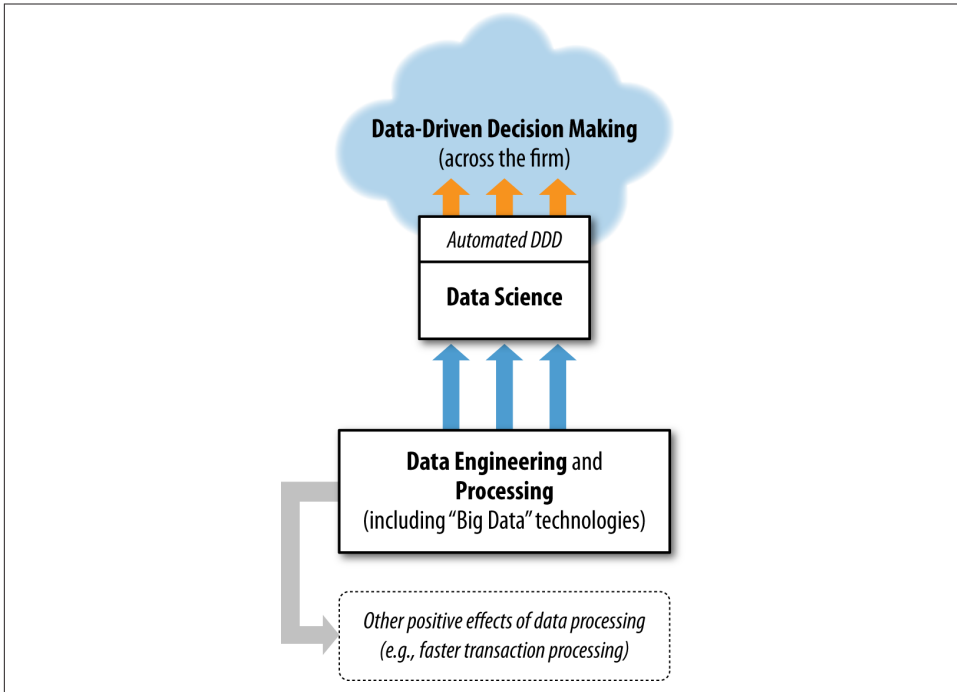


Figure 1-1. Data science in the context of various data-related processes in the organization.

of data science as improving decision making, as this generally is of direct interest to business.

Figure 1-1 places data science in the context of various other closely related and data-related processes in the organization. It distinguishes data science from other aspects of data processing that are gaining increasing attention in business. Let's start at the top.

Data-driven decision-making (DDD) refers to the practice of basing decisions on the analysis of data, rather than purely on intuition. For example, a marketer could select advertisements based purely on her long experience in the field and her eye for what will work. Or, she could base her selection on the analysis of data regarding how consumers react to different ads. She could also use a combination of these approaches. DDD is not an all-or-nothing practice, and different firms engage in DDD to greater or lesser degrees.

The benefits of data-driven decision-making have been demonstrated conclusively. Economist Erik Brynjolfsson and his colleagues from MIT and Penn's Wharton School conducted a study of how DDD affects firm performance (Brynjolfsson, Hitt, & Kim, 2011). They developed a measure of DDD that rates firms as to how strongly they use

data to make decisions across the company. They show that statistically, the more data-driven a firm is, the more productive it is—even controlling for a wide range of possible confounding factors. And the differences are not small. One standard deviation higher on the DDD scale is associated with a 4%–6% increase in productivity. DDD also is correlated with higher return on assets, return on equity, asset utilization, and market value, and the relationship seems to be causal.

The sort of decisions we will be interested in in this book mainly fall into two types: (1) decisions for which “discoveries” need to be made within data, and (2) decisions that repeat, especially at massive scale, and so decision-making can benefit from even small increases in decision-making accuracy based on data analysis. The Walmart example above illustrates a type 1 problem: Linda Dillman would like to discover knowledge that will help Walmart prepare for Hurricane Frances’s imminent arrival.

In 2012, Walmart’s competitor Target was in the news for a data-driven decision-making case of its own, also a type 1 problem (Duhigg, 2012). Like most retailers, Target cares about consumers’ shopping habits, what drives them, and what can influence them. Consumers tend to have inertia in their habits and getting them to change is very difficult. Decision makers at Target knew, however, that the arrival of a new baby in a family is one point where people do change their shopping habits significantly. In the Target analyst’s words, “As soon as we get them buying diapers from us, they’re going to start buying everything else too.” Most retailers know this and so they compete with each other trying to sell baby-related products to new parents. Since most birth records are public, retailers obtain information on births and send out special offers to the new parents.

However, Target wanted to get a jump on their competition. They were interested in whether they could *predict* that people *are expecting* a baby. If they could, they would gain an advantage by making offers before their competitors. Using techniques of data science, Target analyzed historical data on customers who *later* were revealed to have been pregnant, and were able to extract information that could predict which consumers were pregnant. For example, pregnant mothers often change their diets, their wardrobes, their vitamin regimens, and so on. These indicators could be extracted from historical data, assembled into predictive models, and then deployed in marketing campaigns. We will discuss predictive models in much detail as we go through the book. For the time being, it is sufficient to understand that a predictive model abstracts away most of the complexity of the world, focusing in on a particular set of indicators that correlate in some way with a quantity of interest (who will churn, or who will purchase, who is pregnant, etc.). Importantly, in both the Walmart and the Target examples, the

data analysis was not testing a simple hypothesis. Instead, the data were explored with the hope that something useful would be discovered.²

Our churn example illustrates a type 2 DDD problem. MegaTelCo has hundreds of millions of customers, each a candidate for defection. Tens of millions of customers have contracts expiring each month, so each one of them has an increased likelihood of defection in the near future. If we can improve our ability to estimate, for a given customer, how profitable it would be for us to focus on her, we can potentially reap large benefits by applying this ability to the millions of customers in the population. This same logic applies to many of the areas where we have seen the most intense application of data science and data mining: direct marketing, online advertising, credit scoring, financial trading, help-desk management, fraud detection, search ranking, product recommendation, and so on.

The diagram in [Figure 1-1](#) shows data science supporting data-driven decision-making, but also overlapping with data-driven decision-making. This highlights the often overlooked fact that, increasingly, business decisions are being made *automatically* by computer systems. Different industries have adopted automatic decision-making at different rates. The finance and telecommunications industries were early adopters, largely because of their precocious development of data networks and implementation of massive-scale computing, which allowed the aggregation and modeling of data at a large scale, as well as the application of the resultant models to decision-making.

In the 1990s, automated decision-making changed the banking and consumer credit industries dramatically. In the 1990s, banks and telecommunications companies also implemented massive-scale systems for managing data-driven fraud control decisions. As retail systems were increasingly computerized, merchandising decisions were automated. Famous examples include Harrah's casinos' reward programs and the automated recommendations of Amazon and Netflix. Currently we are seeing a revolution in advertising, due in large part to a huge increase in the amount of time consumers are spending online, and the ability online to make (literally) split-second advertising decisions.

Data Processing and “Big Data”

It is important to digress here to address another point. There is a lot to data processing that is not data science—despite the impression one might get from the media. Data engineering and processing are critical to support data science, but they are more general. For example, these days many data processing skills, systems, and technologies often are mistakenly cast as data science. To understand data science and data-driven

2. Target was successful enough that this case raised ethical questions on the deployment of such techniques. Concerns of ethics and privacy are interesting and very important, but we leave their discussion for another time and place.

businesses it is important to understand the differences. Data science needs access to data and it often benefits from sophisticated data engineering that data processing technologies may facilitate, but these technologies are not data science technologies per se. They support data science, as shown in [Figure 1-1](#), but they are useful for much more. Data processing technologies are very important for many data-oriented business tasks that do not involve extracting knowledge or data-driven decision-making, such as efficient transaction processing, modern web system processing, and online advertising campaign management.

“Big data” technologies (such as Hadoop, HBase, and MongoDB) have received considerable media attention recently. *Big data* essentially means datasets that are too large for traditional data processing systems, and therefore require new processing technologies. As with the traditional technologies, big data technologies are used for many tasks, including data engineering. Occasionally, big data technologies are actually used for *implementing* data mining techniques. However, much more often the well-known big data technologies are used for data processing *in support of* the data mining techniques and other data science activities, as represented in [Figure 1-1](#).

Previously, we discussed Brynjolfsson’s study demonstrating the benefits of data-driven decision-making. A separate study, conducted by economist Prasanna Tambe of NYU’s Stern School, examined the extent to which *big data* technologies seem to help firms (Tambe, 2012). He finds that, after controlling for various possible confounding factors, using big data technologies is associated with significant additional productivity growth. Specifically, one standard deviation higher utilization of big data technologies is associated with 1%–3% higher productivity than the average firm; one standard deviation lower in terms of big data utilization is associated with 1%–3% lower productivity. This leads to potentially very large productivity differences between the firms at the extremes.

From Big Data 1.0 to Big Data 2.0

One way to think about the state of big data technologies is to draw an analogy with the business adoption of Internet technologies. In Web 1.0, businesses busied themselves with getting the basic internet technologies in place, so that they could establish a web presence, build electronic commerce capability, and improve the efficiency of their operations. We can think of ourselves as being in the era of Big Data 1.0. Firms are busying themselves with building the capabilities to process large data, largely in support of their current operations—for example, to improve efficiency.

Once firms had incorporated Web 1.0 technologies thoroughly (and in the process had driven down prices of the underlying technology) they started to look further. They began to ask what the Web could do for them, and how it could improve things they’d always done—and we entered the era of Web 2.0, where new systems and companies began taking advantage of the interactive nature of the Web. The changes brought on by this shift in thinking are pervasive; the most obvious are the incorporation of social-

networking components, and the rise of the “voice” of the individual consumer (and citizen).

We should expect a Big Data 2.0 phase to follow Big Data 1.0. Once firms have become capable of processing massive data in a flexible fashion, they should begin asking: “*What can I now do that I couldn’t do before, or do better than I could do before?*” This is likely to be the golden era of data science. The principles and techniques we introduce in this book will be applied far more broadly and deeply than they are today.



It is important to note that in the Web 1.0 era some precocious companies began applying Web 2.0 ideas far ahead of the mainstream. Amazon is a prime example, incorporating the consumer’s “voice” early on, in the rating of products, in product reviews (and deeper, in the rating of product reviews). Similarly, we see some companies already applying Big Data 2.0. Amazon again is a company at the forefront, providing data-driven recommendations from massive data. There are other examples as well. Online advertisers must process extremely large volumes of data (billions of ad impressions per day is not unusual) and maintain a very high throughput (real-time bidding systems make decisions in tens of milliseconds). We should look to these and similar industries for hints at advances in big data and data science that subsequently will be adopted by other industries.

Data and Data Science Capability as a Strategic Asset

The prior sections suggest one of the fundamental principles of data science: *data, and the capability to extract useful knowledge from data, should be regarded as key strategic assets*. Too many businesses regard data analytics as pertaining mainly to realizing value from some existing data, and often without careful regard to whether the business has the appropriate analytical talent. Viewing these as assets allows us to think explicitly about the extent to which one should invest in them. Often, we don’t have exactly the right data to best make decisions and/or the right talent to best support making decisions from the data. Further, thinking of these as assets should lead us to the realization that they are *complementary*. The best data science team can yield little value without the appropriate data; the right data often cannot substantially improve decisions without suitable data science talent. As with all assets, it is often necessary to make investments. Building a top-notch data science team is a nontrivial undertaking, but can make a huge difference for decision-making. We will discuss strategic considerations involving data science in detail in **Chapter 13**. Our next case study will introduce the idea that thinking explicitly about how to invest in data assets very often pays off handsomely.

The classic story of little Signet Bank from the 1990s provides a case in point. Previously, in the 1980s, data science had transformed the business of consumer credit. Modeling

the probability of default had changed the industry from personal assessment of the likelihood of default to strategies of massive scale and market share, which brought along concomitant economies of scale. It may seem strange now, but at the time, credit cards essentially had uniform pricing, for two reasons: (1) the companies did not have adequate information systems to deal with differential pricing at massive scale, and (2) bank management believed customers would not stand for price discrimination. Around 1990, two strategic visionaries (Richard Fairbanks and Nigel Morris) realized that information technology was powerful enough that they could do more sophisticated predictive modeling—using the sort of techniques that we discuss throughout this book—and offer different terms (nowadays: pricing, credit limits, low-initial-rate balance transfers, cash back, loyalty points, and so on). These two men had no success persuading the big banks to take them on as consultants and let them try. Finally, after running out of big banks, they succeeded in garnering the interest of a small regional Virginia bank: Signet Bank. Signet Bank’s management was convinced that modeling profitability, not just default probability, was the right strategy. They knew that a small proportion of customers actually account for *more than* 100% of a bank’s profit from credit card operations (because the rest are break-even or money-losing). If they could model profitability, they could make better offers to the best customers and “skim the cream” of the big banks’ clientele.

But Signet Bank had one really big problem in implementing this strategy. They did not have the appropriate data to model profitability with the goal of offering different terms to different customers. No one did. Since banks were offering credit with a specific set of terms and a specific default model, they had the data to model profitability (1) for the terms they actually have offered in the past, and (2) for the sort of customer who was actually offered credit (that is, those who were deemed worthy of credit by the existing model).

What could Signet Bank do? They brought into play a fundamental strategy of data science: acquire the necessary data at a cost. Once we view data as a business asset, we should think about whether and how much we are willing to invest. In Signet’s case, data could be generated on the profitability of customers given different credit terms by conducting experiments. Different terms were offered at random to different customers. This may seem foolish outside the context of data-analytic thinking: you’re likely to lose money! This is true. In this case, losses are the cost of data acquisition. The data-analytic thinker needs to consider whether she expects the data to have sufficient value to justify the investment.

So what happened with Signet Bank? As you might expect, when Signet began randomly offering terms to customers for data acquisition, the number of bad accounts soared. Signet went from an industry-leading “charge-off” rate (2.9% of balances went unpaid) to almost 6% charge-offs. Losses continued for a few years while the data scientists worked to build predictive models from the data, evaluate them, and deploy them to improve profit. Because the firm viewed these losses as investments in data, they per-

sisted despite complaints from stakeholders. Eventually, Signet's credit card operation turned around and became so profitable that it was spun off to separate it from the bank's other operations, which now were overshadowing the consumer credit success.

Fairbanks and Morris became Chairman and CEO and President and COO, and proceeded to apply data science principles throughout the business—not just customer acquisition but retention as well. When a customer calls looking for a better offer, data-driven models calculate the potential profitability of various possible actions (different offers, including sticking with the status quo), and the customer service representative's computer presents the best offers to make.

You may not have heard of little Signet Bank, but if you're reading this book you've probably heard of the spin-off: Capital One. Fairbanks and Morris's new company grew to be one of the largest credit card issuers in the industry with one of the lowest charge-off rates. In 2000, the bank was reported to be carrying out 45,000 of these “scientific tests” as they called them.³

Studies giving clear quantitative demonstrations of the value of a data asset are hard to find, primarily because firms are hesitant to divulge results of strategic value. One exception is a study by Martens and Provost (2011) assessing whether data on the specific transactions of a bank's consumers can improve models for deciding what product offers to make. The bank built models from data to decide whom to target with offers for different products. The investigation examined a number of different types of data and their effects on predictive performance. Sociodemographic data provide a substantial ability to model the sort of consumers that are more likely to purchase one product or another. However, sociodemographic data only go so far; after a certain volume of data, no additional advantage is conferred. In contrast, detailed data on customers' individual (anonymized) transactions improve performance substantially over just using sociodemographic data. The relationship is clear and striking and—significantly, for the point here—the predictive performance continues to improve as more data are used, increasing throughout the range investigated by Martens and Provost with no sign of abating. This has an important implication: banks with bigger data assets may have an important strategic advantage over their smaller competitors. If these trends generalize, and the banks are able to apply sophisticated analytics, banks with bigger data assets should be better able to identify the best customers for individual products. The net result will be either increased adoption of the bank's products, decreased cost of customer acquisition, or both.

The idea of data as a strategic asset is certainly not limited to Capital One, nor even to the banking industry. Amazon was able to gather data early on online customers, which has created significant switching costs: consumers find value in the rankings and recommendations that Amazon provides. Amazon therefore can retain customers more

3. You can read more about Capital One's story (Clemons & Thatcher, 1998; McNamee 2001).

easily, and can even charge a premium (Brynjolfsson & Smith, 2000). Harrah's casinos famously invested in gathering and mining data on gamblers, and moved itself from a small player in the casino business in the mid-1990s to the acquisition of Caesar's Entertainment in 2005 to become the world's largest gambling company. The huge valuation of Facebook has been credited to its vast and unique data assets (Sengupta, 2012), including both information about individuals and their likes, as well as information about the structure of the social network. Information about network structure has been shown to be important to predicting and has been shown to be remarkably helpful in building models of who will buy certain products (Hill, Provost, & Volinsky, 2006). It is clear that Facebook has a remarkable data asset; whether they have the right data science strategies to take full advantage of it is an open question.

In the book we will discuss in more detail many of the fundamental concepts behind these success stories, in exploring the principles of data mining and data-analytic thinking.

Data-Analytic Thinking

Analyzing case studies such as the churn problem improves our ability to approach problems “data-analytically.” Promoting such a perspective is a primary goal of this book. When faced with a business problem, you should be able to assess whether and how data can improve performance. We will discuss a set of fundamental concepts and principles that facilitate careful thinking. We will develop frameworks to structure the analysis so that it can be done systematically.

As mentioned above, it is important to understand data science even if you never intend to do it yourself, because data analysis is now so critical to business strategy. Businesses increasingly are driven by data analytics, so there is great professional advantage in being able to interact competently with and within such businesses. Understanding the fundamental concepts, and having frameworks for organizing data-analytic thinking not only will allow one to interact competently, but will help to envision opportunities for improving data-driven decision-making, or to see data-oriented competitive threats.

Firms in many traditional industries are exploiting new and existing data resources for competitive advantage. They employ data science teams to bring advanced technologies to bear to increase revenue and to decrease costs. In addition, many new companies are being developed with data mining as a key strategic component. Facebook and Twitter, along with many other “Digital 100” companies (*Business Insider*, 2012), have high valuations due primarily to data assets they are committed to capturing or creating.⁴ Increasingly, managers need to oversee analytics teams and analysis projects, marketers

4. Of course, this is not a new phenomenon. Amazon and Google are well-established companies that get tremendous value from their data assets.

have to organize and understand data-driven campaigns, venture capitalists must be able to invest wisely in businesses with substantial data assets, and business strategists must be able to devise plans that exploit data.

As a few examples, if a consultant presents a proposal to mine a data asset to improve your business, you should be able to assess whether the proposal makes sense. If a competitor announces a new data partnership, you should recognize when it may put you at a strategic disadvantage. Or, let's say you take a position with a venture firm and your first project is to assess the potential for investing in an advertising company. The founders present a convincing argument that they will realize significant value from a unique body of data they will collect, and on that basis are arguing for a substantially higher valuation. Is this reasonable? With an understanding of the fundamentals of data science you should be able to devise a few probing questions to determine whether their valuation arguments are plausible.

On a scale less grand, but probably more common, data analytics projects reach into all business units. Employees throughout these units must interact with the data science team. If these employees do not have a fundamental grounding in the principles of data-analytic thinking, they will not really understand what is happening in the business. This lack of understanding is much more damaging in data science projects than in other technical projects, because the data science is supporting improved decision-making. As we will describe in the next chapter, this requires a close interaction between the data scientists and the business people responsible for the decision-making. Firms where the business people do not understand what the data scientists are doing are at a substantial disadvantage, because they waste time and effort or, worse, because they ultimately make wrong decisions.



The need for managers with data-analytic skills

The consulting firm McKinsey and Company estimates that “there will be a shortage of talent necessary for organizations to take advantage of big data. By 2018, the United States alone could face a shortage of 140,000 to 190,000 people with deep analytical skills as well as 1.5 million managers and analysts with the know-how to use the analysis of big data to make effective decisions.” (Manyika, 2011). Why 10 times as many managers and analysts than those with deep analytical skills? Surely data scientists aren't so difficult to manage that they need 10 managers! The reason is that a business can get leverage from a data science team for making better decisions in multiple areas of the business. However, as McKinsey is pointing out, the managers in those areas need to understand the fundamentals of data science to effectively get that leverage.

This Book

This book concentrates on the fundamentals of data science and data mining. These are a set of principles, concepts, and techniques that structure thinking and analysis. They allow us to understand data science processes and methods surprisingly deeply, without needing to focus in depth on the large number of specific data mining algorithms.

There are many good books covering data mining algorithms and techniques, from practical guides to mathematical and statistical treatments. This book instead focuses on the fundamental concepts and how they help us to think about problems where data mining may be brought to bear. That doesn't mean that we will ignore the data mining techniques; many algorithms are exactly the embodiment of the basic concepts. But with only a few exceptions we will not concentrate on the deep technical details of how the techniques actually work; we will try to provide just enough detail so that you will understand what the techniques do, and how they are based on the fundamental principles.

Data Mining and Data Science, Revisited

This book devotes a good deal of attention to the extraction of useful (nontrivial, hopefully actionable) patterns or models from large bodies of data (Fayyad, Piatetsky-Shapiro, & Smyth, 1996), and to the fundamental data science principles underlying such data mining. In our churn-prediction example, we would like to *take the data* on prior churn and *extract patterns*, for example patterns of behavior, *that are useful*—that can help us to predict those customers who are more likely to leave in the future, or that can help us to design better services.

The fundamental concepts of data science are drawn from many fields that study data analytics. We introduce these concepts throughout the book, but let's briefly discuss a few now to get the basic flavor. We will elaborate on all of these and more in later chapters.

Fundamental concept: *Extracting useful knowledge from data to solve business problems can be treated systematically by following a process with reasonably well-defined stages.* The Cross Industry Standard Process for Data Mining, abbreviated CRISP-DM (CRISP-DM Project, 2000), is one codification of this process. Keeping such a process in mind provides a framework to structure our thinking about data analytics problems. For example, in actual practice one repeatedly sees analytical “solutions” that are not based on careful analysis of the problem or are not carefully evaluated. Structured thinking about analytics emphasizes these often under-appreciated aspects of supporting decision-making with data. Such structured thinking also contrasts critical points where human creativity is necessary versus points where high-powered analytical tools can be brought to bear.

Fundamental concept: *From a large mass of data, information technology can be used to find informative descriptive attributes of entities of interest.* In our churn example, a customer would be an entity of interest, and each customer might be described by a large number of attributes, such as usage, customer service history, and many other factors. Which of these actually gives us information on the customer’s likelihood of leaving the company when her contract expires? How much information? Sometimes this process is referred to roughly as finding variables that “correlate” with churn (we will discuss this notion precisely). A business analyst may be able to hypothesize some and test them, and there are tools to help facilitate this experimentation (see “**Other Analytics Techniques and Technologies**” on page 35). Alternatively, the analyst could apply information technology to automatically discover informative attributes—essentially doing large-scale automated experimentation. Further, as we will see, this concept can be applied recursively to build models to predict churn based on multiple attributes.

Fundamental concept: *If you look too hard at a set of data, you will find something—but it might not generalize beyond the data you’re looking at.* This is referred to as *overfitting* a dataset. Data mining techniques can be very powerful, and the need to detect and avoid overfitting is one of the most important concepts to grasp when applying data mining to real problems. The concept of overfitting and its avoidance permeates data science processes, algorithms, and evaluation methods.

Fundamental concept: *Formulating data mining solutions and evaluating the results involves thinking carefully about the context in which they will be used.* If our goal is the extraction of potentially *useful* knowledge, how can we formulate what is useful? It depends critically on the application in question. For our churn-management example, how exactly are we going to use the patterns extracted from historical data? Should the value of the customer be taken into account in addition to the likelihood of leaving? More generally, does the pattern lead to better decisions than some reasonable alternative? How well would one have done by chance? How well would one do with a smart “default” alternative?

These are just four of the fundamental concepts of data science that we will explore. By the end of the book, we will have discussed a dozen such fundamental concepts in detail, and will have illustrated how they help us to structure data-analytic thinking and to understand data mining techniques and algorithms, as well as data science applications, quite generally.

Chemistry Is Not About Test Tubes: Data Science Versus the Work of the Data Scientist

Before proceeding, we should briefly revisit the engineering side of data science. At the time of this writing, discussions of data science commonly mention not just analytical skills and techniques for understanding data but popular tools used. Definitions of data

scientists (and advertisements for positions) specify not just areas of expertise but also specific programming languages and tools. It is common to see job advertisements mentioning data mining techniques (e.g., random forests, support vector machines), specific application areas (recommendation systems, ad placement optimization), alongside popular software tools for processing big data (Hadoop, MongoDB). There is often little distinction between the science and the technology for dealing with large datasets.

We must point out that data science, like computer science, is a young field. The particular concerns of data science are fairly new and general principles are just beginning to emerge. The state of data science may be likened to that of chemistry in the mid-19th century, when theories and general principles were being formulated and the field was largely experimental. Every good chemist had to be a competent lab technician. Similarly, it is hard to imagine a working data scientist who is not proficient with certain sorts of software tools.

Having said this, this book focuses on the science and not on the technology. You will not find instructions here on how best to run massive data mining jobs on Hadoop clusters, or even what Hadoop is or why you might want to learn about it.⁵ We focus here on the general principles of data science that have emerged. In 10 years' time the predominant technologies will likely have changed or advanced enough that a discussion here would be obsolete, while the general principles are the same as they were 20 years ago, and likely will change little over the coming decades.

Summary

This book is about the extraction of useful information and knowledge from large volumes of data, in order to improve business decision-making. As the massive collection of data has spread through just about every industry sector and business unit, so have the opportunities for mining the data. Underlying the extensive body of techniques for mining data is a much smaller set of fundamental concepts comprising *data science*. These concepts are general and encapsulate much of the essence of data mining and business analytics.

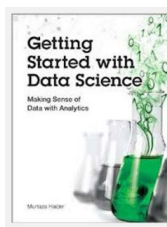
Success in today's data-oriented business environment requires being able to think about how these fundamental concepts apply to particular business problems—to think data-analytically. For example, in this chapter we discussed the principle that data should be thought of as a business asset, and once we are thinking in this direction we start to ask whether (and how much) we should invest in data. Thus, an understanding of these fundamental concepts is important not only for data scientists themselves, but for any-

5. OK: Hadoop is a widely used open source architecture for doing highly parallelizable computations. It is one of the current “big data” technologies for processing massive datasets that exceed the capacity of relational database systems. Hadoop is based on the MapReduce parallel processing framework introduced by Google.

one working with data scientists, employing data scientists, investing in data-heavy ventures, or directing the application of analytics in an organization.

Thinking data-analytically is aided by conceptual frameworks discussed throughout the book. For example, the automated extraction of patterns from data is a process with well-defined stages, which are the subject of the next chapter. Understanding the process and the stages helps to structure our data-analytic thinking, and to make it more systematic and therefore less prone to errors and omissions.

There is convincing evidence that data-driven decision-making and big data technologies substantially improve business performance. Data science supports data-driven decision-making—and sometimes conducts such decision-making automatically—and depends upon technologies for “big data” storage and engineering, but its principles are separate. The data science principles we discuss in this book also differ from, and are complementary to, other important technologies, such as statistical hypothesis testing and database querying (which have their own books and classes). The next chapter describes some of these differences in more detail.



Prescribed Reading: Chapter 1 Pg. 4

Data Science: The Sexiest Job in the 21st Century

In the data-driven world, data scientists have emerged as a hot commodity. The chase is on to find the best talent in data science. Already, experts estimate that millions of jobs in data science might remain vacant for the lack of readily available talent. The global search for skilled data scientists is not merely a search for statisticians or computer scientists. In fact, the firms are searching for well-rounded individuals who possess the subject matter expertise, some experience in software programming and analytics, and exceptional communication skills.

Our digital footprint has expanded rapidly over the past 10 years. The size of the digital universe was roughly 130 billion gigabytes in 1995. By 2020, this number will swell to 40 trillion gigabytes.⁶ Companies will compete for hundreds of thousands, if not millions, of new workers needed to navigate the digital world. No wonder the prestigious *Harvard Business Review* called data science “the sexiest job in the 21st century.”⁷

A report by the McKinsey Global Institute warns of huge talent shortages for data and analytics. “By 2018, the United States alone could face a shortage of 140,000 to 190,000 people with deep analytical skills as well as 1.5 million managers and analysts with the know-how to use the analysis of big data to make effective decisions.”⁸

Because the digital revolution has touched every aspect of our lives, the opportunity to benefit from learning about our behaviors is more so now than ever before. Given the right data, marketers can take sneak peeks into our habit formation. Research in neurology and psychology is revealing how habits and preferences are formed and retailers like Target are out to profit from it. However, the retailers can only do so if they have data scientists working for them. For this reason it is “like an arms race to hire statisticians nowadays,” said Andreas Weigend, the former chief scientist at Amazon.com.”⁹

There is still the need to convince the C-suite executives of the benefits of data and analytics. It appears that the senior management might be a step or two behind the middle management in being informed of the potential of analytics-driven planning. Professor Peter Fader, who manages at the Customer Analytics Initiative at Wharton, knows that executives reach the C-suite without having to interact with data. He believes that the real change will happen when executives are well-versed in data and analytics.¹⁰

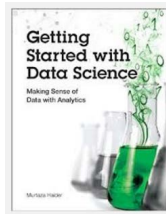
SAP, a leader in data and analytics, reported from a survey that 92% of the responding firms in its sample experienced a significant increase in their data holdings. At the same time, three-quarters identified the need for new data science skills in their firms. Accenture believes that the demand for data scientists may outstrip supply by 250,000 in 2015 alone. A similar survey of 150 executives by KPMG in 2014 found that 85% of the respondents did not know how to analyze data. “Most organizations are unable to connect the dots because they do not fully understand how data and analytics can transform their business,” Alwin Magimay, head of digital and analytics for KPMG UK, said in an interview in May 2015.¹¹

Bernard Marr writing for *Forbes* also raises concerns about the insufficient analytics talent. “There just aren’t enough people with the required skills to analyze and interpret this information—transforming it from raw numerical (or other) data into actionable insights—the ultimate aim of any Big Data-driven initiative,” he wrote.¹² Bernard quotes a survey by Gartner of business leaders of whom more than 50% reported the lack of in-house expertise in data science.

Bernard reported on Walmart, which turned to crowdsourcing for its analytics need. Walmart approached Kaggle to host a competition for analyzing its proprietary data. The retailer provided sales data from a shortlist of stores and asked the competitors to develop better forecasts of sales based on promotion schemes.

Given the shortage of data scientists, the employers are willing to pay top dollars for the talent. Michael Chui, a principal at McKinsey, knows this too well. Data science “has become relevant to every company ... There’s a war for this type of talent,” he said in an interview.¹³ Take Paul Minton, for example. He was making \$20,000 serving tables at a restaurant. He had majored in math at college. Mr. Minton took a three-month programming course that changed everything. He made over \$100,000 in 2014 as a data scientist for a web startup in San Francisco. “Six figures, right off the bat ... To me, it was astonishing,” said Mr. Minton.¹⁴

Could Mr. Minton be exceptionally fortunate, or are such high salaries the norm? Luck had little to do with it; the *New York Times* reported \$100,000 as the average base salary of a software engineer and \$112,000 for data scientists.



Prescribed Reading: Chapter 3 Pg. 52-53

The Final Deliverable

The ultimate purpose of analytics is to communicate findings to the concerned who might use these insights to formulate policy or strategy. Analytics summarize findings in tables and plots. The data scientist should then use the insights to build the narrative to communicate the findings. In academia, the final deliverable is in the form of essays and reports. Such deliverables are usually 1,000 to 7,000 words in length.

In consulting and business, the final deliverable takes on several forms. It can be a small document of fewer than 1,500 words illustrated with tables and plots, or it could be a comprehensive document comprising several hundred pages. Large consulting firms, such as McKinsey⁴ and Deloitte,⁵ routinely generate analytics-driven reports to communicate their findings and, in the process, establish their expertise in specific knowledge domains.

Let's review the *United States Economic Forecast*, a publication by the Deloitte University Press.⁶ This document serves as a good example for a deliverable that builds narrative from data and analytics. The 24-page report focuses on the state of the U.S. economy as observed in December 2014. The report opens with a "grabber" highlighting the fact that contrary to popular perception, the economic and job growth has been quite robust in the United States. The report is not merely a statement of facts. In fact, it is a carefully crafted report that cites Voltaire and follows a distinct theme. The report focuses on the "good news" about the U.S. economy. These include the increased investment in manufacturing equipment in the U.S. and the likelihood of higher consumer consumption resulting from lower oil prices.

The Deloitte report uses time series plots to illustrate trends in markets.⁷ The GDP growth chart shows how the economy contracted during the Great Recession and has rebounded since then. The graphic presents four likely scenarios for the future. Another plot shows the changes in consumer spending. The accompanying narrative focuses on income inequality in the U.S. and refers to Thomas Piketty's book on the same.⁸ The Deloitte report mentions many consumers did not experience an increase in their real incomes over the years, while they still maintained their level of spending. Other graphics focused on housing, business and government sectors, international trade, labor and financial markets, and prices. The appendix carries four tables documenting data for the four scenarios discussed in the report.

Deloitte's *United States Economic Forecast* serves the very purpose that its authors intended. The report uses data and analytics to generate the likely economic scenarios. It builds a powerful narrative in support of the thesis statement that the U.S. economy is doing much better than what most would like to believe. At the same time, the report shows Deloitte to be a competent firm capable of analyzing economic data and prescribing strategies to cope with the economic challenges.

Now consider if we were to exclude the narrative from this report and presented the findings as a deck of PowerPoint slides with eight graphics and four tables. The PowerPoint slides would have failed to communicate the message that the authors carefully crafted in the report citing Piketty and Voltaire. I consider the Deloitte's report a good example of storytelling with data and encourage you to read the report to decide for yourself whether the deliverable would have been equally powerful without the narrative.

Now let us work backward from the Deloitte report. Before the authors started their analysis, they must have discussed the scope of the final deliverable. They would have deliberated the key message of the report and then looked for the data and analytics they needed to make their case. The initial planning and conceptualizing of the final deliverable is therefore extremely important for producing a compelling document. Embarking on analytics, without due consideration to the final deliverable, is likely to result in a poor-quality document where the analytics and narrative would struggle to blend.



Module 4 Prescribed Reading: Chapter 3 Pg. 60-62

The Report Structure

Before starting the analysis, think about the structure of the report. Will it be a brief report of five or fewer pages, or will it be a longer document running more than 100 pages in length? The structure of the report depends on the length of the document. A brief report is more to the point and presents a summary of key findings. A detailed report incrementally builds the argument and contains details about other relevant works, research methodology, data sources, and intermediate findings along with the main results.

I have reviewed reports by leading consultants including Deloitte and McKinsey. I found that the length of the reports varied depending largely on the purpose of the report. Brief reports were drafted as commentaries on current trends and developments that attracted public or media attention. Detailed and comprehensive reports offered a critical review of the subject matter with extensive data analysis and commentary. Often, detailed reports collected new data or interviewed industry experts to answer the research questions.

Even if you expect the report to be brief, sporting five or fewer pages, I recommend that the deliverable follow a prescribed format including the cover page, table of contents, executive summary, detailed contents, acknowledgements, references, and appendices (if needed).

I often find the cover page to be missing in documents. It is not the inexperience of undergraduate students that is reflected in submissions that usually miss the cover page. In fact, doctoral candidates also require an explicit reminder to include an informative cover page. I hasten to mention that the business world sleuths are hardly any better. Just search the Internet for reports and you will find plenty of reports from reputed firms that are missing the cover page.

At a minimum, the *cover page* should include the title of the report, names of authors, their affiliations, and contacts, name of the institutional publisher (if any), and the date of publication. I have seen numerous reports missing the date of publication, making it impossible to cite them without the year and month of publication. Also, from a business point of view, authors should make it easier for the reader to reach out to them. Having contact details at the front makes the task easier.

A *table of contents (ToC)* is like a map needed for a trip never taken before. You need to have a sense of the journey before embarking on it. A map provides a visual proxy for the actual travel with details about the landmarks that you will pass by in your trip. The ToC with main headings and lists of tables and figures offers a glimpse of what lies ahead in the document. Never shy away from including a ToC, especially if your document, excluding cover page, table of contents, and references, is five or more pages in length.

Even for a short document, I recommend an *abstract* or an *executive summary*. Nothing is more powerful than explaining the crux of your arguments in three paragraphs or less. Of course, for larger documents running a few hundred pages, the executive summary could be longer.

An *introductory* section is always helpful in setting up the problem for the reader who might be new to the topic and who might need to be gently introduced to the subject matter before being immersed in intricate details. A good follow-up to the introductory section is a review of available relevant research on the subject matter. The length of the *literature review* section depends upon how contested the subject matter is. In instances where the vast majority of researchers have concluded in one direction, the literature review could be brief with citations for only the most influential authors on the subject. On the other hand, if the arguments are more nuanced with caveats aplenty, then you must cite the relevant research to offer the adequate context before you embark on your analysis. You might use literature review to highlight gaps in the existing knowledge, which your analysis will try to fill. This is where you formally introduce your research questions and hypothesis.

In the *methodology* section, you introduce the research methods and data sources you used for the analysis. If you have collected new data, explain the data collection exercise in some detail. You will refer to the literature review to bolster your choice for variables, data, and methods and how they will help you answer your research questions.

The *results* section is where you present your empirical findings. Starting with descriptive statistics (see [Chapter 4](#), "[Serving Tables](#)") and illustrative graphics (see [Chapter 5](#), "[Graphic Details](#)" for plots and [Chapter 10](#), "[Spatial Data Analytics](#)" for maps), you will move toward formally testing your hypothesis (see [Chapter 6](#), "[Hypothetically Speaking](#)").

In case you need to run statistical models, you might turn to regression models (see [Chapter 7](#), “[Why Tall Parents Don’t Have Even Taller Children](#)”) or categorical analysis (see [Chapters 8](#), “[To Be or Not to Be](#)” and [9](#), “[Categorically Speaking About Categorical Data](#)”). If you are working with time series data, you can turn to [Chapter 11](#), “[Doing Serious Time with Time Series](#).” You can also report results from other empirical techniques that fall under the general rubric of data mining (see [Chapter 12](#), “[Data Mining for Gold](#)”). Note that many reports in the business sector present results in a more palatable fashion by holding back the statistical details and relying on illustrative graphics to summarize the results.

The *results* section is followed by the *discussion* section, where you craft your main arguments by building on the results you have presented earlier. The *discussion* section is where you rely on the power of narrative to enable numbers to communicate your thesis to your readers. You refer the reader to the research question and the knowledge gaps you identified earlier. You highlight how your findings provide the ultimate missing piece to the puzzle.

Of course, not all analytics return a smoking gun. At times, more frequently than I would like to acknowledge, the results provide only a partial answer to the question and that, too, with a long list of caveats.

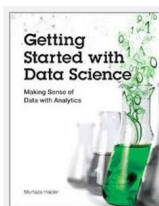
In the *conclusion* section, you generalize your specific findings and take on a rather marketing approach to promote your findings so that the reader does not remain stuck in the caveats that you have voluntarily outlined earlier. You might also identify future possible developments in research and applications that could result from your research.

What remains is housekeeping, including a list of *references*, the *acknowledgement* section (acknowledging the support of those who have enabled your work is always good), and *appendices*, if needed.

Have You Done Your Job as a Writer?

As a data scientist, you are expected to do a thorough analysis with the appropriate data, deploying the appropriate tools. As a writer, you are responsible for communicating your findings to the readers. *Transport Policy*, a leading research publication in transportation planning, offers a checklist for authors interested in publishing with the journal. The checklist is a series of questions authors are expected to consider before submitting their manuscript to the journal. I believe the checklist is useful for budding data scientists and, therefore, I have reproduced it verbatim for their benefit.

1. Have you told readers, at the outset, what they might gain by reading your paper?
2. Have you made the aim of your work clear?
3. Have you explained the significance of your contribution?
4. Have you set your work in the appropriate context by giving sufficient background (including a complete set of relevant references) to your work?
5. Have you addressed the question of practicality and usefulness?
6. Have you identified future developments that might result from your work?
7. Have you structured your paper in a clear and logical fashion?



Prescribed Reading: Chapter 1 Pg. 12-15

What Makes Someone a Data Scientist?

Now that you know what is in the book, it is time to put down some definitions. Despite their ubiquitous use, consensus evades the notions of big data and data science. The question, “who is a data scientist?” is very much alive and being contested by individuals, some of whom are merely interested in protecting their discipline or academic turfs. In this section, I attempt to address these controversies and explain why a narrowly construed definition of either big data or data science will result in excluding hundreds of thousands of individuals who have recently turned to the emerging field.

“Everybody loves a data scientist,” wrote Simon Rogers (2012) in the *Guardian*. Mr. Rogers also traced the newfound love for number crunching to a quote by Google’s Hal Varian, who declared that “the sexy job in the next ten years will be statisticians.”

Whereas Hal Varian named statisticians sexy, it is widely believed that what he really meant were data scientists. This raises several important questions:

- What is data science?
- How does it differ from statistics?
- What makes someone a data scientist?

In the times of big data, a question as simple as, “What is data science?” can result in many answers. In some cases, the diversity of opinion on these answers borders on hostility.

I define *data scientist* as someone who finds solutions to problems by analyzing big or small data using appropriate tools and then tells stories to communicate her findings to the relevant stakeholders. I do not use the data size as a restrictive clause. A data below a certain arbitrary threshold does not make one less of a data scientist. Nor is my definition of a data scientist restricted to particular analytic tools, such as machine learning. As long as one has a curious mind, fluency in analytics, and the ability to communicate the findings, I consider the person a data scientist.

I define *data science* as something that data scientists do. Years ago, as an engineering student at the University of Toronto I was stuck with the question: What is engineering? I wrote my master’s thesis on forecasting housing prices and my doctoral dissertation on forecasting homebuilders’ choices related to what they build, when they build, and where they build new housing. In the civil engineering department, others were working on designing buildings, bridges, tunnels, and worrying about the stability of slopes. My work, and that of my supervisor, was not your traditional garden-variety engineering. Obviously, I was repeatedly asked by others whether my research was indeed engineering.

When I shared these concerns with my doctoral supervisor, Professor Eric Miller, he had a laugh. Dr. Miller spent a lifetime researching urban land use and transportation, and had earlier earned a doctorate from MIT. “Engineering is what engineers do,” he responded. Over the next 17 years, I realized the wisdom in his statement. You first become an engineer by obtaining a degree and then registering with the local professional body that regulates the engineering profession. Now you are an engineer. You can dig tunnels; write software codes; design components of an iPhone or a supersonic jet. You are an engineer. And when you are leading the global response to financial crisis in your role as the chief economist of the International Monetary Fund (IMF), as Dr. Raghuram Rajan did, you are an engineer.

Professor Raghuram Rajan did his first degree in electrical engineering from the Indian Institute of Technology. He pursued economics in graduate studies, later became a professor at a prestigious university, and eventually landed at the IMF. He is currently serving as the 23rd Governor of the Reserve Bank of India. Could someone argue that his intellectual prowess is rooted only in his training as an economist and that the fundamentals he learned as an engineering student played no role in developing his problem-solving abilities?

Professor Rajan is an engineer. So are Xi Jinping, the President of the People’s Republic of China, and

Alexis Tsipras, the Greek Prime Minister who is forcing the world to rethink the fundamentals of global economics. They might not be designing new circuitry, distillation equipment, or bridges, but they are helping build better societies and economies and there can be no better definition of engineering and engineers—that is, individuals dedicated to building better economies and societies.

So briefly, I would argue that data science is what data scientists do.

Others have much different definitions. In September 2015, a co-panelist at a meetup organized by BigDataUniversity.com in Toronto confined data science to machine learning. There you have it. If you are not using the black boxes that make up machine learning, as per some experts in the field, you are not a data scientist. Even if you were to discover the cure to a disease threatening the lives of millions, turf-protecting colleagues will exclude you from the data science club.

Dr. Vincent Granville (2014), an author on data science, offers certain thresholds to meet to be a data scientist.²⁹ On pages 8 and 9 in *Developing Analytic Talent* Dr. Granville describes the new data science professor as a non-tenured instructor at a non-traditional university, who publishes research results in online blogs, does not waste time writing grants, works from home, and earns more money than the traditional tenured professors. Suffice it to say that the thriving academic community of data scientists might disagree with Dr. Granville.

Dr. Granville uses restrictions on data size and methods to define what data science is. He defines a data scientist as one who can “easily process a 50-million-row data set in a couple of hours,” and who distrusts (statistical) models. He distinguishes data science from statistics. Yet he lists algebra, calculus, and training in probability and statistics as necessary background “to understand data science” (page 4).

Some believe that big data is merely about crossing a certain threshold on data size or the number of observations, or is about the use of a particular tool, such as Hadoop. Such arbitrary thresholds on data size are problematic because with innovation, even regular computers and off-the-shelf software have begun to manipulate very large data sets. Stata, a commonly used software by data scientists and statisticians, announced that one could now process between 2 billion to 24.4 billion rows using its desktop solutions. If Hadoop is the password to the big data club, Stata’s ability to process 24.4 billion rows, under certain limitations, has just gatecrashed that big data party.³⁰

It is important to realize that one who tries to set arbitrary thresholds to exclude others is likely to run into inconsistencies. The goal should be to define data science in a more exclusive, discipline- and platform-independent, size-free context where data-centric problem solving and the ability to weave strong narratives take center stage.

Given the controversy, I would rather consult others to see how they describe a data scientist. Why don’t we again consult the Chief Data Scientist of the United States? Recall Dr. Patil told the *Guardian* newspaper in 2012 that a “data scientist is that unique blend of skills that can both unlock the insights of data and tell a fantastic story via the data.” What is admirable about Dr. Patil’s definition is that it is inclusive of individuals of various academic backgrounds and training, and does not restrict the definition of a data scientist to a particular tool or subject it to a certain arbitrary minimum threshold of data size.

The other key ingredient for a successful data scientist is a behavioral trait: curiosity. A data scientist has to be one with a very curious mind, willing to spend significant time and effort to explore her hunches. In journalism, the editors call it having the nose for news. Not all reporters know where the news lies. Only those who have the nose for news get the story. Curiosity is equally important for data scientists as it is for journalists.

Rachel Schutt is the Chief Data Scientist at News Corp. She teaches a data science course at Columbia University. She is also the author of an excellent book, *Doing Data Science*. In an interview with the *New York Times*, Dr. Schutt defined a data scientist as someone who is part computer scientist, part software engineer, and part statistician (Miller, 2013). But that’s the definition of an average data scientist. “The best,” she contended, “tend to be really curious people, thinkers who ask good questions and are O.K. dealing with unstructured situations and trying to find structure in them.”